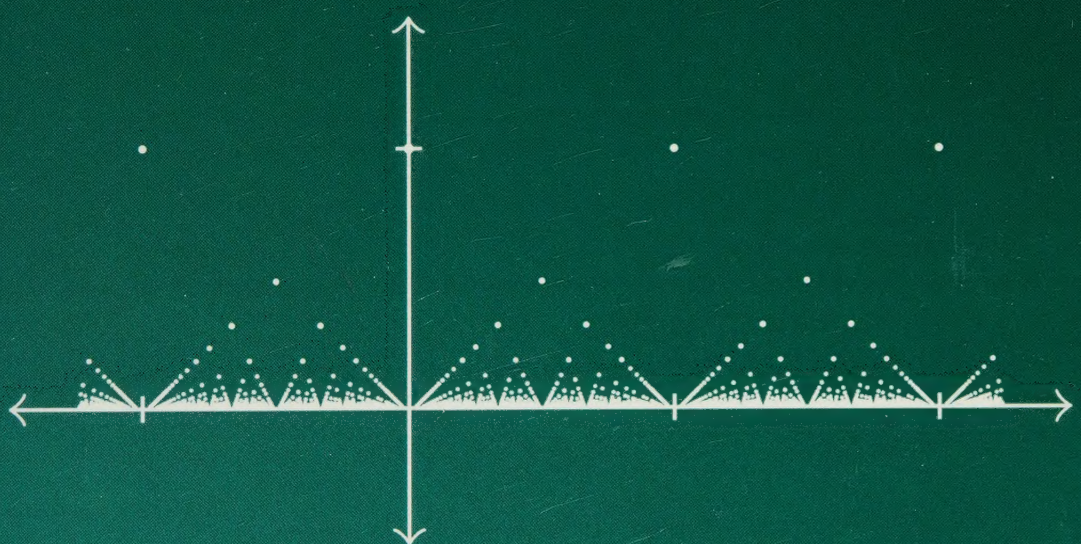
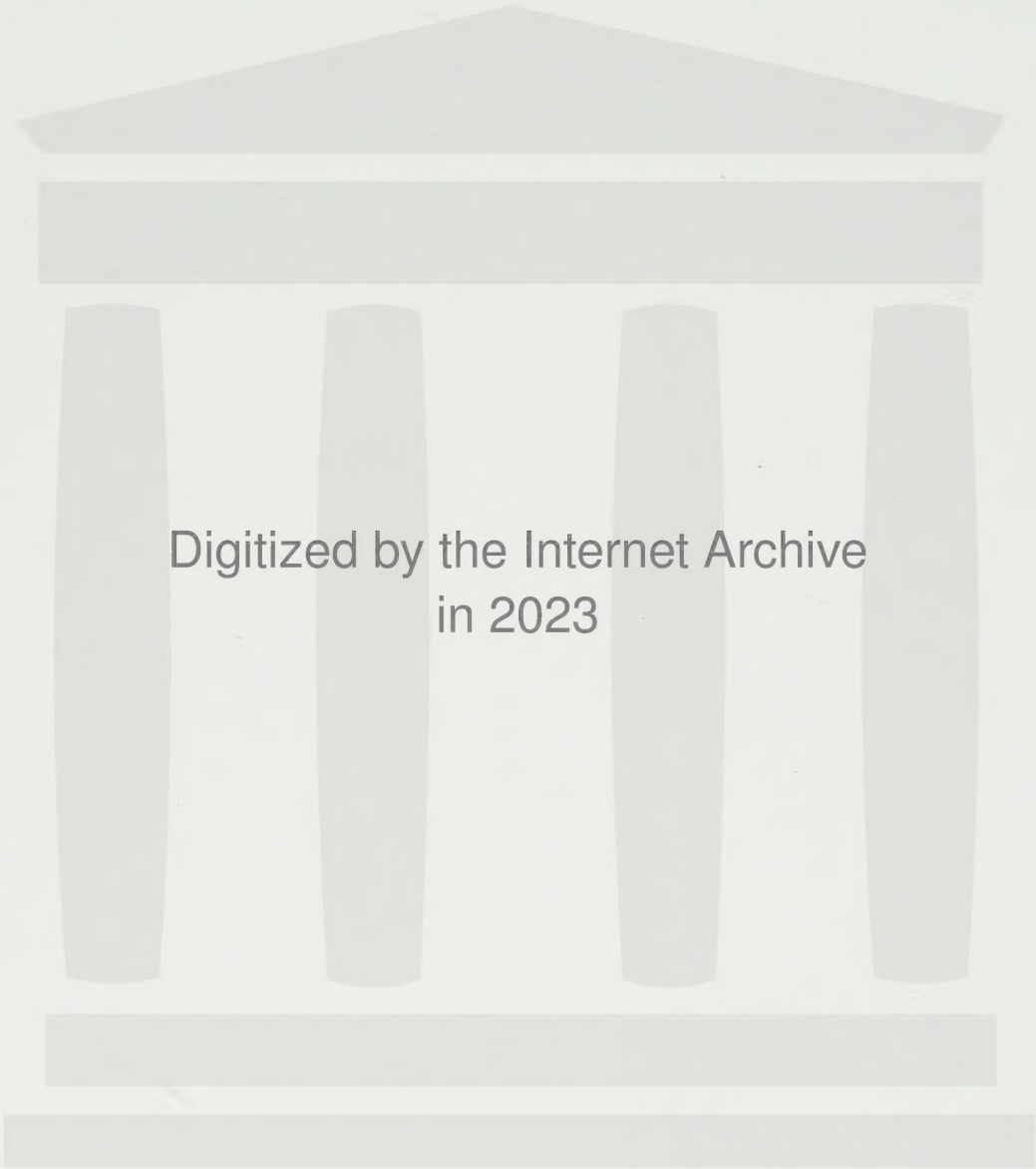
Real Analysis

A Long-Form Mathematics Textbook



Jay Cummings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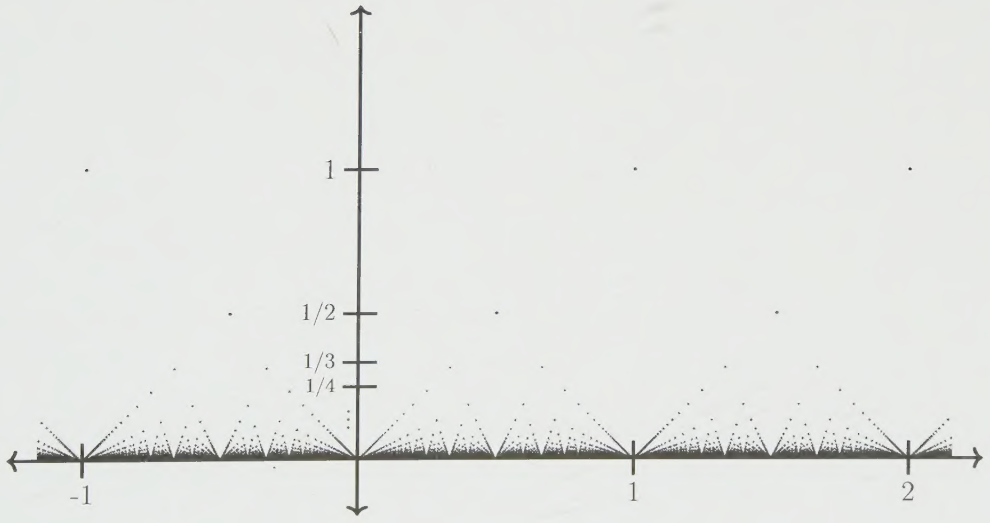
Jay Cummings



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To my students, without whom I would have had to edit this whole damn book myself.

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Chapter 1: The Reals

1.1 Historical Questions

Analysis was developed in the 17^{th} century as a tool to answer some truly fundamental questions from physics, and is now applied throughout the sciences. We begin, however, long before this point, in Greece.

1.2 Zeno's Paradoxes

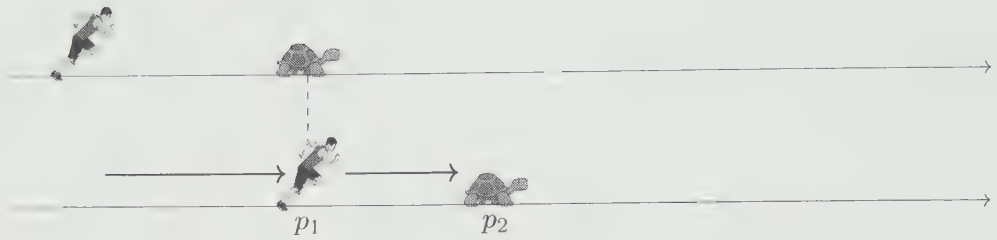
Nearly 2,500 years ago, Zeno of Elea wrote down a set of paradoxes to support the philosophy that *change* is an illusion. Specifically, one type of change is *motion*—the change of an object's position. Zeno set out to show that objects do not actually move, and our senses are lying to us when we perceive motion.[1]

Zeno's First Paradox

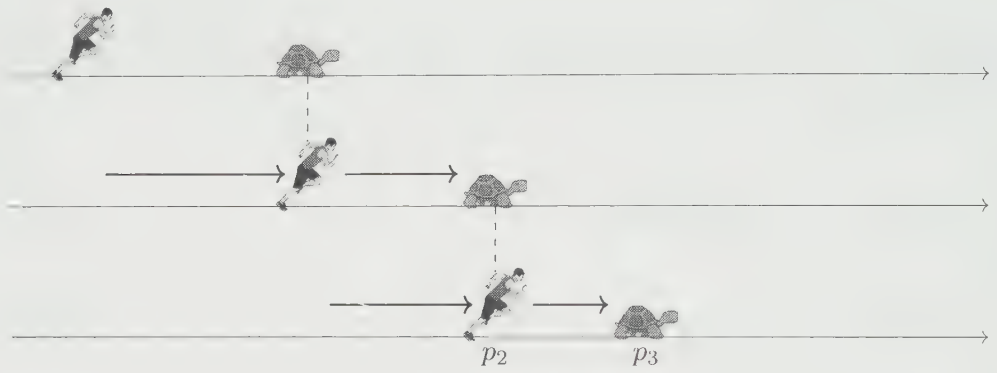
How did he try to prove that motion is an illusion? He created a thought experiment. He said to imagine a race between Achilles (the Greek warrior of legend) and a tortoise (just a regular tortoise). Suppose the race is 1,000 meters long and the tortoise is given a 100 meter head start.

The gun is fired and both begin. Achilles is much faster than the tortoise, so of course you would expect him to overtake the tortoise and win the race, but Zeno said that he can prove that this is impossible. This was his reasoning: For Achilles to catch up to the tortoise, he would first have to reach the 100 meter mark, where the tortoise began the race.

[1] ... And to answer your first question, yes, Greek philosophers were believed to have experimented with psychedelic drugs.



At this point we know that Achilles hasn't yet caught up to the tortoise because in the time it took Achilles to reach the 100 meter mark (call this point p_1), the tortoise has progressed forward some. Let's say the tortoise has made it to the point p_2. And now we repeat our above reasoning: Achilles is still trying to catch the tortoise, but before he can do so he will first have to reach the point p_2.



And you see, once again, while Achilles was reaching p_2 the tortoise progressed to a further point, p_3. So, Zeno says, this will go on forever. Achilles will always be behind the tortoise because he is always at a point that the tortoise has already been to, and the next point he must reach will always be a place that the tortoise is currently at. But by the time he gets to that point, the tortoise will have moved on at least a little bit farther, maintaining his lead. And repeat. By induction, the most recent stage of the picture always looks essentially like this.

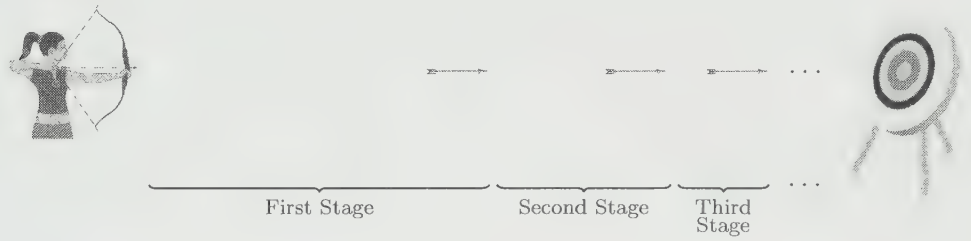
If he can never catch up, then he will always be behind, and therefore is guaranteed to lose the race.

So, what do you think? Does slow and steady win the race? It does seem paradoxical; our senses and life experiences tell us that this is obviously gibberish, but then again Zeno was a pretty smart guy, and his arguments do seem fairly logical and sound... So we have a challenge in front of us. When seemingly-good *reasoning* leads to a false *conclusion*, some part of our thinking must be wrong. In cases like this, it takes very careful thought to diagnose the issue.

Zeno's paradoxes were one of the initial motivations to study limits and develop the theory of real analysis. It will take about a hundred pages of this text before we fully understand why Zeno was mistaken, but we'll get there.

Zeno's Second Paradox

For the interested, Zeno had another similar paradox. Briefly, imagine an archer fires an arrow at a target. Zeno argued that the arrow will never reach the target. He said that for the arrow to reach it, it would first have to reach the halfway point, which takes some amount of time. Then, once there, there is still some distance to go and to get there it would first have to reach the point halfway between its current position and its goal, which again takes some amount of time. Then from its new position it is still not there and getting there again requires that it first gets halfway there, requiring some amount of time.



In general, the arrow will always be short of its goal, and to get to the target it will first have to get halfway there, which always requires some additional amount of time. If there is always some additional amount of time remaining, it will never actually make it... right?

Now, if you remember your calculus you might be able to figure out Zeno's mistake. He was correct that the arrow must first complete the first stage, and then must complete the second stage, and then must complete the third stage, and so on. And he was correct that each stage takes some finite amount of time. And it seems to make sense that if you have to do one thing at a time, and you have infinitely many things to do, that you will never complete all of those things. This is false, though. One *can* do infinitely many things, one after the other, in just a finite amount of time. The reason? Because a series can converge. And if each summand represents an amount of time, then the collection of those infinitely many things can indeed converge to a finite time length.

We will come back to this problem in Chapter 4, but the moral for now is that

when things get smaller and smaller (like the time it takes Achilles to run from p_i to p_{i+1}, for increasingly large i), weird stuff can happen. And it takes careful, detailed thought to avoid getting stuck in paradoxical situations. Indeed, this is essentially our goal for the book/course.

Textbook Goal. By studying the infinite, develop a ground-up understanding of the real numbers and functions on the reals. Also, improve one's mathematical maturity; that is, understand mathematical statements and arguments, construct proofs and find counterexamples.

So, that's what we are going to do. We are going to think carefully about the infinite, and as we progress through this book we will see how that careful reasoning can illustrate some pretty amazing properties of real numbers and functions. And unlike calculus where you spent most of your time studying super nice functions and intuitive situations, and very little time on weird paradoxes like the above, in analysis we will instead spend a significant amount of time on the weird situations, where our intuition may deceive us. We will build analysis from the ground-up, retraining our intuition on solid—and increasingly higher—ground.

So get ready, because I think you will enjoy it.

1.3 Basic Set Theory Definitions

We begin about as basic as possible—with sets. Here is a quick review of basic set theory definitions that you learned in your intro-to-proofs class.

Definition.

Definition 1.1.

- A *set* is an unordered collection of distinct objects, which are called *elements*.[2]
- If x is an element of a set S, we write $x \in S$. This is read as "x in S."
- *Set-builder notation* looks like this:

  $$S = \{\text{elements} \ : \ \text{conditions used to generate the elements}\}.$$

  For example,

  $$\{1, 4, 9, 16, 25, \dots\} = \{n^2 : n \in \mathbb{N}\}.$$

[2] Alternative definition: Everything. Everything is a set. Almost no definition in the world is as general as that of a set.

As a second example, the rationals can be built like so:

$$\mathbb{Q} = \left\{ \frac{p}{q} : p, q \in \mathbb{Z}, q \neq 0 \right\}.$$

- The set containing no elements is called the *empty set* and is denoted $\emptyset$.
- If $x \in B$ for every $x \in A$, then A is a *subset* of B, which is denoted $A \subseteq B$.
- $A \cap B = \{x : x \in A \text{ and}^3 \ x \in B\}$.
- $A \cup B = \{x : x \in A \text{ or } x \in B\}$.
- $A \setminus B = \{x : x \in A \text{ and } x \notin B\}$.
- A and B are *disjoint* if $A \cap B = \emptyset$.
- If $A \subseteq U$ for a *universal set* U (typically $U = \mathbb{R}$), then the *complement* of A in U is $A^c = U \setminus A$.
- The *Cartesian product* $A \times B = \{(a, b) : a \in A \text{ and } b \in B\}$.
- The *power set* of a set A is $\mathcal{P}(A) = \{X : X \subseteq A\}$.
- If $A_1, A_2, A_3, \ldots, A_n$ are all sets, then the union of all of them is

$$\bigcup_{i=1}^{n} A_i = A_1 \cup A_2 \cup \cdots \cup A_n.$$

- If $A_1, A_2, A_3, \ldots, A_n$ are all sets, then the intersection of all of them is

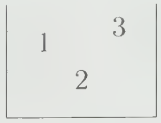
$$\bigcap_{i=1}^{n} A_i = A_1 \cap A_2 \cap \cdots \cap A_n.$$

- The *cardinality* of a set A is the "number" of elements in the set, and is denoted $|A|$. (In Chapter 2 we will investigate what "number" means when A is infinite.)

[3]Example #628 of why English is a confusing language: The word 'and' can mean either a union or[4] an intersection. "Students and faculty are welcome to attend" means the *union* of the students and the faculty are invited. But, "If you are taking algebra and analysis this year, you'll enjoy this lecture" refers the *intersection* of those taking algebra and those taking analysis. Worse still, 'and' can also refer to quantities as well, not just to collections! "Three and five make eight." Mathematicians use symbols to avoid these ambiguities.

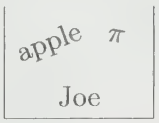
[4]The word 'or' is better.

Another way to think about a set is as a box, possibly with some things inside. When you look into a box, the things inside do not have any particular order; same thing with the elements of a set.



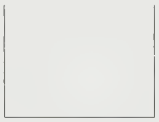
$= \{2,3,1\}$

Of course, the above also corresponds to $\{1, 2, 3\}$, $\{3, 2, 1\}$, and $\{3, 1, 2\}$; with sets, order does not matter. Also, remember that the elements do not have to be numbers. Elements of a set can be *anything*.



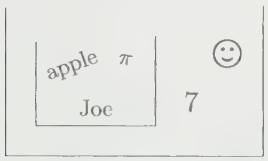
$= \{\text{apple, Joe, } \pi\}$

Also, just as boxes can be empty, so can sets!



$= \emptyset$

Furthermore, it's certainly possible for one box to be inside another box. Likewise, it's certainly possible for one set to be a single element inside another set.



$= \{\{\text{apple, Joe, } \pi\}, 7, \text{☺}\}$

Notice that the above set has three elements in it: (1) a set (containing three specific elements), (2) the number 7, and (3) a smiley face. Your box could also have just one thing in it: a smaller box with nothing inside it. This looks like this:

$= \{\emptyset\}$

Note the difference between the above example and three pictures back: Earlier we had the empty set; now we have a set whose one element is the empty set.

To test your understanding, think about what the union and intersection of two sets would look like, from this box interpretation.[5] Pushing this further, if A and B are two boxes (possibly with things inside), think if you can now describe the following in terms of boxes (Exercise 1.2):

- $A \setminus B$
- $\mathcal{P}(A)$
- $|A|$

Lastly, if $A_1, A_2, A_3, \ldots, A_n$ are all boxes, think if you can now describe the following in terms of boxes (Exercise 1.3):

- $\bigcup_{i=1}^{n} A_i = A_1 \cup A_2 \cup \cdots \cup A_n$.
- $\bigcap_{i=1}^{n} A_i = A_1 \cap A_2 \cap \cdots \cap A_n$.

Functions on sets

Now recall the following, which you also learned in your intro-to-proofs class.

Definition.

Definition 1.2.

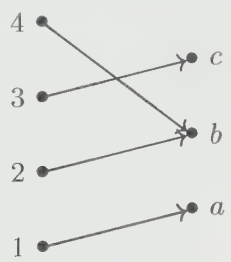
- Given a pair of sets A and B, suppose that each element $x \in A$ is associated, in some way, to one element of B, which we denote $f(x)$. Then f is said to be a *function* from A to B. This is sometimes denoted "$f : A \to B$".[6]
  - Furthermore, A is called the *domain* of f, and B is called the *codomain* of f. The set $\{f(x) : x \in A\}$ is called the *range* of f.
- A function $f : A \to B$ is *injective* (or *one-to-one*) if $f(a) = f(b)$ implies that $a = b$.
- A function $f : A \to B$ is *surjective* (or *onto*) if, for every $b \in B$, there exists some $a \in A$ such that $f(a) = b$.[7]
- A function $f : A \to B$ is *bijective* if it is both injective and surjective.

[5] One answer: The union of two boxes A and B can be obtained by dumping everything in A and everything in B into a new box, and then removing any duplicate items. The intersection can be obtained by identifying everything in A that is also in B, and putting those items into a new box.

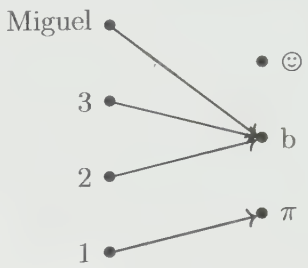
[6] To be completely precise, one can use relations to define a function $f : X \to Y$ as a subset of the cross product $X \times Y$. But that's enough of a pain that even in Baby Rudin that's avoided.

[7] Note that using the contrapositive you can come up with equivalent definitions for *injective* and *surjective*. For the former, the contrapositive gives the (equivalent) definition that f is injective if $a \neq b$ implies $f(a) \neq f(b)$. For the latter, an equivalent definition is that f is surjective if there does not exist a $b \in B$ for which $f(a) \neq b$ for all $a \in A$.

Example 1.3. Here are some examples of functions on finite sets—one which is surjective, and one which is not.



A surjection from
$\{1, 2, 3, 4\}$ to $\{a, b, c\}$



Not a surjection from
$\{1, 2, 3, \text{Miguel}\}$ to $\{\pi, \text{b}, ☺\}$

1.4 What is a Number?

How would you define a number? Some would say this is a philosophical question, while Appendix A demonstrates just how deep of a mathematical question this can be. The German philosopher, logician, and mathematician Gottlob Frege, on the other hand, believed these are one in the same.

> Every good mathematician is at least half a philosopher, and every good philosopher is at least half a mathematician.
>
> – Gottlob Frege

For now we won't get too philosophical, but we do want to discuss how our number system is built up, and what properties are essential at each stage.

Naturals

The natural numbers $\mathbb{N}$ is the set $\{1, 2, 3, \dots\}$. These are the counting numbers, and if anything is a number, they certainly are.[8] A sketch of how to formally, but painfully, construct all real numbers is included in Appendix A.

Integers

The integers $\mathbb{Z}$ is the set $\{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$ where each number is 1 away from the next. The 19^{th} century mathematician Leopold Kronecker said "God made the integers, all else is the work of man." What do you think he meant by this? And do you agree with him? I've found that a class of students will have many different opinions on this question, which I think is great.

[8] Note that 0 is *not* considered a natural number.[9] I will defend this to my grave.

[9] The set $\{0, 1, 2, 3, \dots\}$ is denoted $\mathbb{N}_0$. (Fun fact: '0' was first discovered by an ancient Babylonian who asked how many of his friends wanted to talk about numbers with him.)

Rationals

The next natural set of numbers to consider is the set of *rational numbers*:

$$\mathbb{Q}: \quad \left\{ \frac{p}{q} : p, q \in \mathbb{Z}, q \neq 0 \right\}.$$

They have some nice properties:

- They are *nested*: Between any two rational numbers is another one.[10]
- They are closed under addition, subtraction, multiplication and division:
  - If $p, q \in \mathbb{Q}$ then $p + q \in \mathbb{Q}$.
  - If $p, q \in \mathbb{Q}$ then $p - q \in \mathbb{Q}$.
  - If $p, q \in \mathbb{Q}$ then $p \cdot q \in \mathbb{Q}$.
  - If $p, q \in \mathbb{Q}$ then $p \div q \in \mathbb{Q}$, provided $q \neq 0$.

But they also have some problems:

- It's possible that $p \in \mathbb{Q}$ but yet $\sqrt{p} \notin \mathbb{Q}$.
- It has lots of "holes": It's possible that $p_i \in \mathbb{Q}$ for each $i \in \mathbb{N}$, but yet the sequence $p_1, p_2, p_3, \ldots$ "converges" to some $p \notin \mathbb{Q}$.
- It's not *algebraically closed*: There exist polynomials with rational coefficients whose roots are not rational.[11]
  - e.g. $f(x) = x^2 - 5$
- If we "fill in the holes," we in fact will see that "most" numbers are not rational!

Rational numbers were studied all around the world, but the Greeks' success is typically noted. One of the most influential of these mathematicians was Pythagoras,[12] who is often credited with being the first to prove the Pythagorean theorem, which, as you know, says that if a and b are the lengths of the legs of a right triangle, and c the length of the hypotenuse, then $a^2 + b^2 = c^2$. (Although, it might have been proven at the time by one of his students, and also perhaps earlier by the Babylonians.)

[10] And hence, between any two there are infinitely many others.

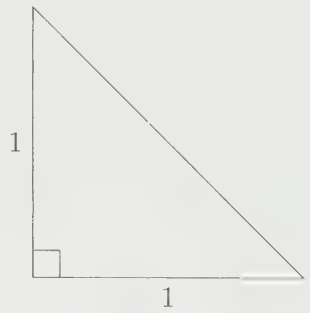
[11] This is the one problem that the reals will only partially solve. One needs the complex numbers to have algebraic closure.

[12] Pythagoras' story is cloaked in legend—but fortunately the legends are all highly amusing. Aristotle wrote that Pythagoras had a golden thigh, was born with a golden wreath upon his head, and that after a deadly snake bit him, he bit the snake back, which killed it; he was supposedly the son of Apollo, and it was said that a priest of Apollo gave him a magic arrow that allowed him to fly; the philosophers Porphyry and Iamblichus both reported that Pythagoras once persuaded a bull not to eat beans, and convinced a notoriously violent bear to swear that it would never harm a living thing again—and the bear was true to his word. What is odd is that none of his own writings have survived, and most of the credible writings about him were done long after his death. Some have even suggested that he was not a real person—but this is certainly a minority opinion among historians.

This theorem is significant not only for its own merits, but also because it is the key to proving that irrational numbers exist. Sadly, despite having the key, Pythagoras lived and died believing that all numbers were rational. After his death, his school of thought, called Pythagoreanism, lived on. About a century after his death, a Pythagorean named Hippasus proved what is now *the* classic proof of one of *the* classic theorems—that $\sqrt{2}$ is irrational. As the legend goes, the other Pythagoreans were so horrified by this theorem that they took Hippasus out to sea and threw him overboard, killing him. They then made a pact to never tell the world of his discovery. This has got to be one of the worst cover-ups in history, as today his proof is probably the second most known proof in the world, only behind Euclid's proof of the infinitude of primes.

In fact, even if only to stick it to the murderous, anti-intellectual Pythagoreans one last time, let's start with Hippasus' proof that $\sqrt{2}$ is irrational.[13]

Example 1.4. The number $\sqrt{2}$ is irrational. That is, the length of the hypotenuse of the following triangle is irrational.



Proof. Assume for a contradiction that $\sqrt{2}$ is rational. Then there must be some non-zero integers p and q where

$$\sqrt{2} = \frac{p}{q}.$$

Moreover, we may assume that this fraction is written in lowest terms, meaning that p and q have no common divisors. That is,

$$\begin{aligned} \sqrt{2}q &= p \\ 2q^2 &= p^2. \end{aligned}$$

This implies[14] that $2 \mid p^2$, and hence $2 \mid p$. So we may write $p = 2k$ for some non-zero

[13] This is typically viewed as a classic *proof by contradiction*. But is it? What it means to be "irrational" is that you are "not rational." So is this in fact a *direct proof* that $\sqrt{2}$ is not rational (hence irrational)? Or is it a proof by contradiction? I'll let you mull that one over.

[14] By the fundamental theorem of arithmetic.

integer k. Plugging this in,

$$\begin{aligned} 2q^2 &= p^2 \\ 2q^2 &= (2k)^2 \\ 2q^2 &= 4k^2 \\ q^2 &= 2k^2. \end{aligned}$$

Therefore $2 \mid q^2$, and hence $2 \mid q$. But this is a contradiction: We had assumed that p and q had no common factors, and yet we proved that 2 divides each. Therefore $\sqrt{2}$ can not be rational, meaning it is irrational.[15] □

So the rationals are not quite enough. That said, they do have *almost* every other fundamental property we would want. To the point: They are what we call an *ordered field*. But first, what's a field? It's a set that satisfies the classic multiplicative and additive properties we know and love.

Definition.

Definition 1.5. A *field* is a nonempty set $\mathbb{F}$, along with two binary operations[16], addition $(+)$ and multiplication $(\cdot)$, satisfying the following axioms.

Axiom 1 (Commutative Law). If $a, b \in \mathbb{F}$, then $a + b = b + a$ and $a \cdot b = b \cdot a$.

Axiom 2 (Distributive Law). If $a, b, c \in \mathbb{F}$, then $a \cdot (b + c) = a \cdot b + a \cdot c$.

Axiom 3 (Associative Law). If $a, b, c \in \mathbb{F}$, then $(a + b) + c = a + (b + c)$ and $(a \cdot b) \cdot c = a \cdot (b \cdot c)$.

Axiom 4 (Identity Law). There are special elements $0, 1 \in \mathbb{F}$, where $a + 0 = a$ and $a \cdot 1 = a$ for all $a \in \mathbb{F}$.

Axiom 5 (Inverse Law). For each $a \in \mathbb{F}$, there is an element $-a \in \mathbb{F}$ such that $a + (-a) = 0$. If $a \neq 0$, then there is also an element $a^{-1} \in \mathbb{F}$ such that $a \cdot a^{-1} = 1$.

Example 1.6. Below are some examples and some non-examples of fields.

- The natural numbers $\mathbb{N}$ do not form a field; they fail the first half of Axiom 4 and both halves of Axiom 5.
- The integers $\mathbb{Z}$ *almost* form a field; they only fail the second half of Axiom 5: For example, given $a = 2$, there is no *integer* a^{-1} such that $2 \cdot a^{-1} = 1$. (The number $\frac{1}{2}$ is the multiplicative inverse of 2, but $\frac{1}{2}$ is not an integer.)
- One can check that the rationals $\mathbb{Q}$ form a field. (Run through the axioms on your own to verify this.)

[15] How does that taste, Pythagoreans? Bitter? Mmmhmm.

[16] Note: These operations are functions from $\mathbb{F} \times \mathbb{F} \to \mathbb{F}$, so implicit in this is that $\mathbb{F}$ is closed under these operations.

1.5 Ordered Fields

While I was a grad student, I once got an email from a student asking me about a homework question involving x, which was given to be a real number. Instead, he wrote: "Assume that x is really a number."

Now, without meaning to, his email did suggest a good question: what really is a number? As we demonstrated in Example 1.4, if we want to include in our number system all possible lengths of triangles, or we want our system to be closed under taking square roots, or we want our system to be closed under taking limits, then the rational numbers alone do not cut it. We need more...

Plug rationals into limits and roots
And you won't like what it reveals...
To apply such functions, and reap their fruits,
We need more... we need... The Reals!

But what essential properties are we missing? What does $\mathbb{R}$ have that a field does not? To get warmed up, think about $\mathbb{Q}$. With the definition of a field, we have *almost* reached the point where we have captured the essence of the rationals. Can you think of what else we need? There are three main properties that we are missing....

First, there are infinitely many rationals[17] (and in some way they are "symmetric" about the 0 element). Second, the rationals have an ordering to them—not only are the elements different from each other, but given any two different ones, one is bigger than the other. So we need to add a "$<$" to the mix. Lastly, we would like *metric properties*; that is, we would like to talk about how big a number is (how far from zero), and the distance between two numbers.

Beautifully, there is a single elegant axiom that we can include to capture *all* of these properties, and we do it without defining any more functions or operations. And this is really good: the simpler and more elegant your axiomatic system, the more general and appealing your theory is.

How can we do this? The beautiful idea is to define a subset of the field which we call the *positive* set P, and which acts just as the positive real numbers do.

Definition

Definition 1.7. An *ordered field* is a field $\mathbb{F}$, along with the following additional axiom.

Axiom 6 (Order Axiom). There is a nonempty subset $P \subseteq \mathbb{F}$, called the *positive elements*, such that

(a) If $a, b \in P$, then $a + b \in P$ and $a \cdot b \in P$;

(b) If $a \in \mathbb{F}$ and $a \neq 0$, then either $a \in P$ or $-a \in P$, but not both.

[17] And for those who have studied abstract algebra, you know that the field axioms do not guarantee an infinite set. Indeed, there are many finite fields, e.g., $\mathbb{Z}/p\mathbb{Z}$ where p is a prime.

Why does this do the job? First, since 1 is positive[18], so are $1+1$, and $1+1+1$, and $1+1+1+1$, and so on, generating infinitely many different elements; likewise for $(-1)+(-1)+\cdots+(-1)$. Second, if you want to say that the number b is larger than the number a, it suffices to say that $b-a$ is positive. Lastly, positive elements are all you need to define the absolute value, and hence enough to give metric properties. We begin the formal treatment of this by defining the inequality.

Definition

Definition 1.8. If $\mathbb{F}$ is an ordered field and $a, b \in \mathbb{F}$, then we say that "$a < b$" if $b - a \in P$. Likewise, $a \leq b$ means that either $a = b$ or $a < b$.

Define "$>$" similarly. Also, note that p is positive if $0 < p$. In many intro-to-proofs classes students work out boring proofs of the basic properties of inequalities. Since students should work through those at most one time, we won't do them here.

As you are probably anticipating, $\mathbb{R}$ is going to be an ordered field. To make this clear when you look back at these pages, I will begin to indicate this fact parenthetically, like in the following note.

Note

Note 1.9. You may use any of the standard properties of inequalities. For a, b, c in an ordered field $\mathbb{F}$ (like $\mathbb{R}$), these include:

(a) If $a < b$, then $a + c < b + c$.

(b) Transitivity: If $a < b$ and $b < c$, then $a < c$.

(c) If $a < b$, then $ac < bc$ if $c > 0$, and $ac > bc$ if $c < 0$.

(d) If $a \neq 0$, then $a^2 > 0$.

Finally, how does defining an ordered field (with positive subset P) give us a distance function? It all comes down to the absolute value function.

[18] *Proof.* If it weren't, then by part (b) of Axiom 6, -1 would be positive. But then, for any positive a, by part (a) we deduce that $a \cdot -1 = -a$ is also positive, contradicting part (b). □

Definition.

Definition 1.10. If $\mathbb{F}$ is an ordered field (like $\mathbb{R}$), define the *absolute value* function $|\cdot| : \mathbb{F} \to \mathbb{F}$ to be

$$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}.$$

It might be a little odd to see "$-x$" when the absolute value function is supposed to make everything positive. But the point is that for those x values, $x < 0$. So x already has a negative sign! And one way to get rid of a negative sign is to add *another* negative sign, canceling out the one that x already has.[19] For example,

$|x| = x$ when $x \geq 0$

- $|7| = 7$
- $|2| = 2$

$|x| = -x$ when $x < 0$

- $|-7| = -(-7) = 7$
- $|-2| = -(-2) = 2$

Note.

Note 1.11. In this course you may use any of the standard properties of absolute values without proof, many of which were proven in your intro-to-proofs class. These include the following, where a and b are elements in an ordered field (like $\mathbb{R}$):

(a) $|a| \geq 0$, with equality if and only if $a = 0$.

(b) $|a| = |-a|$.

(c) $-|a| \leq a \leq |a|$.

(d) $|a \cdot b| = |a| \cdot |b|$.

(e) $1/|a| = |1/a|$, if $a \neq 0$.

(f) $|a/b| = |a|/|b|$, if $b \neq 0$.

(g) $|a| \leq b$ if and only if $-b \leq a \leq b$.

(h) The triangle inequality: $|a + b| \leq |a| + |b|$.

(i) The reverse triangle inequality: $||a| - |b|| \leq |a - b|$.

We will prove the last three now, but as an exercise you may prove the rest.

[19] Sometimes two wrongs *do* make a right! Take that, mom!

Proposition.

Proposition 1.12 (*$|a| \leq b$ iff $-b \leq a \leq b$*). If $\mathbb{F}$ is an ordered field (like $\mathbb{R}$) and $a, b \in \mathbb{F}$, then

$$|a| \leq b \qquad \text{if and only if} \qquad -b \leq a \leq b.$$

Proof. First assume that $|a| \leq b$. Multiplying both sides by -1, we also have that $-|a| \geq -b$, by Note 1.9 part (c). Now, Note 1.11 part (c) says that $-|a| \leq a \leq |a|$. Piecing these three facts together,

$$-b \leq -|a| \leq a \leq |a| \leq b,$$

which implies that $-b \leq a \leq b$, as desired.

To prove the backwards direction, assume that $-b \leq a \leq b$. In Case 1 below we will use the "$a \leq b$" part of this assumption, and in Case 2 we will use the "$-b \leq a$" part of this assumption.

Case 1: $a \geq 0$. If $a \geq 0$, then by the definition of the absolute value $a = |a|$, and so the assumption that $a \leq b$ implies that $|a| \leq b$, as desired.

Case 2: $a < 0$. If $a < 0$, then $-a = |a|$ and hence $a = -|a|$. So the assumption that $-b \leq a$ implies that $-b \leq -|a|$. Multiplying by -1 gives $|a| \leq b$, as desired. □

Theorem.

Theorem 1.13 (*The triangle inequality*). If $\mathbb{F}$ is an ordered field (like $\mathbb{R}$) and if $x, y \in \mathbb{F}$, then

$$|x + y| \leq |x| + |y|.$$

Scratch Work.[20] Want to show:

$$|x + y| \leq |x| + |y|.$$

Which by the last proposition is true if and only if

$$-(|x| + |y|) \leq x + y \leq |x| + |y|.$$

[20] More so than in other math courses, in real analysis it is helpful to do some scratch work before your proof. This is the case for a few reasons, one of which is that in many of our proofs and in many of your homework problems, one discovers the proof by starting with what we are trying to prove, and then working backwards to reach things that we already know (from other theorems, known properties, or the problem's assumptions).

I.e.

$$-|x| - |y| \leq x + y \leq |x| + |y|.$$

And these sums look like they can be broken apart into

$$-|x| \leq x \leq |x|$$

and

$$-|y| \leq y \leq |y|,$$

both of which we know to be true by Note 1.11 part (c). So if we start at the bottom of our scratch work and move upwards, we should get our conclusion... Now here's the actual proof.

Proof. For $x, y \in \mathbb{F}$, by Note 1.11 part (c) we have

$$-|x| \leq x \leq |x| \qquad \text{and} \qquad -|y| \leq y \leq |y|.$$

Adding these two together (technically by applying Exercise 1.8 part (a)) gives

$$-|x| - |y| \leq x + y \leq |x| + |y|.$$

I.e.,

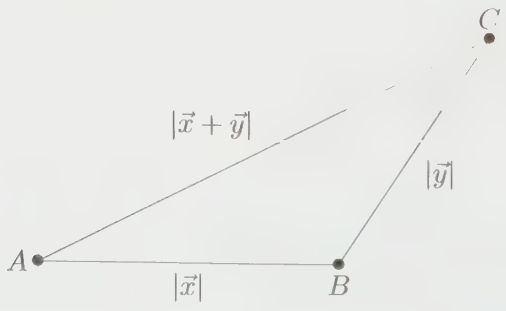
$$-(|x| + |y|) \leq x + y \leq |x| + |y|.$$

And so, by Proposition 1.12,

$$|x + y| \leq |x| + |y|.$$

□

Perhaps surprisingly, the triangle inequality is a very important tool in analysis, and will be used over and over again. Also, it's called the "triangle inequality" because of vectors forming a triangle. A vector $\vec{x}$ has length $|\vec{x}|$. So if you take a pair of vectors $\vec{x}$ and $\vec{y}$, and also consider the vector $\vec{x} + \vec{y}$, then the lengths of each of these can be pictured like this:



The shortest route between two points (on a plane) is a straight line. Therefore, the distance from A to C, which is given by $|\vec{x}+\vec{y}|$, is the shortest route to get from A to C. It's clearly a longer route if you go from A to B, and then from B to C, which is of distance $|\vec{x}|+|\vec{y}|$. Indeed, this is precisely what the triangle inequality says:

$$|\vec{x}+\vec{y}| \leq |\vec{x}|+|\vec{y}|.$$

Consequences of the Triangle Inequality

The triangle inequality has some nice *corollaries*; i.e., other theorems that follow quickly from it.

Corollary.

Corollary 1.14 (*The reverse triangle inequality*)**.** Assume that $\mathbb{F}$ is an ordered field (like $\mathbb{R}$) and $x, y \in \mathbb{F}$. Then,

$$\Big||x|-|y|\Big| \leq |x-y|.$$

Proof. By Proposition 1.12, it suffices to show that

$$-|x-y| \leq |x|-|y| \leq |x-y|.$$

That is, we have to show two separate inequalities. We will first show the right-hand one (that $|x|-|y| \leq |x-y|$), and we will do so by applying a tricky application of the triangle inequality. Let $a = x-y$ and $b = y$. Then by the triangle inequality,

$$|a+b| \leq |a|+|b|.$$

That is,

$$\begin{aligned} |(x-y)+y| &\leq |x-y|+|y| \\ |x| &\leq |x-y|+|y|. \end{aligned}$$

Rearranging,

$$|x|-|y| \leq |x-y|,$$

as desired.

We will use a similar approach to show that $-|x-y| \leq |x|-|y|$. Let $c = y-x$ and $d = x$. By the triangle inequality,

$$|c+d| \leq |c|+|d|.$$

That is,

$$\begin{aligned} |(y-x)+x| &\leq |y-x|+|x| \\ |y| &\leq |y-x|+|x|. \end{aligned}$$

Rearranging,

$$-|y-x| \leq |x| - |y|,$$

which by Note 1.11 part (b) implies that

$$-|x-y| \leq |x| - |y|,$$

as desired. □

Here are two final results that follow quickly from what we have already done.

Corollary.

Corollary 1.15 (*Triangle inequality corollaries*). For both of the following, assume that $\mathbb{F}$ is an ordered field (like $\mathbb{R}$) and $x, y \in \mathbb{F}$.

- $|x-y| \leq |x| + |y|$.

  Proof: Replace y with $-y$ in the triangle inequality.

- $|x+y| \geq \Big||x| - |y|\Big|$.

  Proof: Replace y with $-y$ in the reverse triangle inequality.

And finally, with this, we can define a distance function just how you'd expect.

Definition.

Definition 1.16. Let $\mathbb{F}$ be an ordered field. Then define the *distance* function $d : \mathbb{F} \times \mathbb{F} \to \mathbb{F}$ by

$$d(x,y) = |x-y|.$$

And with that, we have described all of the axioms of the rationals. And in fact, to capture the reals we only need to add a single additional axiom, called the *completeness axiom*. This axiom fills in the "holes" in the rational numbers.

1.6 The Completeness Axiom

What's in a set? That which we call the reals,
By any other name would be as complete.

– Shakespeare in Dimension C-314[21]

We need just one final axiom to obtain the reals, and it is called *the completeness axiom*. Formally, it is an axiom about bounded sets, which requires a little motivation to understand. We begin there.

One of the first problems with the rationals that we mentioned was that if q is rational it need not be the case that $\sqrt{q}$ is rational—we worked out $q = 2$ in detail. Said differently, if q is rational, there may not be a rational number x such that $x^2 = q$.

We want to include $\sqrt{2}$ in what will be the real numbers, but we also want all of the other irrational numbers. We could say "include all the numbers $\sqrt{q}$ where q is rational," but most irrational numbers are not of this form... We want an approach that gets all of them, identifies their ordering, etc. How do we use the rationals to identify *all* of the irrationals? There is a slick way to do this. Turning back to $\sqrt{2}$, consider the set

$$A := \{x \in \mathbb{Q} : x^2 < 2\}.$$

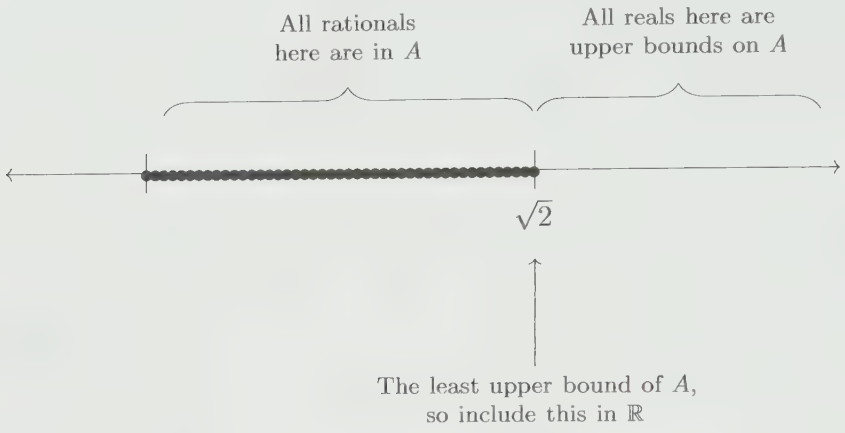
This set looks something like this:

Here are the important ideas:

- This set is *bounded above*. For example, 2 and 6 and 1.5 are all bigger than everything in A, so these are all *upper bounds* on A.
- Among all these upper bounds, there is one that is special—the smallest of them all. This is the *least upper bound*, and as you might imagine, this bound will in fact be exactly $\sqrt{2}$ (which by Example 1.4 is not rational).
- In order to get all of $\mathbb{R}$, we will start with $\mathbb{Q}$ and then we will add in all the least upper bounds from all sets that are bounded above, like A. E.g., due to

[21]The Shakespeare in the parallel universe where all his plays were about math. And Jerry's happy.

the above set A, we include $\sqrt{2}$ into our set $\mathbb{R}$. Doing this for all such A makes $\mathbb{R}$ complete.[22]



That is the blueprint for what we are going to do. Now here it is, formally.

Definition.

Definition 1.17. Let S be an ordered field (like $\mathbb{R}$) and $A \subseteq S$ be nonempty.

(i) The set A is *bounded above* if there exists some $b \in S$ such that $x \leq b$ for all $x \in A$; in this case, b is called an *upper bound* of A.

(ii) The *least upper bound* of A—if it exists—is some $b_0 \in S$ such that (1) b_0 is an upper bound of A, and (2) if b is any other upper bound of A, then $b_0 \leq b$. Such a b_0 is also called the *supremum* of A and is denoted $\sup(A)$.

(iii) Likewise, the set A is *bounded below* if there exists some $b \in S$ such that $x \geq b$ for all $x \in A$; in this case, b is called a *lower bound* of A.

(iv) Again, like above, the *greatest lower bound* of A—if it exists—is some $b_0 \in S$ such that (1) b_0 is a lower bound of A, and (2) if b is any other lower bound of A, then $b_0 \geq b$. Such a b_0 is also called the *infimum* of A and is denoted $\inf(A)$.

(v) If a set is both bounded above and bounded below, then it is simply called *bounded*.

Example 1.18. The below are left without proof for now. We will come back shortly and talk about how to prove these claims.

- The set $\mathbb{N} = \{1, 2, 3, \dots\}$ has no upper bounds. Lower bounds on $\mathbb{N}$ include $-17, 1, 0.123$ and $-\pi$. Note that $\sup(\mathbb{N})$ does not exist,[23] but $\inf(\mathbb{N}) = 1$.
- The set $\mathbb{Q}$ has no upper or lower bounds; consequently, $\sup(\mathbb{Q})$ and $\inf(\mathbb{Q})$ do not exist.
- $\sup\left(\left\{\frac{1}{n} : n \in \mathbb{N}\right\}\right) = 1$; $\inf\left(\left\{\frac{1}{n} : n \in \mathbb{N}\right\}\right) = 0$.

  $\rightarrow$ Note that the supremum here is in the set, while the infimum is not in the set.
- $\sup\left(\left\{\frac{n}{n+1} : n \in \mathbb{N}\right\}\right) = 1$; $\inf\left(\left\{\frac{n}{n+1} : n \in \mathbb{N}\right\}\right) = \frac{1}{2}$.

  $\rightarrow$ This is the set $\{\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \dots\}$. Note that the infimum here is in the set, while the supremum is not in the set. The proof of the above is asked of you in Exercise 1.19.
- As we have discussed, in $\mathbb{Q}$ the set $\{x \in \mathbb{Q} : x^2 < 2\}$ does not have a supremum. In $\mathbb{R}$ it will — in fact, $\sup\left(\{x \in \mathbb{Q} : x^2 < 2\}\right) = \sqrt{2}$.

Whew, ok, so those are the basic definitions plus examples thereof. We are now ready to state the crucial definition which will "fill in the holes" in the rationals and solve nearly all the problems we discussed.

Definition.

Definition 1.19. Let S be an ordered set (like $\mathbb{R}$). Then S has the *least upper bound property* if given any nonempty $A \subseteq S$ where A is bounded above, A has a least upper bound. In other words, $\sup(A) \in S$ for every such A.

Such a set S is also called *complete*.

This definition only explicitly guarantees the existence of least upper bounds, however as a corollary one can prove the existence of greatest lower bounds. Indeed, if S is complete, and $A \subseteq S$ is nonempty and bounded *below*, then $\inf(A) \in S$. This is a consequence of Exercise 1.21, in which you are asked to prove that $\inf(A)$ exists and moreover that $\inf(A) = -\sup(-A)$.

We have already argued that $\mathbb{Q}$ is not complete. To test your understanding, try to convince yourself that $\mathbb{N}$ is complete (this is asked for in Exercise 1.17).

23

– What $\mathbb{R}$ we talkin' about? –

At last, we have reached the big, big idea.

Big idea. The reals are what you get when you start with $\mathbb{Q}$ and then add in just enough numbers so that it satisfies the least upper bound property. This is called *completeness*.

By doing this we will add in numbers like $\sqrt{2}$ and π, but we will not add in "numbers" like the complex number $2+i$ or the ordinal number ω. Thus you see how important it is to carefully specify your axioms before developing a theory like that of real analysis.

All of these definitions have led up to the following huge theorem.

Theorem.

Theorem 1.20 (*Existence and uniqueness of* $\mathbb{R}$). There exists a unique[24] complete ordered field. We call this field *the real numbers*, $\mathbb{R}$.

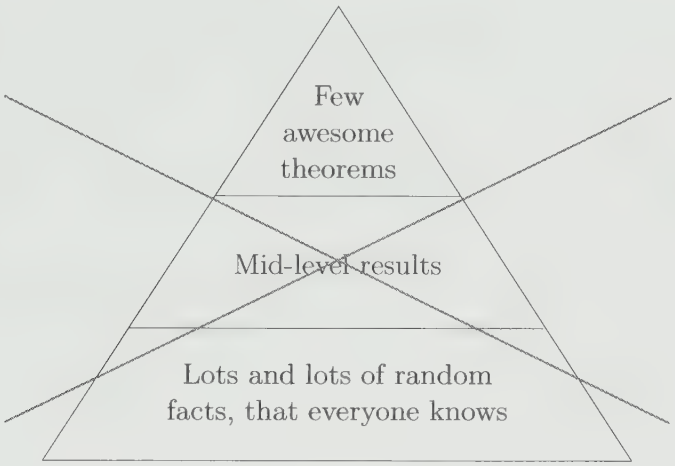
A proof of this using so-called *Dedekind cuts* will be sketched in Appendix A.

It is worthwhile now to sit back and reflect on what we have. If we add a completeness axiom to the six axioms we had for an ordered field, we end up with just seven axioms which collectively give us the real numbers. And everything else that we are going to do during this course will be derived from just these seven axioms. Math is often viewed as a pyramid, where you use a wide base of knowledge and results, along with a smaller number of mid-level theorems, to solve a problem in the "peak" section—specific problems or a new, narrow theorem.

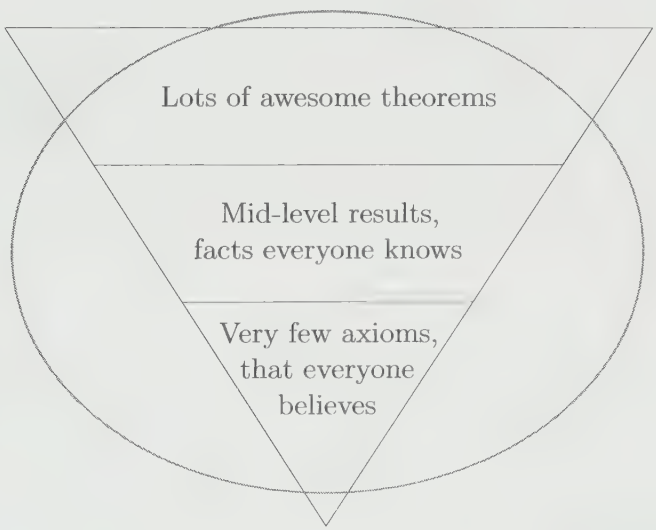
[24] That is, if you had any other complete ordered field, it really is the same as $\mathbb{R}$, you might just need to relabel the elements.

Sometimes this is phrased as "There exists a unique complete ordered field containing $\mathbb{Q}$" or "it is unique up to isomorphism." Conditions like "containing $\mathbb{Q}$" help clarify, but they are unnecessary.

The fact that there is *only one* complete ordered field in entire universe highlights that there is something special about the reals. If we want to properties we do, the complex numbers $\mathbb{C}$, three-dimensional space $\mathbb{R}^3$, the extended reals $\mathbb{R} \cup \{\pm\infty\}$, the hyperreals ${}^*\mathbb{R}$ and all other options fail at least one axiom.



In some narrow sense this is true—usually advancements are focused. But thinking about math as a subject like this is completely upside-down. Math begins with almost nothing—just a handful of structural or self-evident axioms—and from these, an entire universe is built.



How is it that a handful of self-evident axioms can be combined to create startling theorems? It really is amazing. In my opinion, most of the best accomplishments of the (collective) human mind are in mathematics, and we have the rigors and axiomatic structure of mathematics to thank. So throughout this course, when we build derivatives, prove the existence of some remarkable function, or study series of functions, remember where it all started: with almost nothing.

Before moving on, below we collect the axioms which uniquely defined $\mathbb{R}$—for your future reference.

Note.

Note 1.21. The set $\mathbb{R}$ has two binary operations, addition $(+)$ and multiplication $(\cdot)$, and is the smallest such set that contains $\mathbb{Q}$ and satisfies the following axioms.

Axiom 1 (Commutative Law). If $a, b \in \mathbb{R}$, then $a+b = b+a$ and $a \cdot b = b \cdot a$.

Axiom 2 (Distributive Law). If $a, b, c \in \mathbb{R}$, then $a \cdot (b+c) = a \cdot b + a \cdot c$.

Axiom 3 (Associative Law). If $a, b, c \in \mathbb{R}$, then $(a+b)+c = a+(b+c)$ and $(a \cdot b) \cdot c = a \cdot (b \cdot c)$.

Axiom 4 (Identity Law). There are special elements $0, 1 \in \mathbb{R}$, where $a+0 = a$ and $a \cdot 1 = a$ for all $a \in \mathbb{R}$.

Axiom 5 (Inverse Law). For each $a \in \mathbb{R}$, there is an element $-a \in \mathbb{R}$ such that $a + (-a) = 0$. If $a \neq 0$, then there is also an element $a^{-1} \in \mathbb{R}$ such that $a \cdot a^{-1} = 1$.

Axiom 6 (Order Axiom). There is a nonempty subset $P \subseteq \mathbb{R}$, called the *positive elements*, such that

(a) If $a, b \in P$, then $a+b \in P$ and $a \cdot b \in P$;

(b) If $a \in \mathbb{R}$ and $a \neq 0$, then either $a \in P$ or $-a \in P$, but not both.

Axiom 7 (Completeness Axiom). Given any nonempty $A \subseteq \mathbb{R}$ where A is bounded above, A has a least upper bound. In other words, $\sup(A) \in \mathbb{R}$ for every such A.

1.7 Working with Sups and Infs

We now want to develop some properties of supremums and infimums, to understand them better. Here is the first basic result.

Proposition.

Proposition 1.22 (*Supremums are unique*). If the supremum or infimum of $A \subseteq \mathbb{R}$ exists, then it is unique.

Recall that a supremum is the same thing as a least upper bound.

Proof. We will prove that supremums are unique, and leave the proof that infimums are unique as an exercise; it's similar.

Assume for a contradiction that α and β are distinct least upper bounds of A. In particular, both are upper bounds of A, while $\alpha \neq \beta$. On one hand, since α is a least

upper bound and β is an upper bound, we must have $\alpha \leq \beta$. On the other hand, since β is a least upper bound and α is an upper bound, we must have $\beta \leq \alpha$. In summary,

$$\alpha \leq \beta \quad \text{and} \quad \beta \leq \alpha.$$

This implies that $\alpha = \beta$, giving our contradiction. □

Remember that there are always many ways to write up a proof. For example, the below is a second way to write up the same basic idea.

Second proof. Assume for a contradiction that α and β are distinct least upper bounds of A. In particular, both are upper bounds of A, while $\alpha \neq \beta$. Since $\mathbb{R}$ is an ordered field, either $\alpha < \beta$ or $\beta < \alpha$ (technically, this is a consequence of the "positive subset" that we defined); without loss of generality, assume that $\alpha < \beta$. But this contradicts β being the least upper bound: α is an upper bound but yet $\beta \not\leq \alpha$, as is required by the definition of the least upper bound. □

I won't go through all the details, but I do want to include a final theorem to check off one of the first indications that we (and mathematicians over the millennia) should care about more than just $\mathbb{Q}$. It also ties nicely into where we began: with the square root of 2.

Theorem.

Theorem 1.23 (*Square roots exist*). If $a \in \mathbb{R}$ and $a \geq 0$, then $\sqrt{a} \in \mathbb{R}$.

Proof idea. One can show that $\sqrt{a} = \sup\left(\{x \in \mathbb{R} : x^2 < a\}\right)$, which is in $\mathbb{R}$ by completeness. □

Now consider the set $A = \{0.9, 0.99, 0.999, 0.9999, \dots\}$. One upper bound of A is 2.7. This isn't the supremum, though, because it's not the *least* upper bound. One way to think about this is that we can subtract a bit off 2.7 and the result is still an upper bound. For instance, if we subtract 0.3 off then we get to 2.4, which is still an upper bound (but is still not the supremum because it's still not the *least* upper bound).

Now, 1 *does* happen to be the supremum. And like above, one way to think about it is this: First, it is an upper bound; second, if you subtract any bit off of 1, you no longer get an upper bound.

- If you subtract just 0.01 off, you get $1 - 0.01 = 0.99$. And this is not an upper bound because 0.999 is in A and is larger than $1 - 0.01$.
- If you subtract just 0.00012 off, you get $1 - 0.00012 = 0.99988$. And this is again not an upper bound because 0.99999999 is in A and is larger than $1 - 0.00012$.

As you can imagine, if you subtract off *any* positive number from 1, you no longer get an upper bound of A. For this reason, 1 is the *least* upper bound of A. In math: $\sup(A) = 1$.

This type of reasoning suggests the following theorem, which is a very *analytic* way to think about supremums. Essentially, it says that "α is a least upper bound of a set A if, whenever you subtract a little bit off of α (written as "$\alpha - \varepsilon$"), it is no longer an upper bound (meaning that there is an element $x \in A$ that is now bigger than $\alpha - \varepsilon$). The reasoning is, if you could subtract off a little bit and still have an upper bound, then how is α the *least* upper bound? Clearly you could do better!

Theorem.

Theorem 1.24 (*Supremums analytically*). Let $A \subseteq \mathbb{R}$. Then $\sup(A) = \alpha$ if and only if

(i) α is an upper bound of A, and

(ii) Given any $\varepsilon > 0$, $\alpha - \varepsilon$ is *not* an upper bound of A. That is, there is some $x \in A$ for which $x > \alpha - \varepsilon$.

Likewise, $\inf(A) = \beta$ if and only if

(i) β is a lower bound of A, and

(ii) Given any $\varepsilon > 0$, $\beta + \varepsilon$ is *not* a lower bound of A. That is, there is some $x \in A$ for which $x < \beta + \varepsilon$.

Before we prove this, I will note that these ε-terms will be used lots and lots and lots and lots and lots in this course. Always think about them as being some small, positive number. Also, in part (ii) note that the x depends on the ε. In the discussion before Theorem 1.24, for example, $\varepsilon = 0.01$ lead to $x = 0.999$; and $\varepsilon = 0.00012$ lead to $x = 0.99999$. So different values of ε lead to different values of x.

Ok, let's prove the theorem now.

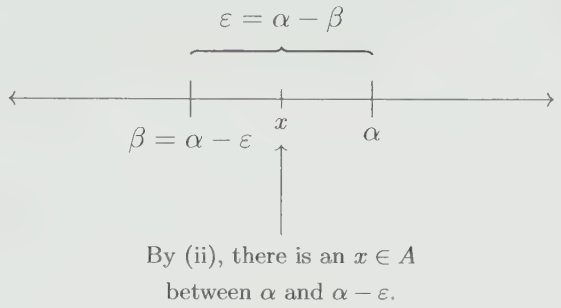
Proof. We will prove the supremum case and leave the infimum case as an exercise; it's similar.

First assume that $\sup(A) = \alpha$. We aim to prove (i) and (ii). The first of these is immediate: Since $\sup(A) = \alpha$, α is the least upper bound of A, which of course also implies that it is an upper bound of A.

Now we will show (ii). Let $\varepsilon > 0$. Since $\alpha - \varepsilon < \alpha$, we know that $\alpha - \varepsilon$ is not an upper bound of A, because if so that would contradict α being the *least* upper bound of A. And so, since $\alpha - \varepsilon$ is not an upper bound, there must be some $x \in A$ which is greater than $\alpha - \varepsilon$, as desired.

Now assume (i) and (ii). We aim to prove that $\sup(A) = \alpha$. That is, we wish to show that α is an upper bound of A (which is implied directly by (i)), and for any

other upper bound β, we have $\alpha \leq \beta$. We have only the latter to prove. Assume that β is some other upper bound of A, and assume for a contradiction that $\beta < \alpha$. Note that $\alpha - \beta > 0$. We will use this value as our ε, and then apply (ii) to contradict β being an upper bound. This will be the picture:



Now we will work it out formally. Let

$$\varepsilon = \alpha - \beta.$$

Since $\varepsilon > 0$, by (ii) there exists some $x \in A$ such that $x > \alpha - \varepsilon = \alpha - (\alpha - \beta) = \beta$. But this is a contradiction, because we assumed that β was an upper bound of A, and yet we found an element $x \in A$ that is larger than β. This completes the proof. □

Note: the forward direction of the proof also works well by contrapositive. The contrapositive of

$$\sup(A) = \alpha \quad \Longrightarrow \quad \begin{array}{c}\text{For all } \varepsilon > 0 \text{ there exists some} \\ x \in A \text{ such that } x > \alpha - \varepsilon\end{array}$$

is

$$\begin{array}{c}\text{There exists some } \varepsilon > 0 \text{ such that}^{27} \\ \text{for all } x \in A \text{ we have } x \leq \alpha - \varepsilon\end{array} \quad \Longrightarrow \quad \sup(A) \neq \alpha.$$

And to prove this, just observe that the left-hand side implies that $\alpha - \varepsilon$ is an upper bound of A, and so $\sup(A) \leq \alpha - \varepsilon$, which of course implies that $\sup(A) \neq \alpha$, as desired.

[27] Does it make sense why you don't have to change "$\varepsilon > 0$" to "$\varepsilon \leq 0$"? A negated statement, if true, falsifies the original statement. Now, the original statement only asserts that positive ε have some property, so proving anything about $\varepsilon \leq 0$ tells you nothing at all about the original statement. Make sense? It's like if you said "Every positive even number is larger than 1," and I said "Nuh-uh, what about -2??" Clearly I have not falsified your statement, because you were only ever referring to the positive case.

Example 1.25. Let $A = \{x \in \mathbb{R} : x < 0\}$. Then $\sup(A) = 0$.

Proof. We will use the supremums analytically theorem (Theorem 1.24). Note that 0 is clearly an upper bound of A because A is defined as the reals less than 0. This proves (i) in the theorem.

To prove (ii), let $\varepsilon > 0$. We need to show that there is some $x \in A$ such that $x > 0 - \varepsilon$; that is, $x > -\varepsilon$. We claim that $x = -\varepsilon/2$ works. The following proves this.[28] We know that $\varepsilon > 0$, and hence

$$\begin{aligned} \varepsilon &> 0 \\ -\varepsilon &> -2\varepsilon && \text{(Subtracting } 2\varepsilon \text{ from both sides)} \\ -\varepsilon/2 &> -\varepsilon. && \text{(Dividing both sides by 2)} \end{aligned}$$

Moreover, $\varepsilon > 0$ implies that $-\varepsilon < 0$, and hence $-\varepsilon/2 < 0$, which proves $-\varepsilon/2 \in A$. In summary, we have shown that $x = -\varepsilon/2$ has the property that $x \in A$ and $x > 0 - \varepsilon$. This proves condition (ii) of Theorem 1.24, and thus concludes the proof. □

1.8 The Archimedean Principle

Earlier we asserted, without proof, that

$$\inf\left(\left\{\frac{1}{n} : n \in \mathbb{N}\right\}\right) = 0 \qquad \text{and} \qquad \sup\left(\left\{\frac{1}{n} : n \in \mathbb{N}\right\}\right) = 1.$$

We aim to prove these precisely. For the former, to show that $\inf\left(\left\{\frac{1}{n} : n \in \mathbb{N}\right\}\right) = 0$, we would need to show that 0 is a lower bound (this is easy), and also that $0 + \varepsilon$ is not a lower bound for any $\varepsilon > 0$. That is, for any $\varepsilon > 0$, there is some n such that $1/n < 0 + \varepsilon$. This may or may not seem obvious, but either way it is something we have to prove ourselves. How we do that is by using the so-called *Archimedean principle.*

Lemma.

Lemma 1.26 (*The Archimedean principle*). If a and b are real numbers with $a > 0$, then there exists a natural number n such that $na > b$.

In particular, for any $\varepsilon > 0$ there exists $n \in \mathbb{N}$ such that $\frac{1}{n} < \varepsilon$.

[28] The scratch work to find this proof: Start with what you want to show, and do algebra until you reach something we do know (like that $\varepsilon > 0$).

$$\begin{aligned} -\varepsilon/2 &> -\varepsilon && \text{(Want to show)} \\ -\varepsilon &> -2\varepsilon && \text{(Multiply both sides by 2)} \\ \varepsilon &> 0 && \text{(Add } 2\varepsilon \text{ to both sides)} \end{aligned}$$

Now if we do this scratch work in the reverse direction, it works for our proof.

If a is already bigger than b, or not too much smaller, then this is easy—a small value of n accomplishes this. The challenge is when a is super small and b is super big. Then n has to be really, *really* big to make up for it. The way this is sometimes phrased is: Even a teaspoon can drain an ocean. What that means is this: Think about the number a as being the volume of water in a teaspoon, and b as the volume of water in the ocean, which is obviously much, much bigger. But according to the Archimedean principle there is some integer n where, if you take a teaspoon of water from the ocean n times in a row, then you will have completely emptied it.

Now let's prove it.

Proof. We aim to show that $na > b$ for some $n \in \mathbb{N}$; by dividing over the a, we aim to prove that there is some $n \in \mathbb{N}$ such that $n > b/a$. Now, the number b/a is just some real number that we know nothing about. In fact, let's just call it x. So, equivalently, we are trying to prove that given any real number x, there is some integer n such that $n > x$.

Assume for a contradiction that there is no integer larger than x. That is, assume that x is an upper bound on the set $\mathbb{N}$. Then $\mathbb{N}$ is a subset of $\mathbb{R}$ that is bounded above, and so by the completeness of $\mathbb{R}$ we deduce that $\sup(\mathbb{N})$ exists. Call this supremum α. Since α is the *least* upper bound of $\mathbb{N}$, we know that $\alpha - 1$ is not an upper bound. That is, there exists some integer $m > \alpha - 1$. Adding 1 to each side,

$$m + 1 > \alpha.$$

But this is a contradiction. If α is the supremum of $\mathbb{N}$, then it is an upper bound on $\mathbb{N}$. But we found $(m + 1) \in \mathbb{N}$ which is larger than α. This concludes the first statement in the principle.

The second part follows directly from the first by letting $a = \varepsilon$ and $b = 1$, and then dividing over the n. □

Example 1.27. Show that $\inf\left(\left\{\frac{1}{n} : n \in \mathbb{N}\right\}\right) = 0$.

Solution. Let $A = \left\{\frac{1}{n} : n \in \mathbb{N}\right\}$. We will use the supremums analytically theorem (Theorem 1.24). We must then show that 0 is a lower bound of A and that, for all $\varepsilon > 0$, $0 + \varepsilon$ is not a lower bound of A.

The first of these is almost immediate: Since 1 and n are positive for each $n \in \mathbb{N}$, so is $1/n$. So $1/n > 0$, and thus 0 is indeed a lower bound for A.

Working toward the second, let $\varepsilon > 0$. Then by the Archimedean principle (Lemma 1.26), there exists some $n \in \mathbb{N}$ such that $\frac{1}{n} < \varepsilon$. This element, $\frac{1}{n}$, is in A and is less than $0 + \varepsilon$. So $0 + \varepsilon$ is not a lower bound of A. □

Example 1.28. Show that $\sup\left(\left\{\frac{1}{n} : n \in \mathbb{N}\right\}\right) = 1$.

Solution. Let $A = \left\{\frac{1}{n} : n \in \mathbb{N}\right\}$. We will use the supremums analytically theorem (Theorem 1.24). We must then show that 1 is an upper bound of A and that, for all $\varepsilon > 0$, $1 - \varepsilon$ is not an upper bound of A.

For the first of these, note that since $n \geq 1$ for all $n \in \mathbb{N}$, and by dividing over the n we have that $1 \geq \frac{1}{n}$ for all $n \in \mathbb{N}$. So 1 is indeed an upper bound for A.

Working towards the second, let $\varepsilon > 0$. We need to show that there is some $x \in A$ such that $1 - \varepsilon < x$. But this is always accomplished by the number 1: Clearly $1 \in A$ and $1 - \varepsilon < 1$. □

Definition.

Definition 1.29. Suppose A and B are ordered field (like $\mathbb{R}$). Then A is *dense* in B if, for any $x, y \in B$, there exists $a \in A$ such that $x < a < y$.

For example, $\mathbb{Q}$ is dense in $\mathbb{Q}$ because given any $x, y \in \mathbb{Q}$, if we let $a = \frac{x+y}{2}$, then one can show that $a \in \mathbb{Q}$ and $x < a < y$. Likewise, $\mathbb{R} \setminus \mathbb{Q}$ is dense in $\mathbb{Q}$ since given $x, y \in \mathbb{Q}$, if we let $a = \frac{x+y}{\sqrt{2}}$, then one can show that $a \in \mathbb{R} \setminus \mathbb{Q}$ and $x < a < y$. Meanwhile, $\mathbb{Z}$ is *not* dense in $\mathbb{Q}$.

We are about to prove an important and far-reaching extension of these results, but first here is a quick lemma that we will use for the proof.

Lemma.

Lemma 1.30 ($y - x > 1 \Rightarrow \exists\, z \in \mathbb{Z}$ *where* $x < z < y$)**.** Let $x, y \in \mathbb{R}$. If $y - x > 1$, then there exists $z \in \mathbb{Z}$ such that $x < z < y$.

For the below proof, recall that $\mathbb{N}_0 = \mathbb{N} \cup \{0\}$.

Proof. First assume that x and y are at least 0, and consider the set

$$A = \{n \in \mathbb{N}_0 : n \leq x\}.$$

Since $x \geq 0$, this set is non-empty, and since it is a set of nonnegative integers which is bounded above by x, this set is finite. So by Exercise 1.16, $\max(A)$ exists and is an element of A. Call this maximum M. We claim $z := M + 1$ works.

Note that since $M \in \mathbb{N}_0$, also $z \in \mathbb{N}_0$. Furthermore, since z is larger than the largest element of A, z is not in A, implying that $x < z$. Finally,

$$M \leq x \qquad \text{implies that} \qquad M + 1 \leq x + 1 < y.$$

So $z < y$. In summary, we have shown that $x < z < y$, as desired.

The cases where x and y are not at least 0 are similar. If both are negative, then by considering $-x$ and $-y$ the above argument gives an integer z where $-y < z < -x$, showing that $-z$ works, since $x < -z < y$. If one is positive and one is negative, then 0 works. □

The next theorem says that between any two real numbers is a rational number. The rationals—which are super nice to describe and work with—are everywhere.[29]

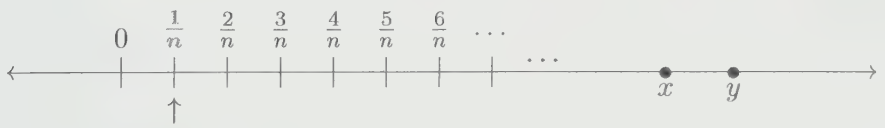
Theorem.

Theorem 1.31 ($\mathbb{Q}$ *is dense in* $\mathbb{R}$)**.** The rational numbers are dense in the real numbers.

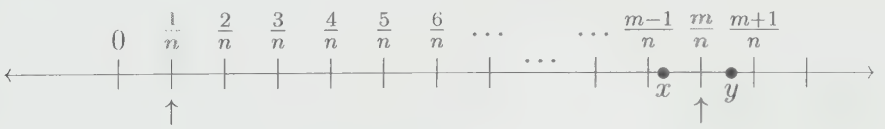
Proof Idea. The result follows from Lemma 1.30, and does so easily enough that it would be easy to miss what is what really going on. Assume that $x, y \in \mathbb{R}$ and assume both are positive with $x < y$. We aim to find a rational number $\frac{m}{n}$ between x and y. That is, with $x < \frac{m}{n} < y$.

Now, $y - x$ is some positive number, and so by Archimedean principle there is some $n \in \mathbb{N}$ with $\frac{1}{n} < y - x$.

Now think about all integer multiples of this $\frac{1}{n}$.



Note that since each one is $\frac{1}{n}$ away from the next one, but $y - x > \frac{1}{n}$, it is impossible for these dashes to completely hop over the interval between x and y. That is, at least one of these must fall between x and y.



This is the idea, although by using Lemma 1.30 the proof hides some of these details. Now here's the proof of Theorem 1.31.

[29] In Chapter 2 when we study the cardinalities of these sets, this result will be even more impressive and important.

Proof. Pick any $x, y \in \mathbb{R}$ where $x < y$. We need to show that there exists some $\frac{m}{n} \in \mathbb{Q}$ (with $m, n \in \mathbb{Z}$) such that

$$x < \frac{m}{n} < y.$$

First note that if $x < 0 < y$ then we are done, since $0 \in \mathbb{Q}$. Furthermore, if we can show the theorem holds in the case that x and y are positive, then it implies that it holds when they are negative $\left(0 < x < \frac{m}{n} < y \text{ implies } -y < \frac{-m}{n} < -x < 0\right)$, so we may assume x and y are positive.

Since $y - x > 0$, by the Archimedean principle (Lemma 1.26) there exists some $n \in \mathbb{N}$ such that $n(y - x) > 1$; i.e., $ny - nx > 1$. And so, by Lemma 1.30, there is some integer m with

$$nx < m < ny.$$

That is,

$$x < \frac{m}{n} < y,$$

which concludes the proof. □

Two final comments

Before concluding the first chapter, the proof of Lemma 1.30 also implies that, for any $x \in \mathbb{R}$, there exists an integer M such that $M \leq x \leq M + 1$. In particular, it implies that the floor and ceiling functions exist.

Definition.

Definition 1.32. Let $x \in \mathbb{R}$.

- The *ceiling* of x, denoted $\lceil x \rceil$, is the integer n such that $x \leq n < x + 1$
- The *floor* of x, denoted $\lfloor x \rfloor$, is the integer n such that $x \leq n < x + 1$.

Lastly, within the formal construction of $\mathbb{R}$ (see Appendix A) contains an assertion of the soundness of the *principle of mathematical induction*, and we will therefore use induction without hesitation. This principle is equivalent to the so-called *well-ordering principle*, which we record now before moving on to cardinality.

Principle.

Principle 1.33 (Well-ordering principle). Every non-empty subset of natural numbers contains a smallest element.

— Notable Exercises —

Exercises test your knowledge and develop your understanding. Many are written simply for this purpose, while others hold a greater significance for one reason or another. Before each chapter's exercises I will comment on a few of the exercises which are in this second camp[30]

- In Exercise 1.4 we prove *De Morgan's laws*, which are quite useful in set theory and beyond, as they describe how the compliment operation distributes over unions and intersections. They say this:

  $$(A \cap B)^c = A^c \cup B^c \qquad \text{and} \qquad (A \cup B)^c = A^c \cap B^c.$$

  They will become useful again when we study topology in Chapter 5, where we will utilize an infinite form of these laws (see Fact 5.11). They are also particularly important in mathematical logic (under the guise of *laws of Boolean algebra*)—but that's a topic for another long-from textbook.

  De Morgan's laws highlight a certain symmetry between unions and intersections, and between a universal set X and the empty set $\emptyset$. This is an example of a *duality* property, which pop up all over mathematics.

- Exercise 1.9 is an early example of an *analytic* statement, in which a property is asserted to hold "for all $\varepsilon > 0$." Properties of this from will soon be filling up the pages of this text, and are ubiquitous in the field of real analysis. Recall that we already have one particular theorem of this form: The supremums analytically theorem (Theorem 1.24). Exercise 1.20 and others provide further work in this direction.

- Exercise 1.14 will be used several times in this book, notably in Chapters 4 and 8.

- Exercise 1.22 is a special case of an important topological property we will study in Chapter 5. You are asked to prove a generalization of it in Exercise 5.4 (b), although the best way to prove Exercise 1.22 at this point utilizes only things we discussed in this chapter.

[30]For instructors: I do not mean to single these out as suggestions as to what to assign for homework, and indeed the others are oftentimes just as good or better suited for assignments.

— Exercises —

- You must prove all of your answers (unless stated otherwise).
- Remember the proof techniques that you learned in your intro-to-proofs class: Direct, by contradiction, by using the contrapositive, by induction, and/or by cases. And you can disprove something by exhibiting a counterexample.

Exercise 1.1. Explain the error in the following "proof" that $2 = 1$.
Let $x = y$. Then

$$\begin{aligned} x^2 &= xy \\ x^2 - y^2 &= xy - y^2 \\ (x+y)(x-y) &= y(x-y) \\ x + y &= y \\ 2y &= y \\ 2 &= 1. \end{aligned}$$

Exercise 1.2. If A and B are two boxes (possibly with things inside), describe the following in terms of boxes:

- $A \setminus B$
- $\mathcal{P}(A)$
- $|A|$

Exercise 1.3. If $A_1, A_2, A_3, \ldots, A_n$ are all boxes (possibly with things inside), describe the following in terms of boxes:

- $\bigcup_{i=1}^{n} A_i = A_1 \cup A_2 \cup \cdots \cup A_n$.
- $\bigcap_{i=1}^{n} A_i = A_1 \cap A_2 \cap \cdots \cap A_n$.

Exercise 1.4. The following are special cases of De Morgan's laws.

(a) Prove that $(A \cap B)^c = A^c \cup B^c$.

(b) Prove that $(A \cup B)^c = A^c \cap B^c$.

Exercise 1.5. Given an ordered field $\mathbb{F}$, recall that we defined the *positive elements* to be a nonempty subset $P \subseteq \mathbb{F}$ that satisfies both of the following conditions:

(i) If $a, b \in P$, then $a + b \in P$ and $a \cdot b \in P$;

(ii) If $a \in \mathbb{F}$ and $a \neq 0$, then either $a \in P$ or $-a \in P$, but not both.

(a) Give an example of some $P_1 \subseteq \mathbb{R}$ that satisfies (i) but not (ii).

(b) Give an example of some $P_2 \subseteq \mathbb{R}$ that satisfies (ii) but not (i).

Exercise 1.6. For each of the following, find all numbers x for which satisfy the expression.

(a) $|x - 4| = 7$

(b) $|x - 4| < 7$

(c) $|x + 2| < 1$

(d) $|x - 1| + |x - 2| > 1$

(e) $|x - 1| + |x + 1| > 1$

(f) $|x - 1| \cdot |x + 1| = 0$

(g) $|x - 1| \cdot |x + 2| = 3$

Exercise 1.7. Let $\max\{x, y\}$ denote the maximum of the numbers x and y, and let $\min\{x, y\}$ denote the minimum. For example, $\min\{-1, 4\} = \min\{-1, -1\} = -1$. Prove that

$$\max\{x, y\} = \frac{x + y + |y - x|}{2} \qquad \text{and} \qquad \min\{x, y\} = \frac{x + y - |y - x|}{2}.$$

Then find a formula for $\max\{x, y, z\}$ and $\min\{x, y, z\}$. Hint: note that $\max\{x, y, z\} = \max\{x, \max\{y, z\}\}$.

Exercise 1.8. Assume that $\mathbb{F}$ is an ordered field and $a, b, c, d \in \mathbb{F}$ with $a < b$ and $c < d$.

(a) Show that $a + c < b + d$.

(b) Prove that it is not necessarily true that $ac < bd$. Hint: prove this by finding a counterexample in the case that $\mathbb{F} = \mathbb{R}$.

Note whenever you use an axiom.

Exercise 1.9. Let a, b and ε be elements of an ordered field.

(a) Show that if $a < b + \varepsilon$ for every $\varepsilon > 0$, then $a \leq b$.

(b) Use part (a) to show that if $|a - b| < \varepsilon$ for all $\varepsilon > 0$, then $a = b$.

Note whenever you use an axiom.

Exercise 1.10. Prove that $\sum_{k=1}^{n} \frac{1}{k(k+1)} = \frac{n}{n+1}$ for every natural number n.

Exercise 1.11. Determine which natural numbers, n, have the property that $\sqrt{n}$ is irrational.

Exercise 1.12. Let $f : X \to Y$, and assume $A_1, A_2 \subseteq X$. Show that

$$f(A_1 \cap A_2) \subseteq f(A_1) \cap f(A_2).$$

Recall that if A is a set, then $f(A) = \{f(x) : x \in A\}$.

Exercise 1.13. Give an example of a function f, and a pair of sets A and B, for which

$$f(A \cap B) \neq f(A) \cap f(B).$$

Recall that if A is a set, then $f(A) = \{f(x) : x \in A\}$.

Exercise 1.14. Assume that $A \subseteq B$ and both are bounded above. Prove that $\sup(A) \leq \sup(B)$.

Exercise 1.15. Suppose $A \subseteq \mathbb{R}$ has a maximal element—that is, there is an element $M \in A$ such that $x \leq M$ for all $x \in A$. Likewise, assume $B \subseteq \mathbb{R}$ has a minimal element m.

(a) Prove that $\sup(A) = M$.

(b) Prove that $\inf(B) = m$.

Exercise 1.16. Suppose that A is a nonempty set containing finitely many elements. Prove by induction that A has a maximal element, and show that $\max(A) \in A$.

Exercise 1.17. Prove that $\mathbb{N}$ is complete.

Exercise 1.18. Prove the infimum case of Theorem 1.24.

Exercise 1.19. Prove that

$$\sup\left(\left\{\frac{n}{n+1} : n \in \mathbb{N}\right\}\right) = 1 \qquad \text{and} \qquad \inf\left(\left\{\frac{n}{n+1} : n \in \mathbb{N}\right\}\right) = \frac{1}{2}.$$

Exercise 1.20. Let $A, B \subseteq \mathbb{R}$, and assume that $\sup(A) < \sup(B)$.

(a) Show that there exists an element $b \in B$ that is an upper bound for A.

(b) Give an example to show that this is not necessarily the case if we instead only assume that $\sup(A) \leq \sup(B)$. You do not need to prove your answer.

Exercise 1.21. For $A \subseteq \mathbb{R}$, we denote $-A$ to be the set obtained by taking the opposite of everything in A. That is,

$$-A := \{-x : x \in A\}.$$

Suppose that $A \neq \emptyset$ and that A is bounded below. Prove that $-A \neq \emptyset$, $-A$ is bounded above, and $\sup(-A) = -\inf(A)$.

Exercise 1.22. For each $n \in \mathbb{N}$, assume we are given a closed interval[31] $I_n = [a_n, b_n]$. Also, assume that each I_{n+1} is contained inside of I_n. This gives a sequence of increasingly smaller intervals,

$$I_1 \supseteq I_2 \supseteq I_3 \supseteq I_4 \supseteq \dots.$$

Prove that $\bigcap_{n=1}^{\infty} I_n \neq \emptyset$. That is, prove that there is some real number x such that $x \in I_n$ for every $n \in \mathbb{N}$.

Exercise 1.23. Give an example showing that the conclusion of Exercise 1.22 need not hold if each I_n is allowed to be an open interval.

Exercise 1.24. Recall that for sets A and B, we define

$$A + B = \{a + b : a \in A \text{ and } b \in B\}.$$

Assume that $A, B \subseteq \mathbb{R}$ and $\sup(A)$ and $\sup(B)$ exist. Prove that

$$\sup(A + B) = \sup(A) + \sup(B).$$

Exercise 1.25. Let $A, B \subseteq \mathbb{R}$. We define

$$A \cdot B = \{a \cdot b : a \in A \text{ and } b \in B\}.$$

Give an example of sets A and B where $\sup(A \cdot B) \neq \sup(A) \cdot \sup(B)$.

Exercise 1.26. Is the subset of rational numbers

$$\left\{\frac{m}{n} : m, n \in \mathbb{Z} \text{ and } 1 \leq n \leq 10\right\}$$

dense in $\mathbb{R}$?

[31] The closed interval is defined as you would expect:

$$[a, b] = \{x \in \mathbb{R} : a \leq x \leq b\}.$$

Likewise the open interval is defined as $(a, b) = \{x \in \mathbb{R} : a < x < b\}$. And half-open intervals and intervals to $\pm\infty$ are again exactly as you would expect.

— Open Questions[32] —

Question 1. Consider the set

$$\left\{\frac{3}{2}, \left(\frac{3}{2}\right)^2, \left(\frac{3}{2}\right)^3, \left(\frac{3}{2}\right)^4, \dots\right\},$$

but for each number remove the integer portion;[33] for example, $(3/2)^3 = 3.375$ would be reduced to 0.375. Is the resulting set dense in $[0, 1]$?

Question 2. Is $e + \pi$ rational?

Question 3. For points x and y in the plane, define $d(x, y)$ to be the distance between x and y. Does there exist a dense subset S of $\mathbb{R}^2$ where, for any $x, y \in S$, $d(x, y) \in \mathbb{Q}$?[34]

Question 4. Can every rational number x be represented as a quotient of *shifted primes*? That is, do there exist primes p and q such that

$$x = \frac{p+1}{q+1}.$$

[32] An "open question" is a question that is unsolved. No one knows the answer to any of these. If you are my student and you solve one of them, I will give you one point of extra credit.

[33] Another way to say this is "$(3/2)^n \pmod 1$." Or: $(3/2)^n - \lfloor (3/2)^n \rfloor$.

[34] That is, if $x = (x_1, x_2)$ and $y = (y_1, y_2)$ are in S, then $d(x, y) = \sqrt{(y_1 - x_1)^2 + (y_2 - x_2)^2} \in \mathbb{Q}$.

Chapter 2: Cardinality

2.1 Bijections and Cardinality

My research is in a field of math called *combinatorics*. One central problem in combinatorics is to count sets of things. The scientific philosopher Ernst Mach went as far as to say "Mathematics may be defined as the economy of counting. There is no problem in the whole of mathematics which cannot be solved by direct counting." It's a beautiful thought,[1] but even I would not go *quite* that far; nevertheless, I do think that the best solutions in math are those that use counting.

We have already defined the number of elements in a set A to be the *cardinality* of that set. For example,

$$|\{a, b, c\}| = 3,$$

and

$$|\{1, 4, 9, 16, 25, \ldots, 100\}| = 10,$$

and

$$|\mathbb{Z}| = \infty.$$

In general, we say that two sets have the same size if there is a way to pair up the elements between the two sets. Equivalently, if there is a bijection between them. This is known as *the bijection principle*.

Principle.

Principle 2.1 (*The bijection principle*). Two sets have the same size if and only if there is a bijection between them.

Example 2.2. One reason that the sets $\{1, 2, 3\}$ and $\{a, b, c\}$ have the same size is that the elements can be paired up like this:

$$1 \leftrightarrow b \qquad 2 \leftrightarrow a \qquad 3 \leftrightarrow c$$

[1] And it's at least semi-faintly-plausible. If he has said, "There is no problem in the whole of mathematics which cannot be solved by integration by parts," now *that* would have been bold.

And one reason that $\{x, y, z\}$ and $\{m, a, t, h\}$ do not have the same size is that the elements can not be paired up. Whenever you try, one element from the second set won't get a pair. □

This next example is one I've given many times. Whenever someone asks what I do, I tell them I'm a math professor. Most then respond saying something like "oh, I was never good at math," or "I never liked math." But sometimes they follow-up and ask what sort of math I study, to which I tell them that I study combinatorics. After repeating that word slowly a few times, I usually tell them that combinatorics is the most fun math, has the most puzzle-like questions, and is the most insight- and problem-driven field. If they seem interested, I oftentimes offer to tell them a problem which I claim conveys what it *feels* like to do combinatorics. If they take the bait, I always give them the following problem.

Example 2.3. In the NCAA's March Madness Tournament there are 64 teams that play. The tournament is single-elimination (once you lose, you're out). How many games are played throughout the tournament?

(Sometimes the person will be able to realize that in the first round 32 games are played (since 64 teams pair up and play a round). I then note that this is indeed a way to solve the problem: if they apply the same reasoning they could find the number of games in the second round, the third round, and so forth, and then just add up the number of games from each round. But this, I tell them, is boring, it can't be easily generalized to other tournaments, and it doesn't give any insight into *why* the answer is what it is.

Furthermore, I and many other combinatorists do not like computational solutions. Computations are boring. Plus, computers can do them better than I can, while also committing fewer errors. And computations often hide what is really going on. I want to know how many games are played not only because the answer might be of interest, but because I want to *understand* how many games are played.

Finally, because people in the wild tend to like when math has applications, I sometimes also mention how the answer has practical value—the number of games played translates directly into the number of times the NCAA has to reserve TV time slots, how much revenue they should expect, the number of times they need to reserve a court, etc.

But usually within a minute, and sometimes within 7 seconds, I give them the following neat solution. Here it is.)

Solution. Note that in each game, precisely 1 team loses, and of course each loss happens in a game. So number of games = number of losses. Now, the champion never loses, but every other team loses one game and then they're out. So with 64 teams, 63 of them lose once, so there must be 63 losses, so there must be 63 games. □

That's it! Don't count games, count losses. There is a bijection between the games and the losses (every game has 1 loss, every loss occurs in 1 game), so counting losses is the same as counting games. And since 63 teams lose one time, it's easy to count those losses! Technically, we can view this solution as an application of the bijection principle, when we paired up each loss with the game it occurred in (and then counted the new set).

The really cool thing, though, is that this definition of the size of a set applies even to infinite sets. And that implies some truly fascinating things.

2.2 Counting Infinities

The ability to "pair up" elements between two sets is what it means for them to have the same size—this is perfectly intuitive for finite sets with nothing too counterintuitive resulting, but with infinite sets... well, some pretty neat stuff pops out. Indeed, the pluralization in this section's title was your sign of the miracles to come.

But first, here is a function-to-inequality fact that you probably learned in your intro-to-proofs class.

Fact

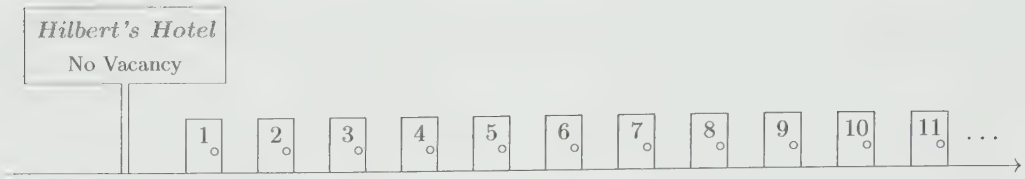
Fact 2.4. Let S and T be sets. Then,

- $|S| = |T|$ if and only if there is a bijection from S to T.
- $|S| \leq |T|$ if and only if there is an injection from S to T.
- $|S| \geq |T|$ if and only if there is a surjection from S to T.

Hilbert's Hotel

> "No other question has ever moved so profoundly the spirit of man; no other idea has so fruitfully stimulated his intellect; yet no other concept stands in greater need of clarification, than that of the infinite."
>
> – David Hilbert

We begin by talking about the set of problems related to the so-called Hilbert's Hotel. Assume that there is a hotel, called Hilbert's Hotel, which has infinitely many rooms in a row.



- Assume every room has someone in it. Then, as with most hotels the "No Vacancy" sign would be turned on and if someone else arrives at the hotel, they will not be given a room. But this isn't the case with Hibert's Hotel. If, for $n \in \mathbb{N}$, the patron in room n moves to room $n+1$, then nobody is left without a room and suddenly room 1 is completely open! So the new customer can go to room 1.[2]

- Now imagine 2 people arrived to the hotel. Can we accommodate them? Certainly! Now, just have everyone move from room n to room $n+2$. This leaves rooms 1 and 2 open to the newcomers, and we are again good-to-go.

- What if, however, we have infinitely many people lined up wanting a room. Can we accommodate *all* of them? Yes! We still can! Just have the person in room n move to room $2n$. Then all of the odd-numbered rooms are vacant and the infinite line of people can take these rooms.[3]

The first point of this exercise is to simply realize that weird stuff can happen when dealing with the infinite. The second point, though, is to realize that each time the people switched rooms, those same exact people got new rooms. So in the first example when they each just moved one room down, that should mean that there are just as many rooms from 1 to ∞ as there are from 2 to ∞... And likewise for the others.

Indeed, with this in mind, let's talk about sizes of specific sets. But first, a ditty:

∞ bottles of beer on the wall,
∞ bottles of beer.
Take one down, pass it around,
∞ bottles of beer on the wall.

(repeat)

Specific Sets

Example 2.5. There are the same number of natural numbers as there are natural numbers larger than 1 (that is, $|\mathbb{N}| = |\{2, 3, 4, \dots\}|$). What's the bijection that shows this? Let

$$f : \mathbb{N} \longrightarrow \{2, 3, 4, \dots\}$$
$$n \longmapsto n + 1.$$

[2]Make sure you take a moment to appreciate just how remarkably, wonderfully weird this is.
[3]Make sure you take a moment to appreciate just how remarkably, wonderfully weird this is.

(Note: this is notation for the function $f(x) = x + 1$ from the naturals to the set $\{2, 3, 4, \dots\}$.) In other numbers, this is the pairing

$$1 \leftrightarrow 2, \quad 2 \leftrightarrow 3, \quad 3 \leftrightarrow 4, \quad 4 \leftrightarrow 5, \ \dots$$

□

The Moral. Two sets can have the same size even though one is a *proper* subset of the other.[4]

Example 2.6. There are the same number of natural numbers as even natural numbers (that is, $|\mathbb{N}| = |2\mathbb{N}|$). What's the bijection that shows this? Let

$$\begin{aligned} f : \mathbb{N} &\longrightarrow 2\mathbb{N} \\ n &\longmapsto 2n. \end{aligned}$$

(Note: this is notation for the function $f(x) = 2x$ from the naturals to the even naturals.) In other numbers, this is the pairing

$$1 \leftrightarrow 2, \qquad 2 \leftrightarrow 4, \qquad 3 \leftrightarrow 6, \qquad 4 \leftrightarrow 8, \ \dots$$

□

The Moral. Two sets can have the same size even though one is a *proper* subset of the other and the larger one even has *infinitely* many more elements than the smaller one.[5]

Likewise:

Example 2.7. $|\mathbb{N}| = |\mathbb{Z}|$. □

The proof of this is Exercise 2.2.

Theorem.

Theorem 2.8 ($|\mathbb{Z}| = |\mathbb{Q}|$). There are the same number of integers as rational numbers.

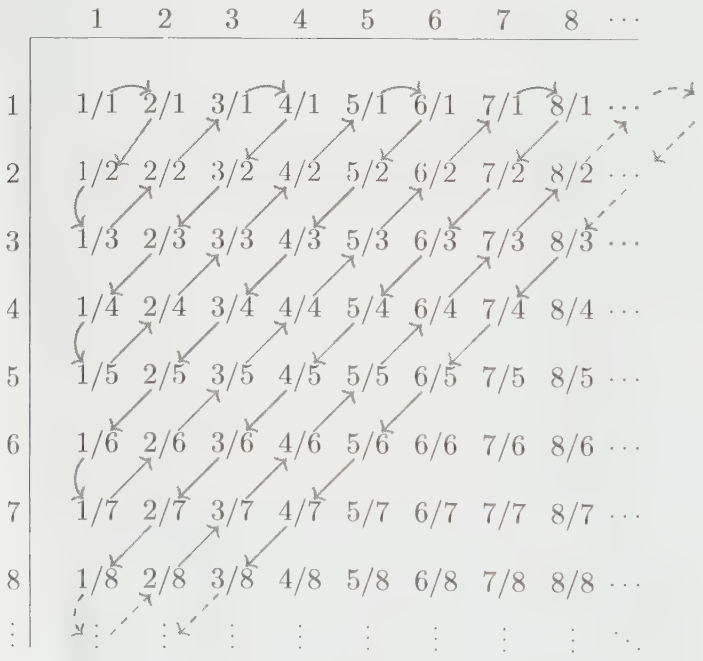
Proof. What's the bijection that shows this? This one is a little complicated... It's difficult to write down the formula, but the formula can be described in a nice way. First note that if we can show that there are the same number of positive integers as positive real rational numbers, then by simply adding a minus sign to everything, we will get a full bijection. So we may focus on just the positive case.

[4] Make sure you take a moment to appreciate just how remarkably, wonderfully weird this is.
[5] Make sure you take a moment to appreciate just how remarkably, wonderfully weird this is.

Consider the rational numbers (with some duplication) written in this way:

	1	2	3	4	5	6	7	8	⋯
1	1/1	2/1	3/1	4/1	5/1	6/1	7/1	8/1	⋯
2	1/2	2/2	3/2	4/2	5/2	6/2	7/2	8/2	⋯
3	1/3	2/3	3/3	4/3	5/3	6/3	7/3	8/3	⋯
4	1/4	2/4	3/4	4/4	5/4	6/4	7/4	8/4	⋯
5	1/5	2/5	3/5	4/5	5/5	6/5	7/5	8/5	⋯
6	1/6	2/6	3/6	4/6	5/6	6/6	7/6	8/6	⋯
7	1/7	2/7	3/7	4/7	5/7	6/7	7/7	8/7	⋯
8	1/8	2/8	3/8	4/8	5/8	6/8	7/8	8/8	⋯
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋱

Our bijection, which we call *the winding bijection*, can be pictured like this:



Weaving through this chart, you are guaranteed to hit every positive rational number. So if you pair up 1 with the first number you hit, 2 with the second number you hit, 3 with the third, and so on, then every positive rational number is in a pair. Now... there's just one small problem: each rational number is actually hit more than once. The number p/q will be written in positions $(p, q), (2p, 2q), (3p, 3q), \ldots$. But the fix is easy: When you come across a number that has already been hit, just skip it. Clearly you won't run out of rational numbers, so this does indeed pair up everything. So

$$f(n) = \text{ the } n^{\text{th}} \text{ new rational number you reach.}$$

And that's it![6] □

The Moral. Even though there are infinitely many rational numbers between *every two consecutive integers*, the two sets *still* have the same size.[7]

Now, at this point you might be tempted to predict that the reason is just that all these sets have infinitely many elements, so maybe it just comes down to all infinities being the same, or something... But amazingly that's not actually the case. Theorem 2.9, which we will get to in a second, demonstrates this. But first, to get a hold on how the proof goes, in class we will play a little game.

– A game of deduction –

Setup. Instructors: Here is a fun in-class game. Write down a set of five length-5 sequences of 0s and 1s. These are kept secret from the students.

Goal. The students' goal is to come up with a length-5 sequence of 0s and 1s which is not one of the instructor's five.

Game Rules. The students can ask the instructor any yes/no questions about the sequences, and receive a truthful reply. When the students make their final guess, they should be 100% confident that this guess is not one of the instructor's sequences. And they should try to figure it out with as few questions as possible.

One solution. One way to solve it is the following. Ask the instructor whether their first sequence has a 1 in the first position. Then ask the instructor whether their second sequence has a 1 in the second position. Then ask if the third sequence has a 1 in the third position, then whether the fourth sequence has a 1 in the fourth position, and finally whether the fifth sequence has a 1 in the fifth position.

---

[6] For the curious, one explicit bijection from $\mathbb{N}$ to $\mathbb{Q}^+$ is $h_+(n) = p_1^{(-1)^{a_1}\left\lfloor \frac{a_1+1}{2} \right\rfloor} \cdot p_2^{(-1)^{a_2}\left\lfloor \frac{a_2+1}{2} \right\rfloor} \cdots$, where p_i is the i^{th} prime and $n = p_1^{a_1} p_2^{a_2} \cdots$ is the prime factorization of n. To get a bijection h from $\mathbb{Z}$ to $\mathbb{Q}$, just define $h(0) = 0$, $h(n) = h_+(n)$ if $n > 0$, and $h(n) = -h_+(-n)$ if $n < 0$.

[7] Make sure you take a moment to appreciate just how remarkably, wonderfully weird this is.

Based on the instructor's yes/no answers, the students can deduce, for each $i \in \{1, 2, 3, 4, 5\}$, what the i^{th} number of the i^{th} sequence is.

Suppose they deduced that the first number has a 0 in the first position; the second number has a 1 in the second position; and the third, fourth and fifth numbers have a 1,0 and 0 in their third, fourth and fifth positions, respectively. Since those numbers, in order, were 0,1,1,0,0, a correct guess would be its opposite: 1,0,0,1,1. You just take everything the instructor has and flip it.

This is guaranteed to work. Consider the first sequence; you know that 1,0,0,1,1 is not the first sequence, because (by design) they differ in the first position! And you know that 1,0,0,1,1 is not the second sequence, because (by design) they differ in the second position! Indeed, each of the instructor's 5 sequences is different from 1,0,0,1,1 in one particular spot, and so you know that this sequence is different from each of the instructor's. Neat!

This clever idea is from Georg Cantor, and is known as "Cantor's diagonalization argument." It is the key idea to a beautiful proof that $|\mathbb{R}| > |\mathbb{N}|$,[8] which implies the amazing fact that there are different sizes of infinity.

Theorem.

Theorem 2.9 ($|\mathbb{R}| > |\mathbb{N}|$). There are more real numbers than natural numbers.

This implies that some infinities are bigger than others.

Proof. Since $\mathbb{N} \subset \mathbb{R}$, clearly $|\mathbb{N}| \leq |\mathbb{R}|$. To show that they are not equal, we must prove that there is no bijection between $\mathbb{R}$ and $\mathbb{N}$. Let's again use the "pairing up" idea. We will prove it by contradiction. In fact, we will prove the stronger statement that there are more real numbers in $(0, 1)$ than there are natural numbers. (This of course would prove the larger statement since then we could say $|\mathbb{R}| \geq |(0, 1)| > |\mathbb{N}|$.)

Assume for a contradiction that there does exist some way to pair up the naturals with the reals in $(0, 1)$. Writing the reals in decimal notation, assume the pairing is this:

$$
\begin{aligned}
1 &\leftrightarrow 0.a_{11}\ a_{12}\ a_{13}\ a_{14}\ a_{15}\ a_{16}\ a_{17}\ a_{18}\ \ldots \\
2 &\leftrightarrow 0.a_{21}\ a_{22}\ a_{23}\ a_{24}\ a_{25}\ a_{26}\ a_{27}\ a_{28}\ \ldots \\
3 &\leftrightarrow 0.a_{31}\ a_{32}\ a_{33}\ a_{34}\ a_{35}\ a_{36}\ a_{37}\ a_{38}\ \ldots \\
4 &\leftrightarrow 0.a_{41}\ a_{42}\ a_{43}\ a_{44}\ a_{45}\ a_{46}\ a_{47}\ a_{48}\ \ldots \\
5 &\leftrightarrow 0.a_{51}\ a_{52}\ a_{53}\ a_{54}\ a_{55}\ a_{56}\ a_{57}\ a_{58}\ \ldots \\
6 &\leftrightarrow 0.a_{61}\ a_{62}\ a_{63}\ a_{64}\ a_{65}\ a_{66}\ a_{67}\ a_{68}\ \ldots \\
7 &\leftrightarrow 0.a_{71}\ a_{72}\ a_{73}\ a_{74}\ a_{75}\ a_{76}\ a_{77}\ a_{78}\ \ldots \\
8 &\leftrightarrow 0.a_{81}\ a_{82}\ a_{83}\ a_{84}\ a_{85}\ a_{86}\ a_{87}\ a_{88}\ \ldots \\
&\vdots
\end{aligned}
$$

[8] By which we mean that the cardinality of $\mathbb{R}$ is strictly larger than the cardinality of $\mathbb{N}$.

So we are assuming that on the left of the arrows is every integer, and on the right of the arrows is every real number, and they are just paired up in some way. Now, each a_{ij} is some digit, from 0 to 9. This proof is due to Georg Cantor and his next idea is quite brilliant. He said, focus now on the "diagonal" of the above. That is, focus on the numbers of the form a_{ii}.

$$
\begin{aligned}
1 &\leftrightarrow 0.\,(a_{11})\, a_{12}\, a_{13}\, a_{14}\, a_{15}\, a_{16}\, a_{17}\, a_{18}\, \ldots \\
2 &\leftrightarrow 0.\, a_{21}\, (a_{22})\, a_{23}\, a_{24}\, a_{25}\, a_{26}\, a_{27}\, a_{28}\, \ldots \\
3 &\leftrightarrow 0.\, a_{31}\, a_{32}\, (a_{33})\, a_{34}\, a_{35}\, a_{36}\, a_{37}\, a_{38}\, \ldots \\
4 &\leftrightarrow 0.\, a_{41}\, a_{42}\, a_{43}\, (a_{44})\, a_{45}\, a_{46}\, a_{47}\, a_{48}\, \ldots \\
5 &\leftrightarrow 0.\, a_{51}\, a_{52}\, a_{53}\, a_{54}\, (a_{55})\, a_{56}\, a_{57}\, a_{58}\, \ldots \\
6 &\leftrightarrow 0.\, a_{61}\, a_{62}\, a_{63}\, a_{64}\, a_{65}\, (a_{66})\, a_{67}\, a_{68}\, \ldots \\
7 &\leftrightarrow 0.\, a_{71}\, a_{72}\, a_{73}\, a_{74}\, a_{75}\, a_{76}\, (a_{77})\, a_{78}\, \ldots \\
8 &\leftrightarrow 0.\, a_{81}\, a_{82}\, a_{83}\, a_{84}\, a_{85}\, a_{86}\, a_{87}\, (a_{88})\, \ldots \\
&\vdots
\end{aligned}
$$

All real numbers were supposed to be paired up, but we are now going to create a real number that was not in that above list, and we will do so using the idea from the Game of Deduction. The new real number will be different than the first number in its 1^{st} position, different than the second number in its 2^{nd} position, and so on. The number will have decimal expansion

$$b \;=\; 0.\, b_1\, b_2\, b_3\, b_4\, b_5\, b_6\, b_7\, b_8\, \ldots$$

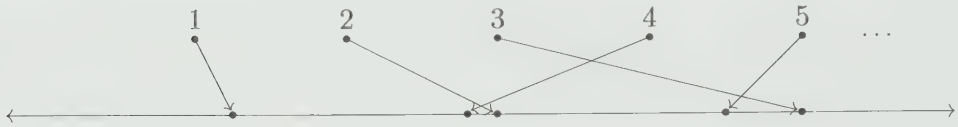
where $b_i \neq a_{ii}$ for all i. To keep it simple, let's just choose

$$b_i : \quad \begin{cases} 1 & \text{if } a_{ii} \neq 1 \\ 2 & \text{if } a_{ii} = 1 \end{cases}.$$

Then notice that, although clearly $b \in (0, 1)$, b is nowhere in our list! We know b is not the number paired up with n_1 because b and that number are different in the first position ($b_1 \neq a_{11}$). We know b is not paired up with n_2 because b and that number are different in the second position ($b_2 \neq a_{22}$). In general, we know b is not paired up with n_k because b and that number are different in the k^{th} position ($b_k \neq a_{kk}$). So this real number b is not anywhere to be found! Thus we have reached a contradiction; clearly we were unable to pair up all the reals, if b got left out.[9] □

I want to quickly mention one other proof of Theorem 2.9 that I particularly enjoy, because it is visual and uses geometry. Imagine you were trying to map $\mathbb{N}$ on to the real number line (which clearly has $|\mathbb{R}|$ points).

[9] Don't feel too bad for b, though; it's in fact the case that "most" real numbers were left out. More on that very soon.



We aim to show that it is impossible for this map to be a bijection—there must be points on the real line that were missed. But instead of mapping just these points, let's make our job slightly harder. Around 1, let's put a little interval of length $\frac{1}{2}$. Around 2, let's put a little interval of length $\frac{1}{4}$. Around 3, let's place a little interval of length $\frac{1}{8}$. And so on. Now, when you map the points in $\mathbb{N}$ to the real line, send over the intervals too (possibly some intervals will overlap; this is ok). We'll now prove that not only are there points on the real line that weren't mapped to, but there are even points that these intervals don't cover!

See what happened? Our intervals' lengths add up to $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots = 1$ (and with overlaps, their collective length when mapped to the real line may even be smaller than 1). But the whole real line has length ∞! So certainly there is no chance that all the points are covered; not only do the points of $\mathbb{N}$ not cover the line, but even if we fatten them up with these intervals, those intervals don't even cover the real line! And any point that's in the "$\infty - 1$" portion of the real line that is not covered by an interval was certainly not mapped to. So we have all sorts of points that were missed, and so the mapping is far from being a bijection.

(And we could instead pick intervals that add up to 0.0001, or any other tiny number. This proof therefore provides a visualization of how we aren't just missing the one point that Cantor's diagonalization argument finds, but "most" points are missed.)

Now here is the moral of Theorem 2.9, and its two proofs above.

The Moral. There are different sizes of infinity, and $|\mathbb{N}|$, $|\mathbb{Z}|$ and $|\mathbb{Q}|$ are all a smaller infinity than $|\mathbb{R}|$.[10]

The smallest infinity is $|\mathbb{N}|$ (Exercise 2.9), and since these are the counting numbers this infinity is called the *countable infinity*.

Definition.

Definition 2.10. If S is an infinite set, then S is *countable* if $|S| = |\mathbb{N}|$. Otherwise S is *uncountable*.

[10] Make sure you take a moment to appreciate just how remarkably, wonderfully weird this is.

Notice that A being a countable set means there is a bijection $f : \mathbb{N} \to A$ where $f(1), f(2), f(3), f(4), \ldots$ is a listing of all elements in A. Therefore an infinite set A being countable is equivalent to being able to write $A = \{a_1, a_2, a_3, a_4, \ldots\}$.

Above we demonstrated a proof of the following.

Theorem.

Theorem 2.11 (*Sizes of infinity*). There are different sizes of infinity, with countable infinity being the smallest. Moreover, $\mathbb{N}$, $\mathbb{Z}$ and $\mathbb{Q}$ are countable while $\mathbb{R}$ is uncountable.

Unprovable statements

We now know that $|\mathbb{N}| < |\mathbb{R}|$. One natural question: is there any infinity between these two? An astounding fact is that, based on the axioms of set theory (called ZFC), whether or not there exists such an infinity is *unprovable*. And what I don't mean is that mathematicians are not smart enough to find the answer; no, I mean that they are smart enough to have shown that *no proof can possibly exist*. That's right, there are statements in math which are impossible to prove and also impossible to disprove (but we *are* able to prove that they are unprovable, amazingly).

In fact, the question above is one of the most famous questions in mathematical history. It was asked by Georg Cantor and is called *the continuum hypothesis*. It is the very first of "Hilbert's problems," which were a collection of 23 problems that David Hilbert published in 1900 and presented in-part to the Paris conference of the International Congress of Mathematicians. They were selected as the most important problems for mathematicians to answer in the 20th century. It wasn't for another 40 years that Kurt Gödel proved his Earth-shaking Incompleteness Theorems—demonstrating that nearly every mathematical theory contains unprovable statements—that Cantor's question was "resolved."[11]

[11]Currently, 9 of the 23 problems have been decisively resolved (woo!), 2 have been resolved provided "it's unprovable" or "it's impossible" counts (see: Gödel's incompleteness theorem), 7 have been partially resolved (woo!), 3 remain completely unresolved (#Goals), and 2 are vaguely stated and hence hard to classify (¯_(ツ)_/¯). Many have driven research mathematics for over a century, and others (including Number 8—better known as the Riemann Hypothesis) may drive mathematical research for centuries longer.

Here is the continuum hypothesis, formally stated.

Unprovable.

Unprovable 2.12 (*The continuum hypothesis*). There is no set whose cardinality is strictly between that of the naturals and the reals.

$$|\mathbb{N}| < |S| < |\mathbb{R}|.$$

But how does one prove that a statement is unprovable? First, we have to be careful with our language here. To say something is *unprovable* means that, within the axioms of our particular theory, the statement can neither be proven nor disproven. And so, since all proofs ultimately can be traced back and deduced from just the axioms, whether something is provable relies fundamentally on the chosen axioms.

That said, we are not picking axioms at random; the fundamental axioms that nearly all of mathematics is built on give unprovable statements. Before we discuss those, though, as a smaller, more concrete example let's consider the axioms for geometry from Euclid's *Elements*:

Axiom 1. Any two points can be joined by a straight line

Axiom 2. Any line segment can be extended to form a line.

Axiom 3. For any point P and choice of radius r we can form a circle centered at P of radius r.

Axiom 4. All right angles are equal to one another.

Axiom 5 (*The parallel postulate*). If ℓ is a straight line and P is a point not on the line ℓ, then there is at most one line that passes through P and is parallel to ℓ.

From these axioms, the first theorems of geometry were rigorously proven. The first four axioms you probably agree are 100% obviously true and are fine to be assumed. However, if you are only 99% sure that the fifth axiom—called the parallel postulate—is obviously true, then you're not alone. The fact that the lines trail off to infinity have left countless mathematicians at least 1% uncomfortable, and over the last two millennia mathematicians have hoped that we could do away with the parallel postulate.

The problem is that it's an important property and many of our theorems rely on it. So we don't want to scrap it; ideally, we would like to deduce it from the other four. If the first four axioms imply the parallel postulate, then the first theorem of geometry can simply be the parallel postulate. Then all of our other theorems would still hold just the same, that 1% doubt would be removed, and our axioms would be rock solid—as they should be.

As a theorem, though, it was shown that the parallel postulate is unprovable. But how was this done? The approach (discovered in the 19th century) is to show both of these:

- The five axioms above are consistent (they don't self-contradict).
- These five axioms are also consistent: Axioms 1-4 from above, along with an axiom that says "If ℓ is a straight line and P is a point not on the line ℓ then there are at least two lines that pass through P and are parallel to ℓ.[12]

Basically, if you can show that the assumption that the parallel postulate is true leads to no contradictions (which means you can't *disprove* it, based on the first four axioms), and the assumption that the parallel postulate is false leads to no contradictions (which means you can't *prove* it, based on the first four axioms), then it must be the case that the parallel postulate is neither provable nor unprovable! It's truth value is completely independent of your axioms.

So, if your "theory" of geometry is based on just the first four axioms, then the above is an approach to show that the parallel postulate is an unprovable statement. In particular, the classic Euclidean geometry you learned in grade school is a consistent theory based on the first bullet above, and what is now called *hyperbolic geometry* is a consistent theory based on the second bullet point above (where we assume that for every line there are multiple other lines parallel to it).[13]

The continuum hypothesis was settled in this same way. As mentioned, ZFC set theory is the most fundamental collection of axioms in mathematics, which nearly all of math is based upon. And Kurt Gödel gave a consistent structure satisfying the axioms of ZFC set theory in which the continuum hypothesis was assumed to be true, while Paul Cohen gave a consistent structure satisfying the axioms of ZFC set theory in which the continuum hypothesis was assumed to be false. Combined, this proves that the continuum hypothesis is neither provable nor disprovable (in ZFC). That is, the continuum hypothesis —which asks what is presumably a basic question about the infinite—is unprovable.

[12]Think about this as: Axioms 1-4 from before, as well as a fifth axiom that directly contradicts the previous Axiom 5.

[13]By the way, when someone asks you whether this crazy theoretical math is good for anything in the real world, remind them that although mathematicians developed their theories of hyperbolic geometry simply because they were curious and thought it was fun, decades later Albert Einstein later asserted in his theory of relativity that the the fabric of spacetime more closely resembles hyperbolic space than Euclidean space, and he was quite grateful that some mathematicians decades ago had gotten curious and thought it would be fun to develop this abstract geometric theory.

2.3 How Many Infinities Are There?

We now know of two distinct infinities, but just how many infinities are there? The first theorem in this direction is the power set theorem. Recall that the power set $\mathcal{P}(A)$ of a set A is the set of all subsets of A.

Theorem.

Theorem 2.13 ($|A| < |\mathcal{P}(A)|$). If A is a set and $\mathcal{P}(A)$ is the power set of A, then

$$|A| < |\mathcal{P}(A)|.$$

Proof. Assume for a contradiction that $|A| \geq |\mathcal{P}(A)|$. That is, assume that there is a surjection f from A to $\mathcal{P}(A)$.[14] Since f is a surjection, for every $T \subseteq A$, there is some element $t \in A$ where $f(t) = T$. To reach our contradiction, we will construct a set $B \subset A$ which is not hit.

For each a there is one special property about the set $f(a)$ that we are going to care about: is $a \in f(a)$ or is $a \notin f(a)$? (In the footnote, for instance, $3 \notin f(3)$ while $4 \in f(4)$.) In general, consider the set of all elements a such that $a \notin f(a)$, and call this set B:

$$B = \{a \in A : a \notin f(a)\}.$$

By the above, if we can show that that there is no b where $f(b) = B$, then we are done; we will have discovered an element of $\mathcal{P}(A)$ that was not hit by f, a contradiction.

Claim. There is no $b \in A$ such that $f(b) = B$.

Proof of claim. Assume for a contradiction that there does exist some $b \in A$ such that $f(b) = B$. Note by the definition of B that

$$b \in B \text{ if and only if } b \notin f(b).$$

But since we assumed that $f(b) = B$, this is equivalent to

$$b \in f(b) \text{ if and only if } b \notin f(b),$$

which is clearly a contradiction. □

[14] For instance, if $A = \mathbb{N}$, then the domain of f is $\mathbb{N}$ and the the range of f is $\mathcal{P}(\mathbb{N})$, which is the collection of all subsets of $\mathbb{N}$. For example, maybe $f(3) = \{2, 5, 756\}$ and $f(4) = \{2, 4, 6, 8, 10, 12, \ldots\}$. In general we only know that, for each $a \in A$, $f(a)$ is some *subset* of A.

Now, returning to the question about how many infinities are there, we can see from the above theorem that there are infinitely many infinities!

Super Cool Corollary.

Corollary 2.14 (*There exist infinitely many infinities*). There exist infinitely many distinct infinite cardinalities.

Proof. By Theorem 2.13, the following is a chain of distinct infinite cardinalities

$$|\mathbb{N}| < |\mathcal{P}(\mathbb{N})| < |\mathcal{P}(\mathcal{P}(\mathbb{N}))| < |\mathcal{P}(\mathcal{P}(\mathcal{P}(\mathbb{N})))| < |\mathcal{P}(\mathcal{P}(\mathcal{P}(\mathcal{P}(\mathbb{N}))))| < \dots. \qquad \square$$

Ok, so we can identify infinitely many infinities. But then here's the next question: we have been talking all day[15] about how there are different infinities, so when we say there are *infinitely many* infinities, which infinity is it? By the above there are at least countably many infinities. But is there an uncountable number of infinities? More?

Similar to the continuum hypothesis, there is not even an answer to this question. In some sense, the "number" of infinities is more than *any* infinity. Why? Assume for a contradiction that A is a set which contains a set of each infinite size (a set of size $|\mathbb{N}|$, a set of size $|\mathbb{R}|$, and all the rest), and let B be the union of all of the sets in A. The contradiction comes when we consider $\mathcal{P}(B)$. By Theorem 2.13,

$$|B| < |\mathcal{P}(B)|. \qquad (✍)$$

Here's the contradiction: The cardinality $|\mathcal{P}(B)|$ can not possibly be in A. And here's why: If it were equal to $|b|$ for some $b \in A$, then since $b \subseteq B$ we would know that $|b| \leq |B|$. But combining this with (✍),

$$|b| \leq |B| < |\mathcal{P}(B)| = |b|,$$

which is impossible. $\square$

So, in fact, there are more infinities than we can imagine. There are more than infinity of them, for any of the infinities! This seems to tear at the very definitions of cardinality. Shouldn't every set have a cardinality? Well, yes. But here's the squeeze: it turns out that there was another contradiction in the above, when referring to A as "the set of all infinities." In fact, such a collection *is not even a set.*

If you find this material a little disquieting, you are not alone. When Cantor's theorems were first published a century and a half ago, many of the great mathematicians of the day responded with disgust. Henri Poincaré called it a "grave disease" infecting mathematics, Leopold Kronecker accused Cantor of being a "corrupter of

[15] *"A day that will live in infinity!"*

youth," and many Christian theologians thought his work against the notion of a unique infinite was an affront to "God's exclusive claim to supreme infinity."[16] Cantor struggled with this for decades.

On the other hand, if this material interests you, then I applaud you and encourage you to read up on *Russell's paradox*, as it is the next step down a fascinating rabbit hole (which, quite literally, is a bottomless pit of mystery). And you should know that Cantor's legacy has been fully restored. The criticisms of the past have been replaced ten times over with praise and accolades. One early defender was the great David Hilbert. Towards the end of Cantor's career he was awarded the highly-prestigious Sylvester Medal by the Royal Society for his mathematical research. Some criticized this move, but Hilbert—characteristically ahead of his time—recognized the brilliance and importance of Cantor's work, saying:

> "No one shall expel us from the paradise that Cantor has created."

[16] Or, less melodramatic:

> There was a young fellow from Trinity,
> Who took $\sqrt{\infty}$.
> But the number of digits
> Gave him the fidgets;
> He dropped Math and took up Divinity.
>
> George Gamow

— Notable Exercises —

- The fact that the rationals are both dense and countable has enormous consequences throughout real analysis. Exercise 2.5 is an example of all of this, and in particular includes ideas that return importantly in the area of real analysis called *measure theory*. If you get a PhD in math there's a good chance that measure theory will be the first topic you'll cover in your real analysis class.
- Exercise 2.6 highlights an important property of what are called *open sets*, which we will discuss in much greater detail in Chapter 5. See Theorem 5.5
- Exercise 2.9 is notable in that there are many mysteries about sizes infinites that remain either very difficult to solve, or impossible to solve. Indeed, work on the continuum hypothesis has told us that it is unprovable whether $|\mathbb{R}|$ is the second smallest infinity. Even the second smallest is unknowable! But as for the smallest infinity — *that* we do know. And this exercise determines the answer.

Exercises[17]

Exercise 2.1.

(a) Prove that if A and B are countable sets, then $A \cup B$ is also a countable set.

(b) Prove that if A_n is a countable set for each $n \in \mathbb{N}$, then $\bigcup_{n=1}^{\infty} A_n$ is also a countable set.

Exercise 2.2. Show that $|\mathbb{N}| = |\mathbb{Z}|$ by finding an explicit bijection from $\mathbb{N}$ to $\mathbb{Z}$. You do not need to prove your bijection works.

Exercise 2.3. Let $A, B \subseteq \mathbb{R}$. We define

$$A \cdot B = \{a \cdot b : a \in A \text{ and } b \in B\}.$$

(a) Give an example of sets A_1 and B_1 where $|A_1 \cdot B_1| < \max\{|A_1|, |B_1|\}$.

(b) Give an example of sets A_2 and B_2 where $|A_2 \cdot B_2| > \max\{|A_2|, |B_2|\}$.

(c) Give an example of sets A_3 and B_3 where $|A_3 \cdot B_3| = \max\{|A_3|, |B_3|\}$.

For which of the above does there exist an example where one or both of the sets are infinite?

Exercise 2.4. Is $|\mathbb{Z} \times \mathbb{N}|$ countable or uncountable?

Exercise 2.5. Let S be the set of sequences (a_n) where, for each n, $a_n \in \{0, 1\}$. Is S countable or uncountable?

Exercise 2.6.

(a) Give an example of a collection of countably many disjoint open intervals, or prove that this does not exist.

(b) Give an example of a collection of uncountably many disjoint open intervals, or prove that this does not exist.

Exercise 2.7. Show that there are uncountably many irrational numbers.

Exercise 2.8. Let $\mathcal{F}$ be the collection of all functions $f : \mathbb{R} \to \mathbb{R}$. Prove that $\mathcal{F}$ is uncountable.

Exercise 2.9. Show that the smallest infinity is $|\mathbb{N}|$. That is, show that if $A \subseteq \mathbb{N}$, then either A is finite or $|A| = |\mathbb{N}|$.

Exercise 2.10. Prove that the set of all finite subsets of $\mathbb{N}$ is countable.

[17] *Mo' chapters, mo' problems*

Exercise 2.11. Show that $|\mathcal{P}(\mathbb{N})| = |\mathbb{R}|$ by finding an explicit bijection from $\mathcal{P}(\mathbb{N})$ to $\mathbb{R}$. You do not need to prove your bijection works.

Exercise 2.12. Prove that the set of points on the unit circle in $\mathbb{R}^2$ (that is, $\{(x, y) : x^2 + y^2 = 1\}$) is uncountable.

Exercise 2.13. A real number x is said to be *algebraic* (over the rationals) if it satisfies some polynomial equation (of positive degree)

$$a_n x^n + a_{n-1} x^{n-1} + a_{n-2} x^{n-2} + \cdots + a_1 x + a_0 = 0$$

where each $a_i \in \mathbb{Q}$. If a real number is not algebraic then it is *transcendental*.

(a) Prove that there are countably many algebraic numbers. Hint: You may use the fundamental theorem of algebra which says that a polynomial with degree n has at most n real roots.

(b) Prove that there are uncountably many transcendental numbers.

— Open Question —

Question. If we assume ZFC and also that $2^{\aleph_n} < \aleph_\omega$ for every n, what restrictions can we prove on $2^{\aleph_\omega}$?

Chapter 3: Sequences

3.1 Basic Sequence Definitions

The next major topic is that of sequences. You probably intuitively think of a sequence as an infinite list of numbers, like

$$1 \ , \ 4 \ , \ 9 \ , \ 16 \ , \ 25 \ , \ \ldots$$

This is a good way to think about them, but formally a sequence is defined to be a function.[1] If you think about mapping 1 to 1, 2 to 4, 3 to 9, and, in general, n to the n^{th} number in the list, then we get the picture:

$$\begin{array}{ccccccccccc} 1 & , & 2 & , & 3 & , & 4 & , & 5 & , & \ldots \\ \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \\ 1 & , & 4 & , & 9 & , & 16 & , & 25 & , & \ldots \end{array}$$

This gives us the formal definition of a sequence.

Definition.

Definition 3.1. A *sequence* of real numbers is a function $a : \mathbb{N} \to \mathbb{R}$.

So in the above example we would have

$$a(1) = 1, \quad a(2) = 4, \quad a(3) = 9, \quad a(4) = 16, \quad a(5) = 25, \ldots$$

and in general $a(n) = n^2$. As you have probably seen, it is more common to write this using subscript notation:

$$a_1 = 1, \quad a_2 = 4, \quad a_3 = 9, \quad a_4 = 16, \quad a_5 = 25, \quad \ldots.$$

[1] From a theoretical standpoint is good, as it means we do not need to formally develop a new mathematical object in order to discuss them.

We will also often write sequences like this, just don't forget that a sequence is really a function from the $\mathbb{N}$ to $\mathbb{R}$ where $a(n)$ equals the n^{th} number in the sequence.

Note.

Note 3.2.

- As mentioned, oftentimes we write a_n instead of $a(n)$.
- Sometimes we write $(a_n)_{n=1}^{\infty}$ or (a_n) or $(a_1, a_2, a_3, \dots)$ or $a_1, a_2, a_3, \dots$ to denote the sequence $a(n)$.
- Sometimes we will give an explicit formula for a_n, like

$$a_n = n^2$$

while other times we may give it a recursive definition, like with the Fibonacci sequence:

$$a_n = a_{n-1} + a_{n-2}$$

with the initial conditions of $a_1 = 1$ and $a_2 = 1$. But of course, *any* function from $\mathbb{N}$ to $\mathbb{R}$ gives a sequence, not just ones with nice formulas. For example we could have

$$\left(14.238, 7, \pi, -\sqrt{2}, e, 0, e^{\pi}, \dots\right).$$

Example 3.3.

(a) Write the sequence $a(n) = 2^{n-1}$ in the form $(a_1, a_2, a_3, \dots)$

(b) For the sequence

$$(3, 6, 11, 18, 27, \dots),$$

determine a formula for a_n.

(I stuck the solution in a footnote, in case you want to try it without accidentally glancing at the solution.[2])

3.2 Bounded Sequences

As is often the case in math, we want to describe how "nice" something is. In calculus, polynomials were quite nice because you knew how to differentiate and integrate them. Oftentimes you could even factor them. Exponential functions like $f(x) = 2^x$ were pretty nice, and logs weren't soooo bad. Likewise, we want to describe what it means for a sequence to be "nice."

We start with the notion of a sequence being *bounded*, which is one "nice" property.

[2] **Solution.** (a) $(1, 2, 4, 8, 16, \dots)$. (b) One answer: $a_n = n^2 + 2$. □

Definition.

Definition 3.4. A sequence (a_n) is *bounded* if the range $\{a_n : n \in \mathbb{N}\}$ is bounded. I.e., if there exists a lower bound $L \in \mathbb{R}$ and an upper bound $U \in \mathbb{R}$ where

$$L \leq a_n \leq U$$

for all n.

Here's a fact that follows quickly from definitions.

Proposition.

Proposition 3.5 (*Bounded* $\Leftrightarrow |a_n| \leq C$)**.** A sequence (a_n) is bounded if and only if there exists some $C \in \mathbb{R}$ for which $|a_n| \leq C$ for all n.

Proof. Recall again that boundedness means that there exists a lower bound $L \in \mathbb{R}$ and an upper bound $U \in \mathbb{R}$ where

$$L \leq a_n \leq U$$

for all n. Now let's prove each direction.

First we will prove the backwards implication. Assume that there exists a $C \in \mathbb{R}$ where

$$|a_n| \leq C.$$

Then

$$-C \leq a_n \leq C.$$

And so, by setting $L = -C$ and $U = C$ we have shown that

$$L \leq a_n \leq U,$$

which means that a_n is bounded. This concludes the backwards direction.

Now we will prove the forwards implication. If a_n is bounded, then there exists such an L and U. Let $C = \max\{|L|, U\}$.[3] Note that this implies that $C \geq U$ and (since $C \geq |L|$, that) $-C \leq -|L|$. Thus, for all n we have

$$-C \leq -|L| \leq L \leq a_n \leq U \leq C.$$

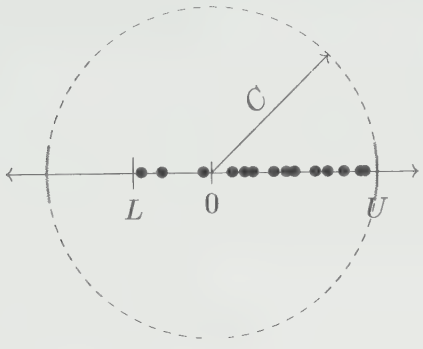
So we see that

$$-C \leq a_n \leq C,$$

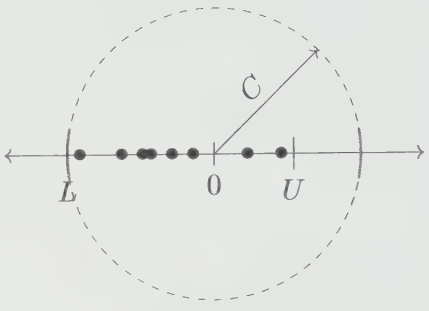
which by Proposition 1.12 is the same as $|a_n| \leq C$. □

[3] See pictures after this example for intuition on where this came from.

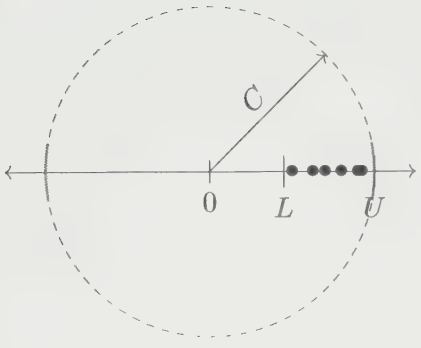
One way to think about the forward direction is to note that what "$|a_n| \leq C$ for all n" means is that you can draw a circle of radius C, center at 0, and catch all of the a_n points inside your circle. Below are examples of this.



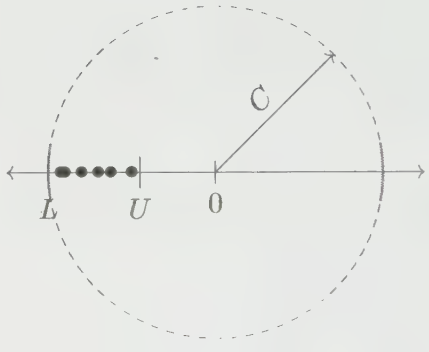
Here, $C = U$



Here, $C = |L|$



Here, $C = U$



Here, $C = |L|$

Example 3.6. Here are a couple examples.

- Let $a_n = (-1)^n \frac{1}{n}$. Then by Proposition 3.5, the sequence

$$(a_n) = (-1, \frac{1}{2}, -\frac{1}{3}, \frac{1}{4}, -\frac{1}{5}, \dots)$$

  is bounded because $|a_n| \leq 1$ for all n.

- Let $a_n = 2 + \sin(n)$. Then the sequence

$$(a_n) = (2 + \sin(1), 2 + \sin(2), 2 + \sin(3), 2 + \sin(4), \dots)$$

  is bounded. To see this, recall that $-1 \leq \sin(n) \leq 1$ for all n, and so $1 \leq 2 + \sin(n) \leq 3$ for all n. At this point we are done by the definition of boundedness. Or we could note that $|a_n| \leq 3$ for all n, and so we are done by Proposition 3.5. □

3.3 Convergent Sequences

"The sequence $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \ldots$ converges to 0" is a true statement, and after taking calculus you probably already have a pretty good understanding of what that statement means. But this is an advanced math course, so we need to be precise about what *convergence* means. The following provides that precision.

Definition.

Definition 3.7. A sequence (a_n) *converges* to $a \in \mathbb{R}$ if for all $\varepsilon > 0$ there exists[4] some N such that $|a_n - a| < \varepsilon$ for all $n > N$.

When this happens, a is called the *limit* of (a_n).

This is an important definition. This is the first (of many) weird-looking definition(s) in this course, and at this stage it is *really* important that you *fully* understand it. So brew a pot of tea, find a comfortable Thinking Chair, and ponder it carefully; don't move on until that pot is emptied. The better you understand it now, the easier the rest of the course will be. Here are a few more comments to help:

Comment 1. The final condition that we are aiming to reach is that $|a_n - a| < \varepsilon$ for appropriate n. Note that this is the same as

$$-\varepsilon < a_n - a < \varepsilon,$$

which is equivalent to

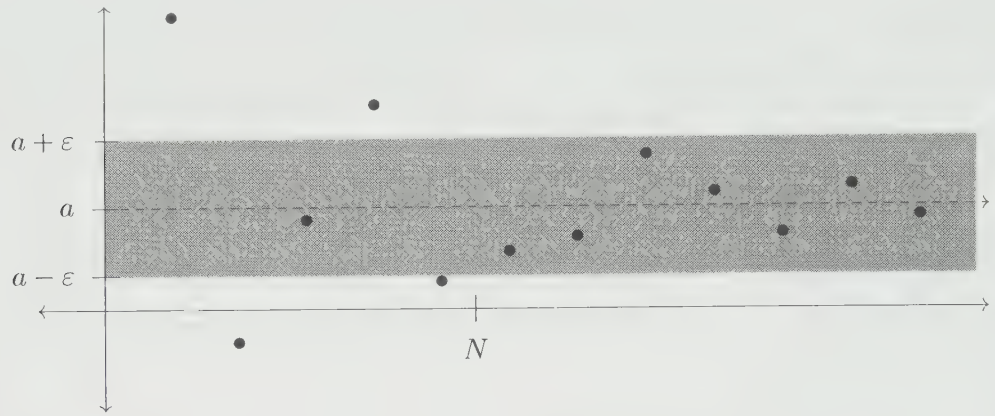
$$a - \varepsilon < a_n < a + \varepsilon.$$

So our sequence has to live inside of this interval after some point N. And we think about ε as being really small. For example, if $a = 6$ and $\varepsilon = 0.01$, then after some point N, we know that *every* term of the sequence satisfies

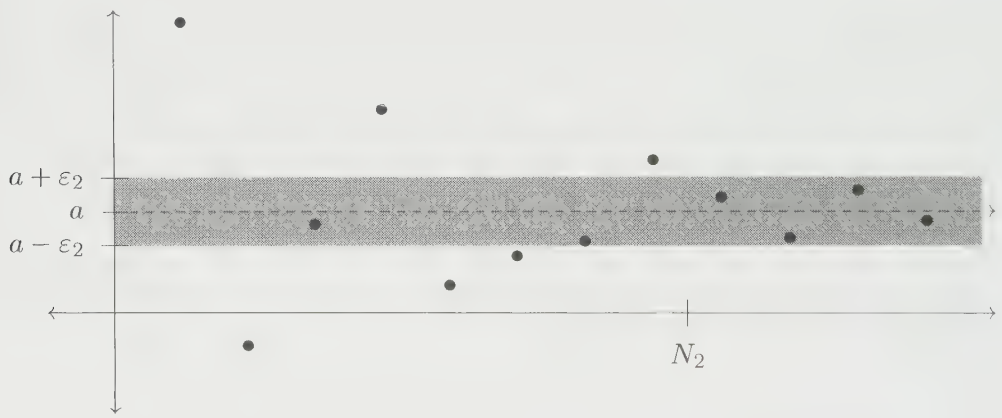
$$6 - 0.01 < a_n < 6 + 0.01$$
$$5.99 < a_n < 6.01.$$

[4]Note: When more convenient, you may assume that $N \in \mathbb{N}$.

Comment 2. Below is the important picture to keep in your head. Note that after the N, all the points are inside the shaded region[5] between $a - \varepsilon$ and $a + \varepsilon$.



Now, if you picked a *smaller* value of ε, which we will call ε_2, you will need a *bigger* value of N (called N_2) to guarantee that all the points after N_2 are inside the shaded region.



So you see, smaller values of ε typically require larger values of N.

Comment 3. You can think about the converging sequence as the tragic story of a rebellious youth. When young, he may be all over the place on any given day, living his life and getting into mischief. But eventually society settles him down. When $\varepsilon = 100$ (miles) he has a car, and with his gang of misfits he can roam wherever he pleases, causing mayhem. But when $\varepsilon = 10$ (miles), the value of N_1 represents the age at which he is placed on at-risk watch and is not allowed to leave the city (maybe $N_1 = 17$ years old). Then $\varepsilon = 0.1$ (miles) represents the age N_2 at which he really gets into trouble and is placed under house arrest. And $\varepsilon = 0.01$ represents the age N_3 at which he is sentenced to life in prison and will forever more be contained within a small wing of prison.[6]

[5] i.e. the shaded region, for those reading in black and white

[6] When I started this paragraph I did not anticipate it getting so dark. I even dropped the solitary confinement stage (or beyond) for that reason. Poor guy. Stay in school, kids.

Comment 4. What the definition means is that eventually the points get "arbitrarily close" to a. Maybe the first few points $(a_1, a_2, \ldots, a_{N-1})$ are not super close to a, but eventually they both get super close to a and stay super close to a.

And we really do mean *super* close, as it has to hold for *all* $\varepsilon > 0$. There is an N for which all the points from a_N onward are within 0.1 of a, but there's also some other N_2 for which all the points from a_{N_2} onward are within 0.0000000000000001 of a. *Any* positive value of ε has to work.

Comment 5. If you haven't spent considerable time thinking about this, or if you do not yet understand it *perfectly*, go brew another pot and settle in. You'll thank me later.

The final comment is about notation:

Notation.

Notation 3.8. The following are notationally equivalent.

- The sequence (a_n) *converges* to a
- $a_n \to a$ as $n \to \infty$
- $a_n \to a$
- $\lim_{n\to\infty} a_n = a$

Example 3.9. Show that the sequence

$$(a_n) = \left(1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \ldots\right)$$

converges to 0.

Solution 1. Fix any $\varepsilon > 0$. By the Archimedean property, there is an N for which $\frac{1}{N} < \varepsilon$. This N has the property that, for all $n > N$, we have $\frac{1}{n} < \frac{1}{N} < \varepsilon$. That is, for all $n > N$,

$$\left|\frac{1}{n} - 0\right| = \frac{1}{n} < \varepsilon.$$

And so, by the definition of sequence convergence (Definition 3.7), we may conclude that $a_n \to 0$. □

Using the Archimedean property will not usually work, though. We need a more general approach. The fundamental idea behind Solution 2 (and most future solutions to this type of question) can be illustrated with the following.

Fingers-on-the-nose game. This is tough to type out, but here goes nothing.[7] Can you find a way to do this: Stick your left arm forward; stick your right arm forward, crossing your left so that your right wrist is in top of your left wrist, hands palm-to-palm; interlock your fingers; move your hands so that your pinkies are against your chest; stick just your two pointer fingers up so that your left pointer finger's fingernail is against the left side of your nose, and your right pointer finger's fingernail is against the right side of your nose. Now: Unwind your hands without your pointer fingers leaving your nose.

The game's lesson: If you don't know how to begin, then oftentimes by starting at the end one can unwind the whole problem and discover the key for how to begin.[8]

Now here is the outline.

Outline

Outline 3.10. To show that $a_n \to a$, begin with preliminary work:

0. Scratch work: Start with $|a_n - a| < \varepsilon$ and unravel to solve for n. This tells you which N to pick for step 2 below.

Now for your actual proof:

1. Let $\varepsilon > 0$.
2. Let N be the final value of n you got in your scratch work, and let $n > N$.
3. Redo scratch work (without ε's), but at the end use N to show that $|a_n - a| < \varepsilon$.

[7] Find me at a conference if you're confused and I'll show you it. It's fun.

[8] From *Alice in Wonderland*: "'Begin at the beginning,' the King said gravely, 'and go on till you come to the end: then stop.'" This is good advice, provided you can find your way from the beginning to the end. But if you find yourself going in loops without reaching the end, instead try to find your way from the end to the beginning. Sometimes that's much easier, just ask an 8 year old trying to solve one of those pencil maze puzzles.[9]

[9] Fun Fact: Essentially all[10] of those mazes can be solved by simply following the wall on your right wherever it goes.

[10] Technical condition is that it must be "simply connected." If it fails, though, then you may have to start over.[8]

Example 3.9, again. Show that the sequence

$$(a_n) = \left(1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \dots\right)$$

converges to 0.

We follow Outline 3.10.

Scratch Work for Solution 2. This second solution is how most of our solutions will go, and it will use Outline 3.10. Given an arbitrary $\varepsilon > 0$, we will find what specific N guarantees that, for every $n > N$, we have $|a_n - 0| < \varepsilon$. For example, if $\varepsilon = \frac{1}{2}$, then $N = 2$ works. If $\varepsilon = \frac{1}{3}$, then $N = 3$ works. You see the pattern, but here is how we might come about it in general:

We want the following:

$$\begin{aligned} |a_n - a| &< \varepsilon \\ \left|\frac{1}{n} - 0\right| &< \varepsilon \\ \frac{1}{n} &< \varepsilon \\ \frac{1}{\varepsilon} &< n. \end{aligned}$$

So as long as we choose $N = \dfrac{1}{\varepsilon}$, then for any $n > N$ we will have $n > \dfrac{1}{\varepsilon}$, which by the above will imply that $\frac{1}{n} \to 0$, as desired. The solution below is how we formally solve it.

(As you saw, in the above we essentially did the important steps of Outline 3.10 in reverse order. We started with Step 3, undoing a bunch of algebra to find an N that will work for Step 2.)

Solution 2. Fix any $\varepsilon > 0$. Set $N = \frac{1}{\varepsilon}$. Then for any $n > N$ (implying $\frac{1}{n} < \frac{1}{N}$),

$$|a_n - a| = \left|\frac{1}{n} - 0\right| = \frac{1}{n} < \frac{1}{N} = \frac{1}{1/\varepsilon} = \varepsilon.$$

That is, $|a_n - a| < \varepsilon$. So by Definition 3.7 we have shown that $\dfrac{1}{n} \to 0$. □

Example 3.11. Let $a_n = 5 - \dfrac{1}{n^2}$. Show that $a_n \to 5$ as $n \to \infty$.

We follow Outline 3.10.

Scratch Work. Again, we first play around. We start with where we want to get to (that $|a_n - a| < \varepsilon$), and then do some algebra to figure out which values of N would give this.

We want the following:

$$\begin{aligned} |a_n - a| &< \varepsilon \\ \left|\left(5 - \frac{1}{n^2}\right) - 5\right| &< \varepsilon \\ \left|-\frac{1}{n^2}\right| &< \varepsilon \\ \frac{1}{n^2} &< \varepsilon \\ \frac{1}{\varepsilon} &< n^2 \\ \frac{1}{\sqrt{\varepsilon}} &< n. \end{aligned}$$

So as long as we choose $N = \dfrac{1}{\sqrt{\varepsilon}}$, then for any $n > N$ we will have $n > \dfrac{1}{\sqrt{\varepsilon}}$, which by the above will imply that $5 - \frac{1}{n^2} \to 5$, as desired. But here's the formal solution:

Solution. Fix any $\varepsilon > 0$. Set $N = \frac{1}{\sqrt{\varepsilon}}$. Then for any $n > N$,

$$|a_n - a| = \left|\left(5 - \frac{1}{n^2}\right) - 5\right| = \frac{1}{n^2} < \frac{1}{N^2} = \frac{1}{1/(\sqrt{\varepsilon})^2} = \frac{1}{1/\varepsilon} = \varepsilon.$$

That is, $|a_n - a| < \varepsilon$. So by Definition 3.7 we have shown that $5 - \dfrac{1}{n^2} \to 5$. □

This next one looks a bit trickier, but the same procedure works.

Example 3.12. Let $a_n = \dfrac{3n+1}{n+2}$. Prove that $\lim\limits_{n\to\infty} a_n = 3$.

Scratch Work. Again, we first play around. We start with where we want to get to (that $|a_n - a| < \varepsilon$), and then do some algebra to figure out which values of n would give this.

We want the following:

$$\begin{aligned}
|a_n - a| &< \varepsilon \\
\left|\frac{3n+1}{n+2} - 3\right| &< \varepsilon \\
\left|\frac{3n+1}{n+2} - \frac{3(n+2)}{n+2}\right| &< \varepsilon \\
\left|\frac{3n+1-3n-6}{n+2}\right| &< \varepsilon \\
\left|\frac{-5}{n+2}\right| &< \varepsilon \\
\frac{5}{n+2} &< \varepsilon \\
\frac{5}{\varepsilon} &< n+2 \\
\frac{5}{\varepsilon} - 2 &< n
\end{aligned}$$

So as long as we choose $N = \dfrac{5}{\varepsilon} - 2$, then for any $n > N$ we will have $n > \dfrac{5}{\varepsilon} - 2$, which by the above will imply that $\frac{3n-1}{n+2} \to 3$, as desired.

Solution. Fix any $\varepsilon > 0$. Set $N = \frac{5}{\varepsilon} - 2$. Then for any $n > N$,

$$\begin{aligned}
|a_n - a| &= \left|\frac{3n+1}{n+2} - 3\right| = \left|\frac{3n+1}{n+2} - \frac{3n+6}{n+2}\right| \\
&= \frac{5}{n+2} < \frac{5}{N+2} = \frac{5}{\left(\frac{5}{\varepsilon} - 2\right) + 2} \\
&= \frac{5}{5/\varepsilon} = \varepsilon.
\end{aligned}$$

That is, $|a_n - a| < \varepsilon$. So by Definition 3.7 we have shown that $\dfrac{3n+1}{n+2} \to 3$. □

Example 3.13. Let $a_n = 5$. Prove that $a_n \to 5$.

Scratch Work. Since a_n is the constant function 5, this means that $a_1 = 5$, $a_2 = 5$, $a_3 = 5$, etc. That is, this is the sequence $5, 5, 5, 5, 5, \ldots$. It makes perfect sense that $a_n \to 5$, but how do we show it? To use Definition 3.7, we need, for each $\varepsilon > 0$, to find an N for which, for every $n > N$, we have $|a_n - 5| < \varepsilon$. Sometimes in math the "easiest" problems are the hardest to work out, because some of the usual steps are missing... We want to figure out for which n we have $|a_n - a| < \varepsilon$. But:

$$\begin{aligned} |a_n - a| &< \varepsilon \\ |5 - 5| &< \varepsilon \\ 0 &< \varepsilon. \end{aligned}$$

And this is always true, since we chose $\varepsilon > 0$! So if we choose any N at all, like $N = 1$ or $N = \pi$ or $N = 17$, we are guaranteed for it to work out.

Solution. Fix any $\varepsilon > 0$. Set $N = 17$. Then for any $n > N$,

$$|a_n - a| = |5 - 5| = 0 < \varepsilon.$$

That is, $|a_n - a| < \varepsilon$. So by Definition 3.7 we have shown that $5 \to 5$. □

ε-neighborhoods

Related to all this is the notion of an *ε-neighborhood* of a point a.

Definition

Definition 3.14. Let $\varepsilon > 0$. The *ε-neighborhood* of a point a is the interval

$$(a - \varepsilon, a + \varepsilon).$$

With this, one could rephrase the definition of convergence like so:

Definition

Definition 3.7. A sequence (a_n) *converges* to $a \in \mathbb{R}$ if for all $\varepsilon > 0$ there exists[11] some N such that a_n is in the ε-neighborhood of a for all $n > N$.

On page 64, an ε-neighborhood is drawn as a shaded horizontal band. Following that figure is a second figure where an ε_2-neighborhood is shaded.

[11] Note: When more convenient, you may assume that $N \in \mathbb{N}$.

3.4 Divergent Sequences

Of course, not all sequences converge.

Definition.

Definition 3.15. If a sequence (a_n) does not converge, then it *diverges*.

Divergence can come in three forms.

- (a_n) *diverges to* ∞ $\left(\text{notation: } \lim_{n\to\infty} a_n = \infty\right)$ if, for all $M > 0$, there exists[12] some N such that $a_n > M$ for all $n > N$.
- (a_n) *diverges to* $-\infty$ $\left(\text{notation: } \lim_{n\to\infty} a_n = -\infty\right)$ if, for all $M < 0$, there exists[12] some N such that $a_n < M$ for all $n > N$.
- Otherwise, (a_n)'s limit *does not exist*.

Typically the first and second types are more straightforward to show, so we will start with an example of one of those.

Example 3.16. Let $a_n = n^2$. Show that $\lim_{n\to\infty} a_n = \infty$.

Scratch Work. We want

$$\begin{aligned} a_n &> M \\ n^2 &> M \\ n &> \sqrt{M}. \end{aligned}$$

So setting $N = \sqrt{M}$ should work.

Solution. Fix any $M > 0$. Set $N = \sqrt{M}$. Then for any $n > N$,

$$a_n = n^2 > N^2 = \left(\sqrt{M}\right)^2 = M.$$

So we have shown that if $n > N$ then $a_n > M$. Therefore $\lim_{n\to\infty} a_n = \infty$. □

What if the sequence's limit does not exist? Then how do we show a sequence diverges?

[12] Note: When more convenient, you may assume that $N \in \mathbb{N}$.

[12] See Footnote 12.

Note.

Note 3.17. One way to show that a_n diverges is to show that $a_n \not\to a$ for any a. Note first, by Definition 3.7, that "$a_n \to a$" means that

- For every $\varepsilon > 0$ there exists some N where for all $n > N$ we have $|a_n - a| < \varepsilon$.

So to show that $a_n \not\to a$, we need to show the *negation* of that statement. That is, we must show that

- There exists some $\varepsilon > 0$ where for all N there exists some $n > N$ such that $|a_n - a| \geq \varepsilon$.

Note: In practice this is usually done with a proof by contradiction. You assume that $a_n \to a$ and then you demonstrate a specific ε where it fails, giving the contradiction.

Make sure this makes sense to you. How do you show that something does not happen for all $\varepsilon > 0$? Answer: You find one ε for which it fails. It's like saying "Every university in America has a student that can dunk a basketball." How would you show that that's not true? All you must do is find a single university where no student can dunk. [13] You don't have to check them all—as soon as you find one that has no student that can dunk, you're done. Conversely, to disprove a "there exists" statement, you do have to demonstrate that all of the possibilities fail.

Now here's an example of showing a sequence diverges (of the "does not exist" variety).

Example 3.18. Let $a_n = (-1)^n$. Prove that (a_n) diverges.

Scratch Work. This is the sequence $-1, 1, -1, 1, -1, 1, \ldots$ It makes sense that there is no a for which $a_n \to a$. It certainly doesn't converge to 1, since half the time it is at -1 which is far away from 1. It likewise can't converge to -1. One might guess 0, since that is halfway between -1 and 1, but that also doesn't make sense since a_n is always of distance 1 from 0, so it's certainly not getting "closer and closer" to 0.

Ok, so we believe that it doesn't converge to anything. We will use Note 3.17 to show that this sequence does not converge to anything. We will use a proof by contradiction. We will assume there is some a for which $a_n \to a$, but then we will follow Note 3.17: we will find a specific ε for which, for every N, there exists some larger n where a_n is *not* within ε of a. In particular, note that if we choose $\varepsilon = \frac{1}{2}$, then whatever we choose for a the sequence will be farther than $\frac{1}{2}$ from a either half of the time or all of the time.

[13] But once you've found such a university, you do have to check all the students, since the negations of "for all universities there exists a student who can dunk" is "there exists a university in which for all students who attend it, they are unable to dunk."

Solution 1. Assume for a contradiction that there is some a for which $a_n \to a$. Let $\varepsilon = \frac{1}{2}$. Since we assumed that $a_n \to a$, there must be some N for which, for all $n > N$, we have $|a_n - a| < \frac{1}{2}$. That is, $|(-1)^n - a| < \frac{1}{2}$ for all $n > N$. We proceed by cases.

Even n. If n is even and $n > N$, then we have that

$$|1 - a| < \frac{1}{2}.$$

Unwinding this:

$$\begin{aligned} -\frac{1}{2} < 1 - a < \frac{1}{2} \\ -\frac{3}{2} < -a < -\frac{1}{2} \\ \frac{1}{2} < a < \frac{3}{2}. \end{aligned}$$

Odd n. If n is odd and $n > N$, then we have that

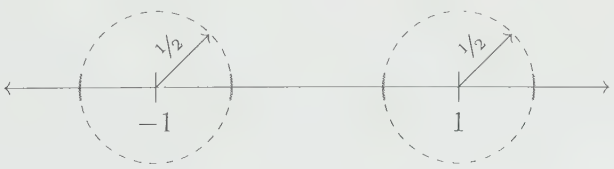
$$|-1 - a| < \frac{1}{2}.$$

Unwinding this:

$$\begin{aligned} -\frac{1}{2} < -1 - a < \frac{1}{2} \\ \frac{1}{2} < -a < \frac{3}{2} \\ -\frac{3}{2} < a < -\frac{1}{2}. \end{aligned}$$

But this is a contradiction; clearly no a can be inside of both $(\frac{1}{2}, \frac{3}{2})$ and $(-\frac{3}{2}, -\frac{1}{2})$. And so we must have $a_n \not\to a$. □

Essentially, focusing on the even case created a ball around 1, of radius $\frac{1}{2}$, which a_n would have to live within for all $n > N$. The odd case created a ball around -1, of radius $\frac{1}{2}$, which a_n would have to live within for all $n > N$. But these two balls are disjoint, so a_n can't live in both, creating a contradiction.



Now here's a second proof of, which is mostly black magic (AKA the triangle inequality).[14]

Solution 2. Again, let $\varepsilon = \frac{1}{2}$. The even case still gives $|1 - a| < \frac{1}{2}$, and the odd case still gives $|-1 - a| < \frac{1}{2}$; although in the odd case we will rewrite $|-1 - a|$ as $|1 + a|$. Then, by the triangle inequality

$$2 = |(1 - a) + (1 + a)| \leq |1 - a| + |1 + a|.$$

But each of the absolute values on the right are assumed to be less than $\frac{1}{2}$. So,

$$2 = |(1 - a) + (1 + a)| \leq |1 - a| + |1 + a| < \frac{1}{2} + \frac{1}{2} = 1.$$

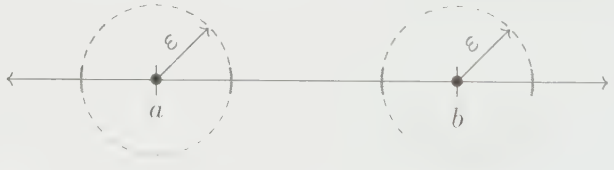
So $2 < 1$? Preposterous! We have our contradiction. □

Next is something that you probably already assumed was true and wouldn't have thought we needed to prove, but when building a theory we shouldn't assume anything.

Proposition.

Proposition 3.19 (*Limits are unique*). A sequence can not have more than one limit.

Scratch Work. The idea in the solution of Example 3.18 should be able to be tweaked to work here. The idea is to assume that you have a sequence a_n that converges to some a and also converges to some other number b. Since it converges to a, after some point all the sequence points are within ε of a, for some really tiny ε. But since it also converges to b, the same thing must happen there too. So if ε is small enough that those two regions are mutually exclusive, we will have a contradiction.



And this is how we will reach a contradiction: there is no way for a_n $(n > N)$ to be in both circles at the same time. Of course, if a and b are really close together we

[14] Ignore this if you like to see magic tricks without knowing how they're done: Intuitively, what the following proof is actually saying is that if $(-1)^n \to a$ (where a is some fixed number), then eventually all the points of the sequence must be within $\varepsilon = 0.5$ of whatever a is. However, the distance from -1 to 1 is equal to 2, and so the distance from -1 to a, plus the distance from a to 1, should be at least 2. But since $\varepsilon = 0.5$, those two latter distances are too small, giving a contradiction.

will need to choose a smaller ε to make sure the intervals remain disjoint, but that's the only difference.[15]

For instance, if the radius of those balls is a third of the distance between a and b, it should work out. With that intuition in mind, here's the proof.

Proof. Suppose for a contradiction that $a_n \to a$ and $a_n \to b$, where $a \neq b$; moreover, without loss of generality let's assume $a < b$. Let $\varepsilon = (b-a)/3 > 0$. Since $a_n \to a$ there exists some N_1 such that for $n > N_1$ we have $|a_n - a| < (b-a)/3$. Likewise, since $a_n \to b$ there exists some N_2 such that for $n > N_2$ we have $|a_n - b| < (b-a)/3$. Let $N = \max\{N_1, N_2\}$. Then for $n > N$ (implying $n > N_1$ and $n > N_2$), we have both $|a_n - a| < (b-a)/3$ and $|a_n - b| < (b-a)/3$. That is, for such n we have

$$-\frac{b-a}{3} < a_n - a < \frac{b-a}{3} \qquad \text{and} \qquad -\frac{b-a}{3} < a_n - b < \frac{b-a}{3}.$$

i.e.,

$$a - \frac{b-a}{3} < a_n < a + \frac{b-a}{3} \qquad \text{and} \qquad b - \frac{b-a}{3} < a_n < b + \frac{b-a}{3}.$$

In particular,

$$a_n < \frac{b+2a}{3} \qquad \text{and} \qquad \frac{2b+a}{3} < a_n.$$

But since $a < b$, we must have $b + 2a < 2b + a$.[16] But this then implies that

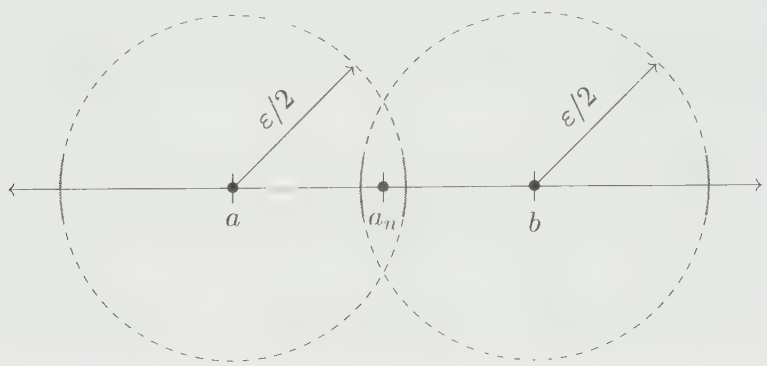
$$a_n < \frac{b+2a}{3} < \frac{2b+a}{3} < a_n,$$

which is clearly a contradiction. □

[15] By now it has probably been ingrained in you that ε is always a small number; this will probably follow you throughout your mathematical career. Paul Erdős was one of the most interesting mathematicians in history. He published over 1,500 papers in his life—more than any mathematician in history—and yet never held a permanent job. He didn't want one. In an age before air travel was common, he would constantly be bouncing between universities and the homes of collaborators. He also had an interesting vocabulary. Alcohol was called "poison." Women were "bosses," and when a man marries a woman he has been "captured." People who retired from research mathematics had "died" (this one caused some terrible miscommunications), while those who actually did die had "left." And me telling you all that is just lead-up to what he called kids. He loved kids, but because they are small, he called them "epsilons."

[16] Since b is larger than a, if you have 2 b's and 1 a, that's larger than if you have 2 a's and 1 b. Alternatively, just add "$a + b$" to both sides of "$a < b$."

Scratch work for second proof. There is a second proof that is even a little shorter. The idea is to show that $|a - b| < \varepsilon$ for all $\varepsilon > 0$, and therefore $|a - b| = 0$ (officially, this is because of Exercise 1.9), implying that $a = b$. Intuitively, the way to show this is to say that if a_n is getting really close to both a and b, then that forces a and b to be close to each other. If we demand that a_n is within $\varepsilon/2$ of both a and b, then the distance between a and b can't be more than ε:



Using the triangle inequality we can then bound the distance from a to b by finding the distance from a to a_n, plus the distance from a_n to b. That's the idea, now here's the formal proof.

Second proof. Let $\varepsilon > 0$. Since $\varepsilon/2 > 0$ and $a_n \to a$, there exists some N_1 such that for $n > N_1$ we have $|a_n - a| < \varepsilon/2$. Since $\varepsilon/2 > 0$ and $a_n \to b$, there exists some N_2 such that for $n > N_2$ we have $|a_n - b| < \varepsilon/2$. Let $N = \max\{N_1, N_2\}$ and pick any $n > N$. Then

$$\begin{aligned}
|a - b| &= |a - a_n + a_n - b| \\
&\leq |a - a_n| + |a_n - b| && \text{(triangle inequality)} \\
&= |a_n - a| + |a_n - b| \\
&< \varepsilon/2 + \varepsilon/2 && (n > N \text{ implies } n > N_1 \text{ and } n > N_2) \\
&= \varepsilon.
\end{aligned}$$

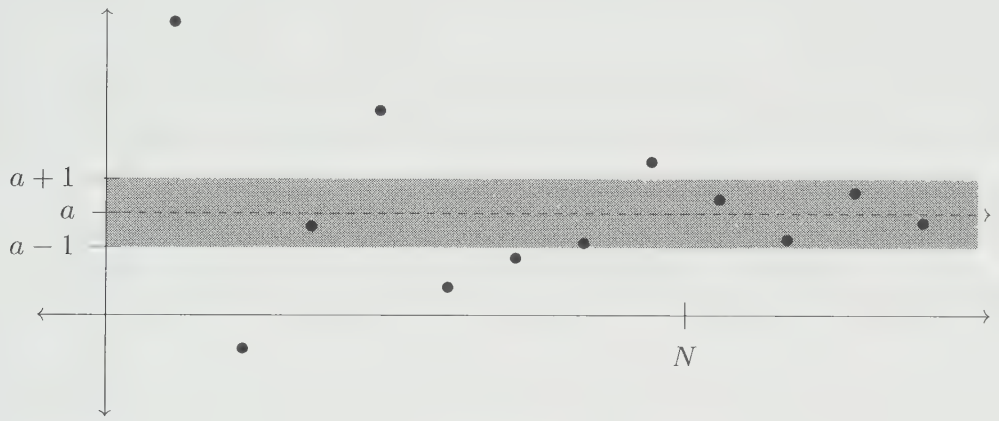
Since this holds for any $\varepsilon > 0$, we have shown that $|a - b| < \varepsilon$ for all $\varepsilon > 0$. Which (by Exercise 1.9) implies that $|a - b| = 0$, and hence $a = b$. So indeed, a_n can converge to only a single point. □

We have been talking about bounded sequences and convergent sequences. The following is a nice proposition which combines these. It also emphasizes the idea that bounded sequences are nice, but convergent sequences are *really* nice. It demonstrates this by showing that the collection of convergent sequences are a (proper) subset of the collection of bounded sequences.

Proposition.

Proposition 3.20 (*Convergent* $\Rightarrow$ *bounded*). If (a_n) is a convergent sequence, then (a_n) is bounded.

Scratch Work. We are told (a_n) converges to some a. By the definition of convergence (Definition 3.7), that means that, for any $\varepsilon > 0$, eventually all the terms of the sequences end up within ε of a. Suppose $\varepsilon = 1$ and N is the value for which $n > N$ implies $|a_n - a| < 1$. Perhaps the picture looks like this:



Now, if we only cared about the points after N, then we are certainly good-to-go, because all those points are bounded by $a-1$ and $a+1$. What about the other points? Well, the only others are $a_1, a_2, a_3, \ldots, a_N$. And since this is a finite collection, there must exist a largest one and the smallest one. The largest one, for instance, is $M := \max\{a_1, a_2, \ldots, a_n\}$. Combined, each a_i either is at most M (if $i \leq N$) or is at most $a + 1$ (if $i > N$). Therefore the larger between M and $a + 1$ is an upper bound on *all* a_i. A succinct way to write this largest value is to just include $a + 1$ in the maximum. That is, $\max\{a_1, a_2, \ldots, a_n, a + 1\}$ will be an upper bound on all a_i. Likewise for the lower bound.

That's the idea, now here's the proof.

Proof. Since (a_n) is convergent, let a be the value it is converging to. By the definition of convergence (with $\varepsilon = 1$), there is some N where

$$|a_n - a| < 1$$

for all $n > N$. That is, $a - 1 < a_n < a + 1$ for all $n > N$. Let

$$U = \max\{a_1, a_2, a_3, \ldots, a_N, a + 1\}$$

and

$$L = \min\{a_1, a_2, a_3, \ldots, a_N, a - 1\}.$$

Note that if $n \leq N$, then $L \leq a_n \leq U$, since each such a_n is included in the sets which we are taking the minimum and maximum of. And if $n > N$ then we already noted that $a - 1 < a_n < a + 1$, which implies that

$$L \leq a - 1 < a_n < a + 1 \leq U,$$

and hence $L < a_n < U$. Combining these cases, we have that

$$L \leq a_n \leq U$$

for all n. And thus, by definition, (a_n) is bounded. □

There is a second, related proof. It's a little shorter by using the absolute value version of boundedness (Proposition 3.5: (a_n) is bounded if and only if there is some C where $|a_n| \leq C$ for all n), which allows you to not worry about lower and upper bounds separately, and instead do them together.

Second Proof. Since (a_n) is convergent, let a be the value it is converging to. By the definition of convergence (with $\varepsilon = 1$), there is some N where

$$|a_n - a| < 1$$

for all $n > N$. That is, $a - 1 < a_n < a + 1$ for all $n > N$. Consider

$$C := \max\{|a_1|, |a_2|, |a_3| \ldots, |a_N|, |a + 1|, |a - 1|\}.$$

Note that if $n \leq N$, then $a_n \leq |a_n| \leq C$. And if $n > N$, then $a - 1 < a_n < a + 1$, which implies that $|a_n| \leq \max\{|a - 1|, |a + 1|\} \leq C$. Combining these cases, we have that

$$|a_n| \leq C$$

for all n, which by Proposition 3.5 means that (a_n) is bounded. □

Whenever you have deduced a result of form "P implies Q," it is worthwhile to think about what the contrapositive says. The contrapositive is "not-Q implies not-P" will look different, but it is logically equivalent. In the present case of Proposition 3.20, the contrapositive of

"If (a_n) is a convergent sequence, then (a_n) is bounded."

is

"If (a_n) is an unbounded sequence, then (a_n) is divergent."

3.5 Limit Laws

When you work with limits, it's good to know what the ground rules are for algebraically manipulating them. These five basic properties are usually called the "limit laws."

Theorem.

Theorem 3.21 (*Sequence limit laws*). Assume that (a_n) and (b_n) are convergent sequences of real numbers such that $a_n \to a$ and $b_n \to b$. Also assume that $c \in \mathbb{R}$. Then

1. $(a_n + b_n) \to a + b$
2. $(a_n - b_n) \to a - b$
3. $(a_n \cdot b_n) \to a \cdot b$
4. $\left(\frac{a_n}{b_n}\right) \to \frac{a}{b}$, provided $b \neq 0$ and each $b_n \neq 0$.
5. $(c \cdot a_n) \to c \cdot a$

Laws 1 and 5 are Exercise 3.7, and we will prove Law 3 below. Once you've proven 1, then 2 is very similar (or follows from 1 and 5 by letting $c = -1$); likewise, once we have proven 3, then 4 is similar. Those two are Exercise 3.8. So we will prove 3 next and leave the rest as exercises.

Scratch Work. Here's the scratch work for Law 3. We want to find an N such that $n > N$ implies $|a_n b_n - ab| < \varepsilon$. Our two big assumptions are that $a_n \to a$ and $b_n \to b$. Here's what we get from that:

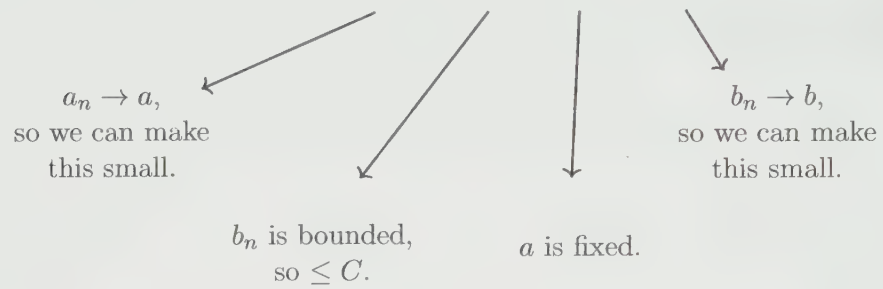
- Since $a_n \to a$, for any $\varepsilon_1 > 0$ there exists some N_1 such that $n > N_1$ implies that $|a_n - a| < \varepsilon_1$.
- Since $b_n \to b$, for any $\varepsilon_2 > 0$ there exists some N_2 such that $n > N_2$ implies that $|b_n - b| < \varepsilon_2$.

Now, going back to what we want to show, there is a very clever trick that makes it all work. The idea is to first rewrite

$$|a_n b_n - ab| \quad \text{as} \quad |a_n b_n - ab_n + ab_n - ab|.$$

The middle terms cancel out, so those are the same. This is clever, though, because if you now use the triangle inequality, you get something we have control over:

$$\begin{aligned} |a_n b_n - ab_n + ab_n - ab| &\leq |a_n b_n - ab_n| + |ab_n - ab| \\ &= |a_n - a| \cdot |b_n| + |a| \cdot |b_n - b|. \end{aligned}$$



To get this all less than ε, we will make both of the products less than $\frac{\varepsilon}{2}$. To get $|a_n - a| \cdot |b_n|$ less than $\frac{\varepsilon}{2}$ we will have to ensure that $|a_n - a|$ is less than $\frac{\varepsilon}{2C}$. To get $|a| \cdot |b_n - b|$ less than $\frac{\varepsilon}{2}$ we will have to ensure $|b_n - b|$ is less than $\frac{\varepsilon}{2a}$. (And then be careful so we aren't dividing by 0.)

Proof. Let $\varepsilon > 0$. Since (b_n) converges, by Proposition 3.20 we know that (b_n) is bounded; that is, there exists some $C \in \mathbb{R}$ such that $|b_n| \leq C$ for all n. Let $\varepsilon_1 = \frac{\varepsilon}{2C+1}$ and $\varepsilon_2 = \frac{\varepsilon}{2|a|+1}$. (Note: we need each "+1" to ensure we are not dividing by 0.)

Since $\varepsilon_1 > 0$ there exists some N_1 such that $|a_n - a| < \varepsilon_1$ for all $n > N_1$. And since $\varepsilon_2 > 0$ there exists some N_2 such that $|b_n - b| < \varepsilon_2$ for all $n > N_2$. Let $N = \max\{N_1, N_2\}$. Then, for any $n > N$,

$$\begin{aligned} |a_n b_n - ab| &= |a_n b_n - ab_n + ab_n - ab| \\ &\leq |a_n b_n - ab_n| + |ab_n - ab| \\ &= |a_n - a| \cdot |b_n| + |a| \cdot |b_n - b| \\ &< \varepsilon_1 \cdot C + |a| \cdot \varepsilon_2 \\ &= \frac{\varepsilon}{2C+1} \cdot C + |a| \cdot \frac{\varepsilon}{2|a|+1} \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\ &= \varepsilon. \end{aligned}$$

That is, for $n > N$ we have $|a_n b_n - ab| < \varepsilon$. Therefore $(a_n b_n) \to ab$, as desired. □

Example 3.22. What is

$$\lim_{n\to\infty} \frac{1}{2} \cdot \left(\frac{\frac{1}{n} + \frac{1}{n^2} + 4}{5 - \frac{1}{n^2}} \right) \cdot \left(\frac{3n+1}{n+2} + \frac{1}{\sqrt{n}} \right) ?$$

Solution. Already in this course we have shown

- $\frac{1}{n} \to 0$
- $5 \to 5$
- $\frac{3n+1}{n+2} \to 3$
- $\frac{1}{n^2} \to 0$
- $5 - \frac{1}{n^2} \to 5$
- $\frac{1}{\sqrt{n}} \to 0$

So, by the limit laws,

$$\lim_{n\to\infty} \frac{1}{2} \cdot \left(\frac{\frac{1}{n} + \frac{1}{n^2} + 4}{5 - \frac{1}{n^2}} \right) \cdot \left(\frac{3n+1}{n+2} + \frac{1}{\sqrt{n}} \right) = \frac{1}{2} \cdot \left(\frac{0+0+4}{5} \right) \cdot (3+0) = \frac{6}{5}. \qquad \square$$

Theorem.

Theorem 3.23 (*Sequence squeeze theorem*). Assume that $a_n \leq x_n \leq b_n$ for all n. Furthermore, assume that

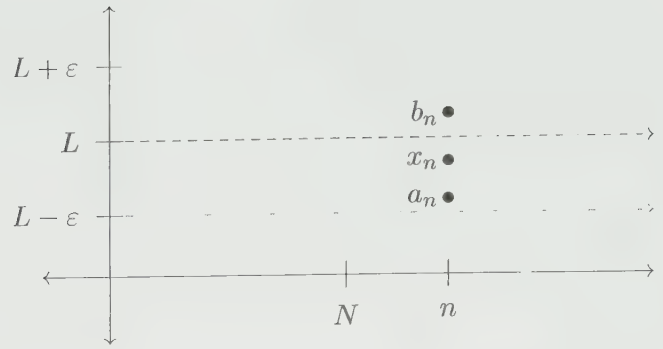
$$a_n \to L \qquad \text{and} \qquad b_n \to L.$$

Then

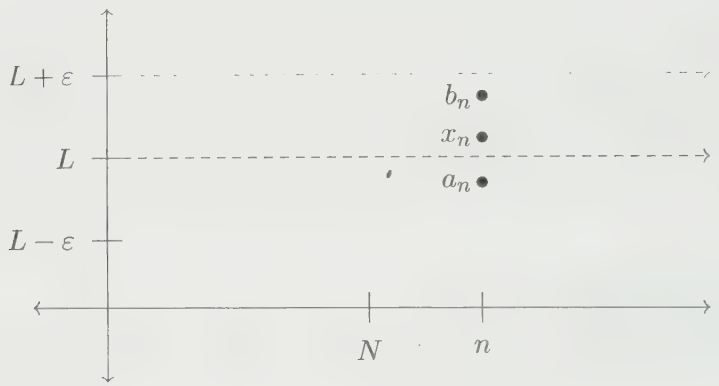
$$x_n \to L.$$

Scratch Work. Just reading the theorem, it does jibe with our intuition of what it means for a sequence to converge.

We will prove it via the definition of convergence, and simply from that definition's vantage point there's good reason to believe it. We wish to show that eventually all the x_n is within ε of L (that is, $|x_n - L| < \varepsilon$ for all $n > N$, for some N). But since a_n and b_n are converging to L, there is some N where, for $n > N$, a_n and b_n are both within ε of L. Combining this with our assumption that $a_n \leq x_n \leq b_n$, it makes perfect sense that x_n would also have to be within ε of L. To formally prove it, it helps to consider two cases: whether x_n is smaller than L or larger than L. Suppose we are past the N at which both a_n and b_n are within ε of L, and suppose that for some $n > N$ we have $x_n \leq L$. Then the picture looks something like this:



You can see that x_n is within ε of L, because a_n is! Similarly, if $L \leq x_n$, then the picture looks something like this:



We again know that x_n is within ε of L, because b_n is!

Combined, the two pictures above show that $L - \varepsilon < x_n < L + \varepsilon$. To make this precise, though, we of course have to get our hands dirty with the definitions of the limits of (a_n) and (b_n).

Proof. Let $\varepsilon > 0$.

- Since $a_n \to L$, there exists some N_1 such that $n > N_1$ implies $|a_n - L| < \varepsilon$. That is, $-\varepsilon < a_n - L < \varepsilon$. Or,

$$L - \varepsilon < a_n < L + \varepsilon. \qquad (☕)$$

- Since $b_n \to L$, there exists some N_2 such that $n > N_2$ implies $|b_n - L| < \varepsilon$. That is, $-\varepsilon < b_n - L < \varepsilon$. Or,

$$L - \varepsilon < b_n < L + \varepsilon. \qquad (🍃)$$

Let $N = \max\{N_1, N_2\}$, and let $n > N$. Combining the inequality $a_n \leq x_n \leq b_n$ with

the left half of (☕) and the right half of (🪐), we get

$$\begin{aligned} L - \varepsilon < a_n \leq x_n \leq b_n < L + \varepsilon \\ L - \varepsilon < x_n < L + \varepsilon \\ -\varepsilon < x_n - L < \varepsilon \\ |x_n - L| < \varepsilon. \end{aligned}$$

□

Here's one example using this theorem.

Example 3.24. Prove that

$$\lim_{n\to\infty} \frac{1}{n^{2.7} + \sqrt{n} + \pi} = 0.$$

Proof. Since

$$0 \leq \frac{1}{n^{2.7} + \sqrt{n} + \pi} \leq \frac{1}{n},$$

and since

$$\lim_{n\to\infty} 0 = 0 \qquad \text{and} \qquad \lim_{n\to\infty} \frac{1}{n} = 0,$$

by the sequence squeeze theorem (Theorem 3.23),

$$\lim_{n\to\infty} \frac{1}{n^{2.7} + \sqrt{n} + \pi} = 0.$$ □

3.6 The Monotone Convergence Theorem

Definition.

Definition 3.25. A sequence (a_n) is *monotone increasing* if $a_n \leq a_{n+1}$ for all n. Likewise, a sequence (a_n) is *monotone decreasing* if $a_n \geq a_{n+1}$ for all n. If it is either monotone increasing or monotone decreasing, it is *monotone*.[17]

Example 3.26.

- (n) is monotone increasing.
- $\left(\frac{1}{n}\right)$ is monotone decreasing.
- (1) is both monotone increasing and monotone decreasing. □

[17]Note: "Monotone" means the same thing as "monotonic." But not "monotonous." *Never* "monotonous."

Note that (n) is the sequence $(1, 2, 3, 4, \dots)$. Likewise, (1) is the sequence $(1, 1, 1, 1, \dots)$; it is the sequence you get when you take $a_n = 1$ for all n, so that $(a_n) = (a_1, a_2, a_3, \dots)$ is just $(1, 1, 1, \dots)$.

Theorem.

Theorem 3.27 (*The monotone convergence theorem*). Suppose (a_n) is monotone. Then (a_n) converges if and only if it is bounded. Moreover,

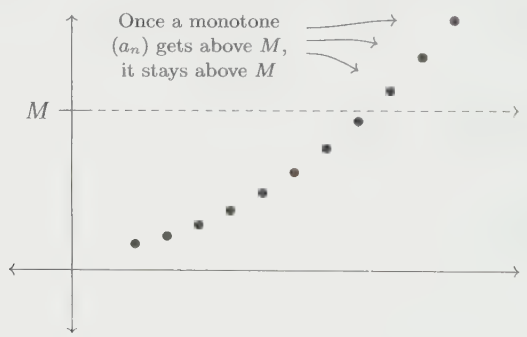
- If (a_n) is increasing then either (a_n) diverges to ∞ or

$$\lim_{n\to\infty} a_n = \sup(\{a_n : n \in \mathbb{N}\}).$$

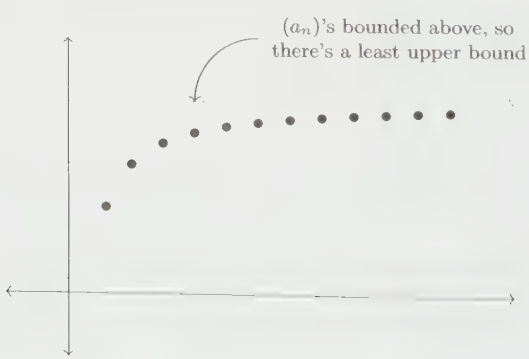
- If (a_n) is decreasing then either (a_n) diverges to $-\infty$ or

$$\lim_{n\to\infty} a_n = \inf(\{a_n : n \in \mathbb{N}\}).$$

Proof Idea. Suppose (a_n) is monotonically increasing. If it's not bounded, then given any $M > 0$, this M is not an upper bound and so eventually (a_n) will get above it. And since it's monotonically increasing, once it gets above a number, it *stays* above that number.



If, on the other hand, (a_n) is bounded, then the sequence must be leveling out (it's monotone, so it can't go up and down).



Proof. Assume that (a_n) is monotonically increasing, and let's suppose first that a_n is not bounded. Then for any $M > 0$ there exists some N such that $a_N > M$. But since (a_n) is monotonically increasing, for $n > N$ we have $a_n \geq a_N > M$. And so, by Definition 3.15, (a_n) diverges to ∞.

Next suppose that (a_n) is bounded. Then we have that $\{a_n : n \in \mathbb{N}\}$ is a subset of $\mathbb{R}$ which is bounded above, which by the completeness of $\mathbb{R}$ implies that $\sup(\{a_n : n \in \mathbb{N}\})$ exists.[18] Call this supremum α. We want to show that $\lim_{n\to\infty} a_n = \alpha$; that is, we want to show that for any $\varepsilon > 0$ there exists some N such that $n > N$ implies $|a_n - \alpha| < \varepsilon$.

To show this, first let $\varepsilon > 0$. Since $\sup(\{a_n : n \in \mathbb{N}\}) = \alpha$, by the supremums analytically theorem (Theorem 1.24) there exists some $a_N > \alpha - \varepsilon$. And since (a_n) is monotonically increasing, we see that for any $n > N$ we have that $a_n \geq a_N > \alpha - \varepsilon$. And of course, $a_n \leq \alpha$ due to the fact that α is the supremum (and hence an upper bound) of $\{a_n : n \in \mathbb{N}\}$. And so, for $n > N$, we have $\alpha - \varepsilon < a_n < \alpha$. This implies $\alpha - \varepsilon < a_n < \alpha + \varepsilon$, and hence $-\varepsilon < a_n - \alpha < \varepsilon$, which, at last, gives

$$|a_n - \alpha| < \varepsilon,$$

completing the proof in the case when (a_n) is monotonically increasing.

The case where (a_n) is monotonically decreasing is *very* similar. It is asked of you in Exercise 3.16. □

This theorem is nice for several reasons, but one of which is the fact that you do not have to know what the limit of (a_n) is in order to show that (a_n) is convergent. This is notable since in the definition of sequence convergence—which, until now, we have heavily relied on to show a sequence converges—requires that you already know (and can write down) what the limit is going to be. Below is an example where it would be quite a challenge to write down the limit of the sequence in a beneficial way. Yet by the monotone convergence theorem, we will be able to conclude that the limit exists.

Example 3.28. Let (a_n) be the sequence where $a_1 = 0.1$, $a_2 = 0.12$, $a_3 = 0.123$, $a_4 = 0.1234$, and so on. (And, to be clear, this pattern does not change when you reach double digits. For example, $a_{12} = 0.123456789101112$.) Prove that (a_n) converges.

Solution. Note that a_{n+1} and a_n match exactly until the last digits of a_{n+1}, which are the digits of $n + 1$. Therefore, $a_{n+1} - a_n$ is a number with a bunch of zeros followed the digits of $n + 1$. In particular,

$$a_{n+1} - a_n > 0$$

for all n. We have shown that $a_{n+1} > a_n$ for all n, proving that (a_n) is monotone

[18] Remember that so far only sets have been defined to have supremums. Sequences don't. So we have to turn the sequence $(a_1, a_2, a_3, \dots)$ into the set $\{a_n : n \in \mathbb{N}\}$ in order to talk about their supremum.

increasing. Furthermore note that since all the terms begin with 0.1, we have that $a_n \leq 1$ for all n.

We have shown that (a_n) is monotone increasing and bounded above, therefore by the monotone convergence theorem (Theorem 3.27) the sequence converges, completing the proof. □

Proposition.

Proposition 3.29 (*Bounded S contains $a_n \to \sup(S)$*). Suppose $S \subseteq \mathbb{R}$ is bounded above. Then there exists a sequence (a_n) where $a_n \in S$ for each n and

$$\lim_{n \to \infty} a_n = \sup(S).$$

Likewise, if S is bounded below then there exists a sequence (b_n) where $b_n \in S$ for each n and

$$\lim_{n \to \infty} b_n = \inf(S).$$

Proof. Suppose $S \subseteq \mathbb{R}$ is bounded above, implying that $\sup(S)$ exists. Let $\sup(S) = \alpha$ (implying $a_n \leq \alpha$ for all n). By the supremums analytically theorem (Theorem 1.24), for any $\varepsilon > 0$ there exists some $x \in S$ such that $x > \alpha - \varepsilon$. In particular, there exists some $a_1 \in S$ such that $a_1 > \alpha - 1$; and there exists some $a_2 \in S$ such that $a_2 > \alpha - \frac{1}{2}$; and there exists some $a_3 \in S$ such that $a_3 > \alpha - \frac{1}{3}$; and, in general, for each $n \in \mathbb{N}$ there exists some $a_n \in S$ such that $a_n > \alpha - \frac{1}{n}$.

Thus we obtain a sequence (a_n) from S. Moreover, this sequence does indeed converge to α: To see this, recall that in Example 3.9 we showed that $\frac{1}{n} \to 0$. Also, clearly (as in Example 3.13) we have the sequence $(\alpha, \alpha, \alpha, \dots)$ converges to α. Therefore, by Limit Law 2 we have that

$$\alpha - \frac{1}{n} \to \alpha - 0 = \alpha.$$

So we have an inequality

$$\alpha - \frac{1}{n} \leq a_n \leq \alpha,$$

where both the upper and lower bounds converge to α. By the sequence squeeze theorem (Theorem 3.23) we see that $\lim_{n \to \infty} a_n = \alpha$, as desired.

The infimum case is *very* similar. It's asked of you in Exercise 3.17. □

3.7 Subsequences

Consider the sequence

$$(a_n) = (n^2).$$

Remember what this means: It means that when you plug in $n = 1$ you get the first term of the sequence, when you plug in $n = 2$ you get the second term, and so on. Therefore (a_n) is the sequence

$$(1, 4, 9, 16, 25, 36, 49, 64, 81, 100, \dots).$$

Now, a *subsequence* of (a_n), as you would expect, is something like

$$\begin{array}{l} (4, 16, 36, 64, 100, \dots) \\ \uparrow\ \uparrow\ \uparrow\ \uparrow\ \uparrow \\ a_2, a_4, a_6, a_8, a_{10}, \dots \end{array} = (a_{2n})$$

Or,

$$\begin{array}{l} (1, 25, 81, 169, 289, \dots) \\ \uparrow\ \uparrow\ \uparrow\ \uparrow\ \uparrow \\ a_1, a_5, a_9, a_{13}, a_{17}, \dots \end{array} = (a_{4n-3})$$

Of course, a subsequence does not have to be perfectly spaced like this; $(1, 121, 144, 484, 10201, \dots)$ also is a subsequence. As long as you pick out one term after another in order, then you have a subsequence. This reasoning suggests the following definition.

Definition.

Definition 3.30. Let (a_n) be a sequence of real numbers and let

$$n_1 < n_2 < n_3 < \dots$$

be an increasing sequence of integers. Then

$$a_{n_1}, a_{n_2}, a_{n_3}, \dots$$

is called a *subsequence* of (a_n), and is denoted (a_{n_k}).

Here are some examples.

Example 3.31. Let $(a_n) = \left(1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots\right)$. Then

- $\left(1, \frac{1}{3}, \frac{1}{5}, \frac{1}{7}, \dots\right)$ is a subsequence where

$$n_1 = 1, \quad n_2 = 3, \quad n_3 = 5, \quad n_4 = 7, \quad \dots,$$

- $\left(\frac{1}{10}, \frac{1}{100}, \frac{1}{1{,}000}, \frac{1}{10{,}000}, \dots\right)$ is a subsequence where

$$n_1 = 10, \quad n_2 = 100, \quad n_3 = 1{,}000, \quad n_4 = 10{,}000, \quad \dots,$$

- And $\left(\frac{1}{2}, \frac{1}{25}, \frac{1}{16{,}129}, \frac{1}{100{,}007}, \dots\right)$ is a subsequence where

$$n_1 = 2, \quad n_2 = 25, \quad n_3 = 16{,}129, \quad n_4 = 100{,}007, \quad \dots.$$

- But $\left(1, \frac{1}{5}, \frac{1}{3}, \frac{1}{7}, \frac{1}{2}, \dots\right)$ is a *not* a subsequence (because we don't have $n_1 < n_2 < n_3 < \dots$),
- And $\left(\frac{1}{2}, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots\right)$ is a *not* a subsequence (because $n_1 = n_2$).

□

Notice that in this example, $a_n \to 0$. Also, each of the three actual subsequences had the property that $a_{n_k} \to 0$, as well. As you can imagine, this is true in general.

Proposition.

Proposition 3.32 ($a_n \to \alpha \Leftrightarrow$ *every* $a_{n_k} \to \alpha$). A sequence (a_n) converges to α if and only if every subsequence of (a_n) also converges to α.

Proof. Assume that (a_n) converges to α and (a_{n_k}) is a subsequence of (a_n). Let $\varepsilon > 0$. Then there exists some N such that

$$|a_n - \alpha| < \varepsilon \qquad (\text{⚽})$$

for all $n > N$.

We want to show that $a_{n_k} \to \alpha$. That is, we want to show that there exists some N_1 such that $|a_{n_k} - \alpha| < \varepsilon$ for all $k > N_1$.

Notice that since each $n_i \in \mathbb{N}$ and

$$n_1 < n_2 < n_3 < \dots.$$

that we have[19] $n_k \geq k$. Therefore by letting $N_1 = N$, for any $k > N_1$ we then know that $n_k > N$, so by (⚽) we have

$$|a_{n_k} - \alpha| < \varepsilon.$$

This completes the forward direction. The backward direction is asked of you[20] in Exercise 3.22. □

[19] For example, one subsequence is $(a_{n_\ell}) = (a_1, a_3, a_5, a_7, \dots)$, giving $n_1 = 1$, $n_2 = 3$, $n_3 = 5$, $n_4 = 7$, and so on. And so, $1 \leq n_1$, and $2 \leq n_2$, and $3 \leq n_3$, and $4 \leq n_4$, and so on.

[20] Hint: There is a 1-line proof for this exercise.

(It is also true that if (a_n) diverges to ∞, then every subsequence of (a_n) will also diverge to ∞. This is asked of you in Exercise 3.23.)

The contrapositive of Proposition 3.32 is that if there are two subsequences which don't converge to the same thing, then the original sequence diverges. Said differently, it is required that *all* subsequences converge to some α for (a_n) to converge as well. Can you see why it is not enough to know that just a single (a_{n_k}) converges? Try to think up an example on your own before reading the one below.

Example 3.33. Let $(a_n) = ((-1)^n)$. This sequence diverges despite many of its subsequences converging. For example, $(a_{2n}) = ((-1)^{2n}) = (1) = (1, 1, 1, 1, \dots)$ converges to 1.[21] Likewise, (a_{4n}) and (a_{28n}) converge to 1, while (a_{2n+1}) converges to -1. □

The problem is that there are some subsequences which converge to one thing, and others which converge to something else. This suggests the following.

Corollary.

Corollary 3.34 (*Different subsequential limits $\Rightarrow$ diverges*). If (a_n) has a pair of subsequences converging to different limits, then (a_n) diverges.

This result is a special case of the contrapositive of the forward direction of Proposition 3.32. But it is perhaps easier to see via a proof by contradiction.

Proof. Assume for a contradiction that (a_n) converges. Say, $a_n \to L$. By Proposition 3.32, this implies that every subsequence also converges to L. This contradicts the assumption that there are subsequences converging to different limits. □

We showed in Example 3.33 that a subsequence of (a_n) can converge while (a_n) itself diverges. However, there is one important special class of sequences where having a single subsequence converge *does* imply that (a_n) also converges. Can you think of what sort of sequences have this property? Try to think it up on your own below looking at the footnote.[22]

Proposition.

Proposition 3.35 (*Monotone (a_n) has $a_{n_k} \to \alpha \Rightarrow a_n \to \alpha$*). If a monotone sequence (a_n) has a convergent subsequence, then (a_n) converges too, and has the same limit.

[21] In the notation of Defintion 3.30, here we have $n_k = 2k$.

[22] If (a_n) is monotone, it has this property. We prove this in Proposition 3.35.

Proof Idea. First recall that by Proposition 3.32, a convergent sequence always has the same limit as any of its subsequences. So by this, our task reduces to showing that (a_n) converges.

A monotone sequence is convergent if and only if it is bounded (by the monotone convergence theorem). We are told that (a_n) is monotone and to show that it's bounded, we use the assumption that it has a convergent subsequence, (a_{n_k}). Here's the progression: (a_n) being monotone implies that its subsequence is too, and (a_{n_k}) being convergent (and now monotone) will mean that (a_n) is bounded too. Now here's the formal proof.

Proof. Assume that (a_n) is monotone increasing; the case where (a_n) is monotone decreasing is *very* similar.

Let (a_{n_k}) be a convergent subsequence of (a_n) and observe that since (a_n) is monotone increasing, being a subsequence means that (a_{n_k})'s terms comes from (a_n) and do so in the same order as they appear in (a_n), implying that (a_{n_k}) is also monotone increasing. That is, (a_{n_k}) was assumed to be convergent and was shown to be monotone—by the monotone convergence theorem this means (a_{n_k}) converges, and in particular is converging to $\sup(\{a_{n_k} : k \in \mathbb{N}\})$.

We now show that (a_n) is convergent, again by using the monotone convergence theorem. Since (a_n) is monotone increasing by assumption, we need only show that (a_n) is bounded above. To see this, observe that since the subsequence (a_{n_k}) is an infinite list of elements from (a_n), any particular element a_n from the sequence has an element a_{n_k} from this subsequence that comes after it.[23] And since both a_n and a_{n_k} are terms of (a_n), which is monotone increasing, we have $a_n \leq a_{n_k}$. So we have

$$a_n \leq a_{n_k} \leq \sup(\{a_{n_k} : k \in \mathbb{N}\}),$$

which proves that (a_n) is bounded above (by this supremum), and so by the monotone convergence theorem we have proven that (a_n) converges.

Finally, since we have shown that the sequence (a_n) is convergent, by Proposition 3.32 it has the same limit as any of its subsequences. This means that (a_n) must also be converging to $\sup(\{a_{n_k} : k \in \mathbb{N}\})$, completing the proof. □

There is a second proof which, if you're a master of indices, is pretty snappy.

Second proof. Without loss of generality, assume that (a_n) is monotone increasing. There is a quick technical property to note first. Observe that since (a_n) is increasing, if $s < t$, then $a_s \leq a_t$. Next, note that for any subsequence (a_{n_t}), we have $t \leq n_t$ for

[23] Probably not *immediately* after it, but certainly at some point after it. For instance, $k = n$ always works. To see this, recall that $\ell \leq n_\ell$ for any ℓ (See Footnote 19), which implies the headache-inducing expression $n \leq n_n$ (meaning the n^{th} index of the subsequence is at least n), and hence $a_n \leq a_{n_n}$ (meaning the n^{th} term of the subsequence is at least as large as the n^{th} term of the original sequence, or equivalently that the $n_n{}^{\text{th}}$ term of the original sequence is at least as large as the n^{th} term of the original sequence).

all t.[24] Therefore, again since (a_n) is increasing, $a_t \leq a_{n_t}$. Putting this all together,

$$a_s \leq a_t \leq a_{n_t} \qquad (🚲)$$

whenever $s < t$.

Now we begin the main argument. Suppose that (a_{n_k}) is a convergent subsequence of (a_n). Let

$$\alpha = \sup\left(\{a_{n_k} : k \in \mathbb{N}\}\right).$$

By the monotone convergence theorem and the fact that (a_{n_k}) converges, we know that $(a_{n_k}) \to \alpha$. That is, there exists some $M \in \mathbb{N}$ such that

$$\alpha - \varepsilon < a_{n_k} < \alpha + \varepsilon$$

for all $k > M$. In particular, this holds when $k = M + 1$. Moreover, since α is in fact a supremum and hence an upper bound on the a_{n_k} terms, we can even say

$$\alpha - \varepsilon < a_{n_{M+1}} \leq \alpha.$$

Combining this with (🚲) gives

$$\alpha - \varepsilon < a_{n_{M+1}} \leq a_\ell \leq a_{n_\ell} \leq \alpha$$

for all $\ell > n_{M+1}$. So there does indeed exist an N (in particular, $N = n_{M+1}$) for which

$$\alpha - \varepsilon < a_\ell \leq \alpha$$

for all $\ell > N$. Therefore $a_\ell \to \alpha$, as desired. □

3.8 The Bolzano-Weierstrass Theorem

We are now gearing up for a big theorem called the Bolzano-Weierstrass[25] theorem. The basic idea is to ask: Given an arbitrary sequence, how nice of a subsequence is that sequence guaranteed to have? And for which types of sequences are we guaranteed some *really* nice subsequence? One of the best properties a subsequence can have is being convergent. But after a moment's reflection we realize that not every sequence has a convergent subsequence. Try to think of an example on your own, and then you can check the footnote to see one answer.[26]

The Bolzano-Weierstrass theorem will tell us one class of sequences which are guaranteed to contain a convergent subsequence. But we begin in a more general setting. We begin with an arbitrary sequence—which we obviously have very little

[24] See Footnote 19.

[25] Pro-tip: "Weierstrass" is pronounced with a hard 'v' at the start, like *vie-er-strahs*. It's German. It's also more fun to say this way.

[26] The sequence (n) works. This is the sequence $1, 2, 3, 4, 5, \ldots$, which diverges to ∞ and every subsequence of it also diverges to ∞.

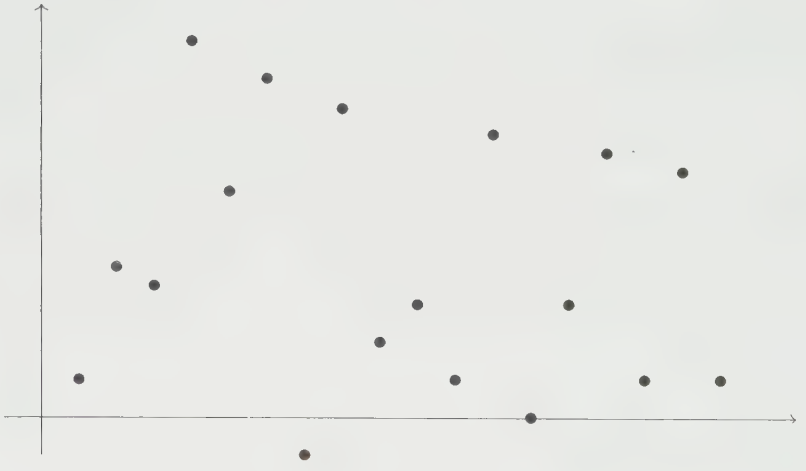
control over—but which we will prove must contain one particularly nice subsequence: a monotone subsequence. We begin here, and we begin with a lemma.

Lemma.

Lemma 3.36 (*(a_n) has monotone a_{n_k}*)**.** Every sequence has a monotone subsequence.

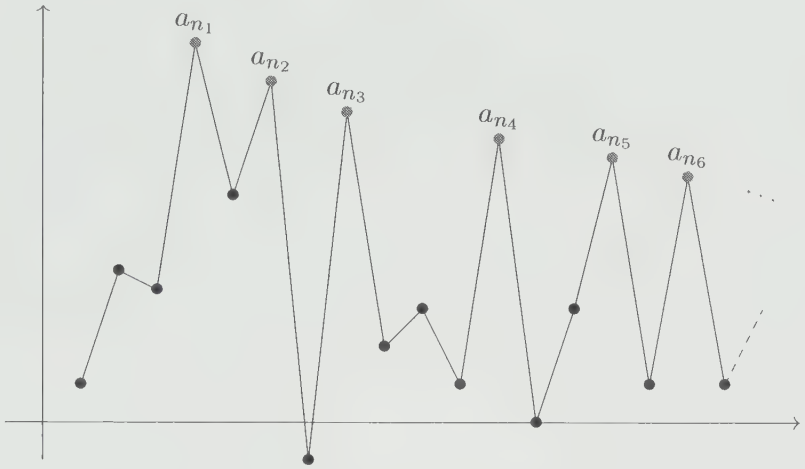
Scratch Work. The proof of this is just straight-up snappy. Once you know what to look for, it falls through perfectly. We just need to define precisely what we want to look for. First, here's the idea behind it:

Usually we draw a sequence in the xy-plane by just plotting the points (e.g. if $a_3 = 6$ then we put a point at $(3, 6)$). This gives a picture like this:



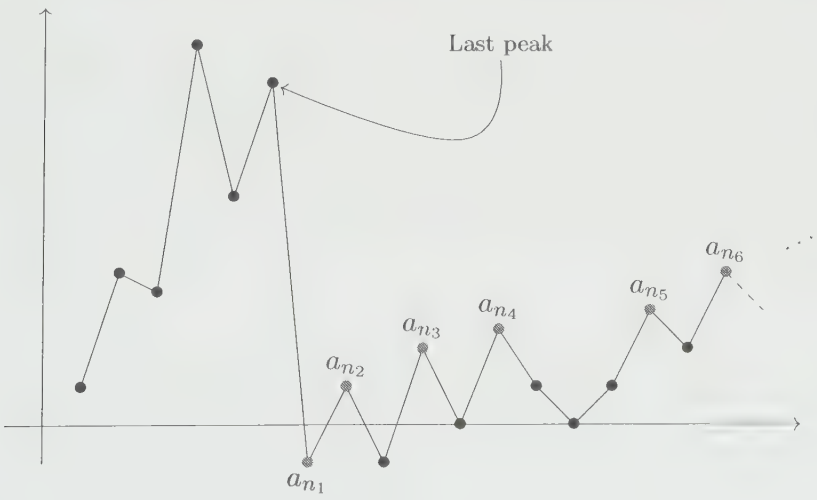
This time, though, we will connect the dots to make a zig-zagged line:

From this picture you can maybe spot a nice *decreasing* sequence, which is of course monotone:



The magical definition that will solve everything is that of a *peak*; we want those labeled points to form *peaks*, and we want to be able to say that if we have a(n infinite) sequence of peaks then we do indeed have a decreasing sequence. The definition that does this is this: define a *peak* to be a point a_n which is larger than every later point; that is, a_n is a peak if $a_n \geq a_m$ for all $m > n$.

So if we have infinitely many peaks then we obtain a sequence like the one above which will be decreasing. So what if we don't? Then we only have finitely many peaks. In this case, we can find an *increasing* sequence, which is again monotone. To see how, just note that if you're past the last peak then any point you pick is not a peak, which means there is some point after it which is larger. So one at a time you can pick larger and larger points, giving an increasing sequence.



Let's now, at last, prove Lemma 3.36.

Proof. Call a_n a *peak* if a_n is larger than every later point. That is, if

$$a_n \geq a_m$$

for all $m > n$. Either there are infinitely many peaks or finitely many peaks. Assume first that (a_n) has infinitely many peaks. Then let a_{n_k} be the k^{th} peak. Then, by the definition of a peak, $a_{n_k} \geq a_{n_{k+1}}$, implying that (a_{n_k}) is a decreasing subsequence.

Now assume that (a_n) has finitely many peaks, and let a_N be the last one. Then let $a_{n_1} = a_{N+1}$. Since a_N was the last peak, a_{N+1} is *not* a peak, implying that there is some later point a_{n_2} that is larger than it. Likewise, since a_{n_2} is after the last peak, it is also not a peak, and so there must be some later point a_{n_3} that is larger than it. Continuing in this way we construct an increasing subsequence a_{n_k}.

In either case we found a monotone subsequence, so we are done. □

The lemma showed that any sequence has a monotone subsequence. It would be really nice if every sequence had a *convergent* subsequence, although unfortunately this is not the case (e.g., $(a_n) = (n)$). Indeed, any sequence which diverges to $\pm\infty$ will have no convergent subsequence. There are also examples of sequences whose limit does not exist, and none of whose subsequences converge—but every such example must be unbounded. Indeed, this is what the Bolzano-Weierstrass theorem says.

Theorem.

Theorem 3.37 (*The Bolzano-Weierstrass theorem*)**.** Every bounded sequence has a convergent subsequence.

Proof. Assume (a_n) is a bounded sequence. Then by Lemma 3.36 it has a monotone subsequence, (a_{n_k}). Also, since (a_n) is bounded, so is (a_{n_k}).

We have shown that (a_{n_k}) is both bounded and monotone, which by the monotone convergence theorem implies that (a_{n_k}) converges. □

3.9 The Cauchy Criterion

Consider the following objective.

Objective: Show (a_n) converges without knowing its limit.

We have been talking a lot about what it means for a sequence (a_n) to "converge to a point a." The definition was:

Definition 3.7. A sequence (a_n) converges to a point a if, for all $\varepsilon > 0$, there exists some N where

$$|a_n - a| < \varepsilon$$

for all $n > N$.

So what does it mean to just say that (a_n) "converges" (without knowing what a is)? It means that *there exists* some a for which $a_n \to a$. That is:

> A sequence (a_n) converges if there exists some a such that, for all $\varepsilon > 0$, there exists some N where
>
> $$|a_n - a| < \varepsilon$$
>
> for all $n > N$.

The important thing to note here is that as of now, to show that a sequence (a_n) converges, we need to already know its limit, a, in order to do the calculation. But what if we don't know the limit? Is there any way to show a sequence converges to *something* without knowing what that something is?

Proposed Solution 1: Maybe convergence means $|a_{n+1} - a_n|$ gets arbitrarily small.

One thing to note is that whenever a sequence converges, each term is deviating less and less from the previous term; the "jumps" are tending towards 0. So if consecutive terms of that sequence are getting closer and closer together, does that mean the sequence converges? Maybe. But with this intuition, one possible condition would be this:

Possibly it's the case that a sequence (a_n) converges if, for all $\varepsilon > 0$, there exists some N where

$$|a_{n+1} - a_n| < \varepsilon$$

for all $n > N$.

Is this condition necessary for convergence? The answer seems like it should be 'yes.' If a sequence is convergent, even if we don't know what the limit is, there still must be some a which the terms are getting closer and closer to, and if the terms are getting close to a then they must also be getting closer to each other. So yes, saying "the terms are getting closer and closer to each other" does seem necessary.

Is it sufficient for convergence? Is it possible that the terms are getting closer and closer but yet the sequence diverges anyways? If your intuition says 'no,' then that's quite reasonable. However, counterexamples do exist.

Example 3.38. The sequence $(a_n) = (\sqrt{n})$ diverges, and yet we can show that it also has the property that, for all $\varepsilon > 0$, there exists some N such that

$$|a_{n+1} - a_n| < \varepsilon$$

for all $n > N$.

Solution. Note that

$$\begin{aligned} |a_{n+1} - a_n| &= |\sqrt{n+1} - \sqrt{n}| \\ &= \sqrt{n+1} - \sqrt{n} \\ &= \left(\sqrt{n+1} - \sqrt{n}\right) \cdot \frac{\sqrt{n+1} + \sqrt{n}}{\sqrt{n+1} + \sqrt{n}} \\ &= \frac{(n+1) - n}{\sqrt{n+1} + \sqrt{n}} \\ &= \frac{1}{\sqrt{n+1} + \sqrt{n}} \\ &< \frac{1}{\sqrt{n}}. \end{aligned}$$

Let $\varepsilon > 0$ and let $N = 1/\varepsilon^2$. Then by the above, for any $n > N$ we have

$$|a_{n+1} - a_n| < \frac{1}{\sqrt{n}} < \frac{1}{\sqrt{N}} = \frac{1}{\sqrt{1/\varepsilon^2}} = \varepsilon.$$

□

So simply having the terms get close to each other is not enough. What can go wrong? Well, even if $|a_{n+1} - a_n|$ is small, and $|a_{n+2} - a_{n+1}|$ is small, and $|a_{n+3} - a_{n+2}|$ is small, and so on, when you add up enough small things the result could still be big. For instance, possibly $|a_{n+10000} - a_n|$ is big. How do we fix this?

Proposed Solution 2: Maybe convergence means $|a_m - a_n|$ gets arbitrarily small.

We fix the above shortcoming by making not just consecutive terms (past some N) close to each other, but by making *all* terms close to each other (past some N). This is so important that it has a name; it is known as *the Cauchy criterion.*

This criterion is named after 18th century French mathematician Augustin-Louis Cauchy, a brilliant man who just happens to have looked remarkably like Vladimir Putin. Seriously, you be the judge:

Cauchy, age 49

Putin, age 46

Definition

Definition 3.39. A sequence (a_n) is *Cauchy* if for all $\varepsilon > 0$ there exists[27] some N such that

$$|a_m - a_n| < \varepsilon$$

for all $m, n > N$.

Example 3.40. Show that $\left(7 + \frac{1}{n}\right)$ is Cauchy.

Scratch Work. Looking back at the definition, we need to think about $a_n = 7 + \frac{1}{n}$ and $a_m = 7 + \frac{1}{m}$. For an $\varepsilon > 0$ we want to show that there exists some N such that

$$|a_n - a_m| < \varepsilon \qquad n, m > N.$$

We now work backwards. We start with this conclusion and unwind it, trying to get

[27] Note: When more convenient, you may assume that $N \in \mathbb{N}$.

to something that we can control.

$$\begin{aligned} |a_n - a_m| &< \varepsilon \\ \left|\left(7 + \frac{1}{n}\right) - \left(7 + \frac{1}{m}\right)\right| &< \varepsilon \\ \left|\frac{1}{n} + \frac{-1}{m}\right| &< \varepsilon \end{aligned}$$

Moving to the next line, we are applying the triangle inequality. The left-hand side below at least as big as the left-hand side above. So, if the below inequality holds, then the above will as well. This is what we want, since for the actual solution we will be doing this process in reverse.

$$\begin{aligned} \left|\frac{1}{n}\right| + \left|\frac{-1}{m}\right| &< \varepsilon \\ \left|\frac{1}{n}\right| + \left|\frac{1}{m}\right| &< \varepsilon \end{aligned}$$

So if we make each of these two terms less than $\varepsilon/2$, we will be good to go. The Archimedean principle should allow us to do that.

We now do this in reverse for the actual solution.

Solution. Let $\varepsilon > 0$. Since $\varepsilon/2 > 0$, by the Archimedean principle there exists some $N > 0$ such that $\frac{1}{N} < \varepsilon/2$. Then, for any $m, n > N$,

$$\begin{aligned} |a_n - a_m| &= \left|\left(7 + \frac{1}{n}\right) - \left(7 + \frac{1}{m}\right)\right| \\ &= \left|\frac{1}{n} + \frac{-1}{m}\right| \\ &\leq \left|\frac{1}{n}\right| + \left|\frac{-1}{m}\right| && \text{(Triangle inequality)} \\ &= \left|\frac{1}{n}\right| + \left|\frac{1}{m}\right| \\ &< \frac{1}{N} + \frac{1}{N} \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\ &= \varepsilon. \end{aligned}$$

□

In words: A sequence is Cauchy if, after some point N, *any* two terms of the sequences differ by at most ε. That is, after some point, all the terms of the sequences are close to each other. Notice how this is different than the definition of convergence: (a_n) converges to a point a if after some point all the terms are getting close to a; for

a Cauchy sequence, though, you do not need to know what the limit is, the definition depends only on the terms of the sequence. That said, as the next theorem shows, these two concepts are actually the same. To prove it, though, we will need the following lemma.

Lemma.

Lemma 3.41 (*Cauchy* $\Rightarrow$ *bounded*). If (a_n) is Cauchy, then (a_n) is bounded.

Proof Idea. This proof is very similar to that of Proposition 3.20, where we showed that convergent sequences are bounded. The reasoning there was: if $a_n \to \alpha$, then after some N, all the terms are bounded between $\alpha - 1$ and $\alpha + 1$. And before N there are only finitely many points, so those are bounded too.

A very similar idea works here.

Proof. Assume (a_n) is Cauchy. Then (for $\varepsilon = 1$) there exists some $N \in \mathbb{N}$ such that

$$|a_m - a_n| < 1$$

for all $m, n > N$. In particular, for all $n > N$,

$$|a_n - a_{N+1}| < 1.$$

Therefore for $n > N$, we know that the terms are bounded:

$$a_{N+1} - 1 < a_n < a_{N+1} + 1.$$

And there are only finitely many points before a_N, so these are certainly bounded. Consequently, we can find a general bound on all a_n. Indeed, if we let

$$L = \min\left(\{a_1, a_2, a_3, \ldots, a_N, a_{N+1} - 1\}\right)$$

and

$$U = \max\left(\{a_1, a_2, a_3, \ldots, a_N, a_{N+1} + 1\}\right),$$

then

$$L \leq a_n \leq U$$

for all $n \in \mathbb{N}$. So (a_n) is bounded. □

Theorem.

Theorem 3.42 (*Cauchy criterion for convergence*). A sequence converges if and only if it is Cauchy.

Proof Sketch. For the forward direction, suppose a_n converges to a. If a_n and a_m are both within $\varepsilon/2$ of a, then they must be within ε of each other. We can formally show this using the triangle inequality.

For the backwards direction, the idea is this: It will be helpful to identify what the Cauchy sequence is converging to. And by Bolzano-Weierstrass we can find a subsequence that is converging to some a. Our goal will then be to prove that a is in fact the sequence's limit. How? Well, that subsequence gets super close to a. What about the terms in (a_n) which are not in the subsequence? By the Cauchy criterion, they get super close to the elements of the subsequence! And if a term in the sequence is super close to a term of the subsequence, which is in turn super close to the limit... then the sequence must be close to the limit, too.

Proof. First we will prove the forwards direction. Assume that (a_n) converges to some $a \in \mathbb{R}$. Let $\varepsilon > 0$. Since $\varepsilon/2 > 0$, there exists some $N \in \mathbb{N}$ such that for any $n > N$ we have

$$|a_n - a| < \frac{\varepsilon}{2}.$$

Then, for any $n, m > N$,

$$\begin{aligned} |a_n - a_m| &= |a_n - a + a - a_m| \\ &\leq |a_n - a| + |a - a_m| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\ &= \varepsilon. \end{aligned}$$

Next, we will prove the backwards direction. Assume that (a_n) is Cauchy, and note that by Lemma 3.42, (a_n) is bounded. Combining these two facts, and applying the Bolzano-Weierstrass theorem (Theorem 3.37), we conclude that some subsequence of (a_n) converges. Say, (a_{n_j}) converges to a. Our goal is to show that (a_n) converges; we will in fact prove that $a_n \to a$.

Let $\varepsilon > 0$. Since $\varepsilon/2 > 0$ and (a_n) is Cauchy, there exists some N_1 such that

$$|a_n - a_m| < \varepsilon/2 \qquad (\star)$$

for all $n, m > N_1$. Since $\varepsilon/2 > 0$ and (a_{n_j}) converges to a, there exists some N_2 such that $j > N_2$ implies

$$|a_{n_j} - a| < \varepsilon/2. \qquad (\bigstar)$$

We want to choose some J so that, for subscripts past this point, both $(\star)$ and $(\bigstar)$ hold. Choose $J = \max\{N_1, N_2\}$. Notice, from the definition of a subsequence,

that $n_j \geq j$.[28] In particular, $n_{J+1} > J$. And so, for any $j > J$,

$$\begin{aligned} |a_j - a| &= |a_j - a_{n_{J+1}} + a_{n_{J+1}} - a| \\ &\leq |a_j - a_{n_{J+1}}| + |a_{n_{J+1}} - a| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\ &= \varepsilon. \end{aligned}$$

□

Back to Axioms

In this course we assumed as an axiom that the reals were complete (AoC). Using AoC we proved, in this order, the monotone convergence theorem (MCT) in Theorem 3.27, the Bolzano-Weierstrass theorem (B-W) in Theorem 3.37, and then the Cauchy criterion (CC) in Theorem 3.42. That is,

$$\text{AoC} \Rightarrow \text{MCT} \Rightarrow \text{B-W} \Rightarrow \text{CC}.$$

- In fact, AoC and MCT are equivalent! If we had started by supposing as an axiom that the reals were an ordered field containing $\mathbb{Q}$ which satisfies the MCT, then we could have proved AoC and everything else and ended up right where we are now.
- Also, AoC and B-W are equivalent! So by supposing B-W instead of AoC, we again could have developed our entire theory.
- CC is almost enough, but not quite. Surprisingly, with CC you can not prove the Archimedean principle. But if you assume both CC and the Archimedean principle (AP), then you can prove everything.

That is,

$$\text{AoC} \Leftrightarrow \text{MCT} \Leftrightarrow \text{B-W} \Leftrightarrow (\text{CC} + \text{AP}).$$

Neat stuff.

[28] For example, one subsequence is $(a_{n_j}) = (a_1, a_3, a_5, a_7, \dots)$, giving $n_1 = 1$, $n_2 = 3$, $n_3 = 5$, $n_4 = 7$, and so on. And so, $n_1 \geq 1$, and $n_2 \geq 2$, and $n_3 \geq 3$, and $n_4 \geq 4$, and so on.

— Notable Exercises —

- The definition of sequence convergence (Definition 3.7) is tough, with lots of qualifiers, there exists/for all statements, and inequalities flying around. And with many more similar definitions on the horizon, it is important to understand it *fully*. Exercise 3.4 probes your understanding of the quantifiers by asking for examples satisfying related definitions, while Exercises 3.3 and 3.6 ask for specific examples about the actual convergence definition, in special circumstances.

- A *subsequential limit* of a sequence (a_n) is any number L for which there is some subsequence (a_{n_k}) converging to L. Some books work a lot with subsequential limits, whereas I do not. But Exercises 3.5, 3.13 and 3.21 are indirect examples of this concept.

- We finish proving the limit laws (Theorem 3.21) in Exercises 3.6 and 3.7. These proofs also contain some important ideas which will be used many more times in this course.

- Exercise 3.28 is a (disguised) first example of *series*, which is the focus of Chapter 4.

— Exercises —

Exercise 3.1. Suppose that a sequence (a_n) converges to 0.001. Prove that finitely many values of a_n are negative.

Exercise 3.2. Prove the following using the definition of sequence convergence.

(a) Let $a_n = 7 - \frac{1}{\sqrt{n}}$. Show that $a_n \to 7$ as $n \to \infty$.

(b) Let $a_n = \frac{2n-2}{3n+1}$. Show that $\lim_{n\to\infty} a_n = \frac{2}{3}$.

(c) Let $a_n = 7 - \frac{1}{\sqrt{n+\sqrt{n}+13}}$. Show that $a_n \to 7$ as $n \to \infty$.

Exercise 3.3. Give an example of a sequence (a_n) where a_n is negative for all n, and yet $a_n \to 0$.

Exercise 3.4. This problem is to help you get a feel for the quantifiers in the definition of convergence (Definition 3.7). Your task: For each of the following definitions of *Non*verges, give an example a sequence (a_n) and a value a for which

- (a_n) does not converge to a (based on the real definition),
- (a_n) does *Non*verge to a based on the definition given below.

Give a different example for each problem. For each of them, explain why your example works in a few sentences (no need to prove it completely). Your example for *Non*verges-type-4 should not work for *Non*verges-type-3.

(a) Definition 1: The sequence (a_n) *Non*verges-type-1 to a if for all $\varepsilon > 0$ there exists some $n \in \mathbb{N}$ such that $|a_n - a| < \varepsilon$.

(b) Definition 2: The sequence (a_n) *Non*verges-type-2 to a if for all $\varepsilon > 0$ there exists some $N \in \mathbb{N}$ such that, for some $n > N$, we have $|a_n - a| < \varepsilon$.

(c) Definition 3: The sequence (a_n) *Non*verges-type-3 to a if there exists some $\varepsilon > 0$ such that for all $N \in \mathbb{N}$ there exists some $n > N$ such that $|a_n - a| < \varepsilon$.

(d) Definition 4: The sequence (a_n) *Non*verges-type-4 to a if there exists some $\varepsilon > 0$ and there exists some $N \in \mathbb{N}$ such that for some $n > N$ we have $|a_n - a| < \varepsilon$.

Exercise 3.5. Consider the sequence

$$\frac{1}{2}, \frac{1}{3}, \frac{2}{3}, \frac{1}{4}, \frac{2}{4}, \frac{3}{4}, \frac{1}{5}, \frac{2}{5}, \frac{3}{5}, \frac{4}{5}, \frac{1}{6}, \frac{2}{6}, \frac{3}{6}, \frac{4}{6}, \frac{5}{6}, \frac{1}{7}, \ldots.$$

For which numbers L does the above have a subsequence converging to L?

Exercise 3.6. Each of the following is an independent question, but for each suppose that the sequence (a_n) has the property that $a_n \in \mathbb{Z}$ for all n.

(a) Is it possible that $a_n \to 3.5$?

(b) If $a_n \neq a_m$ for all $n \neq m$, prove that (a_n) does not converge.

(c) If (a_n) converges, what can be said about this sequence?

Exercise 3.7. Assume that (a_n) converges to some $a \in \mathbb{R}$ and (b_n) converges to some $b \in \mathbb{R}$. Also assume $c \in \mathbb{R}$.

(a) Prove that $(a_n + b_n)$ converges to $a + b$.

(b) Prove that $(c \cdot a_n)$ converges to $c \cdot a$.

Exercise 3.8. Assume that (a_n) converges to a and (b_n) converges to b. Also assume $c \in \mathbb{R}$.

(a) Prove that $(a_n - b_n)$ converges to $a - b$.

(b) Prove that $\left(\frac{a_n}{b_n}\right) \to \frac{a}{b}$, provided $b \neq 0$ and each $b_n \neq 0$.

Exercise 3.9. Give an example of two divergent sequences (a_n) and (b_n) for which $(a_n + b_n)$ converges.

Exercise 3.10. Give an example of two divergent sequences (a_n) and (b_n) for which $(a_n \cdot b_n)$ converges.

Exercise 3.11. Is it possible for a sequence (a_n) to converge, a sequence (b_n) to diverge, and for $(a_n + b_n)$ to converge? Prove that it is impossible or give an example showing that it is possible.

Exercise 3.12. Assume that (a_n) is bounded and (b_n) converges to 0. Prove that $(a_n \cdot b_n)$ also converges to 0.

Exercise 3.13. Give an example of a sequence (a_n) where $6 < a_n < 7$ for all n, and which has a subsequence converging to 6 and also one converging to 7.

Exercise 3.14. Assume a sequence (a_n) has a bounded subsequence. Must (a_n) have a convergent subsequence?

Exercise 3.15. Suppose a sequence (a_n) has the property that for any $\varepsilon > 0$ there exists some N such that $m, n > N$ implies

$$|a_m - a_n| < \varepsilon.$$

Prove that (a_n) is bounded.

Exercise 3.16. Prove the decreasing case of the monotone convergence theorem (Theorem 3.27).

Exercise 3.17. Prove the infimum case of Proposition 3.29.

Exercise 3.18. Give an example of an unbounded divergent sequence whose terms are all positive and whose limit does not exist. (Recall, there are three types of divergence: (1) diverging to ∞, (2) diverging to $-\infty$, or (3) does not exist. Your sequence should be of this third type.). You do not need to prove your answer.

Moved to be an example:

Exercise 3.19. Suppose (a_n) is a sequence for which $a_n \to a$. Define a new sequence by

$$b_n := \frac{a_1 + a_2 + \cdots + a_n}{n}.$$

Prove that $b_n \to a$.

Exercise 3.20. Let (a_n) be a bounded sequence, and consider a second sequence (b_n) defined by

$$b_n := \sup\{a_n, a_{n+1}, a_{n+2}, \ldots\}.$$

Prove that (b_n) converges.

Exercise 3.21. Give an example of a sequence (a_n) which has:

- A subsequence converging to 1,
- Another subsequence converging to 17,
- And another subsequence converging to $-\pi$.

Give a brief explanation for why your example works.

Exercise 3.22. Let (a_n) be a sequence of real numbers. Prove that if *every* subsequence of (a_n) converges, then (a_n) converges too.

Exercise 3.23. Let (a_n) be a sequence of real numbers. Prove that if (a_n) diverges to ∞, then every subsequence of (a_n) diverges to ∞ as well.

Exercise 3.24. Define a sequence (a_n) recursively by $a_1 = \sqrt{2}$ and $a_{n+1} = \sqrt{2 + a_n}$ for all $n \geq 1$.

(a) Show that $a_n \leq 2$ for every n.

(b) Show that (a_n) is an monotone increasing sequence, and use this to conclude that (a_n) converges.

(c) Show that $a_n \to 2$.

Exercise 3.25. Define a sequence (a_n) recursively as follows. Let a_1 and a_2 be a pair of real numbers, and recursively define

$$a_n := \frac{a_{n-1} + a_{n-2}}{2}$$

for all $n \geq 3$. Does (a_n) necessarily converge?

Exercise 3.26. Give an example of a monotone sequence that is not Cauchy.

Exercise 3.27. Assume (a_n) and (b_n) are Cauchy sequences, and let $c_n = |a_n - b_n|$. Use a triangle inequality argument to prove that (c_n) is Cauchy.

Exercise 3.28. Let (a_n) be the sequence where $a_1 = 1$ and, for each $n > 1$, $a_n = a_{n-1} + \frac{1}{n^2}$. In 1734, Leonhard Euler famously proved that (a_n) converges to $\frac{\pi^2}{6}$. Now let (b_n) be the sequence where $b_1 = 1$ and, for each $n > 1$, $b_n = b_{n-1} + \frac{1}{n^3}$. Use Euler's result to prove that (b_n) converges.

— Open Questions —

Question 1. For a natural number input x_0, construct a sequence recursively by

$$x_{n+1} = \begin{cases} \frac{x_n}{2} & \text{if } x_n \text{ is even} \\ 3x_n + 1 & \text{if } x_n \text{ is odd} \end{cases}$$

Is such a sequence always bounded, regardless of what x_0 is? Will this sequence eventually reach 1, regardless of what x_0 is?

Question 2. Consider the sequence where $a_1 = 2$ and, for $n > 1$, a_n is the smallest prime factor of

$$\left(\prod_{i=1}^{n-1} a_i\right) + 1.$$

(This product plays a starring role in Euclid's proof of the infinitude of the primes.) The first 47 elements of this sequence are

2, 3, 7, 43, 13, 53, 5, 6221671, 38709183810571, 139, 2801, 11, 17, 5471, 52662739, 23003, 30693651606209, 37, 1741, 1313797957, 887, 71, 7127, 109, 23, 97, 159227, 643679794963466223081509857, 103, 1079990819, 9539, 3143065813, 29, 3847, 89, 19, 577, 223, 139703, 457, 9649, 61, 4357, 87991098722552272708281251793312351581099392851768893748012603709343, 107, 127, 3313

Does every prime number appear in this sequence? If not, is the problem of testing whether a given prime appears a computable problem?

Chapter 4: Series

4.1 Sequences of Partial Sums

A(n infinite) series is something like $\sum_{k=1}^{\infty} \frac{1}{2^k}$. This one you have probably seen before: $\sum_{k=1}^{\infty} \frac{1}{2^k} = 1$. Now consider the sequence

$$\frac{1}{2}, \frac{3}{4}, \frac{7}{8}, \frac{15}{16}, \ldots.$$

This sequence seems to be converging to 1. But note that we could rewrite the terms of this sequence as so:

$$\frac{1}{2}, \left(\frac{1}{2}+\frac{1}{4}\right), \left(\frac{1}{2}+\frac{1}{4}+\frac{1}{8}\right), \left(\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\frac{1}{16}\right), \ldots.$$

So even though we wrote the terms as a sum, this is still a sequence of numbers. Now notice that this is the same as

$$\sum_{k=1}^{1} \frac{1}{2^k}, \sum_{k=1}^{2} \frac{1}{2^k}, \sum_{k=1}^{3} \frac{1}{2^k}, \sum_{k=1}^{4} \frac{1}{2^k}, \ldots,$$

so this is again sequence. So you see, an infinite *series* can just be viewed as the limit of a *sequence*! And when, in Calc II, you guys found the value of an infinite series like

$$\sum_{k=1}^{\infty} \frac{1}{2^k},$$

what you were actually doing is finding the limit of a sequence! This is an important point, because we have already developed all these tools to help us work with sequences, and so if you think of a series as a *sequence of partial sums*, then we can use all of our theorems on sequences to help us understand series. We will come back to this later in the course, but this chapter will serve as a quick introduction to series, while knowledge of sequences is still fresh in your mind.

Basic definitions

Definition.

Definition 4.1. Given a(n infinite) series $\sum_{k=1}^{\infty} a_k$,

- The numbers a_k are the *terms* of the series;
- The *sequence of partial sums* is the sequence $\left(\sum_{k=1}^{n} a_k\right)_{n=1}^{\infty}$. That is, it's the sequence (s_n) where[1]

$$\begin{aligned} s_1 &= a_1 \\ s_2 &= a_1 + a_2 \\ s_3 &= a_1 + a_2 + a_3 \\ s_4 &= a_1 + a_2 + a_3 + a_4 \\ &\vdots \end{aligned}$$

- The series *converges* to $L \in \mathbb{R}$ (or $\sum_{k=1}^{\infty} a_k = L$) if $s_n \to L$, and *diverges* (to ∞, to $-\infty$ or does not exist) if (s_n) does. Likewise, the series *bounded* or is *monotone* if (s_n) is.

In Chapter 3 we defined what is means for a sequence (s_n) to converge (in terms of ε's and N's) and to diverge (in terms of M's and N's); we can therefore use these and the above to write out an alternative definition for a *series* to converge or diverge. For example, $\sum_{k=1}^{\infty} a_k = \infty$ if, for every $M > 0$ there exists an N such that $\sum_{k=1}^{n} a_k > M$ for all $n > N$.

Example 4.2. Since the sequence

$$\frac{1}{2}, \frac{3}{4}, \frac{7}{8}, \frac{15}{16}, \ldots, \tag{🦇}$$

converges to 1, we also have that

$$\sum_{k=1}^{\infty} \frac{1}{2^k} = 1,$$

since (🦇) is the sequence of partial sums of this series.[2] □

[1] Remember, to get the first term of a sequence you plug in $n = 1$, to get the second you plug in $n = 2$, and so on. It just so happens that, in this sequence, each term is itself a sum of numbers.

[2] Riddle: If I offered you either $\sum_{n=0}^{\infty} \frac{1}{2^{n+1}}$ dollars or $\sum_{n=0}^{\infty} \frac{n}{2^{n+1}}$ dollars, which would you choose?

Corollary.

Corollary 4.3 (*Series (limit) laws*). Assume that $\sum_{k=1}^{\infty} a_k = \alpha$ and $\sum_{k=1}^{\infty} b_k = \beta$. Also assume that $c \in \mathbb{R}$. Then

(i) $\sum_{k=1}^{\infty}(a_k + b_k) = \alpha + \beta$,

(ii) $\sum_{k=1}^{\infty}(a_k - b_k) = \alpha - \beta$, and

(iii) $\sum_{k=1}^{\infty} c \cdot a_k = c \cdot \alpha$.

Proof. Let

$$s_n = \sum_{k=1}^{n} a_k \qquad \text{and} \qquad t_n = \sum_{k=1}^{n} b_k.$$

Now, (s_n) is the sequence of partial sums of the series $\sum_{k=1}^{\infty} a_k$ and (t_n) is the sequence of partial sums of the series $\sum_{k=1}^{\infty} b_k$. Therefore, by assumption, we have that

$$s_n \to \alpha \qquad \text{and} \qquad t_n \to \beta.$$

By the Sequence Limit Laws (Theorem 3.21), we know that

- $(s_n + t_n) \to \alpha + \beta$,
- $(s_n - t_n) \to \alpha - \beta$, and
- $(c \cdot s_n) \to c \cdot \alpha$.

Now let $v_n = \sum_{k=1}^{n}(a_k + b_k)$ and note that

$$\begin{aligned} s_n + t_n &= (a_1 + a_2 + \cdots + a_n) + (b_1 + b_2 + \cdots + b_n) \\ &= (a_1 + b_1) + (a_2 + b_2) + \cdots + (a_n + b_n) \\ &= v_n, \end{aligned}$$

Collecting everything,

$$\begin{aligned} \sum_{k=1}^{\infty}(a_k + b_k) &= \lim_{n\to\infty} \sum_{k=1}^{n}(a_k + b_k) && \text{(Definition of series)}, \\ &= \lim_{n\to\infty} (v_n) && \text{(Definition of } v_n\text{)}, \\ &= \lim_{n\to\infty} (s_n + t_n) && \text{(Showed above)}, \\ &= \alpha + \beta, && \text{(First bullet point)} \end{aligned}$$

as desired.

In almost exactly the same way we can prove that

$$\sum_{k=1}^{\infty}(a_k - b_k) = \alpha - \beta \qquad \text{and} \qquad \sum_{k=1}^{\infty} c \cdot a_k = c \cdot \alpha.$$

□

Question 4.4. There were two other limit laws, one for multiplication and one for division. However, we can *not* conclude that $\sum_{k=1}^{\infty} a_k \cdot b_k = \alpha \cdot \beta$. Why not?

Answer. Remember, to determine what any series equals, it all comes back to *sequences of partial sums.* The first sequences involved are

$$(s_n) = \left(\sum_{k=1}^{n} a_k\right) \qquad \text{and} \qquad (t_n) = \left(\sum_{k=1}^{n} b_k\right),$$

and so we have $s_n \to \alpha$ and $t_n \to \beta$. We can then apply limit laws to the limit

$$\lim_{n\to\infty} \left(\sum_{k=1}^{n} a_k\right)\left(\sum_{k=1}^{n} b_k\right),$$

to conclude that

$$\lim_{n\to\infty} \left(\sum_{k=1}^{n} a_k\right)\left(\sum_{k=1}^{n} b_k\right) = \alpha \cdot \beta,$$

but since, in general,

$$\left(\sum_{k=1}^{n} a_k\right)\left(\sum_{k=1}^{n} b_k\right) \neq \sum_{k=1}^{n} a_k \cdot b_k,$$

the limit laws have next-to-nothing to say about

$$\sum_{k=1}^{n} a_k \cdot b_k.$$

□

As a first example of this, note that

$$\left(\sum_{k=1}^{2} k\right)\left(\sum_{k=1}^{2} k^2\right) = (1+2)(1+4) = 15 \neq 9 = (1+8) = \sum_{k=1}^{2} k \cdot k^2.$$

4.2 Series Convergence Tests

This section is a lot of fun because here we begin proving theorems that you learned in calculus. Calculus is truly a paramount accomplishment of the human mind, and yet in just 113 pages we are already set to prove results important enough that just about every STEM major in the world learns them. Let's get after it.

Proposition.

Proposition 4.5 (*k^{th}-term test*).[3] If $a_k \not\to 0$, then $\sum_{k=1}^{\infty} a_k$ diverges.

By the contrapositive, this is equivalent to saying that if $\sum_{k=1}^{\infty} a_k$ converges, then $a_k \to 0$. And this seems like an easier way to proceed: The assumption that $a_k \not\to 0$ could mean that it is converging to something else, diverging to $\pm\infty$, or that the limit simply does not exist; but assuming that $\sum_{k=1}^{\infty} a_k$ converges has a specific definition, and hence is a more solid place to begin.

Proof. We will prove the contrapositive: if $\sum_{k=1}^{\infty} a_k$ converges, then $a_k \to 0$. To that end, assume that $\sum_{k=1}^{\infty} a_k$ converges and let $s_n = \sum_{k=1}^{n} a_k$. Then we are assuming that the sequence (s_n) converges, and hence (s_n) is Cauchy (by Theorem 3.42). So for any $\varepsilon > 0$ there exists some N such that for all $m, n > N$ we have

$$|s_n - s_m| < \varepsilon.$$

And without loss of generality, let's assume that $n \geq m$. Then, going back to what s_n was defined to be, this means that

$$\left| \sum_{k=1}^{n} a_k - \sum_{k=1}^{m} a_k \right| < \varepsilon$$

$$\left| \sum_{k=m+1}^{n} a_k \right| < \varepsilon$$

[3] Also known as the *divergence test*.

This holds *for all* values of $m, n > N$. In the special case where $m = n - 1$, this gives

$$\begin{aligned} \left| \sum_{k=n}^{n} a_k \right| &< \varepsilon \\ |a_n| &< \varepsilon \\ |a_n - 0| &< \varepsilon. \end{aligned}$$

That is, we have shown that for any $\varepsilon > 0$ there exists some N such that for all $n > N$,

$$|a_n - 0| < \varepsilon.$$

That is, by definition, $a_n \to 0$, completing the proof. □

One particular class of series has a very nice, natural form: the *geometric series*.

Definition.

Definition 4.6. A *geometric series* is a series of the form

$$\sum_{k=0}^{\infty} ar^k = a + ar + ar^2 + ar^3 + \dots,$$

where $a, r \in \mathbb{R}$.

Example 4.7. Here are a couple quick examples of *geometric series*.

- $\displaystyle\sum_{k=0}^{\infty} 2 \cdot \left(\frac{3}{5}\right)^k$
- $\displaystyle\frac{\pi}{2} + \frac{3\pi^2}{4e} + \frac{9\pi^3}{8e^2} + \frac{27\pi^4}{16e^3} + \dots$

As is often the case in mathematics, good questions have good answers. Here, we ask when this nice class of series converges—and the answer is likewise nice.

Lemma.

Lemma 4.8 ($|r|^n$ *diverges iff* $|r| > 1$). The sequence (r^n) converges to 0 if $r \in (-1, 1)$, converges to 1 if $r = 1$, and diverges otherwise.

Proof. This is Exercise 4.3. □

The *geometric series test* gives the detailed version of the very nice answer.

Proposition.

Proposition 4.9 (*Geometric series test*). Assume that a and r are non-zero real numbers. Then

$$\sum_{k=0}^{\infty} a \cdot r^k = \begin{cases} \dfrac{a}{1-r} & \text{if } |r| < 1 \\ \text{diverges}^4 & \text{if } |r| \geq 1 \end{cases}$$

Notice that a is always the first term of the series, and r is the common ratio (what you get whenever you divide any term by its previous term).

Proof. The case where $|r| > 1$ follows from Lemma 4.8 and the k^{th}-term test. When $r = 1$, the series is $a + a + a + \dots$ which clearly diverges, and when $r = -1$ the series is $a - a + a - a + a - \dots$ whose sequence of partial sums, $(a, 0, a, 0, \dots)$, is clearly not converging. We therefore turn to the case that $|r| < 1$. Note that

$$\begin{aligned} (1-r)(1 + r + r^2 + r^3 + \cdots + r^n) & \\ &= 1 + r + r^2 + r^3 + \cdots + r^n \\ &\quad - r - r^2 - r^3 - \cdots - r^{n-1} - r^{n+1} \\ &= 1 + 0 + 0 + \cdots + 0 - r^{n+1} \\ &= 1 - r^{n+1}, \end{aligned}$$

which, by dividing over the $1 - r$, shows that

$$1 + r + r^2 + r^3 + \cdots + r^n = \frac{1 - r^{n+1}}{1 - r},$$

and hence

$$s_n = a + ar + ar^2 + ar^3 + \cdots + ar^n = \frac{a(1 - r^{n+1})}{1 - r},$$

where s_n is the n^{th} partial sum of $\displaystyle\sum_{k=0}^{\infty} a \cdot r^k$. Thus,

$$\begin{aligned} \sum_{k-0}^{\infty} a \cdot r^k &= \lim_{n \to \infty} s_n \\ &= \lim_{n \to \infty} \frac{a(1 - r^{n+1})}{1 - r} \\ &= \frac{a(1-0)}{1-r}. && \text{(Lemma 4.8)} \\ &= \frac{a}{1-r}. \end{aligned}$$

□

[4] Exercise 4.8 asks you to classify the type of divergence.

Example 4.10. In Example 4.2, we noted—without proof—that the sequence

$$\frac{1}{2}, \frac{3}{4}, \frac{7}{8}, \frac{15}{16}, \cdots$$

converges to 1. Using Proposition 4.9, we can now prove this.

Note that $\sum_{k=1}^{\infty} \frac{1}{2^k}$ has a first term of $a = 1/2$ and a common ratio of $r = 1/2$. Therefore

$$\sum_{k=1}^{\infty} \frac{1}{2^k} = \frac{1/2}{1 - \frac{1}{2}} = \frac{1/2}{1/2} = 1.$$

□

An alternative approach to finding r and a is this: First write the series in the form given in Proposition 4.9. That would be this:

$$\sum_{k=1}^{\infty} \frac{1}{2^k} = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} \cdots = \sum_{k=0}^{\infty} \frac{1}{2} \cdot \left(\frac{1}{2}\right)^k.$$

And now it is clear that $a = 1/2$ and $r = 1/2$.

Proposition.

Proposition 4.11 ($a_k \geq 0 \Rightarrow \sum a_k$ *converges or* $= \infty$). If $a_k \geq 0$ for all k, then $\sum_{k=1}^{\infty} a_k$ either converges or it diverges to ∞.

The series is either bounded (hence converges) or is not bounded (hence diverges).

Proof. Since each a_k is nonnegative, observe that the sequence of partial sums is monotone increasing. Applying the Monotone Convergence Theorem (Theorem 3.27) then gives the result. □

Proposition.

Proposition 4.12 (*Comparison test*). Assume $0 \leq a_k \leq b_k$ for all k.

(i) If $\sum_{k=1}^{\infty} b_k$ converges, then $\sum_{k=1}^{\infty} a_k$ converges.

(ii) If $\sum_{k=1}^{\infty} a_k$ diverges, then $\sum_{k=1}^{\infty} b_k$ diverges.

The converse of these statements is not true, and Exercise 4.12 asks for example demonstrating this.

Proof. Let (s_n) be the sequence of partial sums of $\sum_{k=1}^{\infty} a_k$, and let (t_n) be the sequence of partial sums of $\sum_{k=1}^{\infty} b_k$. And here is a fact that will be useful: note that since $a_k \leq b_k$ for all k,

$$s_n = a_1 + a_2 + \cdots + a_n \leq b_1 + b_2 + \cdots + b_n = t_n$$

for all n. That is,

$$s_n \leq t_n \qquad \text{for all } n. \tag{$\clubsuit$}$$

To see (i), first note[5] that by Proposition 4.11 that all we need to show is that (s_n) is bounded above. If we can do this we know that $s_n \not\to \infty$, which by that proposition implies that it must converge.

Recall that in Proposition 3.20 we showed that if a sequence converges, then it is bounded. And so, since by assumption $\sum_{k=1}^{\infty} b_k$ converges (meaning that (t_n) converges), we deduce that this series is bounded above by some value M; that is $t_n \leq M$ for all n (this is Definition 3.4). And so by ($\clubsuit$), $s_n \leq t_n \leq M$ for all n, showing that (s_n) is also bounded above by M, completing the proof of (i).

The proof of (ii) is very similar. Let $M > 0$. Assume that $\sum_{k=1}^{\infty} a_k$ diverges, meaning that (s_n) diverges (and hence by Proposition 4.11 this means it diverges to ∞). And so by the definition of diverging to ∞ (Definition 3.15), there exists some N for which $M < s_n$ for all $n > N$. And so by ($\clubsuit$), $M < s_n \leq t_n$ for all $n > N$. Therefore, again by Definition 3.15, (t_n) also diverges to ∞. □

Observe that if, say, $0 \leq a_k \leq b_k$ for all but the first 10 terms, then the conclusion of the theorem would still hold. Indeed, if $\sum b_k$ converges and $0 \leq a_k \leq b_k$ for all k, then the comparison test says that $\sum a_k$ converges. But if you now change the first

[5]The rest of the proof is a mile wide and an inch deep. It's almost more memory than math– basically just a piecing together of a bunch of old definitions and propositions.

10 terms of a_k so that they are larger than b_k, there is no way that those ten terms now cause $\sum a_k$ to diverge.

In general, as long as $a_k \leq b_k$ holds for *all but finitely many* terms, the theorem still applies, as the following note says.

Note.

Note 4.13. Changing finitely many terms of a sequence or a series does not affect whether or not the sequence or series converges.

Example 4.14. Suppose

$$a_k = \begin{cases} k^2 + 7 & \text{if } k \leq 1,000, \\ \left(\frac{1}{2}\right)^k & \text{if } k > 1,000. \end{cases}$$

Then $\sum_{k=0}^{\infty} a_k$ converges. To see this, simply note that by changing finitely many terms we can reach $\sum_{k=0}^{\infty} \left(\frac{1}{2}\right)^k$, which converges by the geometric series test (where $r = \frac{1}{2}$). Now apply Note 4.13. □

Harmonic Series and the Series p-Test

The *harmonic series* is famous. It is the series

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \cdots.$$

The terms are getting smaller and and smaller, and yet:

Proposition.

Proposition 4.15 ($\sum \frac{1}{k}$ *diverges*). The harmonic series $\sum_{k=1}^{\infty} \frac{1}{k}$ diverges.

Proof. Observe that

$$\begin{aligned}\sum_{k=1}^{\infty}\frac{1}{k} &= 1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+\frac{1}{5}+\frac{1}{6}+\frac{1}{7}+\frac{1}{8}+\ \dots\\ &= 1+\left(\frac{1}{2}\right)+\left(\frac{1}{3}+\frac{1}{4}\right)+\left(\frac{1}{5}+\frac{1}{6}+\frac{1}{7}+\frac{1}{8}\right)+\ \dots\\ &\geq 1+\left(\frac{1}{2}\right)+\left(\frac{1}{4}+\frac{1}{4}\right)+\left(\frac{1}{8}+\frac{1}{8}+\frac{1}{8}+\frac{1}{8}\right)+\ \dots\\ &= 1+\left(\frac{1}{2}\right)+\left(\frac{1}{2}\right)+\left(\frac{1}{2}\right)+\ \dots.\end{aligned}$$

In particular, if s_n is the n^{th} partial sum of the harmonic series, then (s_n) is monotonically increasing and, by the above,

$$s_{2^n} \geq 1+n\cdot\frac{1}{2}.$$

And since $(1+n\cdot\frac{1}{2})$ diverges to ∞, by the comparison test (Proposition 4.12) the subsequence (s_{2^n}) diverges to ∞. And for a monotonically increasing sequence, if a subsequence diverges to ∞—implying that (s_n) is unbounded—the entire sequence, (s_n), is also diverging to ∞ by the monotone convergence theorem (Theorem 3.27). That is, the harmonic series diverges to ∞. □

The above proof is the classic proof that the harmonic series diverges, and it was discovered way back in the *Middle Ages*, predating almost all other results in this text. It was proved by French philosopher Nicole Oresme around 1350 (who somehow found time to study math while his country fought England in the Hundred Years' War, and the Black Death was becoming perhaps the worst pandemic in human history). Due to the significance of this theorem, though, many later mathematicians worked to find their own proofs of the result.[6]

[6]Check out the survey *The Harmonic Series Diverges Again and Again* by Kifowit and Stamps, in which they collect 20 more proofs of the theorem. It includes a 17^{th} century proof by Jacob Bernoulli utilizing partial fractions, an 18^{th} century proof by Leonhard Euler using power series, and many more, from using the integral test to using binomial coefficients. A proof that I particularly like, which is by contradiction but otherwise uses similar ideas to the proof above, is this: Suppose for a contradiction that the harmonic series converges to S. Then

$$\begin{aligned}S &= 1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+\frac{1}{5}+\frac{1}{6}+\frac{1}{7}+\frac{1}{8}+\dots\\ &= \left(1+\frac{1}{2}\right)+\left(\frac{1}{3}+\frac{1}{4}\right)+\left(\frac{1}{5}+\frac{1}{6}\right)+\left(\frac{1}{7}+\frac{1}{8}\right)+\dots\\ &> \left(\frac{1}{2}+\frac{1}{2}\right)+\left(\frac{1}{4}+\frac{1}{4}\right)+\left(\frac{1}{6}+\frac{1}{6}\right)+\left(\frac{1}{8}+\frac{1}{8}\right)+\dots\\ &= 1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+\frac{1}{5}+\frac{1}{6}+\frac{1}{7}+\frac{1}{8}+\dots\\ &= S.\end{aligned}$$

We've shown $S > S$, a contradiction. □

So we know that $\sum_{k=1}^{\infty} \frac{1}{k} = \infty$. That is, for any $M > 0$, eventually the sum $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \ldots$ gets above M. This sum diverges suuuuuuper slowly, though. In fact, in 1968, way before the age of the supercomputer, J.W. Wrench Jr. showed that $\sum_{k=1}^{n} \frac{1}{k} > 100$ for the first time when

$$n = 15,092,688,622,113,788,323,693,563,264,538,101,449,859,497$$

That is, summing the first 15,092,688,622,113,788,323,693,563,264,538,101,449,859,496 integer reciprocals gives a result that is less than 100. But when you add

$$\frac{1}{15,092,688,622,113,788,323,693,563,264,538,101,449,859,497}$$

to that sum, you get above 100. He showed it in just 6 pages[7] and a few (very weak) computer calculations. Amazing.[8]

So yes, the Harmonic series just *barely* diverges; in a precise sense, it is right on the borderline between convergence and divergence, as the next result shows.

Proposition.

Proposition 4.16 (*The series p-test*). The series $\sum_{k=1}^{\infty} \frac{1}{k^p}$ converges if and only if $p > 1$.

Assume $k \geq 1$. In the proof we will use the fact that k^p increases if you fix k and increase p, or if you fix p and increase k.

Proof. If $p \leq 1$, then $\frac{1}{k} \leq \frac{1}{k^p}$ for all $k \in \mathbb{N}$. And since $\sum_{k=1}^{\infty} \frac{1}{k}$ diverges, by the comparison test, $\sum_{k=1}^{\infty} \frac{1}{k^p}$ diverges too.

[7] He used something called the Euler-Maclaurin formula, which gives an integral expression for how far off the n^{th} partial sum of the harmonic series is from the natural log function.

[8] Nowadays, even WolframAlpha can verify that Wrench was correct. If you have the patience, check his solution on your own. It's pretty amazing to see.

Now assume that $p > 1$. Then

$$\begin{aligned}
\sum_{k=1}^{\infty} \frac{1}{k^p} &= 1 + \frac{1}{2^p} + \frac{1}{3^p} + \frac{1}{4^p} + \frac{1}{5^p} + \frac{1}{6^p} + \frac{1}{7^p} + \dots \\
&= 1 + \left(\frac{1}{2^p} + \frac{1}{3^p}\right) + \left(\frac{1}{4^p} + \frac{1}{5^p} + \frac{1}{6^p} + \frac{1}{7^p}\right) + \dots \\
&< 1 + \left(\frac{1}{2^p} + \frac{1}{2^p}\right) + \left(\frac{1}{4^p} + \frac{1}{4^p} + \frac{1}{4^p} + \frac{1}{4^p}\right) + \dots \\
&= 1 + \left(\frac{2}{2^p}\right) + \left(\frac{4}{4^p}\right) + \dots \\
&= 1 + \frac{1}{2^{p-1}} + \frac{1}{4^{p-1}} + \dots \\
&= 1 + \frac{1}{2^{p-1}} + \frac{1}{2^{2(p-1)}} + \dots \\
&= 1 + \frac{1}{2^{p-1}} + \left(\frac{1}{2^{p-1}}\right)^2 + \dots \\
&= \sum_{k=0}^{\infty} \left(\frac{1}{2^p}\right)^k.
\end{aligned}$$

Finally, note that $\displaystyle\sum_{k=0}^{\infty} \left(\frac{1}{2^p}\right)^k$ is a geometric series with $|r| = \frac{1}{2^p} < 1$, so by the geometric series test (Proposition 4.9) implies that this series converges. And the above also shows that, term-by-term, this geometric series is larger than $\displaystyle\sum_{k=1}^{\infty} \frac{1}{k^p}$. So by the comparison test, $\displaystyle\sum_{k=1}^{\infty} \frac{1}{k^p}$ converges too. □

The $p = 2$ case is particularly interesting; that is, the sum $1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \dots$. It's called the *Basel Problem* after Basel, Switzerland where both Leonhard Euler and his mentor Johann Bernoulli were born. Bernoulli and many other prominent world-class mathematicians tried and failed to determine the sum. But Euler, at age 24, solved the problem, thrusting himself onto the world stage. It was a significant problem, but the most remarkable part of all of this is its incredible answer:

$$\sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6}.$$

The number π is defined by circles. What the hell do sums of inverse squares have to do with circles?![9] And why squared? It seems utterly unreasonable that a π

[9] As far as I can tell, Euler's original proof provided almost no intuition as to what that π is doing there. For some intuition based on a 2010 paper by Johan Wästlund, watch the excellent 3blue1brown video at `https://youtu.be/d-o3eB9sfls`.

should be involved in the answer, and I hope Euler fell off his seat when he discovered it. I wouldn't say there's too much direct evidence that there's a God. But... I think we need to put this one in God's column.

4.3 Absolute Convergence

Suppose each $a_k \geq 0$. Then $\sum a_k$ may or may not converge, but note that the series $\sum(-1)^k a_k$ is, if anything, *more* likely to converge. The original series, $\sum a_k$ can only converge or diverge to ∞, so changing half of the terms to a negative should only help us cancel more things out and cause the series to converge.

Indeed, we have seen examples that show that the converse to the k^{th}-term test (Proposition 4.5) does not hold. That is, it is possible to have the terms of a series to converge to 0 while the series still diverges. If your series is alternating, though, this is not true, as the following theorem states.

Proposition.

Proposition 4.17 (*Alternating series test*). Assume that (a_k) is a monotonically decreasing sequence and $a_k \to 0$. Then $\displaystyle\sum_{k=1}^{\infty}(-1)^{k+1}a_k$ converges.

Note: Since (a_k) is monotonically decreasing and converging to zero, this implies $a_k \geq 0$ for all k.

Proof. Let (s_n) be the sequence of partial sums of this series. First we will show that the subsequence (s_{2n}) is monotonically increasing. To see this, observe that by grouping the terms of the subsequence (s_{2n}),

$$s_{2n} = (a_1 - a_2) + (a_3 - a_4) + \cdots + (a_{2n-1} - a_{2n}),$$

and simply observe that the fact that (a_n) is monotonically decreasing implies that $(a_i - a_{i+1}) \geq 0$ for all i. But not only is (s_{2n}) monotonically increasing, by grouping the terms differently we can see that it is bounded above by a_1:

$$a_1 \geq a_1 - (a_2 - a_3) - (a_3 - a_4) - \cdots - (a_{2n-2} - a_{2n-1}) - a_{2n}.$$

In summary, (s_{2n}) is a monotonically increasing sequence that is bounded above by a_1, and hence converges by the Monotone Convergence Theorem. Call its limit L.

Through a very similar argument to the above, one can show that (s_{2n+1}) is monotonically decreasing and bounded below by $a_1 - a_2$, and hence also converges. And moreover, it must also be converging to L. The reason is, if we call its limit U, then

$$L - U = \lim_{n\to\infty} s_{2n} - \lim_{n\to\infty} s_{2n-1} = \lim_{n\to\infty} (s_{2n} - s_{2n-1}) = \lim_{n\to\infty} a_{2n} = 0.$$

And so $U = L$, as asserted.

What we still need to prove is that if (s_n) is a sequence in which (s_{2n}) and (s_{2n+1}) both converge to L, it must also be the case that (s_n) converges to L. To prove this last claim, let $\varepsilon > 0$ and note that for the first subsequence there is an N_1 for which $|s_n - L| < \varepsilon$ for all even $n > N_1$, and for the second there is an N_2 for which $|s_n - L| < \varepsilon$ for all odd $n > N_2$. Now let $N = \max\{N_1, N_2\}$ and observe that since every $n > N$ was an index in one of the two subsequences, we have that

$$|s_n - L| < \varepsilon.$$

Thus $s_n \to L$. And since s_n is the sequence of partial sums of the series, this means that the series also converges to L, completing the proof. □

Why is it necessary for the sequence to be monotonically decreasing? If we only assumed that it was an alternating sequence of numbers converging to zero, why is that not enough? The idea is to have a series where the positive terms converge to zero but their sum diverges to ∞; the negative terms converge to zero and their sum also converges, and hence are not big enough to "slow down" the divergence of the positive terms. For example, the series

$$\sum_{k=1}^{\infty} \left(\frac{1}{k} - \frac{1}{k^2} \right)$$

would have this property. In order to get a series $\sum_{k=1}^{\infty} a_k$ which produces this (where the odd-indexed terms are positive and the even-indexed terms are negative), let

$$a_k = \begin{cases} \frac{2}{k+1} & \text{if } k \text{ is odd;} \\ \frac{4}{k^2} & \text{if } k \text{ is even.} \end{cases}$$

Then

$$\begin{aligned} \sum_{k=1}^{\infty} (-1)^{k-1} a_k &= \frac{1}{1} - \frac{1}{1^2} + \frac{1}{2} - \frac{1}{2^2} + \frac{1}{3} - \frac{1}{3^2} + \frac{1}{4} - \frac{1}{4^2} + \cdots \\ &= \left(\frac{1}{1} - \frac{1}{1^2} \right) + \left(\frac{1}{2} - \frac{1}{2^2} \right) + \left(\frac{1}{3} - \frac{1}{3^2} \right) + \left(\frac{1}{4} - \frac{1}{4^2} \right) + \cdots \\ &= \sum_{k=1}^{\infty} \left(\frac{1}{k} - \frac{1}{k^2} \right). \end{aligned}$$

Indeed, if your sum is alternates between positive and negative, and if either

- The sum of your positive terms diverge to ∞ while the sum of your negative terms converges, *or*
- The sum of your negative terms diverge to $-\infty$ while the sum of your positive terms converges,

then your series must diverge. On the other hand, if both the positive terms and the negative terms converge on their own, then your series must converge. One way to think about this is that if you took the absolute value of all the terms, and the resulting sum converges, then the original sum must converge too. We codify this idea in the following proposition.

Proposition.

Proposition 4.18 ($\sum |a_k|$ *converges* $\Rightarrow \sum a_k$ *converges*). If $\sum_{k=1}^{\infty} |a_k|$ converges, then $\sum_{k=1}^{\infty} a_k$ converges too.

Proof. This is Exercise 4.2. □

The converse, though, is not true. For example, in Proposition 4.15 we showed that the harmonic series $\sum_{k=1}^{\infty} \frac{1}{k}$ diverges. Moreover, by the alternating series test we see that the *alternating harmonic series* $\sum_{k=1}^{\infty} (-1)^{k+1} \frac{1}{k}$ converges. Indeed, here are the first few partial sums of this series:

$$\begin{aligned} 1 &= 1 \\ 1 - \frac{1}{2} = \frac{1}{2} &= 0.5 \\ 1 - \frac{1}{2} + \frac{1}{3} = \frac{5}{6} &= 0.8333\ldots \\ 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} = \frac{7}{12} &= 0.58333\ldots \\ 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} = \frac{23}{60} &= 0.78333\ldots \\ &\vdots \end{aligned}$$

It certainly seems to be converging, and to something around 0.7, but just like with the Basel problem, what it is converging to is quite surprising:

Fact.

Fact 4.19.

$$\sum_{k-1}^{\infty}(-1)^{k+1}\frac{1}{k} = \ln(2).$$

Unlike some of the crazy mention throughout this text, this one can be demonstrated just with Calculus II material.[10]

Now, some *alternating series* (that is, series which alternate between positive and negative) are like the alternating harmonic series—they converge, but if you take away the $(-1)^k$ component they won't converge. Other alternating series, though, would converge even without alternating. This distinction is important.

Definition.

Definition 4.20. Consider the series $\sum_{k=1}^{\infty} a_k$.

- If $\sum_{k=1}^{\infty} |a_k|$ converges, then we say $\sum_{k=1}^{\infty} a_k$ *converges absolutely.*
- If $\sum_{k=1}^{\infty} a_k$ converges but $\sum_{k=1}^{\infty} |a_k|$ diverges, then $\sum_{k=1}^{\infty} a_k$ *converges conditionally.*

We are now ready for what I think is by-far the coolest theorem about this topic.

[10]We won't reach this until Chapter 9, but if you recall your Calculus II well enough, you might recall a way to see that this is true. The geometric series test says that (for $x \in (-1, 1)$),

$$1 - x + x^2 - x^3 + x^4 - \cdots = \frac{1}{1+x}.$$

Integrating term-by-term is allowed, and doing so gives the Taylor series for $\ln(x+1)$:

$$x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5} - \cdots = \ln(1+x).$$

Now just plug in $x = 1$. □

But note: The above only argues *that* the fact is true. But *why* is it true?? Why is a logarithm with a base of e so intimately related to inverses of integers? That's a *much* harder question to answer.

4.4 Rearrangements

Definition.

Definition 4.21. A *rearrangement* of a series $\sum_{k=1}^{\infty} a_k$ is a series $\sum_{k=1}^{\infty} b_k$ for which there is a bijection $f : \mathbb{N} \to \mathbb{N}$ such that $b_{f(k)} = a_k$.

It is precise to define a rearrangement in terms of bijections (especially if the series has repeated terms) but, intuitively, a rearrangement of a series is simply the same series where you add up the terms in a different order.

Example 4.22. The alternating harmonic series is

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \frac{1}{7} - \frac{1}{8} + \cdots.$$

A rearrangement of this series is

$$\frac{1}{5} - \frac{1}{2} - \frac{1}{152} + \frac{1}{57} - \frac{1}{1847604} - \frac{1}{8} + \cdots.$$

A different rearrangement is

$$1 + \frac{1}{3} + \frac{1}{5} - \frac{1}{2} + \frac{1}{7} + \frac{1}{9} + \frac{1}{11} - \frac{1}{4} + \cdots.$$

If you have *finitely* many numbers that you are adding/subtracting, then of course the order in which you add/subtract them does not matter—you get the same answer either way. Amazingly, though, if you have *infinitely* many numbers, then it *can* make a difference. It is possible to add up infinitely many numbers and have it equal one thing, but then by simply changing the order that you are adding up the numbers it is possible for the sum to equal something different! The order that you add the numbers can change what it equals!

But if that's not freaky enough, it gets worse/better: in some cases, the sum can equal *anything at all.* This is one of my favorite theorems of this entire book, so pay attention.

Super Cool Theorem.

Theorem 4.23 (*Rearrangement theorem*). If a series $\sum_{k=1}^{\infty} a_k$ converges conditionally, then for any L ($L \in \mathbb{R}$ or $L = \pm\infty$) there exists some rearrangement of $\sum_{k=1}^{\infty} a_k$ which converges to L.

Proof Sketch. Let p_k be the k^{th} positive term of (a_n) and let n_k be the k^{th} negative term of (a_n). Here's the basic idea of the proof:

- Since $\sum_{k=1}^{\infty} a_k$ converges, by the k^{th}-term test we must have $a_k \to 0$.
- Since $\sum_{k=1}^{\infty} a_k$ converges conditionally, we must have:
  - $\sum_{k=1}^{\infty} p_k = \infty$ (while $p_k \to 0$), and
  - $\sum_{k=1}^{\infty} n_k = -\infty$ (while $n_k \to 0$).

  This is the case since, if just one were infinite, then $\sum_{k=1}^{\infty} a_k$ would diverge. And if both were finite, then $\sum_{k=1}^{\infty} a_k$ would converge absolutely.
- Assume you are trying to converge to $L \in \mathbb{R}$ (the case $L = \infty$ is Exercise 4.15), and without loss of generality assume $L > 0$.
- Since $\sum_{k=1}^{\infty} p_k = \infty$, there exists some P_1 such that $\sum_{k=1}^{P_1} p_k > L$. Choose P_1 to be the smallest such P_1 that works.
- Since $\sum_{k=1}^{\infty} n_k = -\infty$, there exists some N_1 such that $\sum_{k=1}^{P_1} p_k + \sum_{k=1}^{N_1} n_k < L$. Choose N_1 to be the smallest such N_1 that works.
- Since $\sum_{k=P_1+1}^{\infty} p_k = \infty$, there exists some $P_2 > P_1$ such that $\sum_{k=1}^{P_1} p_k + \sum_{k=1}^{N_1} n_k + \sum_{P_1+1}^{P_2} p_k > L$. Choose P_2 to be the smallest such P_2 that works.
- Continuing in this way, we construct a sum that gets above L and then below L and then above L and then below L, and so on forever.
- And since we always chose P_i and N_i to be the smallest possible, each time we hopped over L we did so by no more than the size of[11] p_{P_i} or n_{N_i}. And since $p_k \to 0$ and $n_k \to 0$, this "hop over distance" also converges to 0, which forces the sum to converge to L. □

Therefore, there exists a way to rearrange the alternating harmonic series to converge to π, to $-e^{103}$, to LeBron James' phone number, to 8675309, to $-\infty$, and to any other real number you'd like... which I think is just awesome.[12]

[11] It looks funny because of the size differences, but what's written next is p with a subscript of P_i, and then n with a subscript of N_i. Meaning the P_i^{th} positive term, or the N_i^{th} negative term.

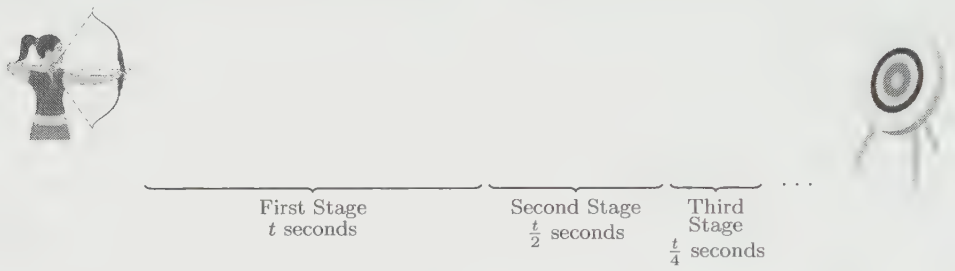
[12] In fact, even more is true. There's also a rearrangement whose limit does not exist. And for those who know what *limit supremums* and *limit infimums* are, for any $\alpha, \beta \in \mathbb{R} \cup \{\pm\infty\}$ with $\alpha \leq \beta$, there is a rearrangment whose lim sup is β and lim inf is α.

Back to Zeno

This book began with a pair of paradoxes from Zeno of Elea, which we can at last settle. We begin with Zeno's second paradox—feel free to flip back to the start of the book to recall what it said.

Example 4.24. For simplicity let's assume the arrow is traveling at a constant rate. In particular, let's suppose Zeno's arrow takes t seconds to get halfway to the target, which means it will take $\frac{t}{2}$ seconds to travel the next quarter of the distance, $\frac{t}{4}$ seconds to travel the following eighth of the distance, and so on.

In Zeno's thought experiment, the arrow's first stage was when it traveled from the archer to the point halfway to the target. The arrow's second stage begins at this point, and lasts until the arrow cuts its distance in half again (i.e. when the arrow is three quarters of the way from the archer to the target). And so on.



Then by the geometric series test the total amount of time (in seconds) to complete all the stages is

$$\sum_{k=0}^{\infty} \frac{t}{2^k} = \frac{t}{1-\frac{1}{2}} = 2t.$$

So the the arrow completes all the stages in just 2t seconds. □

What was Zeno's error? He was correct that the arrow must first complete the first stage, and then must complete the second stage, and then must complete the third stage, and so on. And he was correct that each stage takes some finite amount of time. And it seems to make sense that if you have to do one thing at a time, and you have infinitely many things to do, that you will never complete all of those things. This is false though. One *can* do infinitely many things, one after the other, in just a finite amount of time. The reason? Because a positive series can converge. And if each summand represents a unit of time, then the collection of those infinitely many things can indeed converge to a finite time length.

Zeno's error in his first paradox was also a faulty assumption that an infinite sum of positive terms must diverge. Just to rub it in, let's work out that one too.

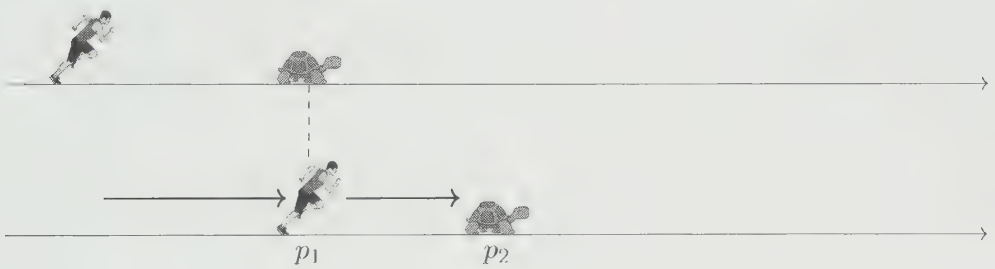
Example 4.25. To make the problem concrete. First, let's be excessively generous[13] and assume the tortoise can move at 1 m/s. Achilles—the greatest warrior of his age—can certainly move quickly. He was easily 10 times faster than even our exceptionally

[13]The Guinness Book of World Records says the fastest tortoise was Bertie, who clocked in at an impressive 0.28 m/s.

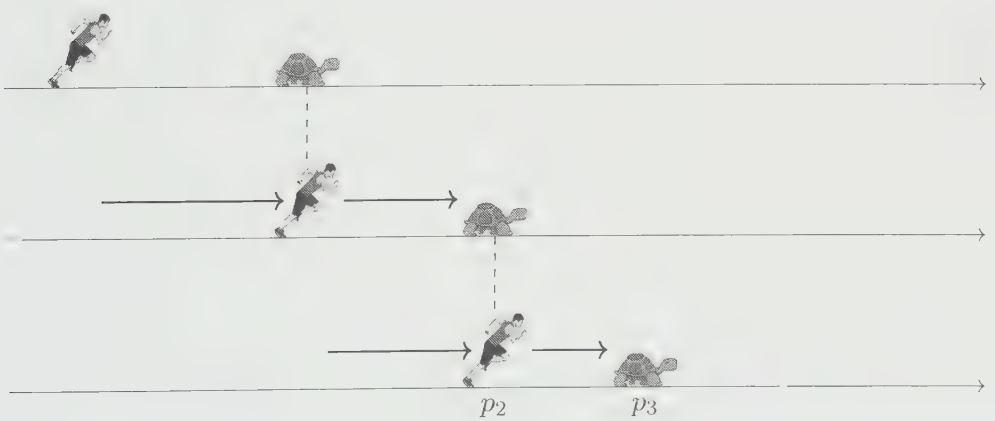
speedy tortoise, so we may assume he can run n m/s, for some $n \geq 10$. (For simplicity, we ignore their acceleration.)[14]

We will look at each stage and determine how long it takes to complete each. We will also keep a *running tally*[15] so that we can find how long it takes to complete all the stages.

- Here, $p_0 = 0$ and $p_1 = 100$.
- So it takes Achilles $\frac{p_1 - p_0}{n} = \frac{100}{n}$ seconds to get from p_0 to p_1.
- Total time this far: $\frac{100}{n}$ seconds.



- Here, $p_1 = 100$ and $p_2 = 100 + \frac{100}{n}$.
- So it takes Achilles $\frac{p_2 - p_1}{n} = \frac{100}{n^2}$ seconds to get from p_1 to p_2.
- Total time this far: $\frac{100}{n} + \frac{100}{n^2}$ seconds.



- Here, $p_2 = 100 + \frac{100}{n}$ and $p_3 = 100 + \frac{100}{n} + \frac{100}{n^2}$.

[14] As the famous joke goes: In response to a question about the milk production of cows, the scientist begins, "For simplicity, assume the cow is a perfect sphere and lives in a vacuum."

[15] Ba dum chhh.

- So it takes Achilles $\frac{p_3-p_2}{n} = \frac{100}{n^3}$ seconds to get from p_2 to p_3.
- Total time this far: $\frac{100}{n} + \frac{100}{n^2} + \frac{100}{n^3}$.

This pattern will continue. So by the geometric series test, the total amount of time (in seconds) to complete all the stages is

$$\sum_{k=0}^{\infty} \frac{100}{n} \cdot \left(\frac{1}{n}\right)^k = \frac{100/n}{1-\frac{1}{10}} = \frac{1000}{9n}.$$

And since $n \geq 10$, the two will pass before $\frac{1000}{90} < 12$ seconds into the race when the tortoise (and hence also Achilles) is only at the 112 meter mark. For the last 888 meters of the race the tortoise will be eating dust. □

— Notable Exercises —

The exercises contain two more convergence tests that you probably learned in calculus—the ratio and the root tests. These tests are not as important as the others to this text's development of real analysis, but are nevertheless important results with particular utility in determining the convergence of specific series.

- In Exercise 4.9 we prove the *ratio test*. That is, given a series $\sum_{k=1}^{\infty} a_k$ with $a_k \neq 0$, we will show that if $\lim_{k\to\infty} \left|\frac{a_{k+1}}{a_k}\right|$ if less than 1 then the series converges.

- In Exercise 4.10 we prove the *root test*. That is, if $\sum_{k=1}^{\infty} a_k$ is a series where each $a_k \geq 0$ and the limit $\lim_{k\to\infty} (a_k)^{1/k}$ exists, we call this limit ρ. Then this series converges if $\rho < 1$ and diverges if $\rho > 1$. (The test is inconclusive if $\rho = 1$.)

The rearrangement theorem (Theorem 4.23) stated that any conditionally convergent series can be arranged to converge to anything at all—to any $L \in \mathbb{R}$ or to either ∞ or $-\infty$. After stating the theorem we sketched a proof for the case when $L \in \mathbb{R}$ and $L > 0$.

- In Exercise 4.15 we prove that there exists a rearrangement of this sum which diverges to ∞.

This roughly completes the proof of Theorem 4.23, and in the below we extend this result.

- In Exercise 4.16 we show that there exists a rearrangement of this sum whose limit does not exist.

- In Exercise 4.17 we prove a generalization of Theorem 4.23 called *Riemann's series theorem* in which in a rearrangement can be found whose *limit supremum* and *limit infimum* can be arbitrarily chosen.

- Notice that Theorem 4.23 says nothing about what happens for series which are not conditionally convergent. In Exercise 4.18 we prove that if a series converges absolutely, then every rearrangement converges to the same thing. And in Exercise 4.19 we prove that if each $a_k \geq 0$ and $\sum_{k=1}^{\infty} a_k = \infty$, then any rearrangement of this sum also diverges to ∞.

— Exercises —

Exercise 4.1. Determine whether each of the following converges conditionally, converges absolutely, or diverges. You do not need to prove your answers, but state which of the following tests gives the answers: the k^{th}-term test, the geometric series test, and the alternating series test.

- $\displaystyle\sum_{k=1}^{\infty}(-1)^k\frac{1}{\sqrt{k}}$
- $\displaystyle\sum_{k=1}^{\infty}\frac{1}{(\ln(4))^k}$
- $\displaystyle\sum_{k=1}^{\infty}\left(\sqrt{n+1}-\sqrt{n}\right)$
- $\displaystyle\sum_{k=1}^{\infty}(-1)^k\frac{k}{k+7}$
- $\displaystyle\sum_{k=1}^{\infty}\frac{1}{k!}$
- $\displaystyle\sum_{k=1}^{\infty}\frac{1}{2}$

Exercise 4.2. Prove Proposition 4.18. That is, prove that if $\displaystyle\sum_{k=1}^{\infty}|a_k|$ converges, then $\displaystyle\sum_{k=1}^{\infty}a_k$ converges too.

Exercise 4.3. Prove Lemma 4.8. That is, prove that the sequence (r^n) converges to 0 if $r \in (-1,1)$, converges to 1 if $r = 1$, and diverges otherwise.

Exercise 4.4. Give an example of a series with nonnegative terms where $\displaystyle\sum_{k=1}^{\infty}a_k$ diverges, but $\displaystyle\sum_{k=1}^{\infty}a_k^2$ converges.

Exercise 4.5.

(a) Prove that if $\displaystyle\sum_{k=1}^{\infty}a_k$ converges where each $a_k > 0$, then $\displaystyle\sum_{k=1}^{\infty}a_k^2$ converges.

(b) Show by example that this is not true if we do not insist that each $a_k > 0$.

Exercise 4.6. Give an example of a series $\sum_{k=1}^{\infty}a_k$ where

- $\displaystyle\sum_{k=1}^{\infty}a_k$ converges,
- $\displaystyle\sum_{k=1}^{\infty}a_k^2$ diverges, and
- $\displaystyle\sum_{k=1}^{\infty}a_k^3$ converges.

Exercise 4.7. Find an estimate for $\sum_{k=1}^{\infty}(-1)^k \frac{1}{7k}$ that is accurate to 0.01.

Exercise 4.8. The geometric series test (Proposition 4.9) says that a geometric series diverges if $r \geq 1$. Recall that a series can diverge to ∞, to $-\infty$, or can be "does not exist." Which form of divergence is it for a geometric series? Note that your answer may depend on r.

Exercise 4.9. Prove the *ratio test* via the following steps. Given a series $\sum_{k=1}^{\infty} a_k$ with $a_k \neq 0$, assume that

$$\lim_{k\to\infty} \left|\frac{a_{k+1}}{a_k}\right| =: r < 1.$$

We will prove that the series converges absolutely.

(a) Let q be such that $r < q < 1$. Explain why there is some N such that $n \geq N$ implies that $|a_{k+1}| \leq |a_k| \cdot q$.

(b) Explain why $\sum_{k=1}^{\infty} |a_N| \cdot q^k$ necessarily converges.

(c) Finally, use part (b) to prove that $\sum_{k=1}^{\infty} |a_k|$ converges.

Exercise 4.10. Prove the *root test* via the following steps. Given a series $\sum_{k=1}^{\infty} a_k$ where each $a_k \geq 0$, assume that the limit $\lim_{k\to\infty} (a_k)^{1/k}$ exists. Call this limit ρ. Then this series converges if $\rho < 1$ and diverges if $\rho > 1$. (The test is inconclusive if $\rho = 1$.)

(a) Suppose $\rho < 1$. Let $\varepsilon = \frac{1-\rho}{2}$ and $\rho_1 = \rho + \varepsilon$. Prove that there is some N for which $(a_n)^{1/n} < \rho_1$ for all $n > N$.

(b) Prove that $\sum_{k=N}^{\infty} a_k$ converges by comparing it to a geometric series. Then conclude that $\sum_{k=1}^{\infty} a_k$ also converges.

(c) Suppose $\rho > 1$. Let $\varepsilon = \frac{\rho-1}{2}$ and $\rho_2 = \rho - \varepsilon$. Prove that there is some N for which $(a_n)^{1/n} > \rho_2$ for all $n > N$.

(d) Use this to argue that $\sum_{k=N}^{\infty} a_k$ diverges by using the k^{th}-term test (Proposition 4.5). Then conclude that $\sum_{k=1}^{\infty} a_k$ also diverges.

Exercise 4.11. Give an example of a divergent series $\sum_{k=1}^{\infty} a_k$ for which $\lim_{k\to\infty} (a_{k+1} - a_k) = 0$.

Exercise 4.12. The converses of (i) and (ii) of the Comparison test (Proposition 4.12) are both false. Give a pair of examples demonstrating this.

Exercise 4.13. Give an example of a divergent series $\sum_{k=1}^{\infty} a_k$ and a convergent series $\sum_{k=1}^{\infty} b_k$ where $a_k \leq b_k$ for all k.

Exercise 4.14. Consider the sum $\sum_{k=1}^{\infty} a_k$, and define s_n to be this series' n^{th} partial sum; that is, $s_n = \sum_{k=1}^{n} a_k$. The series $\sum_{k=1}^{\infty} a_k$ is called *Cesaro summable* if

$$\lim_{n\to\infty} \frac{s_1 + s_2 + \cdots + s_n}{n}$$

converges.

(a) Prove that if $\sum_{k=1}^{\infty} a_k$ converges, then this series is Cesaro summable.

(b) Prove by example that if $\sum_{k=1}^{\infty} a_k$ is Cesaro summable this does *not* imply that $\sum_{k-1}^{\infty} a_k$ converges.

Exercise 4.15. Show that if $\sum_{k-1}^{\infty} a_k$ is conditionally convergent, then there exists a rearrangement of this sum which diverges to ∞.

Exercise 4.16. Show that if $\sum_{k=1}^{\infty} a_k$ is conditionally convergent, then there exists a rearrangement of this sum whose limit does not exist.

Exercise 4.17. Assume that $\sum_{k=1}^{\infty} a_k$ is conditionally convergent. Define the *limit superior* of a sequence (s_n) (Notation: "$\limsup_{n\to\infty} s_n$") to be

$$\limsup_{n\to\infty} s_n = \lim_{n\to\infty} \left(\sup_{m\geq n} s_m \right).$$

And define the *limit inferior* of sequence (s_n) (Notation: "$\liminf_{n\to\infty} s_n$") to be

$$\liminf_{n\to\infty} s_n = \lim_{n\to\infty} \left(\inf_{m\geq n} s_m \right).$$

Prove that for any $\alpha, \beta \in \mathbb{R} \cup \{\pm\infty\}$ with $\alpha \leq \beta$, there is a rearrangement of $\sum_{k=1}^{\infty} a_k$ whose sequence of partial sums, (s_n), have

$$\limsup_{n\to\infty} s_n = \beta \qquad \text{and} \qquad \liminf_{n\to\infty} s_n = \alpha.$$

Exercise 4.18. Prove that if each $\sum_{k=1}^{\infty} a_k$ converges absolutely to L, then any rearrangement of this sum also converges to L.

Hint: Consider the partial sums (s_n) of the given series, and the partial sums (s'_n) of a rearrangement of the given series, and show that $|s_n - s'_n| \to 0$.

Exercise 4.19. Prove that if each $a_k \geq 0$ and $\sum_{k=1}^{\infty} a_k = \infty$, then any rearrangement of this sum also diverges to ∞.

— Open Questions —

Question 1. Does $\displaystyle\sum_{k=1}^{\infty} \frac{1}{k^3 \sin^2(k)}$ converge?

Question 2. Is it true that $\displaystyle\sum_{k=0}^{\infty} \frac{1+14k+76k^2+168k^3}{2^{20k}} \binom{2k}{k}^7 = \frac{32}{\pi^3}$?

Question 3. Is $\displaystyle\sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)^2}$ irrational?

Question 4. Is $\displaystyle\sum_{k=1}^{\infty} \frac{1}{k^3}$ transcendental?

Question 5. One can show that[16]

$$\sum_{k=1}^{\infty} \left(\frac{1}{k} \times \frac{1}{k+1} \right) = 1.$$

We now reinterpret this algebraic fact geometrically. Notice that the left-hand side is the area of a $\frac{1}{1} \times \frac{1}{2}$ rectangle, plus the area of a $\frac{1}{2} \times \frac{1}{3}$ rectangle, plus the area of a $\frac{1}{3} \times \frac{1}{4}$ rectangle, and so on. The right-hand side is the area of a 1×1 square. This suggests that it is *possible* that you can tile a 1×1 square with this infinite collection of rectangles. Prove or disprove that such a tiling exists.[17]

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[16] A super neat way to show this is to first use a discrete version *partial fractions*—something you might remember from calculus as a technique to solve integrals. Indeed,

$$\frac{1}{k(k+1)} = \frac{A}{k} + \frac{B}{k+1} \quad \Rightarrow \quad 1 = (A+B)k + A \quad \Rightarrow \quad A = 1,\ B = -1.$$

Applying this, the sum then telescopes!

$$\begin{aligned} \sum_{k=1}^{\infty} \frac{1}{k(k+1)} &= \sum_{k=1}^{\infty} \left(\frac{1}{k} + \frac{-1}{k+1} \right) = \lim_{n\to\infty} \left[\left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \cdots + \left(\frac{1}{n} - \frac{1}{n+1}\right) \right] \\ &= \lim_{n\to\infty} \left[1 + \left(-\frac{1}{2} + \frac{1}{2}\right) + \left(-\frac{1}{3} + \frac{1}{3}\right) + \cdots + \left(-\frac{1}{n} + \frac{1}{n}\right) - \frac{1}{n+1} \right] \\ &= \lim_{n\to\infty} \left[1 - \frac{1}{n+1} \right] = 1 - 0 = 1. \end{aligned}$$

[17] It's known this infinite collection of rectangles can be packed in the 1.002×1.002 square. See *Two packing problems* by Vojtech Bálint and *An algorithm for packing squares* by Marc Paulhus.

Chapter 5: The Topology of $\mathbb{R}$

5.1 Open Sets

We are approaching the study of continuous functions.[1] This is a surprisingly subtle task. In fact, there is a whole area of math which is the study of sets and continuous functions on those sets; this area is called *topology*. One could easily create a full-year undergraduate math course just on topology, so our treatment, totaling a mere 11 (long-form) pages, will certainly not be comprehensive.

Nevertheless, this brief introduction will aim to do two things. The first is practical: it lays the foundation for our study of continuous functions by introducing special classes of sets which, when we ask for a function's behavior on these sets, will *determine* whether that function is continuous. The second is more philosophical. Although math is traditionally taught by first partitioning it into discrete[2], non-overlapping areas like real analysis, abstract algebra, linear algebra, combinatorics and topology, mathematics is in reality much more interconnected than this, and one's mathematical education should include examples thereof.

This chapter will be focused on three fundamental objects—those of open sets, closed sets and compact sets—which in turn help explain real analysis' place in the sprawling ecosystem of mathematics. We begin with the idea of an *open set*.

Definition.

Definition 5.1. A set $U \subseteq \mathbb{R}$ is *open* if, for every $x \in U$, there is a number $\delta > 0$ such that

$$(x - \delta, x + \delta) \subseteq U.$$

That is, if for all $x \in U$ there exists some *δ-neighborhood of x*, denoted $V_\delta(x)$, such that $V_\delta(x) \subseteq U$.

Note that this is equivalent to saying $U \subseteq \mathbb{R}$ is *open* if, for each $x \in U$, there is a number $\delta > 0$ such that if y satisfies $|x - y| < \delta$, then $y \in U$.

[1] Believe it or not, this is a (highly nerdy) joke. After Chapter 6 you may find it mildly amusing.
[2] (Or continuous.)

Example 5.2.

(i) The set $\mathbb{R}$ is open. To see this, note that for any $x \in \mathbb{R}$, if you let $\delta = 17$ then $(x - 17, x + 17) \subseteq \mathbb{R}$.

(ii) The empty set $\emptyset$ is open. To see this, note that, because there are no elements in $\emptyset$, it's true to say "for any $x \in \emptyset$, x is a purple elèphant that speaks Spanish." It's vacuously[3] true! You certainly can't disprove it, right? You can't present to me any element in $\emptyset$ that is *not* a purple elephant that speaks Spanish. Likewise, for any $x \in \emptyset$ there exists a δ such that $(x - \delta, x + \delta) \subseteq U$. That is again true, as you certainly cannot find an $x \in \emptyset$ that does not satisfy this.

(iii) The open interval (a, b) is open. To see this, pick any $x \in (a, b)$. Let $\delta = \min\{x - a, b - x\}$. Then $(x - \delta, x + \delta) \subseteq (a, b)$.

Here, $\delta = x - a$

Here, $\delta = b - x$

(iv) The intervals (a, ∞) and $(-\infty, b)$ are open. To see this, pick any $x \in (a, \infty)$ and let $\delta = x - a$; or pick any $x \in (-\infty, b)$ and let $\delta = b - x$.

(v) A non-example: The set $[3, 7]$ is *not* an open set. To see this, note that $3 \in [3, 7]$, but for any $\delta > 0$ we have

$$(3 - \delta, 3 + \delta) \not\subseteq [3, 7].$$

That is, every δ-neighborhood of $x = 3$ is *not* a subset of $[3, 7]$. So we dissatisfied the definition "for all $x \in U$, there is..." by exhibiting a single x for which it fails. Therefore, $[3, 7]$ is *not* open.

□

[3] A vacuum in physics is a container in which the air inside has been sucked out, leaving nothing left. Likewise in math, saying something is *vacuously true* means that the set of elements that the statement is referring to is empty; there is therefore nothing to prove, and it's automatically true.

Proposition.

Proposition 5.3 (*Open sets via arbitrary unions and finite intersections*).

(i) If $\{U_\alpha\}$ is any collection of open sets, then $\bigcup_\alpha U_\alpha$ is also an open set.

(ii) If $\{U_1, U_2, \ldots, U_n\}$ is a finite collection of open sets, then $\bigcap_{k=1}^{n} U_k$ is also an open set.

When you see "$\bigcup_\alpha U_\alpha$," what this means is that you are taking the union of all the sets in $\{U_\alpha\}$, whether that is finitely many sets, countably many sets, $|\mathbb{R}|$ many sets, or $|\mathcal{P}(\mathbb{R})|$ many sets. Below is a quick table summarizing what this union would look like in each case.

Note.

If $\{U_\alpha\}$ contains	Then $\bigcup_\alpha U_\alpha$ is just
Finitely many sets	$\bigcup_{k=1}^{n} U_k$
Countably many sets	$\bigcup_{k=1}^{\infty} U_k$
$\lvert\mathbb{R}\rvert$ many sets	$\bigcup_{x\in\mathbb{R}} U_x$
$\lvert\mathcal{P}(\mathbb{R})\rvert$ many sets	$\bigcup_{x\in\mathcal{P}(\mathbb{R})} U_x$

Proof of Proposition 5.3. For (i), assume that $x \in \bigcup_\alpha U_\alpha$. By the definition of an open set we need to find some δ-neighborhood of x contained inside this union. Since x is in this union, we know $x \in U_{\alpha_0}$ for some α_0. By definition of U_{α_0} being open, this implies that there is some δ-neighborhood $V_\delta(x) \subseteq U_{\alpha_0}$. But since $U_{\alpha_0} \subseteq \bigcup_\alpha U_\alpha$, we have that $V_\delta(x) \subseteq \bigcup_\alpha U_\alpha$, completing part (i).

To prove part (ii), first assume that we have some $x \in \bigcap_{k=1}^{n} U_k$. We need to show that there is a δ-neighborhood[4] of x that is contained inside this intersection. Now, since x is in the intersection, we know $x \in U_k$ for each $k \in \{1, 2, \ldots, n\}$. And since each of these sets is open, that implies that for each k there is a δ_k-neighborhood of x, $V_{\delta_k}(x)$, such that $V_{\delta_k}(x) \subseteq U_k$. Now, these δ_k-neighborhoods may be different sizes, and so if, say, $\delta_1 > \delta_2$, it might not be the case that $V_{\delta_1}(x) \subseteq U_2$. But if we choose $\delta = \min\{\delta_1, \delta_2, \ldots, \delta_n\}$, then $V_\delta(x) \subseteq V_{\delta_k}(x) \subseteq U_k$ for all k, and so $V_\delta(x) \subseteq \bigcap_{k=1}^{n} U_k$, completing the proof. □

Example 5.4.

(i) We already showed in Example 5.2 that all intervals (a, b) are open, so by Proposition 5.3, also $(a, b) \cup (c, d)$ is open, and $(a, b) \cup (c, d) \cup (e, f)$ is open, and $(a_1, b_1) \cup (a_2, b_2) \cup \ldots$ is open, and any other union of open intervals is open.

(ii) Likewise, $(a, b) \cap (c, d)$ is open. For instance, $(0, 2) \cap (1, 3) = (1, 2)$ is open, as is $(0, 2) \cap (4, 6) = \emptyset$. (But note that such finite intersections are always of the original form: either a single open interval (a, b), or the empty set.)

(iii) We also showed in Example 5.2 that intervals $(-\infty, a)$ and (b, ∞) are open, and hence their union $(-\infty, a) \cup (b, \infty)$ is open, too.

• Note that this third example actually follows from the first, because $(-\infty, a) \cup (b, \infty)$ is also a union of open intervals:

$$(-\infty, a) \cup (b, \infty) = \bigcup_{n=1}^{\infty} (a - n, a) \cup (b, b + n).$$

(iv) Likewise,

$$\Big[(-\infty, a) \cup (b, \infty)\Big] \cap \Big[(-\infty, c) \cup (d, \infty)\Big] = (-\infty, \min\{a, c\}) \cup (\max\{b, d\}, \infty)$$

is open. (But again, this is always of the original form.)

Based on the discussion in the bullet point after (iii), we have not yet seen an example where an open set is anything other than a union of open intervals. This makes even more sense when you combine (1) the fact that every point in an open set is contained inside a δ-neighborhood (which is an open interval) and (2) the fact that if you union together any collection of open sets then the result is still open. You can therefore imagine that it is true that every open set is a union of open intervals. In fact, even more is true: every open set is a union of *countably many* disjoint open intervals.

[4]Confession: It's difficult for me to do topology without Mister Rogers playing in my head... ♫ *It's a beautiful day in this neighborhood* ♫

Theorem.

Theorem 5.5 (*Each open set* $= \bigcup_{k=1}^{\infty}(a_k, b_k)$). Every open set is a countable union of disjoint open intervals.

Note that open intervals can be of the form (a, b), $(-\infty, a)$, or (b, ∞).

Proof. Let A be an open set. Then each $x \in A$ is contained inside some open interval $(x - \delta, x + \delta)$. There may be a larger open interval that contains x, though, so let's let I_x be the largest open interval containing x. Formally, $I_x = (\alpha, \beta)$ where

$$\alpha = \inf\{a : (a, x) \subseteq A\} \qquad \text{and} \qquad \beta = \sup\{b : (x, b) \subseteq A\}.$$

Note that, given any pair x and y, either $I_x = I_y$ or $I_x \cap I_y = \emptyset$. The reason for this is that, if instead you had two intervals overlapping but not equal, then you could expand at least one of them, contradicting the fact that we had chosen the *largest* intervals.

We claim that these I_x intervals are the ones we are looking for. To see this, first note that since each $x \in A$ is in $I_x \subseteq A$, the union of all the I_x intervals does indeed equal A. Moreover, we can see that there are only countably many distinct such intervals because by the density of $\mathbb{Q}$ in $\mathbb{R}$, every open interval contains a rational number, so there can not be more intervals than rationals, which is a countable set.[5] □

5.2 Closed Sets

Definition.

Definition 5.6. A set $A \subseteq \mathbb{R}$ is *closed* if A^c is open.

Note that[6] since $(A^c)^c = A$, the above is equivalent to saying that A is closed if there exists some open set B such that $B^c = A$.

Example 5.7.

- The closed interval $[a, b]$ is closed. To see this, note that we have already shown that $(-\infty, a) \cup (b, \infty)$ is open, and its complement is $[a, b]$, and so by definition $[a, b]$ is closed.
- $\mathbb{R}$ is closed. To see this, note that we have already shown that $\emptyset$ is open, and $\emptyset^c = \mathbb{R}$.

[5] The details of this final step were asked for in Exercise 2.6.

[6] Recall that if $A \subseteq \mathbb{R}$, then the complement $A^c = \mathbb{R} \setminus A$.

- $\emptyset$ is closed.[7] To see this, note that we have already shown that $\mathbb{R}$ is open, and $\mathbb{R}^c = \emptyset$.

Intuitively, one difference between the open interval $(0, 1)$ and the closed interval $[0, 1]$ is that in the open interval there is a limit of terms from *inside* the set which is converging to something *outside* the set; one example: $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots$ is a sequence of numbers from $(0, 1)$ which converge to 0, which is not in $(0, 1)$. The closed interval does not have this property; if a sequence of terms from the set $[0, 1]$ is converging, then we know that it converges to something inside the set. This intuition jives with our definitions of open and closed sets, as the following definition and proposition say.

Definition.

Definition 5.8. A point x is a *limit point* of a set A if there is a sequence of points $a_1, a_2, a_3, \ldots$ from $A \setminus \{x\}$ such that $a_n \to x$.

In Exercise 5.5 you're asked to show this: a point x is a limit point of A if and only if every ε-neighborhood of x intersects[8] A at some point other than x.

Example 5.9. The point 0 is a limit point of the interval $(0, 3)$ because, for each $n \in \mathbb{N}$, we see that $a_n = \frac{1}{n} \in (0, 1)$, while $a_n \to 0$. The point 0.24 is also a limit point of $[0, 1]$, as it is the limit of the sequence $0.23, 0.239, 0.2399, 0.23999, \ldots$.

Being closed in fact means that you contain your limit points. This is what the next theorem states.

Theorem.

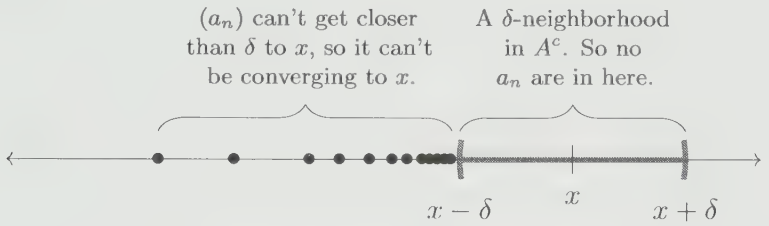
Theorem 5.10 (*Closed iff contains limit points*). A set A is closed if and only if it contains all of its limit points.

Scratch Work. The "contains all of its limit points" condition can be tough to work with. It's much easier to handle a specific limit point (which, by the definition of limit point, gives a specific sequence converging to that point). Therefore for the forward direction we will proceed by contradiction (giving us a specific sequence from A that converges to an $x \notin A$), and for the reverse direction we will use the contrapositive (meaning that we are in search of just a single sequence from A that converges outside A).

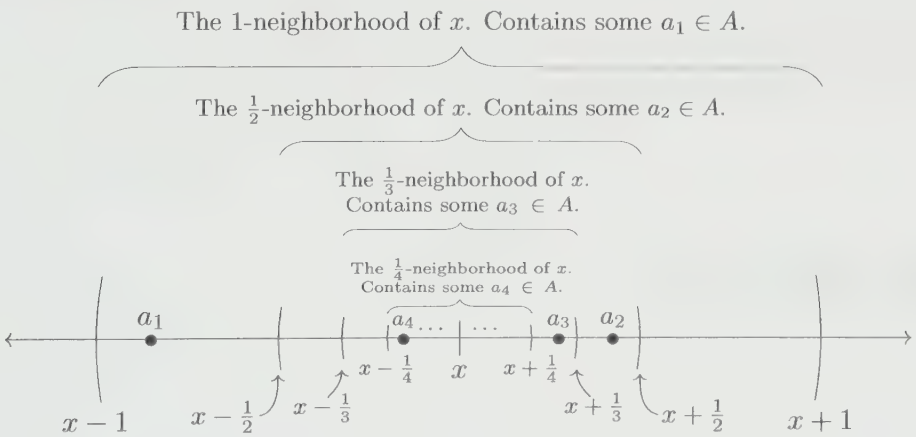
[7] Yes, $\mathbb{R}$ and $\emptyset$ are open *and* closed. Yes, this is silly. See the YouTube video "Hitler Learns Topology" (`www.youtu.be/SyD4p8_y8Kw`) for a dramatic response. These sets are sometimes called *clopen*. In Exercise 5.11 you are asked to prove that these are the only clopen subsets. (For general topologies—beyond the topology of $\mathbb{R}$, which we are studying—there often many more clopen sets.)

[8] Saying a set X *intersects* a set Y means $X \cap Y \neq \emptyset$.

For the forward direction, when we assume for a contradiction that A does not contain all its limit points, this gives us an $x \notin A$ for which there is a sequence (a_n) converging to x, where each $a_n \in A$. Where's the contradiction? Well, converging to x means these points a_n from A are getting closer and closer to x. But since $x \notin A$, that means $x \in A^c$; and since A is closed, A^c is open. And being open means there is a δ-neighborhood of x which is entirely contained inside A^c; i.e., is disjoint from A. That's the contradiction: it's impossible for a sequence (a_n) be getting closer and closer to x, if all these points have to be from A, and hence have to stay outside this δ-neighborhood of x.



As for the reverse direction, by using the contrapositive we may assume that A is not closed, which is to say that A^c is not open. And being "not open" means that for some $x \in A^c$, every δ-neighborhood of x contains a member of A. So if we let $\delta = 1$, then this 1-neighborhood of x contains a member of A, call it a_1. Likewise, if we let δ be $1/2$, then $1/3$, then $1/4$, and so on, we get an a_2, a_3, a_4, and so on, each from A and contained inside these respective neighborhoods. These elements comprise a sequence from A, which converges to x.



Proof. For the forward direction, suppose that A is closed, and assume for a contradiction that A does not contain all of its limit points. That is, suppose that there is a sequence $a_1, a_2, a_3, \ldots$ from A where $a_n \to x$, but $x \notin A$. Observe that this implies that A^c is open and $x \in A^c$. And so, by openness, there is some δ-neighborhood $(x - \delta, x + \delta) \subseteq A^c$. We now show that this is a contradiction to the claim that $a_n \to x$.

Since $a_n \to x$ where each $a_n \in A$, for all $\varepsilon > 0$ there exists some N such that $|a_n - x| < \varepsilon$ for all $n > N$. In particular, if we let $\varepsilon = \delta$, then such an N exists, which

in turn implies that $|a_{N+1} - x| < \delta$. That is, $a_{N+1} \in (x-\delta, x+\delta)$ and $a_{N+1} \in A$. But this is a contradiction, since we showed in the last paragraph that $(x-\delta, x+\delta) \subseteq A^c$, meaning that $a_{N+1} \notin A$.

For the backwards direction, we will prove the contrapositive. Assume that A is not closed; that is, A^c is not open. Being not open means that there is some $x \in A^c$ such that every δ-neighborhood of x contains a point that is not in A^c, meaning that it is in A. In particular, for each $n \in \mathbb{N}$ there is some point $a_n \in A$ such that a_n is in the $\frac{1}{n}$-neighborhood of A. We claim that $a_n \to x$. Note that if we can prove this then we are done, since we will have constructed a sequence of points from A converging to a point outside of A, proving that A does not contain all of its limit points.

To prove the claim, let $\varepsilon > 0$. By the Archimedean principle there is some N such that $\frac{1}{N} < \varepsilon$. And so for any $n > N$ we have $\frac{1}{n} < \varepsilon$, and hence a_n is in the ε-neighborhood of x. That is,

$$|a_n - x| < \varepsilon$$

for all $n > N$, implying that $a_n \to x$ and completing the proof. □

The final proposition of this section follows fairly directly from Proposition 5.3 provided you recall the following fact, known as *De Morgan's laws* (whose proof in the finite case was also asked of you in Exercise 1.4).

Fact.

Fact 5.11. De Morgan's laws state that for an arbitrary union or intersection,

$$\left(\bigcup_\alpha U_\alpha\right)^c = \bigcap_\alpha {U_\alpha}^c \qquad \text{and} \qquad \left(\bigcap_\alpha U_\alpha\right)^c = \bigcup_\alpha {U_\alpha}^c.$$

Proposition.

Proposition 5.12 (*Closed sets via finite unions and arbitrary intersections*).

(i) If $\{U_1, U_2, \ldots, U_n\}$ is a collection of closed sets then $\bigcup_{k=1}^{n} U_k$ is also a closed set.

(ii) If $\{U_\alpha\}$ is a collection of closed sets then $\bigcap_\alpha U_\alpha$ is also a closed set.

Proof. This is Exercise 5.4. I suggest you make use of De Morgan's laws when you prove it. □

5.3 Open Covers

Definition

Definition 5.13. Let A be a set.

- The sets $\{U_\alpha\}_{\alpha \in S}$ are a *cover* of A if
$$A \subseteq \bigcup_{\alpha \in S} U_\alpha.$$
- If each U_α is open, then $\{U_\alpha\}_{\alpha \in S}$ is an *open cover* of A.
- If $\{U_\alpha\}_{\alpha \in S}$ has a finite subset $\{U_\alpha\}_{\alpha \in F}$ (meaning $S \subseteq F$) which is still a cover of A, then $\{U_\alpha\}_{\alpha \in F}$ is called a *finite subcover* of A.

This idea may seem odd. It's not at all clear that this should be anything that we should care about. Bear with me, though, because in the grand scheme it is amazing how much this is *exactly* something that we should care about.

Example 5.14.

- An open cover of $(0, 4)$ is
$$\left\{\left(\frac{1}{k}, 4 - \frac{1}{k}\right)\right\}_{k=1}^{\infty} = \left\{(1, 3), \left(\frac{1}{2}, 4 - \frac{1}{2}\right), \left(\frac{1}{3}, 4 - \frac{1}{3}\right), \dots\right\}.$$
Clearly each set in the above is open, as each is an open interval. Moreover, for any $x \in (0, 4)$ note that for large enough n, one of those open intervals will contain x.
- Another open cover of $(0, 4)$ is
$$\{(-\infty, 2), (1, 3), (1, 2), (0.5, 3.5), (3, 7)\}.$$
Again, all of the above intervals are clearly open sets, as they are open intervals. Note, though, that not all of the above sets are necessary in order to cover $(0, 4)$; we could remove $(1, 2)$, for example. This is fine, though, and in fact it will be a staple of what's to come.

  Also note that not all the intervals above are subsets of $(0, 4)$. This is also fine. Oftentimes sets in our covers will go outside the set we are covering.

□

Here is a final example.

Example 5.15. An open cover of $[0, 4]$ is
$$\left\{\left(\frac{1}{k}, 4 - \frac{1}{k}\right)\right\}_{k=1}^{\infty} \cup \{(-0.2, 0.2)\} \cup \{(3.9, 4.1)\}.$$ □

5.4 The Greatest Definition in Mathematics

Without further ado,[9] here it is:

Definition.

Definition 5.16. A set A is *compact* if every open cover of A contains a finite subcover of A.

It may seem crazy that this is one of the great definitions in mathematics, but it is.[10] Indeed, one of the reasons it is so significant is that it is not at all obvious. It took great insight to realize that this topological concept is crucial. Indeed, from the topological perspective, compact sets are sets which are *essentially* finite. If your properties only care about what's happening on open sets, and there are essentially only finitely many of them at a time, then things are simple and nice. This course illustrates how messy the infinite is, but compactness illustrates when things are simple(r).

Let's return to our previous two examples.

Example 5.17. The open cover

$$\left\{\left(\frac{1}{k}, 4-\frac{1}{k}\right)\right\}_{k=1}^{\infty}$$

of $(0,4)$ has *no* finite subcover Exercise 5.6. So $(0,4)$ is *not* compact. □

Example 5.18. The open cover

$$\left\{\left(\frac{1}{k}, 4-\frac{1}{k}\right)\right\}_{k=1}^{\infty} \cup \{(-0.2, 0.2)\} \cup \{(3.9, 4.1)\}$$

of $[0,4]$ *does* have a finite subcover. For instance, here's one:

$$\left\{(-0.2, 0.2), (3.9, 4.1), \left(\frac{1}{11}, 4-\frac{1}{11}\right)\right\}.$$

So $[0,4]$ *may* be compact (this cover has a finite subcover, but to be compact *every* open cover must have a finite subcover.) □

In fact, the above set is indeed compact. This is a consequence of a very important theorem called the Heine-Borel theorem.[11]

[9] Ok, maybe not the *absolute* greatest, but definitely up there. It's simple to state and it's not obvious at all that it is a property we should care about, but it turns out that it is *exactly* what we should care about. If you have other great definitions in mind, send me an email to let me know.

[10] That said, great definitions do not always come out of nowhere. For an interesting historical look at its development, see *A pedagogical history of compactness* by Manya Raman-Sundström, viewable here: `https://arxiv.org/pdf/1006.4131.pdf`

[11] Fair warning: The proof is quite long and arduous. But *[footnote continues on next page]*

Theorem.

Theorem 5.19 (*The Heine-Borel theorem*). A set $S \subseteq \mathbb{R}$ is compact if and only if S is closed and bounded.

Proof. This is a long proof, but if you break it down into chunks it is manageable. We will first prove the forward direction.

— Forward direction —

For the forward direction, assume that S is compact. That is, that *every* open cover of S contains a finite subcover of S. To prove that S is closed and bounded, the idea is to pick a specific open cover of S for which the existence of a finite subcover proves the property.

Proof that S is bounded. To prove that S is bounded, we will consider the following open cover. Let $I_n = (-n, n)$; clearly I_n is an open set. Moreover, since

$$\bigcup_{n=1}^{\infty} I_n = \mathbb{R},$$

and $S \subseteq \mathbb{R}$, we see that

$$S \subseteq \bigcup_{n=1}^{\infty} I_n.$$

So we have seen that $\{I_n\}_{n=1}^{\infty}$ is an open cover of S, and since by assumption S is compact this means that there is some finite subcover

$$\{I_{n_1}, I_{n_2}, \ldots, I_{n_k}\}$$

of S. Assume that the sets were written in increasing order so that $n_1 < n_2 < \cdots < n_k$. Notice that this implies that

$$I_{n_1} \subseteq I_{n_2} \subseteq \cdots \subseteq I_{n_k},$$

and hence

$$S \subseteq \bigcup_{\ell=1}^{k} I_{n_\ell} = I_{n_k}.$$

But saying that $S \subseteq I_{n_k}$ precisely means that S is bounded between $-n_k$ and n_k, completing this part of the proof. $\square_{(S \text{ bounded})}$

surprisingly when I surveyed my last real analysis class on what their favorite theorem was throughout the course, quite a few said the Heine-Borel theorem, and they specifically cited the toughness of the proof as a reason! Now, if you don't believe you have that sort of character yet... then I've got proof right here to help you build it right up! Also, to read a good discussion of the theorem's history, see *An Analysis of the First Proofs of the Heine-Borel Theorem—Borel's Proof* by Andre, Engdahl and Parker.

Proof that S is closed. Suppose for a contradiction that S is not closed. Then there exists some $x \notin S$ and a sequence $a_1, a_2, a_3, \ldots$ where each $a_k \in S$ and $a_k \to x$.

We again are going to find an open cover of S, and it will turn out that the existence of a finite subcover will give us our contradiction. To this end, for each $n \in \mathbb{N}$ let

$$U_n = \left(-\infty, x - \frac{1}{n}\right) \cup \left(x + \frac{1}{n}, \infty\right).$$

Notice that each U_n is an open set and moreover that

$$\bigcup_{n=1}^{\infty} U_n = \mathbb{R} \setminus \{x\}.$$

And this is great, since all we know about S is that $S \subseteq \mathbb{R}$ and $x \notin S$. So since $\{U_n : n \in \mathbb{N}\}$ covers *everything* except x, certainly it will cover S:

$$S \subseteq \bigcup_{n=1}^{\infty} U_n.$$

Now we use that S is compact. By compactness, this open cover of S must contain a finite subcover of S:

$$\{U_{n_1}, U_{n_2}, \ldots, U_{n_k}\}.$$

And once again if we assume that the subscripts are increasing and hence n_k is the largest, then all of the other U_{n_ℓ} are subsets of U_{n_k}, and hence

$$S \subseteq \bigcup_{\ell=1}^{k} U_{n_\ell} = U_{n_k}.$$

But saying that $S \subseteq U_{n_k}$ means that

$$S \subseteq \left(-\infty, x - \frac{1}{n_k}\right) \cup \left(x + \frac{1}{n_k}, \infty\right).$$

Which in particular implies that

$$S \cap \left(x - \frac{1}{n_k}, x + \frac{1}{n_k}\right) = \emptyset.$$

But this is a contradiction. We had assumed that there was a sequence (a_k) of numbers from S for which $a_k \to x$. But if we let $\varepsilon = \frac{1}{n_k} > 0$, then by the above there are *no* a_k for which

$$|a_k - x| < \varepsilon,$$

clearly contradicting the definition of convergence of a_k to x. $\square_{(S \text{ closed})}$

— Backwards direction —

Now assume that S is closed and bounded. We will show that S is compact. To that end, assume that $\mathcal{F}$ is an open cover of S. We will show that $\mathcal{F}$ contains a finite subcover of S.

For each x, define the set

$$S_x = S \cap (-\infty, x].$$

And let

$$B = \{x : \mathcal{F} \text{ contains a finite subcover of } S_x\}.$$

Intuitively, S_x is the set S up to the point x, and as x increases we are slowly revealing more and more of it. Since S is bounded, both $\inf(S)$ and $\sup(S)$ exist. Moreover, recall that such a set always contains a sequence of points converging to its infimum, and a sequence converging to its supremum (Proposition 3.29). And since S is also closed, this means that those limits —$\inf(S)$ and $\sup(S)$—must in fact be elements of S.

Let $L = \inf(S)$ and $M = \sup(S)$. In the last paragraph we showed that $L, M \in S$.

Now, returning to the sets S_x and B, note that L being the minimum of S implies that $S_L = \{L\}$. And certainly the cover $\mathcal{F}$ of S contains a finite subcover of $\{L\}$: Since $L \in S$ we know that some open set $U \in \mathcal{F}$ covers $\{L\}$, and then $\{U\}$ is a finite subcover of $\{L\}$. We have therefore shown that $L \in B$. So B is nonempty.

Note that since $S_M = S$, if $M \in B$ then we are done: that would imply that S has a finite subcover and hence is compact. Therefore assume for a contradiction that $M \notin B$. Then $x < M$ for all $x \in B$, implying that B is bounded above. In summary, B is a nonempty subset of $\mathbb{R}$ that is bounded above, and hence by completeness $\sup(B)$ exists. Let $T = \sup(B)$.

We will now show that this leads to a contradiction. In fact, we will consider two cases and show that in either case we can reach a contradiction.

Case 1: $T \in S$. Since $\mathcal{F}$ is a cover of S, we know that the sets in $\mathcal{F}$ collectively cover all of S. In particular, since in this case we are assuming that $T \in S$, there must be some open set $U \in \mathcal{F}$ that covers the number T. That is, $T \in U$ for some open set $U \in \mathcal{F}$. But by the definition of U being an open set, this means that there is some δ-neighborhood around T which is also contained in U:

$$(T - \delta, T + \delta) \subseteq U.$$

In particular, this implies that

$$\left(T - \delta, T + \frac{\delta}{2}\right] \subseteq U.$$

We will now show that T was not in fact $\sup(B)$. In particular, we will show that $(T + \delta/2) \in B$.

Note that since $T - \delta < T = \sup(B)$, this means that $T - \delta \in B$. I.e., there exists

a finite subcover $\{F_1, F_2, \ldots, F_k\}$ from $\mathcal{F}$ of the set $S \cap (-\infty, T - \delta]$.

But notice that $\{F_1, F_2, \ldots, F_k, U\}$ is also a finite subset of $\mathcal{F}$ and this set covers $S \cap \left(-\infty, T + \frac{\delta}{2}\right]$. That is, we have found a finite subcover of $S_{T+\delta/2}$, implying that $(T + \delta/2) \in B$, contradicting our choice that $\sup(B) = T$. So in Case 1, we did find a contradiction. $\square_{\text{(Case 1)}}$

Case 2: $T \notin S$. The idea behind this proof is that if $T \notin S$ then we could have increased T by a bit and literally not changed the set. That is, there is some ε for which $S_T = S_{T+\varepsilon}$, and so if S_T can be finitely subcovered, then certainly $S_{T+\varepsilon}$ can be too, since they are literally the same set. This would contradict the claim that $T = \sup(B)$, since $T + \varepsilon$ must then also be in B. The trick is finding a way to increase T by some small amount.

Since by assumption $T \notin S$ and S is closed, this implies that $T \in S^c$ and S^c is open. And to be an element of an open set means that there is some δ-neighborhood

$$(T - \delta, T + \delta) \subseteq S^c.$$

In particular,

$$\left[T - \frac{\delta}{2}, T + \frac{\delta}{2}\right] \cap S = \emptyset.$$

And so

$$S \cap \left(-\infty, T - \frac{\delta}{2}\right] = S \cap \left(-\infty, T + \frac{\delta}{2}\right]. \qquad (☢)$$

But since $T - \delta/2 < T$ and $\sup(B) = T$, we know that there is a finite subcover of $S \cap (-\infty, T - \delta/2]$. But then, by (☢), this is also a finite subcover of

$$S \cap \left(-\infty, T + \frac{\delta}{2}\right].$$

So $(T + \delta/2) \in B$, contradicting our choice that $\sup(B) = T$. $\square_{\text{(Case 2)}}$ $\square$

Theorem.

Theorem 5.20 (*Heine-Borel, expanded*). Let $A \subseteq \mathbb{R}$. Then the following are equivalent.

(i) A is compact.

(ii) A is closed and bounded.

(iii) If (a_n) is a sequence of numbers in A, then there is a subsequence (a_{n_k}) that converges to a point in A.

Proof. The Heine-Borel theorem (Theorem 5.19) shows that (i) holds if and only if (ii) holds. In Exercise 5.7 you will prove that (ii) holds if and only (iii) holds. $\square$

— Notable Exercises —

- A snappy solution to Exercise 5.4 utilizes de Morgan's laws (Fact 5.11). In fact, they are so intimately connected that if you didn't know of de Morgan's laws but independently proved this exercise, then this exercise (and the fact that the complement of an open set is a closed set) might lead you to believe that something like de Morgan's law should hold.

  A special case of this exercise was asked of you in Exercise 1.22.

— Exercises —

Exercise 5.1. For each of the following, determine whether the set is open, whether it is closed, and whether it is compact. (It might be more than one, or none of these.) You do not need to prove your answers.

(a) $\mathbb{Z}$

(b) $\{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots\} \cup \{0\}$

(c) $\mathbb{R}$

(d) $(0,1) \cup [3,4]$

(e) $\mathbb{Q}$

(f) $\{17\}$

Exercise 5.2. For each of the following, provide a proof or a counterexample.

(a) If A and B are compact, must $A \cup B$ be compact?

(b) If A and B are compact, must $A \cap B$ be compact?

Exercise 5.3. For each of the following, you should state which sets you are choosing and what their intersection/union is, but you do not need to prove your examples work.

(a) Give an example of an infinite collection of open sets whose intersection is *not* open.

(b) Give an example of an infinite collection of closed sets whose union is *not* closed.

Exercise 5.4. Prove the following

(a) If $\{U_1, U_2, \dots, U_n\}$ is a collection of closed sets, then $\bigcup_{k=1}^{n} U_k$ is also a closed set.

(b) If $\{U_\alpha\}$ is a collection of closed sets, then $\bigcap_{\alpha} U_\alpha$ is also a closed set.

Exercise 5.5. Prove that a point x is a limit point of a set A if and only if every ε-neighborhood of x intersects A at some point other than x.

Exercise 5.6. Prove that

$$\left\{\left(\frac{1}{k}, 4 - \frac{1}{k}\right)\right\}_{k=1}^{\infty}$$

is an open cover of of $(0,4)$, but that this cover has *no* finite subcover. This implies that $(0,4)$ is *not* compact.

Exercise 5.7. Let $A \subseteq \mathbb{R}$. Prove that A is closed and bounded if and only if every sequence of numbers from A has a subsequence that converges to a point in A.

Exercise 5.8. Let A be compact and $U \subseteq A$ be closed. Prove that U is compact.

Exercise 5.9. Assume that A is compact and B is bounded. Must $A \cap B$ be compact? Prove or give a counterexample.

Exercise 5.10. Prove that if $\{U_\alpha\}$ is a collection of compact sets, then $\bigcap_\alpha U_\alpha$ is also a compact set.

Exercise 5.11. Call a subset A of real numbers *clopen* if it is both open and closed. Prove that the only clopen sets are $\emptyset$ and $\mathbb{R}$.

Exercise 5.12. Give an example of a set which is neither open nor closed.

— Open Question —

In the area of topology called *knot theory*,[12] a piece of rope that is tied at the ends is called a *knot*. If you took an untangled piece of rope and tied the ends, what you get is called the *unknot*.

If the rope is tangled up before the ends are tied, though, it may or may not be possible to untangle it to resemble the unknot. The below, for instance, is the *trefoil knot*, and as you can see it is impossible to untangle it to make it look like the unknot.

Given a knot, if you can untangle it to resemble the unknot, then you essentially had (a messy version of) the unknot the whole time. Therefore if your knot can be completely untangled, then we say that your knot *is* the unknot. Now here's the open question: If knots are all drawn on the paper like the one above, does there exist a polynomial-time algorithm to determine whether a knot is the unknot?

[12] A mathematician walks into a bar with a dog and a cow. The bartender says, "Hey, you can't bring wild animals in here!"

The mathematician replies, "You don't understand, these are very special animals."

"How so?"

"They're knot theorists," he says. And, seeing the bartender's look of disbelief, he adds, "here, I'll prove it to you—ask them a question."

The bartender turns the dog, "Name a knot invariant."

"Arf! Arf!" barks the dog.

Scowling, the bartender faces the cow. "Name a topological invariant."

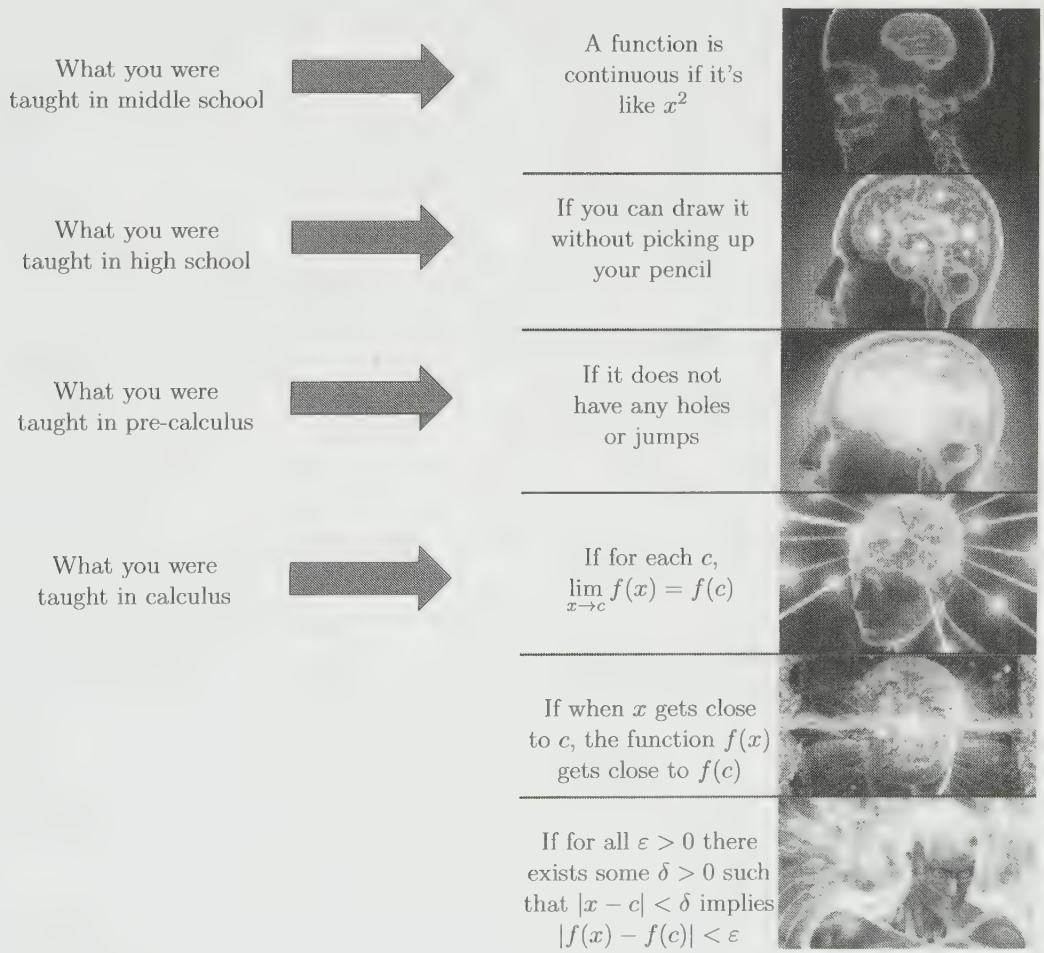
"Mu! Mu!" responds the cow.

Irate, the bartender throws all three out. The dog, bothered at being yelled at, turns to the mathematician and asks, "Should have said the Jones polynomial?"

Chapter 6: Continuity

6.1 Approaching Continuity

I made a meme describing your discrete approach towards understanding continuity.[1]



[1] Here's to hoping that memes have a longer half-life than I fear!

6.2 Weird Examples

What should it mean to say a function is continuous? Or to say that the function is continuous at the point $x = c$?

When you were young you might have just imagined some basic examples, which were all probably polynomials. At some point you may have been told that it means that you can draw it without picking up your pencil. Then that there weren't any holes or gaps in the graph —that might have been where you were when you got your high school diploma. (And at this point, you must admit that all of these definitions are imprecise and informal; they do not provide the rigor that mathematicians demand.)

When taking calculus you might have learned that it's a more localized property, and f being continuous at some point c means that as x gets close to c, you need to have $f(x)$ get close to $f(c)$. This was then formalized by saying that f is continuous at c if[2]

$$\lim_{x \to c} f(x) = f(c).$$

This is now getting pretty good, although you were likely given little to no explanation for why this is how it is. In fact, at this point we have not even defined what it means to take a limit of a function. But intuitively it should mean something like this: as x gets *near* c, $f(x)$ will be getting *near* $f(c)$.

We have two goals for the next few sections:

- Understand what is meant by "$\lim_{x \to c} f(x)$."
- Understand what is meant by "$f(x)$ is continuous at c."

In 15 pages time we will have ironed these out. There is still some preliminary work to go, but for now let's just look at some interesting examples that highlight the complexity and subtleties of these goals.

Important Examples

Question 6.1. Do there exist functions which are continuous nowhere? That is, can we find a function f which is not continuous at any point c?

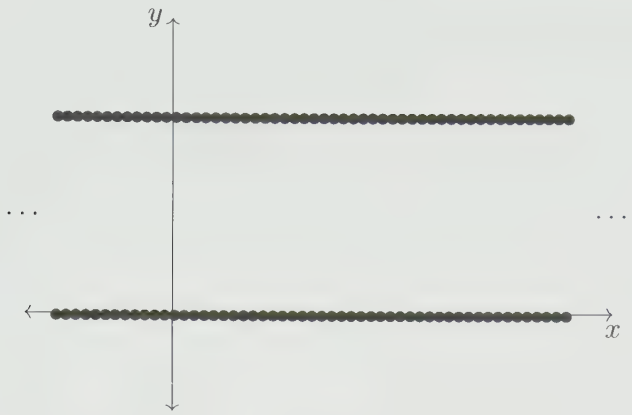
The first example is known as the *Dirichlet's function*. Dirichlet's idea was to consider a piecewise function f in which $f(x)$ was equal to 1 whenever x was a rational number, and $f(x) = 0$ whenever x was irrational. Here's that:

Example 6.2. Define the *Dirichlet function* to be

$$f(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q} \\ 0 & \text{if } x \notin \mathbb{Q}. \end{cases}$$

Recall that the irrationals are dense in the rationals, and the rationals are dense in the irrationals. Therefore the graph looks pretty wacky:

[2]The notation "$\lim_{x \to c} f(x)$" will be formally defined soon.



Pick any point, like $c = 3$. Is f continuous at c? Based on our intuition of "f is continuous at c if x being close to c implies that $f(x)$ is close to $f(c)$," it is not, right? Since $c = 3$ is rational, $f(c) = 1$. But you can find irrational numbers x as close to c as you want, and for those numbers $f(x) = 0$, which is not near $f(c) = 1$.

In calculus, we talked about continuity in terms of limits. So what in this situation should $\lim_{x\to 3} f(x)$ be equal to? At this point in this book, the only limits we know how to take are limits of sequences. So, for example, $(a_n) = \left(3 - \frac{1}{n}\right)$ is a sequence converging to 3. And for each $n \in \mathbb{N}$, note that $\left(3 - \frac{1}{n}\right) \in \mathbb{Q}$. So each term of this sequence has $f(a_n) = 1$, and so for this sequence we do have $f(a_n) \to f(c)$, since all this is saying is that a sequence of 1s is converging to 1. So this suggests that maybe $\lim_{x\to 3} f(x) = 1$.

However, not all sequences behave like this. For instance, note that, for each $n \in \mathbb{N}$, $b_n := 3 + \frac{\sqrt{2}}{n}$ is irrational, and also $b_n \to 3$. So we have a sequence (b_n) which is converging to 3, but $(f(b_n))$ is just a sequence of 0s, and hence converges to 0, which is not $f(3)$.

(Worse yet, you could find a third sequence (c_n) for which $\lim_{n\to\infty} f(c_n)$ does not even exist, by having the terms alternate between being rational and irrational.)

So we can not in general find $\lim_{x\to 3} f(x)$ by simply finding a sequence (a_n) converging to 3 and applying f to it and checking what $f(a_n)$ is converging to, because it might depend on which sequence we pick! Indeed, our eventual definition of these functional limits will label the above limit as "does not exist."

And of course, there was nothing special about $c = 3$. By the same reasoning, any $c \in \mathbb{R}$ will have the property that $\lim_{x\to c} f(x)$ does not exist. We are therefore going to say that this function f is *continuous nowhere.* □

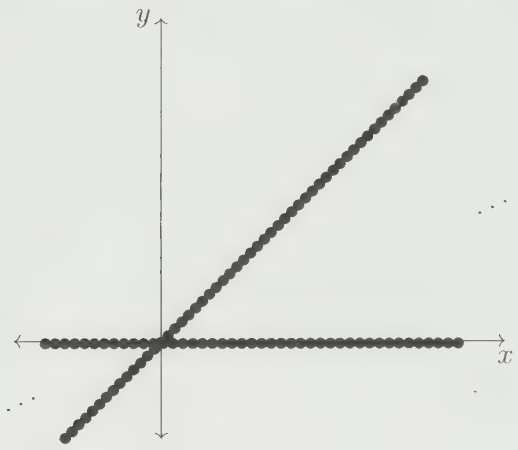
Question 6.3. Can a function be continuous at only one point? Or does being continuous somewhere mean that there is a region around that point which is also continuous? (Think about these before jumping to the answer.)

The next example is similar, but highlights an important point: just because a function is not "smooth"-looking, does not mean that it is discontinuous everywhere. This destroys the "continuous means you can draw it without picking up your pencil" claim.

Example 6.4. Define the *modified Dirichlet function* to be

$$g(x) = \begin{cases} x & \text{if } x \in \mathbb{Q} \\ 0 & \text{if } x \notin \mathbb{Q}. \end{cases}$$

Again, recall that the irrationals are dense in the rationals, and the rationals are dense in the irrationals. Therefore the graph again looks pretty wacky:



Pick any $c \neq 0$. Like with the Dirichlet function, there is a gap between the dots on the x-axis near c, and those on the line $y = x$ near c. And if (a_n) is a sequence of rational numbers converging to c and b_n is a sequence of irrational numbers converging to c, then

$$\lim_{n\to\infty} g(a_n) = c \neq 0 = \lim_{n\to\infty} g(b_n).$$

Therefore, just like with the Dirichlet function, g is not going to be continuous at any such c.

What about at $c = 0$? Well, here things seem to be different. We can no longer construct two sequences converging to 0 which have different limits when we apply g to them; all such sequences also converge to 0. Sure, the function is still jumping all over the place, so it doesn't look anything like what we imagine it means to be continuous at a point, but it does seem to satisfy our guiding principle of "$f(x)$ is continuous at c if x being close to c implies that $f(x)$ is close to $f(c)$." Indeed, as x gets close to 0, all the $g(x)$-values are close to (or equal to) $g(c) = 0$. When we eventually write down a precise definition, it will agree that g is continuous at 0 (and discontinuous at every other point).

So yes, this will be an example of a function which is continuous at only one point![3] □

[3]Make sure you take a moment to appreciate just how remarkably, wonderfully weird this is.

Question 6.5. Can a function be continuous at every irrational number and discontinuous at every rational number?

Now we are asking for all sorts of craziness. If this is possible then you should stop trusting all of your prior intuition about continuity. Well, get ready, because the next example does just this, and is called *Thomae's function.*

Example 6.6. Define *Thomae's function*[4] to be

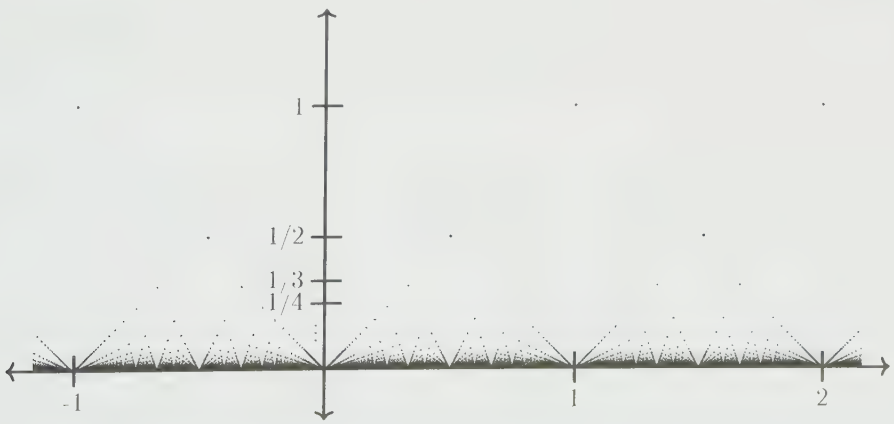
$$h(x) = \begin{cases} 1 & \text{if } x = 0 \\ 1/n & \text{if } x \in \mathbb{Q} \text{ and } x = m/n \text{ in lowest terms with } n > 0 \\ 0 & \text{if } x \notin \mathbb{Q}. \end{cases}$$

Notice that at any rational point c, we have $h(c) > 0$; meanwhile, there are irrational numbers x arbitrarily close to c which have $h(x) = 0$. So h will indeed be discontinuous at every rational number.

At any irrational number c, though, we can show that h is actually continuous. We will prove this in detail later, but the basic idea is this: let $\varepsilon > 0$, and find an N for which $\frac{1}{N} < \varepsilon$. We can show that, in some region around c, all x-values have $h(x) < \frac{1}{N} < \varepsilon$. To see this, note that there are only finitely many rational numbers which have a denominator of N or smaller and are also within, say, distance 1 of c. Find the one that is closest to c, and suppose it has distance $\delta > 0$ from c. Then, all rational numbers in $(c - \delta, c + \delta)$ have denominators that are larger than N. So within this range, all function values $h(x)$ are smaller than $1/N$. Awesome!

And, as you can imagine, as N gets large this forces all the function values to be converging to 0 in these shrinking regions around c, kind of like what happened at $c = 0$ in the Modified Dirichlet function. So yes, this will be an example of a function which is discontinuous on $\mathbb{Q}$, and continuous on $\mathbb{R} \setminus \mathbb{Q}$![5] □

This function's graph looks like this:[6]



[4]AKA the *popcorn function*, AKA the *raindrop function*, AKA the *countable cloud function*, AKA the *ruler function*, AKA the *Riemann function*, AKA the *Stars over Babylon.*

[5]Make sure you take a moment to appreciate just how remarkably, wonderfully weird this is.

[6]So cool that it made the cover.

Its graph also explains why it's called the popcorn function—it somewhat resembles popcorn kernels popping up off a pan.

So yes, with these functions in mind, we certainly are in need of a careful definition of continuity. But first, as we have discussed, we need a careful definition of a functional limit.

6.3 Functional Limits

Continuing with our discussion, we think about "$\lim_{x \to c} f(x) = L$" as meaning "as x gets arbitrarily close to c, $f(x)$ is getting arbitrarily close to L." We will state this in a fairly general way. Rather than requiring that f be a function from $\mathbb{R}$ to $\mathbb{R}$, we will allow the domain to be some subset $A \subseteq \mathbb{R}$. Moreover, c does not necessarily have to be a member of A; if c is a limit point of A, then we can still pick x-values from A that approach it, and everything goes through just fine.

Definition.

Definition 6.7. Let $f : A \to \mathbb{R}$ and let c be a limit point of A. Then we say that

$$\lim_{x \to c} f(x) = L$$

if for all $\varepsilon > 0$ there exists some $\delta > 0$ such that for every $x \in A$ for which $0 < |x - c| < \delta$, we have

$$|f(x) - L| < \varepsilon.$$

We also say $\lim_{x \to c} f(x)$ *converges* to L in such a situation.

This definition is both important and confusing, so some comments are in order.

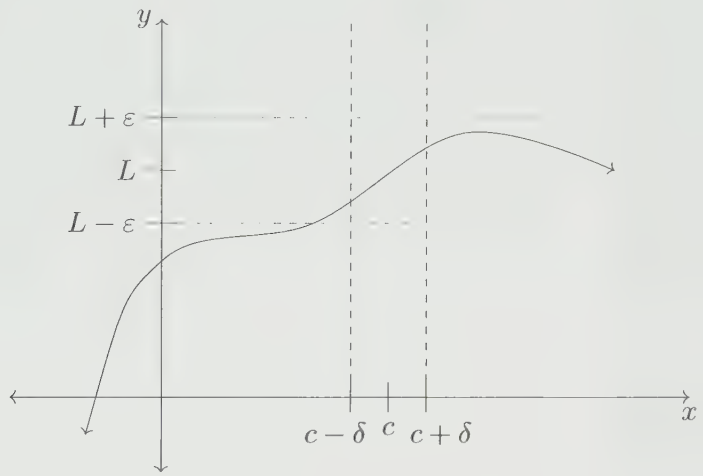
Comment 1. The reason we require that $x \in A$ is just so that $f(x)$ makes sense. Remember, $f : A \to \mathbb{R}$, so if $x \notin A$ then $f(x)$ is not even defined. (Note that we never take $f(c)$, so that's why it's ok if $c \notin A$.)

Comment 2. The reason why we require $0 < |x - c|$ (i.e. $x \neq c$) is because limits only care about what happens as you *approach* the point c, not what actually happens *at* the point c. Indeed, $f(c)$ could be changed to anything at all—*or could not even exist*—and the limit would be unaffected.

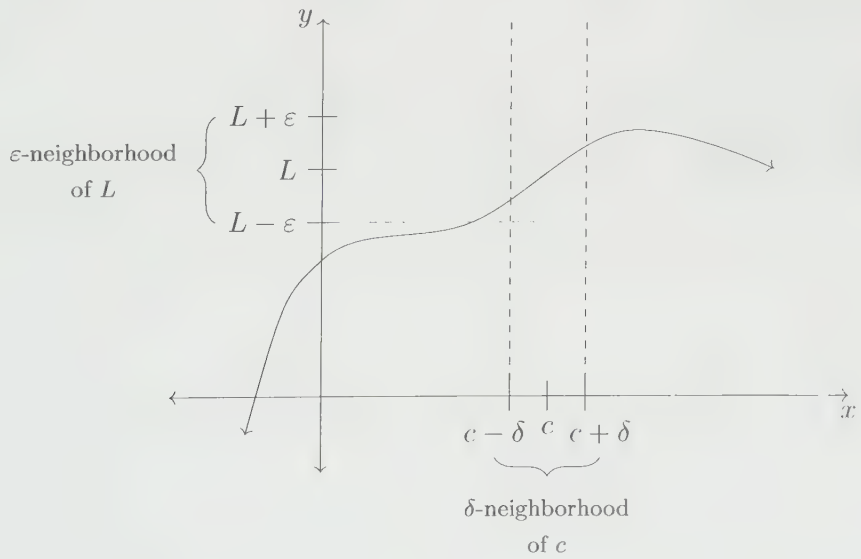
Comment 3. When the context is clear, for convenience we do not always explicitly state conditions $x \in A$ and $0 < |x - c|$.

Comment 4. Below is the important picture to keep in mind, even though as we will and have seen, it is overly simplistic.

Suppose f, c, and L are as pictured below, and we are interested in showing that $\lim_{x \to c} f(x) = L$.



The picture shows that if x has the property that $0 < |x - c| < \delta$ (meaning that the x-values are between the vertical dashed lines), then $f(x)$ has the property that $|f(x) - L| < \varepsilon$ (meaning that the graph of $f(x)$ has y-values inside of the horizontal dashed lines). Indeed, there is another way to think about this using the neighborhood terminology from topology.



So we could restate the definition: $\lim_{x \to c} f(x) = L$ if for all $\varepsilon > 0$ there exists a $\delta > 0$ such that if x is in the δ-neighborhood of c, then $f(x)$ is in the ε-neighborhood of L.

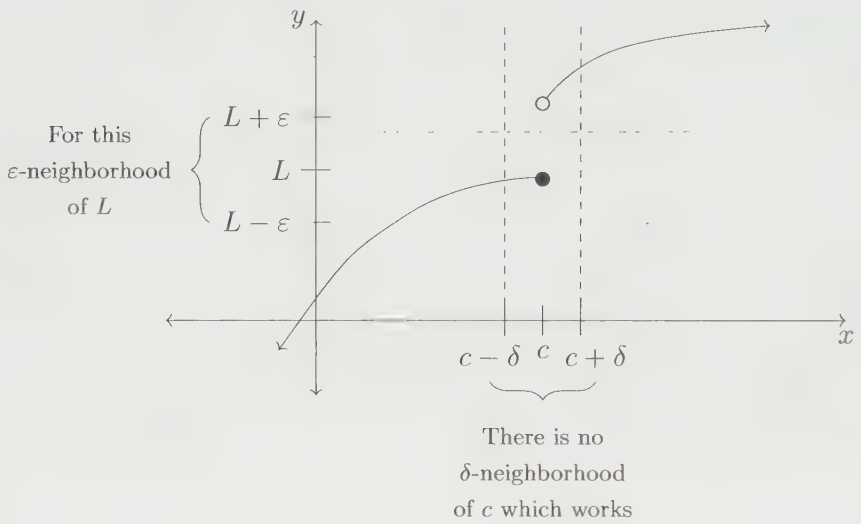
Comment 5. To show something is *not* continuous, you must prove the negation of the definition. The definition of f being continuous at c is this:

> For all $\varepsilon > 0$ there exists some $\delta > 0$ such that for all $x \in A$ for which $0 < |x - c| < \delta$, we have $|f(x) - L| < \varepsilon$.

By negating the above, here is what it means for f to *not* be continuous at c:

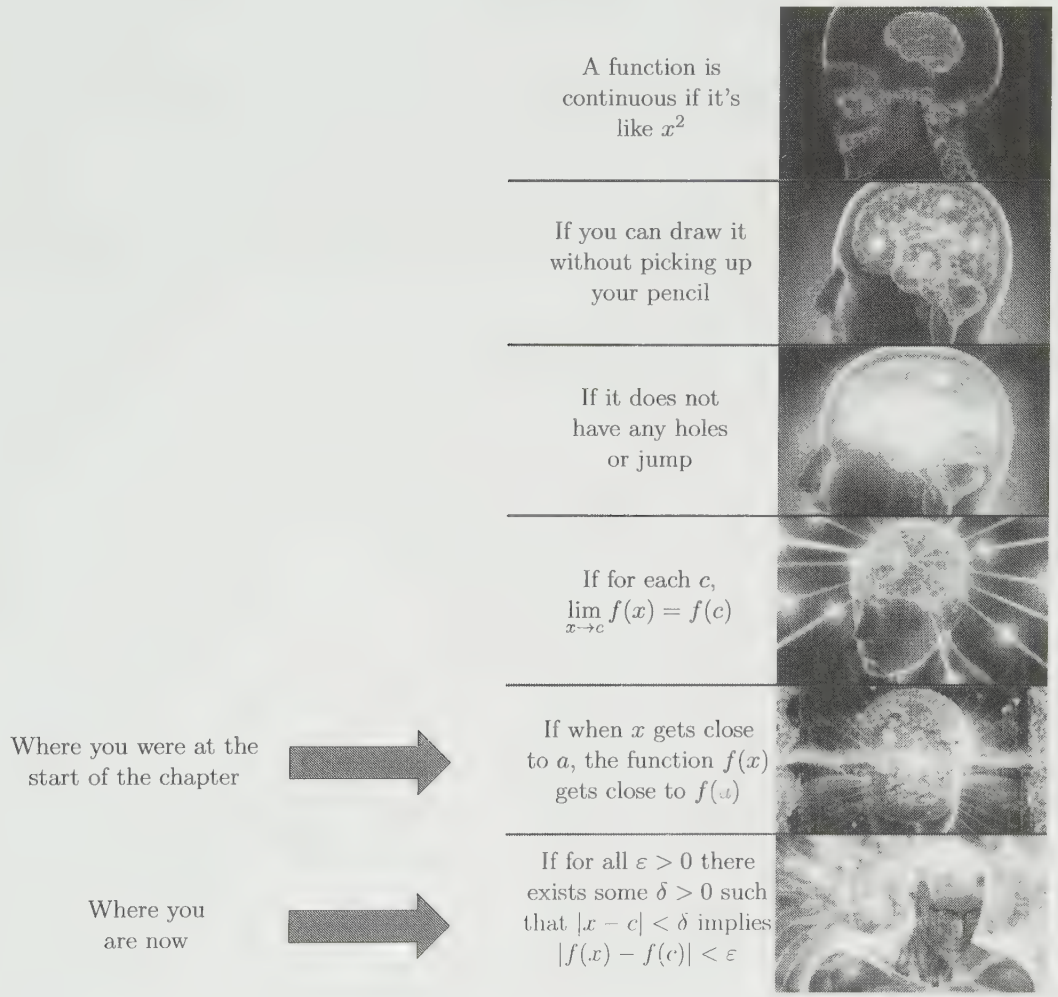
> There exists $\varepsilon > 0$ such that for all $\delta > 0$ there exists some $x \in A$ satisfying $0 < |x - c| < \delta$ for which $|f(x) - L| \geq \varepsilon$.

That is, we must find just a single ε which fails. The (still simplistic) picture for $\lim_{x \to c} f(x) \neq L$ is this:



For that particular ε, there is no possible $\delta > 0$ that we could choose to have the limit criterion hold. In particular, for any $\delta > 0$ that we choose, the x-values between c and $c + \delta$ are guaranteed to have $|f(x) - L| \geq \varepsilon$. And since we found a single ε for which the criterion fails, f is *not* continuous at c.

And now here's an update on your brain expansion status:[7]



[7] I'm really doubling down on this meme still being a thing...

Examples

Let's now do some examples of this definition.

Example 6.8. Let $f(x) = 5x + 2$. Prove $\lim_{x \to 3} f(x) = 17$.

Scratch Work. Let $\varepsilon > 0$. Recall, $|x - 3| < \delta$ is what we will assume, for whichever $\delta > 0$ that we choose. So in this scratch work we will assume that our conclusion holds,

$$|f(x) - L| < \varepsilon,$$

and then we unwind this until we get just an $|x - 3|$ on one side, which then tells us which δ will make it work.

$$\begin{aligned} \text{WTS:} \quad |f(x) - L| &< \varepsilon \\ |5x + 2 - 17| &< \varepsilon \\ |5x - 15| &< \varepsilon \\ 5 \cdot |x - 3| &< \varepsilon \\ |x - 3| &< \frac{\varepsilon}{5} \quad \checkmark \\ \text{So let } \delta &= \frac{\varepsilon}{5} \end{aligned}$$

We were able to get down to just $|x - 3|$ on the left, which is the thing that we can control. So if we choose δ to be $\varepsilon/5$, the above can be redone to provide a proof that $\lim_{x \to 3} f(x) = 17$.

Solution. Let $\varepsilon > 0$. And let $\delta = \frac{\varepsilon}{5} > 0$. Then for any x where $0 < |x - 3| < \delta$,

$$\begin{aligned} |f(x) - L| &= |5x + 2 - 17| \\ &= |5x - 15| \\ &= 5 \cdot |x - 3| \\ &< 5 \cdot \delta \\ &= 5 \cdot \frac{\varepsilon}{5} \\ &= \varepsilon. \end{aligned}$$

□

The next example is that of a rational function—a polynomial divided by another polynomial.

Example 6.9. Let $f : \mathbb{R} \setminus \{2\} \to \mathbb{R}$ where $f(x) = \dfrac{3x^2 - 12}{x - 2}$. Prove $\lim\limits_{x \to 2} f(x) = 12$.

Scratch Work. Let $\varepsilon > 0$. Recall, $|x - 2| < \delta$ is what we will assume, for whichever $\delta > 0$ that we choose. So in our scratch work we will assume that our conclusion holds,

$$|f(x) - L| < \varepsilon,$$

and then we unwind this until we get just an $|x - 2|$ on one side, which then tells us which δ will make it work. Here's that:

$$\begin{aligned}
\text{WTS:} \quad |f(x) - L| &< \varepsilon \\
\left| \frac{3x^2 - 12}{x - 2} - 12 \right| &< \varepsilon \\
\left| \frac{3(x-2)(x+2)}{x - 2} - 12 \right| &< \varepsilon \\
|3(x + 2) - 12| &< \varepsilon \\
3 \cdot |(x + 2) - 4| &< \varepsilon \\
3 \cdot |x - 2| &< \varepsilon \\
|x - 2| &< \frac{\varepsilon}{3} \quad \checkmark \\
\text{So let } \delta &= \frac{\varepsilon}{3}.
\end{aligned}$$

We were able to get down to just $|x - 2|$ on the left, which is the thing that we can control. So if we choose δ to be $\varepsilon/3$, the above can be redone to provide a proof that $\lim\limits_{x \to 3} f(x) = 12$.

Solution. Let $\varepsilon > 0$. And let $\delta = \frac{\varepsilon}{3} > 0$. Then for any x where $0 < |x - 2| < \delta$,

$$\begin{aligned}
|f(x) - L| &= \left| \frac{3x^2 - 12}{x - 2} - 12 \right| \\
&= \left| \frac{3(x-2)(x+2)}{x - 2} - 12 \right| \\
&= |3(x + 2) - 12| \\
&= 3 \cdot |(x + 2) - 4| \\
&= 3 \cdot |x - 2| \\
&< 3 \cdot \delta \\
&= 3 \cdot \frac{\varepsilon}{3} \\
&= \varepsilon.
\end{aligned}$$

□

6.4 Properties of Functional Limits

Proposition.

Proposition 6.10 (*Functional limits are unique*). A limit $\lim_{x \to c} f(x)$ can converge to at most one value.

Proof. Suppose $A \subseteq \mathbb{R}$, c is a limit point of A, and $f : A \to \mathbb{R}$. Suppose

$$\lim_{x \to c} f(x) = L_1 \qquad \text{and} \qquad \lim_{x \to c} f(x) = L_2.$$

We will show that $|L_1 - L_2| < \varepsilon$ for all $\varepsilon > 0$, which would imply that $L_1 = L_2$, as desired.

To this end, let $\varepsilon > 0$. Since $\lim_{x \to c} f(x) = L_1$, there exists some $\delta_1 > 0$ such that $0 < |x - c| < \delta_1$ implies

$$|f(x) - L_1| < \frac{\varepsilon}{2}.$$

Likewise, since $\lim_{x \to c} f(x) = L_2$, there exists some $\delta_2 > 0$ such that $0 < |x - c| < \delta_2$ implies

$$|f(x) - L_2| < \frac{\varepsilon}{2}.$$

Let $\delta = \min\{\delta_1, \delta_2\}$. Then $0 < |x - c| < \delta$ implies

$$\begin{aligned} |L_1 - L_2| &= |L_1 - f(x) + f(x) - L_2| \\ &\leq |L_1 - f(x)| + |f(x) - L_2| \qquad \text{(Triangle inequality)} \\ &= |f(x) - L_1| + |f(x) - L_2| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\ &= \varepsilon. \end{aligned}$$

In our discussion in Section 6.2 on the three interesting examples, we mentioned that one reason a functional limit does not exist is because there are two sequences a_n and b_n, both of which converge to c, for which $\lim_{n \to \infty} f(a_n) \neq \lim_{n \to \infty} f(b_n)$. We will show now, though, that if all such sequences *do* converge to the same thing, then the functional limit *does* exist, and equals that common value.

Theorem.

Theorem 6.11 ($\lim_{x\to c} f(x) = L$ *iff every such* $f(a_n) \to L$)**.** Assume that $A \subseteq \mathbb{R}$, $f : A \to \mathbb{R}$, and c is a limit point of A. Then $\lim_{x\to c} f(x) = L$ if and only if, for every sequence a_n from A for which each $a_n \neq c$ and $a_n \to c$, we have $f(a_n) \to L$.

Proof. Throughout the proof, assume that $x \neq c$ and $a_n \neq c$ for all n.

First we will prove the forward direction. Recall that the definition of "$\lim_{x\to c} f(x) = L$" is that for any $\varepsilon > 0$ there is some $\delta > 0$ such that if $0 < |x - c| < \delta$, then $|f(x) - L| < \varepsilon$. Let a_n be an arbitrary sequence from A converging to c. We will now show that $f(a_n)$ is converging to L.

Let $\varepsilon > 0$. We wish to show that there exists some N for for which $|f(a_n) - L| < \varepsilon$ for all $n > N$. To see this, simply note that since $a_n \to c$ and $\delta > 0$, by definition[8] there exists some N for which $|a_n - c| < \delta$ for all $n > N$. But by the last paragraph, this then implies that $|f(a_n) - L| < \varepsilon$, as desired.

We will now prove the reverse direction. Let $\varepsilon > 0$. Assume that for every sequence a_n from A for which each $a_n \neq c$ and $a_n \to c$, we have $f(a_n) \to L$. And assume for a contradiction that $\lim_{x\to c} f(x) \neq L$. That is, we are assuming this:

> This is *not* true: For all $\varepsilon > 0$ there exists some $\delta > 0$ such that for all $x \in A$ for which $0 < |x - c| < \delta$, we have $|f(x) - L| < \varepsilon$.

Therefore, by negating the above,

> This *is* true: There exists $\varepsilon > 0$ such that for all $\delta > 0$ there exists some $x \in A$ satisfying $0 < |x - c| < \delta$ for which $|f(x) - L| \geq \varepsilon$.

In particular, by letting $\delta_n = \frac{1}{n}$, we see that there exists some $x_n \in A$ satisfying $0 < |x_n - c| < \frac{1}{n}$ for which $|f(x_n) - L| \geq \varepsilon$. Note that $|x_n - c| < \frac{1}{n}$ implies that $x_n \to c$. So we have found a sequence (x_n) converging to c for which $|f(x_n) - L| \geq \varepsilon$ for all n, and hence $f(x_n) \not\to L$. This contradicts our original assumption, completing the proof. □

Note.

Note 6.12. Theorem 6.11 gives a criterion to determine that a functional limit does not exist: If you can find two such sequences a_n and b_n which both converge to c but for which

$$\lim_{n\to\infty} f(a_n) \neq \lim_{n\to\infty} f(b_n),$$

then we can conclude that $\lim_{x\to c} f(x)$ does not exist.

[8] Here, δ is taking the role of ε in the definition of a convergent sequence (Definition 3.7).

With Theorem 6.11, any result that holds for all sequences converging to a point c will also apply to functional limits! Indeed, this theorem quickly gives us function analogues to the sequence limit laws and the sequence squeeze theorem.

Corollary.

Corollary 6.13 (*Func-y limit laws*). Let f and g be functions from $A \subseteq \mathbb{R}$ to $\mathbb{R}$ and let c be a limit point of A. Lastly, assume that

$$\lim_{x \to c} f(x) = L \qquad \text{and} \qquad \lim_{x \to c} g(x) = M.$$

Then,

(i) $\lim_{x \to c}[k \cdot f(x)] = k \cdot L$ for any $k \in \mathbb{R}$,

(ii) $\lim_{x \to c}[f(x) + g(x)] = L + M$,

(iii) $\lim_{x \to c}[f(x) \cdot g(x)] = L \cdot M$, and

(iv) $\lim_{x \to c} \dfrac{f(x)}{g(x)} = \dfrac{L}{M}$, provided $M \neq 0$ and $g(x) \neq 0$ for any $x \in A$.

Proof. By the sequence limit laws (Theorem 3.21), each of the asserted properties hold for all sequences $a_n \to c$. Therefore by Theorem 6.11 they also hold for the functional limit. □

Corollary.

Corollary 6.14 (*Func-y squeeze theorem*). Let f, g and h be functions from $A \subseteq \mathbb{R}$ to $\mathbb{R}$ and let c be a limit point of A, and suppose that

$$f(x) \leq g(x) \leq h(x)$$

for all $x \in A$. Lastly, assume that

$$\lim_{x \to c} f(x) = L \qquad \text{and} \qquad \lim_{x \to c} h(x) = L.$$

Then

$$\lim_{x \to c} g(x) = L.$$

Proof. By the sequence squeeze theorem (Theorem 3.23) the result holds for all sequences $a_n \to c$. So by Theorem 6.11, it also holds for the functional limit. □

6.5 Continuity

We have already mentioned the definition of continuity, but let's state it here formally. One way to state it is this: a function is continuous at a point c (where c is a limit point of $A \setminus \{c\}$) if $\lim_{x \to c} f(x) = f(c)$. This definition is sometimes hard to apply; furthermore it doesn't really explain, at a nuts-and-bolts level, the reason why a function is continuous at a point. So we are going to unwind this functional-limit definition, and write it in terms of epsilons and deltas. This will additionally clarify the case where c is an isolated point.

Definition.

Definition 6.15. A function $f : A \to \mathbb{R}$ is *continuous at a point* $c \in A$ if for all $\varepsilon > 0$ there exists some $\delta > 0$ such that for all $x \in A$ where $|x - c| < \delta$ we have

$$|f(x) - f(c)| < \varepsilon.$$

If f is continuous at every point in its domain, then f is called *continuous*.

With functional limits, we proved that its ε-δ definition was equivalent to a topological and sequential definition. The same holds with continuity, as the next theorem shows.

Theorem.

Theorem 6.16 (*Continuity topologically and sequentially*). Let $f : A \to \mathbb{R}$ and $c \in A$. Then the following are equivalent.

(i) f is continuous at c,

(ii) For all $\varepsilon > 0$ there exists some $\delta > 0$ such that for all $x \in A$ where $|x - c| < \delta$ we have $|f(x) - f(c)| < \varepsilon$.

(iii) For any ε-neighborhood of $f(c)$, denoted $V_\varepsilon(f(c))$, there exists some δ-neighborhood of c, denoted $V_\delta(c)$, with the property that for any $x \in A$ for which $x \in V_\delta(c)$, we have $f(x) \in V_\varepsilon(f(c))$.

(iv) For all sequences (a_n) from A which converge to c, we have $f(a_n) \to f(c)$.

(v) $\lim_{x \to c} f(x) = f(c)$ if c is a limit point of A.

Proof Sketch. The fact that (i) is true if and only if (ii) is simply Definition 6.15. And (ii) being equivalent to (iii) is again just due to a definition: Applying Definition 3.14 we can simply replace the absolute values with ε- or δ-neighborhoods.

The proof that (iv) is equivalent to (ii) is essentially identical to that of Theorem 6.11. The proof that (v) is equivalent to (iv) is due to Theorem 6.11. □

Poincaré once said that "mathematics is the art of giving the same name to different things." This is certainly true for most of Theorem 6.16. Most lines in this theorem are simply rephrasing the original definition using other terms that we have learned. The exception is (iv)—this is substantially different, and it will be useful to keep it in mind as an equivalent condition. Indeed, this sequential formulation is often easiest to work with, especially to show that a function is *not* continuous. In fact, we will now formally note this.

Note.

Note 6.17. By Theorem 6.16 part (iv), if you can find a single sequence (a_n) from the domain for which $a_n \to c$ while $f(a_n) \not\to f(c)$, then you may conclude that f is not continuous at c.

In particular, if you can find two sequences (a_n) and (b_n) from the domain, both of which converge to c, but $\lim_{n\to\infty} f(a_n) \neq \lim_{n\to\infty} f(b_n)$, then at least one of these is not equal to $f(c)$ and so f is not continuous at c.

Example 6.18. Our argument in Example 6.2 that the Dirichlet function is continuous nowhere now works. We showed that every c has a pair of sequences converging to it with different limits when you apply f. By Note 6.17, that's enough. □

Next are more results which are easily obtained from previous results.

Proposition.

Proposition 6.19 (*Continuity limit laws*). Let $f : A \to \mathbb{R}$ and $g : A \to \mathbb{R}$ be continuous at c, and let $c \in A$. Then,

(i) $k \cdot f(x)$ is continuous at c, for any $k \in \mathbb{R}$;

(ii) $f(x) + g(x)$ is continuous at c;

(iii) $f(x) \cdot g(x)$ is continuous at c; and

(iv) $\dfrac{f(x)}{g(x)}$ is continuous at c, provided $g(x) \neq 0$ for all $x \in A$.

Proof. By Theorem 6.16 part (v), we can rephrase each of these in terms of a functional limit, and after doing so we exactly have the statement of Corollary 6.13, and hence this completes the proof. □

Examples

On this page, we begin with one of the simplest continuous functions, and then we repeatedly apply Proposition 6.19 to it, to see what other functions we can deduce are continuous.

Example 6.20. The function $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = x$ is continuous. Here's the proof of that: Let c be any real number. Let $\varepsilon > 0$, and then let $\delta = \varepsilon$. Then $|x - c| < \delta$ implies $|x - c| < \varepsilon$. And so, by the definition of f, this is equivalent to $|f(x) - f(c)| < \varepsilon$. And hence by the definition of a continuous function, f is continuous at c. And since c was arbitrary, f is continuous. □

Example 6.21. By the previous example and Proposition 6.19, this then implies that every polynomial is continuous.

Indeed, since by n applications of Proposition 6.19 part (iii) where each function is $f(x) = x$, this implies that $f(x) = x^n$ is also continuous. Then by (i) we see that for any $a_n \in \mathbb{R}$ we have the monomial $f(x) = a_n x^n$ being continuous. Now apply (ii) for different functions of this form, we get a polynomial

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0,$$

which is then continuous. □

If I gave you the limit of a polynomial (from $\mathbb{R}$ to $\mathbb{R}$), like $\lim_{x \to 2}(x^3 - 4x^2 + 1)$, how would you find the limit? From calculus you probably are dying to just plug the limit in. Well by the above, now you can.

Example 6.22. Define a *rational function* to be a polynomial divided by a polynomial, with some domain for which the denominator is never zero. We see that rational functions are always continuous, since by the previous example we know that any two polynomials are continuous, and by Proposition 6.19 part (iv) we then know that their quotient is as well, as we have already noted that the domain is properly restricted so that the denominator is never zero. □

This approach has now ran dry, though. Any additional applications of Proposition 6.19 will continue to yield rational functions. We next investigate another way to new continuous functions from known continuous functions: compositions.

Recall that given two functions f and g, that $f \circ g$ is notation for the composition function. That is,

$$(f \circ g)(x) = f(g(x)).$$

It might look funny that we have parentheses around the "$f \circ g$" and around the "x", but the ones around the "$f \circ g$" are there to be clear that the "$f \circ g$" is its own thing,

so the parentheses are just grouping them together—it is a function and we should be thinking of it as a unit. The parentheses around the "x" are the usual ones around the argument of a function, like $f(x)$ or $h(x)$.

Proposition.

Proposition 6.23 (*f, g continuous $\Rightarrow$ $f \circ g$ is too.*)**.** Suppose $A, B \subseteq \mathbb{R}$, $g : A \to B$ and $f : B \to \mathbb{R}$. If g is continuous at $c \in A$, and f is continuous at $g(c) \in B$, then $f \circ g : A \to \mathbb{R}$ is continuous at c.

Proof. We will use the sequential formulation of continuity, as per Theorem 6.16 part (iv). Consider an arbitrary sequence (a_n) from A converging to c. We aim to show that $(f \circ g)(a_n)$ converges to $(f \circ g)(c)$.

Since each a_n is in g's domain, and $a_n \to c$, and g is continuous at c, by Theorem 6.16 part (iv) we know that $g(a_n) \to g(c)$.

Now, note that since g is a function from A to B, each $g(a_n)$ is an element of B. So $g(a_n)$ is a sequence in B! Likewise, since $c \in A$, $g(c) \in B$. Now, the last paragraph concluded that $g(a_n) \to g(c)$, so we can again apply Theorem 6.16 part (iv). We have that each $g(a_n)$ is in f's domain, and $g(a_n) \to g(c)$, and f is continuous. So by Theorem 6.16 part (iv), we know that $f(g(a_n)) \to f(g(c))$.

We have shown that for an arbitrary sequence a_n in $(f \circ g)$'s domain, we have that $(f \circ g)(a_n)$ converges to $(f \circ g)(c)$, as desired. □

Now here are a few more standard functions. In Exercise 6.4 you are asked to check that they are continuous.

Fact.

Fact 6.24.

- $f : \mathbb{R} \to \mathbb{R}$, $f(x) = e^x$ is continuous.
- $f : (0, \infty) \to \mathbb{R}$, $f(x) = \log(x)$ is continuous
- $f : \mathbb{R} \to \mathbb{R}$, $f(x) = \sin(x)$ is continuous.
- $f : \mathbb{R} \to \mathbb{R}$, $f(x) = \cos(x)$ is continuous.

6.6 Topological Continuity

Recall that not every functions has an inverse. For example, $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = x^2$ does not have an inverse—each input of a function needs to give exactly one output, but if $f^{-1}(x)$ were a function, then is $f^{-1}(4)$ equal to 2 or -2? So we can not simply define the general inverse for an arbitrary function. We can, however, always define the inverse of a *set*.

Definition.

Definition 6.25. Let $X, Y \subseteq \mathbb{R}$ and $f : X \to Y$. For $B \subseteq Y$, define the *inverse image* (or *pre-image*)

$$f^{-1}(B) = \{x \in X : f(x) \in B\}.$$

So the inverse of a set is itself a set.

Example 6.26.

- If $f : \mathbb{R} \to \mathbb{R}$ is given by $f(x) = 5x$ and B is the open interval $(1, 20)$, then
$$f^{-1}(B) = (1/5, 4).$$
- If $f : \mathbb{R} \to \mathbb{R}$ is again given by $f(x) = 5x$, then
$$f^{-1}(\mathbb{R}) = \mathbb{R}.$$
- If $f : \mathbb{R} \to \mathbb{R}$ is given by $f(x) = x^2$ and B is the open interval $(1, 49)$, then
$$f^{-1}(B) = (-7, -1) \cup (1, 7).$$
- If $f : \mathbb{R} \to \mathbb{R}$ is again given by $f(x) = x^2$, then
$$f^{-1}((1, 4) \cup (25, \infty)) = (-\infty, -5) \cup (-2, -1) \cup (1, 2) \cup (5, \infty).$$

□

Note that in all of these examples, we were taking the inverse image of an open set, and what we got out was itself an open set. This is not an accident, and it is *almost* a general property; there is just one way to prevent this from holding, which the next example illustrates.

Example 6.27.

- If $f : [2, 10] \to \mathbb{R}$ is given by $f(x) = 5x$ and B is the open interval $(1, 20)$, then

$$f^{-1}(B) = [2, 4).$$

  You see? It would have been an open set, but was just forced to be restricted because of the domain. Another way we could have written the above is that

$$f^{-1}(B) = (1/5, 4) \cap [2, 10].$$

- Likewise, with the same f as above,

$$f^{-1}(\mathbb{R}) = \mathbb{R} \cap [2, 10].$$

□

What these examples suggest is that if $f : X \to \mathbb{R}$ is continuous and B is open, then $f^{-1}(B)$ is either itself an open set, or is equal to $A \cap X$, where A is some open set. This is further supported if you recall the fact from your intro-to-proofs class that $f^{-1}(\bigcup_\alpha B_\alpha) = \bigcup_\alpha f^{-1}(B_\alpha)$ (also asked of you in Exercise 6.6).

Indeed, as the next theorem states, this is an if-and-only-if condition.

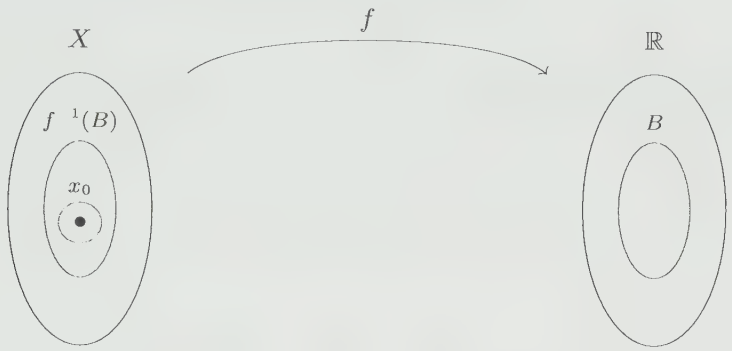
Theorem.

Theorem 6.28 (*f continuous $\Leftrightarrow$ B open implies $f^{-1}(B)$ = open $\cap$ X*)**.** Let $f : X \to \mathbb{R}$. Then f is continuous if and only if for every open set B, $f^{-1}(B) = A \cap X$ for some open set A.

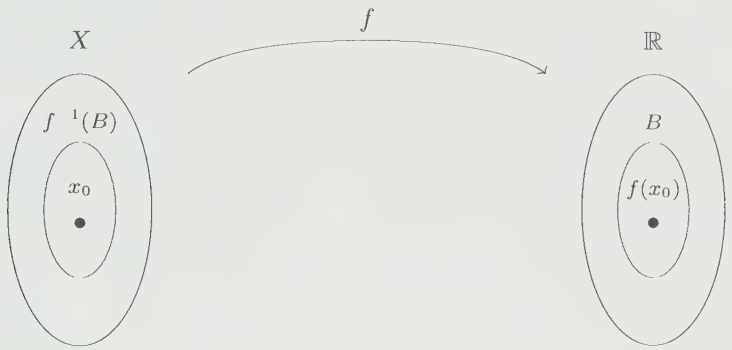
Proof Sketch. For the forward direction, we begin with the following picture, where we have a domain set X, a codomain set $\mathbb{R}$ (each drawn as a blob), a function f going between them, a subset $B \subseteq \mathbb{R}$, and its inverse image under f:

We want to show that $f^{-1}(B) = A \cap X$ where A is open. As you'll see, getting $f^{-1}(B)$ to be in X will be easy. The thing we need to show is that it behaves like

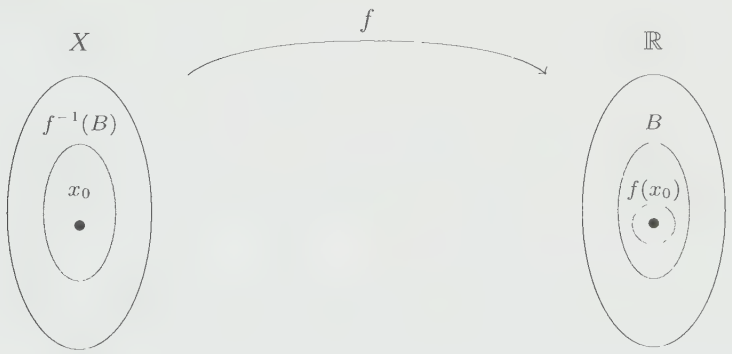
an open set. That is, we want to show that if $x_0 \in f^{-1}(B)$, then there is some open neighborhood around x_0 that is also contained inside $f^{-1}(B)$. That is, we want to be able to draw a circle like this:



How do we do it? Well, we don't know anything about $f^{-1}(B)$, but we do know something about B—it's open. So any point over there we *can* find a region around. Seems like a start, so let's use that. To do it, let's take x_0 and apply f to it. What we get, $f(x_0)$, will be a point in B, and hence subject to B's openness property.



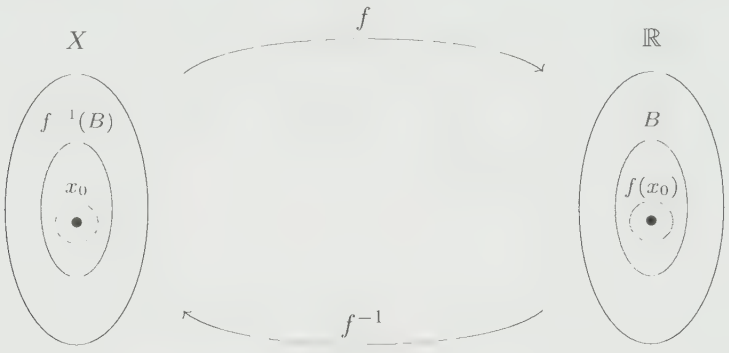
Since B is open we can find some open neighborhood around $f(x_0)$ which is contained inside B:



How do we use this to get an open set around x_0? The idea is to use the fact that f is continuous, which we haven't used up to this point. Recall the topological definition of continuity from Theorem 6.16 part (iii):

For $f : A \to \mathbb{R}$ and $c \in A$, f is continuous at c if and only if, for any ε-neighborhood $V_\varepsilon(f(c))$ of $f(c)$, there exists a δ-neighborhood $V_\delta(c)$ of c with the property that for any $x \in A$ for which $x \in V_\delta(c)$, we have $f(x) \in V_\varepsilon(f(c))$.

Now, in our problem our domain was called X, not A. But really, this is almost exactly what we want to say.



We will have to show why this dashed (red, for those reading in color) set is inside of $f^{-1}(B)$, plus a few other details, but this is the big picture idea. Now let's prove it formally.

***Proof* of Theorem 6.28.** First assume that f is continuous and let B be an open set in $\mathbb{R}$. We aim to show that $f^{-1}(B) = A \cap X$ for some open set A. If we were only trying to show that $f^{-1}(B)$ equaled some open set A, then (by the definition of an open set) what we would want to show is that for any $x_0 \in f^{-1}(B)$, there is some open interval $(x_0 - \delta, x_0 + \delta)$ which is contained entirely inside of A. But since we are trying to show that $f^{-1}(B) = A \cap X$ for some open set A, we only care about this property holding for $x \in X$. So we just have to amend the openness property like so:

Want to show: For any $x_0 \in f^{-1}(B)$, there is some δ-neighborhood $V_\delta(x_0)$ of x_0 with the property that for any $x \in X$ where $x \in V_\delta(x_0)$, we have $x \in f^{-1}(B)$.

Now, to prove this, we will use the fact that f is continuous and B is open. Since $x_0 \in f^{-1}(B)$, we know that $f(x_0) \in B$. So we have a point inside of an open set. By the definition of openness, there must exist some $\varepsilon > 0$ such that $(f(x_0) - \varepsilon, f(x_0) + \varepsilon) \subseteq B$. That is, we have an ε-neighborhood $V_\varepsilon(f(x_0))$ of $f(x_0)$ inside of B.

Now we use that f is continuous. By the topological definition of continuity (Theorem 6.16 part (iii)), there does indeed exist some δ-neighborhood $V_\delta(x_0)$ of x_0 with the property that for any $x \in X$ where $x \in V_\delta(x_0)$, we have $f(x) \in V_\varepsilon(f(x_0))$; i.e., for all these x, we have $x \in f^{-1}(V_\varepsilon(f(x_0))) \subseteq f^{-1}(B)$, as desired.

To prove the reverse direction, assume that for every open set B, $f^{-1}(B) = A \cap X$ for some open set A. We aim to show that f is continuous, and we will do so by

showing that it satisfies the topological definition that we have been using. Choose any $x_0 \in X$ and $\varepsilon > 0$. Since $V_\varepsilon(f(x_0))$ is an open set, the hypothesis gives that

$$f^{-1}\left(V_\varepsilon(f(x_0))\right) = A \cap X$$

for some open set A. And of course, $x_0 \in A \cap X$. Since A is open and $x_0 \in A$, there must exist some δ-neighborhood $V_\delta(x_0)$ of x_0 contained inside A. Therefore, if $x \in X$ and $x \in V_\delta(x_0)$, then $x \in A \cap X$, and hence $f(x) \in V_\varepsilon(x_0)$, as desired. □

Note that we now have another characterization of continuity. The previous were all local conditions that demonstrated continuity at a point, while this is a broader condition guaranteeing continuity of the entire function.

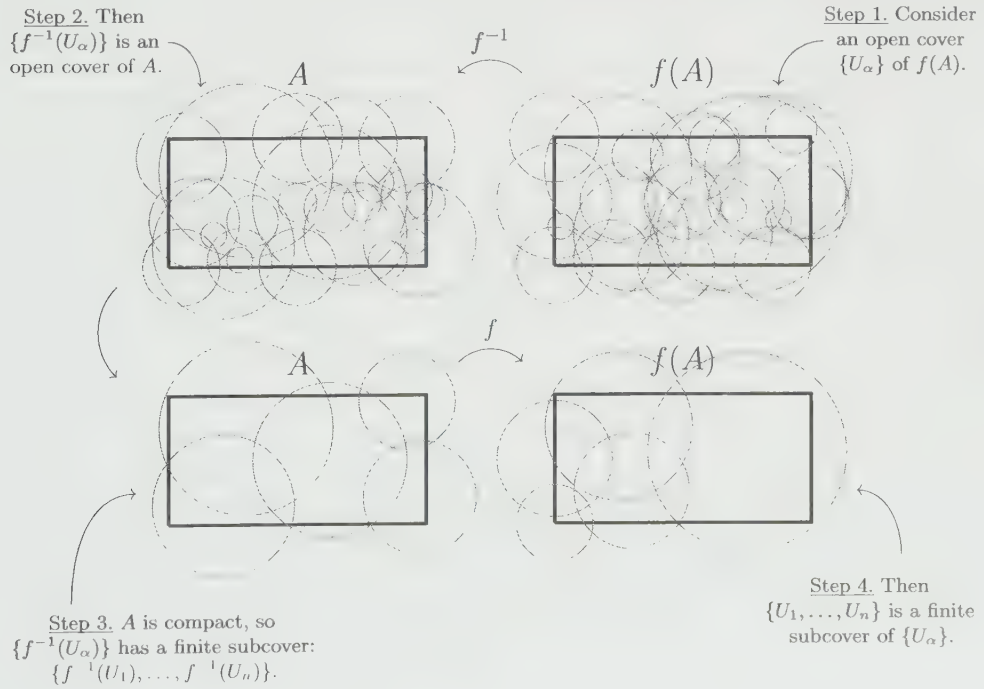
6.7 The Extreme Value Theorem

This next section begins with one more connection between continuity and one of the fundamental objects of topology: compact sets.

Theorem.

Theorem 6.29 (*The continuous image of a compact set is compact*). Let $f : X \to \mathbb{R}$ be continuous. If $A \subseteq X$ is compact, then $f(A)$ is compact.

Proof Sketch. The idea behind the proof is contained in the following picture.



The only additional justification that is needed is in Step 2. Theorem 6.28 does not technically guarantee that $f^{-1}(U_\alpha)$ is open, but that $f^{-1}(U_\alpha) = V_\alpha \cap X$ for some open set V_α. So technically our open cover of A will be $\{V_\alpha\}$—but the above sketch will still essentially work.

Proof. Suppose A is compact. We aim to show that $f(A)$ is also compact, which we will show via the definition. To that end, let $\{U_\alpha\}$ be an open cover of $f(A)$. We will find a finite subcover of this cover, which will complete the proof.

By Theorem 6.28, each $f^{-1}(U_\alpha) = V_\alpha \cap X$ for some open set V_α. In particular, note that $f^{-1}(U_\alpha) \subseteq V_\alpha$.

We will show that $\{V_\alpha\}$ is an open cover of A. By the above, each V_α is open. To see that $\{V_\alpha\}$ is a cover of A, we will instead show that $\{f^{-1}(U_\alpha)\}$ is a cover. This is sufficient since we already noted that each $f^{-1}(U_\alpha) \subseteq V_\alpha$. Consider any $x_0 \in A$; we will show that some $f^{-1}(U_{\alpha_0})$ covers x_0. Consider $f(x_0)$. Since $f(x_0) \in f(A)$ and $\{U_\alpha\}$ is a cover of $f(A)$, there exists some U_{α_0} which covers $f(x_0)$; that is, $f(x_0) \in U_{\alpha_0}$. But then $x_0 \in f^{-1}(U_{\alpha_0})$, as desired.

So indeed, $\{f^{-1}(U_\alpha)\}$ covers A, and hence $\{V_\alpha\}$ is an open cover of A. And hence, by the compactness of A, there must exist some finite subcover $\{V_1, V_2, \ldots, V_n\}$ of A. Recall that each V_i has a corresponding set $f^{-1}(U_i)$ for which $f^{-1}(U_i) = V_i \cap X$. We claim $\{U_1, U_2, \ldots, U_n\}$ is a finite subcover of $f(A)$. Doing so would complete the proof.

The sets $U_1, U_2, \ldots, U_n$ are from the open cover $\{U_\alpha\}$, so clearly these are open. So all we need to show is that they do indeed cover $f(A)$. To this end, choose any $y \in f(A)$. To be in $f(A)$ means that there is some $x \in A$ such that $f(x) = y$. We showed above that $f^{-1}(U_1), f^{-1}(U_2), \ldots, f^{-1}(U_n)$ covers A, and so since $x \in A$ this in particular means that there is some $f^{-1}(U_k)$ that covers x. But $x \in f^{-1}(U_k)$ implies that $y = f(x) \in U_k$. So indeed, given an arbitrary $y \in f(A)$, one of the open sets $U_1, U_2, \ldots, U_k$ did cover it. So our open cover does have a finite subcover, completing the proof. □

This has a nice corollary. Notice that $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = x$ is a continuous function that is also *unbounded*; meaning, the range of f, which is $\mathbb{R}$, is an unbounded set. You see, if the domain is an unbounded set (like $\mathbb{R}$), then it is easy to have a range that is also unbounded (like $\mathbb{R}$). That said, you don't *need* the domain to be unbounded in order to get an unbounded range. For instance, $f : (0, 1) \to \mathbb{R}$ given by $f(x) = \frac{1}{x}$ is a continuous function on the bounded set $(0, 1)$, but yet f is unbounded.

However, if $f(x)$ is continuous on $[0, 1]$, then we *can* guarantee that f is *bounded*; meaning, the range of f is a bounded set. And on any closed interval, this same property will hold. But not any closed set, though—for example, $\mathbb{R}$ is a closed set, so the first example in the last paragraph shows that guaranteeing a closed domain is not sufficient. You also need the domain to be bounded. But that's enough—if the domain is closed and bounded, then any continuous f will be bounded. Said differently, if f is a continuous function on a *compact* set, then f is bounded.

Corollary.

Corollary 6.30 (*Continuous on compact set ⇒ bounded*). A continuous function on a compact set is bounded.

Proof. If A is compact and $f : A \to \mathbb{R}$ is continuous, then by Theorem 6.28, $f(A)$ is also compact. And hence by the Heine-Borel theorem (Theorem 5.19), $f(A)$ is closed and bounded. So f is bounded. □

Here's another corollary, and you know it's important because it has a name. It's called the extreme value theorem.

Corollary.

Corollary 6.31 (*The extreme value theorem*). A continuous function on a compact set attains its supremum and infimum.

Proof. Let A be a compact set; so A is closed and bounded. By Theorem 6.29, $f(A)$ is also closed and bounded. Since it is bounded above and below, the supremum and infimum of $f(A)$ exist. Let $M = \sup\{f(a) : a \in A\}$ and $L = \inf\{f(a) : a \in A\}$ (that is, $M = \sup(f(A))$ and $L = \inf(f(A))$).

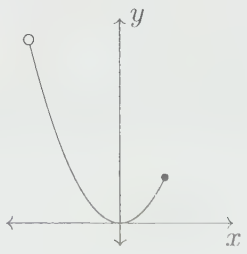
There then exists a sequence $f(a_n)$ from $f(A)$ converging to M, and a sequence $f(b_n)$ from $f(A)$ converging to L. Since $f(A)$ is closed, they both converge to some $f(a_0)$ and $f(b_0)$ inside the set $f(A)$. And so f achieves its supremum and infimum. □

Because this theorem is so important, let's do a few examples to make sure we understand it.

Example 6.32. If $f : [-2, 1] \to \mathbb{R}$ is given by $f(x) = x^2$, then $f([-2, 1])$'s supremum is 4 and its infimum is 0. And moreover, these are achieved. That is, we can find points in the domain whose images equal these values. In particular, $-2 \in [-2, 1]$ and $f(-2) = 4$. And $0 \in [-2, 1]$ and $f(0) = 0$. □

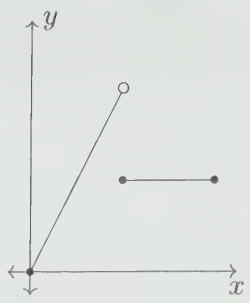
The two assumptions in the extreme value theorem were that f is continuous and f's domain is compact. Now we will give two examples to show that both assumptions are necessary.

Example 6.33. If $f : (-2, 1] \to \mathbb{R}$ is given by $f(x) = x^2$, then its supremum is 4 and its infimum is 0. The supremum is *not* achieved, though. That is, there is no $c \in (-2, 1]$ where $f(c) = 4$. This shows that the assumption that the domain be compact is indeed necessary. □



Example 6.34. Suppose $f : [0, 4] \to \mathbb{R}$ is given by

$$f(x) = \begin{cases} 2x & \text{if } x \in [0, 2), \\ 2 & \text{if } x \in [2, 4]. \end{cases}$$



Note that here the supremum is 4 but the supremum is *not* achieved. That is, there is no $c \in [2, 4]$ where $f(c) = 4$. This shows that the assumption that the function be continuous is indeed necessary. □

6.8 The Intermediate Value Theorem

Lemma.

Lemma 6.35 (f *continuous,* $f(c) > 0 \Rightarrow f(x) > 0 \ \forall \ |x - c| < \delta$). If f is continuous and $f(c) > 0$, then there exists some $\delta > 0$ such that $f(x) > 0$ for all x in $(c - \delta, c + \delta)$.

Likewise, if f is continuous and $f(c) < 0$, then there exists some $\delta > 0$ such that $f(x) < 0$ for all x in $(c - \delta, c + \delta)$.

Proof. Assume we are in the first situation and let $\varepsilon = \dfrac{f(c)}{2} > 0$. Then since f is continuous there exists some $\delta > 0$ such that $|x - c| < \delta$ implies $|f(x) - f(c)| < \varepsilon$. With a little algebra, we can show that this implies that $f(x) > 0$. To that end, first we unpack the the inequality $|f(x) - f(c)| < \varepsilon$:

$$\begin{aligned} |f(x) - f(c)| &< \varepsilon \\ -\varepsilon < f(x) - f(c) &< \varepsilon \\ f(c) - \varepsilon < f(x) &< f(c) + \varepsilon. \end{aligned}$$

And now the left inequality, combined with $\varepsilon = \frac{f(c)}{2}$, implies that

$$\begin{aligned} f(c) - \frac{f(c)}{2} &< f(x) \\ 0 < \frac{f(c)}{2} &< f(x), \end{aligned}$$

as desired.

In the second situation let $\varepsilon = \dfrac{-f(c)}{2} > 0$, and apply the same reasoning. □

Proposition.

Proposition 6.36 (*f continuous, $f(a) > 0 > f(b) \Rightarrow \exists\, c \in (a,b),\ f(c) = 0$*). If f is continuous on $[a,b]$ and $f(a)$ and $f(b)$ have different signs, then there is some $c \in (a,b)$ for which $f(c) = 0$.

Proof. Without loss of generality, assume that $f(a)$ is positive and $f(b)$ is negative. Let

$$A = \{t : f(x) > 0 \text{ for all } x \in [a,t]\}.$$

Since $f(a) > 0$, we know $a \in A$ and hence $A \neq \emptyset$. Moreover, since $f(b) < 0$, we know $b \notin A$ and hence A is bounded above by b. By completeness, these imply that $\sup(A)$ exists. Let $c = \sup(A)$, and note that $c \in (a,b)$. If $f(c) = 0$ then we are done, so assume for a contradiction that $f(c) \neq 0$.

If $f(c) > 0$ then by Lemma 6.35 there exists some $\delta > 0$ for which $f(x) > 0$ for all $x \in (c-\delta, c+\delta)$; in particular, $f(x) > 0$ for all $x \in (c-\delta, c+\delta/2]$. And certainly $(c-\delta) \in A$ since otherwise $c-\delta$ would have been an upper bound of A, contradicting our claim that c was the *least* upper bound of A. And so by definition, $(c-\delta) \in A$ means that $f(x) > 0$ for all $x \in [a, c-\delta]$. But at the start of this paragraph we also showed that $f(x) > 0$ for all $x \in (c-\delta, c+\delta/2]$. Putting these together, we see that $f(x) > 0$ for all $x \in [a, c+\delta/2]$, which means that $(c+\delta/2) \in A$, contradicting our claim that c is an upper bound on A.

By very similar reasoning, if $f(c) < 0$ then there is some $\delta > 0$ for which $f(c-\delta/2) < 0$, contradicting c being the *least* upper bound of A. This completes the proof. □

With that, we are now ready for this section's titular theorem.

Theorem.

Theorem 6.37 (*The intermediate value theorem*). If f is continuous on $[a,b]$ and α is any number between $f(a)$ and $f(b)$, then there is some $c \in (a,b)$ for which $f(c) = \alpha$.

Note that "α is any number between $f(a)$ and $f(b)$" means either "$f(a) < \alpha < f(b)$" or "$f(b) < \alpha < f(a)$." But we don't know whether $f(a)$ or $f(b)$ is larger, so it's easier to say it the way we did.

Proof. If $f(a) = f(b)$ then there are no numbers between $f(a)$ and $f(b)$, so there is nothing to show. So assume that $f(a) \neq f(b)$. Without loss of generality, we may assume $f(a) > f(b)$. So $f(a) > \alpha > f(b)$.

Define $g(x) = f(x) - \alpha$ and note that g is continuous on $[a,b]$. Then note that since $f(a) > \alpha$, we have $g(a) > 0$. And since $f(b) < \alpha$, we have $g(b) < 0$. So by

Proposition 6.36, there is some $c \in (a, b)$ such that

$$g(c) = 0$$

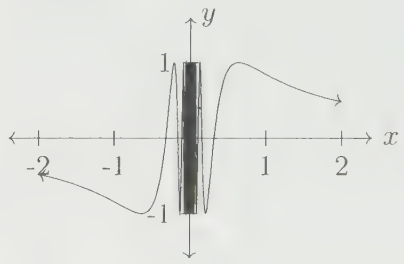
That is,

$$\begin{aligned} f(c) - \alpha &= 0 \\ f(c) &= \alpha. \end{aligned}$$

So we have found a $c \in (a, b)$ such that $f(c) = \alpha$, as desired. □

Interestingly, in some sense the converse to Theorem 6.37 is not true. That is, suppose you know that a function has the property that f has this property: Given any pair of points a and b, and any value α between $f(a)$ and $f(b)$, there exists some $c \in (a, b)$ for which $f(c) = \alpha$. Must such an f be continuous? Surprisingly, the answer is 'no.' The below, called the *topologist's sine curve*, is such an example.

$$f(x) = \begin{cases} \sin(1/x) & \text{for } x \neq 0, \\ 0 & \text{for } x = 0. \end{cases}$$



6.9 Uniform Continuity

Consider the function $f(x) = x^2$ on the domain $(0, \infty)$. We know this is a continuous function: Let $\varepsilon > 0$. For any $c \in (0, \infty)$ we can find some $\delta > 0$ such that, if $|x - c| < \delta$, we have $|f(x) - f(c)| < \varepsilon$. In particular, if you wanted to write a proof that f is continuous, you might begin with the following scratch work:

$$\begin{aligned} |f(x) - f(c)| &< \varepsilon \\ |x^2 - c^2| &< \varepsilon \\ |x - c| \cdot |x + c| &< \varepsilon \\ |x - c| &< \frac{\varepsilon}{|x + c|} \\ \delta &< \frac{\varepsilon}{|x + c|} \end{aligned}$$

If we insist $\delta < 1$, then $|x - c| < 1$ implies $|x + c| < 2c + 1$. So by choosing

$$\delta < \frac{\varepsilon}{2c + 1}$$

we can guarantee the previous line holds.

Then, to write a formal proof that $f(x) = x^2$ is continuous on $(0, \infty)$, you would let $\varepsilon > 0$, $c > 0$ and $\delta = \min\{1, \frac{\varepsilon}{2c+1}\}$. Then for any $x \in (0, \infty)$ for which $|x - c| < \delta$, we have

$$\begin{aligned} |f(x) - f(c)| &= |x^2 - c^2| \\ &= |x - c| \cdot |x + c| \\ &< \delta \cdot |x + c| \\ &\leq \frac{\varepsilon}{2c+1} \cdot |x + c| \end{aligned}$$

And since $|x - c| < 1$ implies $|x + c| \leq (c + 1) + c = 2c + 1$, the above is

$$\begin{aligned} &\leq \frac{\varepsilon}{2c+1} \cdot (2c + 1) \\ &= \varepsilon. \end{aligned}$$

Notice how δ depends on c: we had set $\delta = \min\{1, \frac{\varepsilon}{2c+1}\}$, and so as c gets larger and larger, δ in turn gets smaller and smaller. Remember that for a fixed $\varepsilon > 0$, the steeper the function is, the smaller the δ you need. The fact that δ is doing this then makes sense, as larger points of $f(x) = x^2$ give steeper slopes. And therefore for a fixed ε, the required δ should indeed get smaller and smaller as c gets larger and larger.

The important point from this is that for $f(x) = x^2$ with domain $(0, \infty)$, we certainly can not hope, for a fixed ε, to choose just a single δ and have it work for all of the c-values.

For some curves, though, we can pick just a single δ. Consider $f(x) = x^2$ on $[1, 4]$. The steepest this curve gets is at the very end, where its slope is about 8 (we haven't covered derivatives yet, but I'm sure you remember that the slope at the end is $f'(4) = 2(4) = 8$).

So if the slope is never more than 8, then for any $\varepsilon > 0$, setting $\delta = \varepsilon/8$ will work. Indeed, pick any $c \in [1, 4]$, let $\varepsilon > 0$, and let $\delta = \varepsilon/8 > 0$. Then for any $x \in [1, 4]$ for which $|x - c| < \delta$, we have

$$\begin{aligned} |f(x) - f(c)| &= |x^2 - c^2| \\ &= |x - c| \cdot |x + c| \\ &< \delta \cdot 8 \\ &= \frac{\varepsilon}{8} \cdot 8 \\ &= \varepsilon. \end{aligned}$$

So given any ε, a single δ worked no matter which c we were at. (Indeed, when x and c are close, the "$|x + c|$" being about $2x$ gives reason to see why the derivative of $f(x) = x^2$ is going to be $f'(x) = 2x$.) Let's summarize these observations.

- Let $\varepsilon > 0$. Let $f(x) = x^2$ on $(0, \infty)$ and let $c \in (0, \infty)$. It is true that f is continuous at c, but in the proof of this it is impossible to find a single δ which works no matter which c you choose. The reason is that for c-values getting larger and larger, it requires an increasingly smaller δ. In fact, the δ would be approaching 0. (Since δ depends on both on ε and c, this is sometimes written "$\delta = \delta(\varepsilon, c)$," as δ is in fact a function of ε and c.)

- Let $\varepsilon > 0$. Let $f(x) = x^2$ on $[1, 4]$ and let $c \in [1, 4]$. It is true that f is continuous at c, and moreover, this time, in the proof of this it *is* possible to find a single δ which works no matter which c you choose. Indeed, $\delta = \varepsilon/8$ worked. (Since δ depends just on ε, this is sometimes written "$\delta = \delta(\varepsilon)$," as δ is only a function of ε.)

We are going to distinguish between these cases. We will call functions in the second case *uniformly continuous*. Note also from this that the domain is super important. Functions always have domains, and saying a function is uniformly continuous means that it only has to satisfy the condition for the chosen domain.

The first way to state the definition of uniform continuity is this:

> Let $f : A \to \mathbb{R}$. We say f is *uniformly continuous* if for all $\varepsilon > 0$ there exists some $\delta > 0$ such that for all $c \in A$ and for every $x \in A$ for which $|x - c| < \delta$, we have
>
> $$|f(x) - f(c)| < \varepsilon.$$

But note that since c is completely unanchored, there is no difference between that x and that c; they are just any two points from A. Therefore a second way to state it is this:

Definition.

Definition 6.38. Let $f : A \to \mathbb{R}$. We say f is *uniformly continuous* if for all $\varepsilon > 0$ there exists some $\delta > 0$ such that, for all $x, y \in A$,

$$|x - y| < \delta \qquad \text{implies} \qquad |f(x) - f(y)| < \varepsilon.$$

Indeed,[9] below is a table of continuous functions (on a domain), and the δ that you would get if you tried to prove it was continuous at c. For each, we noted whether it was uniformly continuous or not.

[9]Unrelated side note: if you look at the definition, it almost looks like a continuous, point-centric version of being Cauchy.

$f(x)$ on a domain	δ	Uniformly continuous?
x^2 on $(0,\infty)$	$\frac{\varepsilon}{2c+1}$	No
x^2 on $[0,4]$	$\frac{\varepsilon}{2c+1}$	Yes[10]
$\sqrt{x}$ on $(0,\infty)$	ε^2	Yes
e^x on $(0,\infty)$	$\frac{\varepsilon}{e^c}$	No
$5x+2$ on $\mathbb{R}$	$\frac{\varepsilon}{5}$	Yes
$1/x$ on $(0,\infty)$	$\min\left\{\frac{c^2\varepsilon}{2}, \frac{\varepsilon}{2}\right\}$	No
$\sin(x)$ on $\mathbb{R}$	ε	Yes
$\cos(x)$ on $\mathbb{R}$	ε	Yes
$\tan(x)$ on $(-\pi/2, \pi/2)$	???	No

So being uniformly continuous is a strictly stronger condition than being continuous. So how do we identify when a continuous function is in fact uniformly continuous? Well, here's one nice result: If your continuous function is on a *compact* set, then you are guaranteed that it is uniformly continuous.

Proposition.

Proposition 6.39 (*Continuous f on compact $A \Rightarrow$ f uniformly continuous*). If $f : A \to \mathbb{R}$ is continuous and A is compact, then f is uniformly continuous on A.

Pre-proof trick. There is a trick that we have used a number of times, but because the proof below uses it twice, I thought I'd just highlight it now. Recall that if $|a-b| < \beta/2$ and $|b-c| < \beta/2$, then $|a-c| < \beta$. This follows quickly from applying the triangle inequality: $|a-c| = |a-b+b-c| \leq |a-b| + |b-c| = \frac{\beta}{2} + \frac{\beta}{2} = \beta$.

Proof. Suppose that f is continuous and A is compact. Let $\varepsilon > 0$. For each $c \in A$, since f is continuous at c, there exists some $\delta(c) > 0$ for which $x \in A$ and $|x-c| < \delta(c)$ implies that $|f(x) - f(c)| < \varepsilon/2$. Note then that the collection

$$\left\{ \left(c - \frac{\delta(c)}{2}, c + \frac{\delta(c)}{2} \right) \ : \ c \in A \right\}$$

of open intervals is then an open cover of A, and hence by the compactness of A

[10]The first two rows have the same δ but one is uniformly continuous while the other is not. What's the difference? It is in the domain. For the first, c can be arbitrarily big, and so $\frac{\varepsilon}{2c+1}$ will get arbitrarily small. For the second, $c \in [0,4]$. If the biggest c is 4, then the smallest δ is $\frac{\varepsilon}{2(4)+1} = \frac{\varepsilon}{9}$. So for any $\varepsilon > 0$, letting $\delta = \frac{\varepsilon}{9}$ will work, regardless of what c you chose

there exists some finite subcover,

$$\left\{\left(c_1 - \frac{\delta(c_1)}{2}, c_1 + \frac{\delta(c_1)}{2}\right), \left(c_2 - \frac{\delta(c_2)}{2}, c_2 + \frac{\delta(c_2)}{2}\right), \ldots, \left(c_n - \frac{\delta(c_n)}{2}, c_n + \frac{\delta(c_n)}{2}\right)\right\},$$

of A.

Pick any $x, y \in A$. By the definition of uniform continuity we aim to find a $\delta > 0$ such that if $|x - y| < \delta$ we have $|f(x) - f(y)| < \varepsilon$. Set

$$\delta = \min\left\{\frac{\delta(c_1)}{2}, \ldots, \frac{\delta(c_n)}{2}\right\}.$$

Since $x \in A$ and we found a finite subcover of A, we know that

$$x \in \left(c_k - \frac{\delta(c_k)}{2}, c_k + \frac{\delta(c_k)}{2}\right),$$

for some $k \in \{1, 2, \ldots, n\}$. Therefore $|c_k - x| < \frac{\delta(c_k)}{2}$. And also, since $|x - y| < \delta \leq \frac{\delta(c_k)}{2}$, by the pre-proof trick we see that

$$|c_k - y| < \delta(c_k).$$

That is, we have seen that both

$$|x - c_k| < \delta(c_k) \qquad \text{and} \qquad |y - c_k| < \delta(c_k).$$

And therefore by the original definition of $\delta(c_k)$, we see that

$$|f(x) - f(c_k)| < \varepsilon/2 \qquad \text{and} \qquad |f(y) - f(c_k)| < \varepsilon/2.$$

Applying the pre-proof trick one more time, this implies that

$$|f(x) - f(y)| < \varepsilon,$$

completing the proof. □

Note.

Note 6.40. Proposition 6.39 in particular implies that any function that is continuous on a closed interval $[a, b]$ is uniformly continuous on that interval. This is the form of the proposition that is most commonly used.

— Notable Exercises —

- In Exercise 6.4 you will prove the continuity of some fundamental functions. The sine and cosine functions, in conjunction with the material of Chapter 9, are particularly important in the branch of analysis called *Fourier analysis*. A teaser straight from the 1822 mouth of Joseph Fourier: "There is no function $f(x)$, or part of a function, which cannot be expressed by a trigonometric series." That certainly caught the math world's attention!

- It is often useful in mathematics to decompose something down into simpler parts. In number theory we talk about prime factorizations, and in abstract algebra we talk about the extent to which general $\mathbb{N}$-like algebraic structures can be uniquely factored. In group theory in particular, an important question is which groups can be expressed has a direct product of cyclic groups. As we just mentioned in the above bullet point, Fourier claimed that all functions can be decomposed in some way into sums of trig functions. In Chapter 9 we will see similar decompositions into sums of polynomials. Exercises 6.18 and 6.19—although a fairly tame form of this—is yet another useful decomposition (of a function, in this case) into a form which is sometimes easier to handle.

— Exercises —

Exercise 6.1. Use the definition of the functional limit to prove the following:

(a) $\lim_{x \to -2} (4x + 3) = -5$.

(b) $\lim_{x \to 1} \left(\frac{x^3}{x - 1} - \frac{1}{x - 1} \right) = 3$.

Exercise 6.2. This is a *false* statement:

> If $|f(x) - L| < \varepsilon$ for all x for which $0 < |x - a| < \delta$, then $|f(x) - L| < \varepsilon/2$ for all x for which $0 < |x - a| < \delta/2$.

Prove that this is false by finding a function f and numbers a and L which are a counterexample. (Make sure you also state f's domain and codomain,)

Exercise 6.3. For each of the following, give an example or prove that no such example exists.

(a) If $\lim_{x \to a} f(x)$ and $\lim_{x \to a} g(x)$ do not exist, can $\lim_{x \to a} [f(x) + g(x)]$ exist?

(b) If $\lim_{x \to a} f(x)$ and $\lim_{x \to a} g(x)$ do not exist, can $\lim_{x \to a} f(x)g(x)$ exist?

(c) If $\lim_{x \to a} f(x)$ and $\lim_{x \to a} [f(x) + g(x)]$ exist, must $\lim_{x \to a} g(x)$ exist?

(d) If $\lim_{x \to a} f(x)$ exists and $\lim_{x \to a} [f(x) + g(x)]$ does not exist, can $\lim_{x \to a} g(x)$ exist?

(e) If $\lim_{x \to a} f(x)$ and $\lim_{x \to a} f(x)g(x)$ exist, must $\lim_{x \to a} g(x)$ exist?

Exercise 6.4. Use the ε-δ definition of continuity to prove that each of the following functions is continuous on its domain.

(a) $f : (0, \infty) \to \mathbb{R}$, $f(x) = \sqrt{x}$.

(b) $g : \mathbb{R} \to \mathbb{R}$, $g(x) = e^x$.

(c) $h : (0, \infty) \to \mathbb{R}$, $h(x) = \log(x)$.

(d) $s : \mathbb{R} \to \mathbb{R}$, $s(x) = \sin(x)$.

(e) $t : \mathbb{R} \to \mathbb{R}$, $t(x) = \cos(x)$.

Exercise 6.5. Let $a, b \in \mathbb{R}$, and let $f : \mathbb{R} \to \mathbb{R}$ be given by $f(x) = ax + b$. Use the ε-δ definition of continuity to prove that f is continuous.

Exercise 6.6. Suppose that $f : X \to Y$ and $\{B_\alpha\}$ is a collection of subsets of Y. Prove that for $f^{-1}(\bigcup_\alpha B_\alpha) = \bigcup_\alpha f^{-1}(B_\alpha)$.

Exercise 6.7. Define

$$f(x) = \begin{cases} 1 & \text{if } x \geq 0 \\ -1 & \text{if } x < 0. \end{cases}$$

Prove that f is discontinuous at $c = 0$.

Exercise 6.8. Let $g(x) = x^2$ and let f be *Dirichlet function*:

$$f(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q} \\ 0 & \text{if } x \notin \mathbb{Q}. \end{cases}$$

Prove the following.

(a) For which x is $f(x) \leq x$?

(b) For which x is $f(x) \leq g(x)$?

(c) What is $g(f(x)) - f(x)$?

Exercise 6.9. Let

$$k(x) = \begin{cases} x & \text{if } x \in \mathbb{Q} \\ -x & \text{if } x \notin \mathbb{Q}. \end{cases}$$

Prove that $\lim_{x \to 0} k(x)$ exists, but for any $a \neq 0$, $\lim_{x \to a} k(x)$ does not exist.

Exercise 6.10. Let h be *Thomae's function*:

$$h(x) = \begin{cases} 1 & \text{if } x = 0 \\ 1/n & \text{if } x \in \mathbb{Q} \text{ and } x = m/n \text{ in lowest terms with } n > 0 \\ 0 & \text{if } x \notin \mathbb{Q}. \end{cases}$$

Prove that h is continuous at every irrational number and discontinuous at every rational number.

Exercise 6.11. Suppose that, for each $n \in \mathbb{N}$, A_n is a finite set of numbers from $[0, 1]$. Further, assume that $A_n \cap A_m = \emptyset$ whenever $n \neq m$. Define $f : [0, 1] \to \mathbb{R}$ to be

$$f(x) = \begin{cases} 1/n & \text{if } x \in A_n \\ 0 & \text{if } x \notin A_n \text{ for any } n. \end{cases}$$

Prove that $\lim_{x \to a} f(x) = 0$ for all $a \in [0, 1]$.

Exercise 6.12. Suppose $f : \mathbb{R} \to \mathbb{R}$ is a continuous function with the property that $f(x) = 0$ for every $x \in \mathbb{Q}$. Prove that $f(x) = 0$ for every $x \in \mathbb{R}$.

Exercise 6.13. Suppose that a function f satisfies $|f(x)| \leq |x|$ for all x. Show that f is continuous at 0.

Exercise 6.14. Suppose that a function g is continuous at 0 and $g(0) = 0$, and f is some function which satisfies $|f(x)| \leq |g(x)|$ for all x. Show that f is also continuous at 0.

Exercise 6.15.

(a) Give an example of a function $f : \mathbb{R} \to \mathbb{R}$ which is discontinuous at $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots$, but is continuous at every other point.

(b) Give an example of a function $f : \mathbb{R} \to \mathbb{R}$ which is discontinuous at $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots$, and at 0, but is continuous at every other point.

Exercise 6.16. Give an example of a function on the interval $[0, 1]$ that does not attain its supremum.

Exercise 6.17.

(a) Prove that if f is continuous, then $|f|$ is continuous.

(b) Give an example where $|f|$ is continuous, but f is not continuous.

Exercise 6.18. A function E is said to be *even* if $E(-x) = E(x)$ for all x. A function O is said to be *odd* if $O(-x) = -O(x)$ for all x. Prove that if $f : \mathbb{R} \to \mathbb{R}$ is continuous, then $f(x) = E(x) + O(x)$ for some even function $E(x)$ and odd function $O(x)$.

Exercise 6.19. Let f be continuous. Prove that f may be written as $f(x) = g(x) - h(x)$, for some continuous functions g are h which are also nonnegative (meaning $g(x) \geq 0$ and $h(x) \geq 0$ for all x). Hint: First show that if f_1 and f_2 are continuous, then so is $\max\{f_1, f_2\}$.

Exercise 6.20. How many continuous functions f are there which satisfy $(f(x))^2 = x^2$ for all x?

Exercise 6.21. In the below, by "finite open interval" we mean an interval (a, b) where $a, b \in \mathbb{R}$ and $a < b$. And by "finite closed interval" we mean an interval $[a, b]$ where $a, b \in \mathbb{R}$ and $a < b$.

(a) Let $f : A \to B$ be a continuous function where $f(A) = B$. Is it possible for A to be a finite open interval while B is a finite closed interval? Either provide an example showing it is possible, or prove that it is not possible.

(b) Let $f : A \to B$ be a continuous function where $f(A) = B$. Is it possible for A to be a finite closed interval while B is a finite open interval? Either provide an example showing it is possible, or prove that it is not possible.

Exercise 6.22. Let $f : [0,1] \to \mathbb{R}$ be continuous with $f(0) = f(1)$. Show that there must exist $x, y \in [0,1]$ which are of distance $1/2$ apart (i.e. $|x - y| = 1/2$) for which $f(x) = f(y)$.

Exercise 6.23. Let S be a dense subset of $\mathbb{R}$, and assume that f and g are continuous functions on $\mathbb{R}$. Prove that if $f(x) = g(x)$ for every $x \in S$, then $f(x) = g(x)$ for all $x \in \mathbb{R}$. Hint: First prove it in the case that $g(x) = 0$ for all x.

Exercise 6.24.

(a) Prove that if $f : A \to \mathbb{R}$ is uniformly continuous and A is bounded, then $f(A)$ is bounded.

(b) Give an example of an unbounded continuous function on $(0,1)$.

(c) Give an example of a bounded continuous function on $(0,1)$ that is not uniformly continuous. You do not need to prove your answer.

Exercise 6.25. Prove that if $f : \mathbb{Z} \to \mathbb{R}$, then f is uniformly continuous.

Exercise 6.26. Let $f : [a,b] \to \mathbb{R}$ be continuous and suppose that $f(x) > 0$ for all x. Show that there is a positive constant L such that $f(x) \geq L$ for all $x \in [a,b]$.

Exercise 6.27. Prove that there does not exist a continuous function which maps the closed interval $[0,1]$ onto the open interval $(0,1)$.

Exercise 6.28.

(a) Give an example of a continuous function $f : (0,1) \to \mathbb{R}$ and a Cauchy sequence (a_n) where $(f(a_n))$ is *not* a Cauchy sequence.

(b) Prove that if f is uniformly continuous and (a_n) is a Cauchy sequence, then $(f(a_n))$ is a Cauchy sequence.

Exercise 6.29. Let f and g be continuous function on $[a,b]$, and suppose that $f(a) < g(a)$ while $f(b) > g(b)$. Prove that $f(x) = g(x)$ for some $x \in [a,b]$.

Exercise 6.30. Let $f : [0,1] \to [0,1]$ be continuous. Prove that $f(x) = x$ for some x.

Exercise 6.31. For each of the following, prove that there is a number x satisfying the expression.

(a) $x^{83} + x^{17} + \dfrac{242}{1 + x^4 + \sin^2(x)} = 201.$

(b) $\cos(x) = x$

Exercise 6.32.

(a) Give an example of a continuous function defined on $\mathbb{R}$ which obtains every value exactly once. (That is, for each $b \in \mathbb{R}$ there is exactly one $a \in \mathbb{R}$ for which $f(a) = b$.)

(b) Prove that there does not exist a continuous function defined on $\mathbb{R}$ which obtains every value exactly *twice*.

(c) Give an example of a continuous function which obtains every value exactly three times. (If your function's graph is simple and clear, just sketching its graph is sufficient.)

Exercise 6.33. Is it possible for a continuous function $f : \mathbb{R} \to \mathbb{R}$ to have the property that $f(\mathbb{R}) = \mathbb{Q}$?

Exercise 6.34. Assume that $f : \mathbb{R} \to \mathbb{R}$ and $g : \mathbb{R} \to \mathbb{R}$ are both uniformly continuous functions.

(a) Prove that $f + g$ is uniformly continuous.

(b) If f and g are both bounded, prove that fg is uniformly continuous.

(c) Give an example showing that if g is not bounded then it is possible for fg to not be uniformly continuous.

(d) Prove that $g \circ f$ is uniformly continuous.

Exercise 6.35. Define a function $f : A \to B$ to be *monotone increasing* if for all $a, b \in A$ for which $a < b$, we have $f(a) \leq f(b)$. Is it possible for a monotone increasing function to have uncountably many discontinuities?

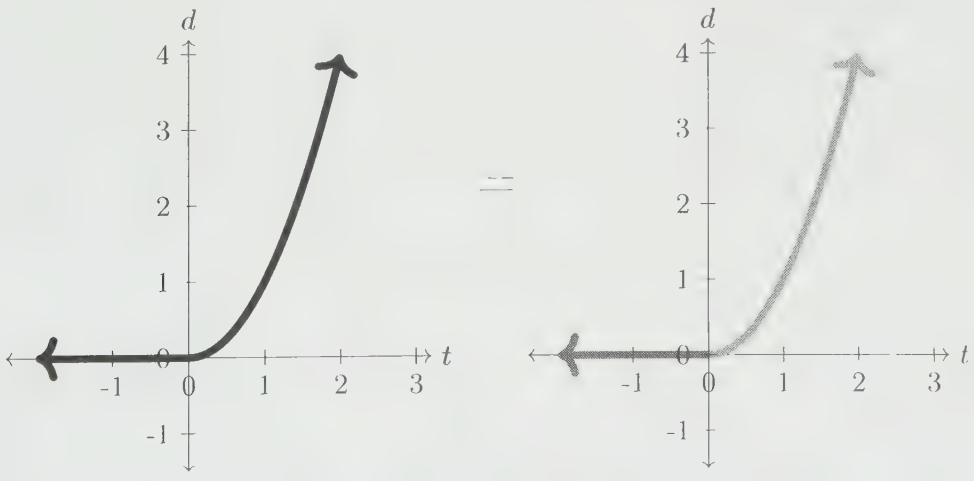
Exercise 6.36. A clock has been made with perfect precision: the hour and minute hands move continuously around the clock, and you are able to measure their positions with perfect accuracy. There's only one problem: the clocksmith accidentally put on identically-looking hands for the minute hand and the hour hand. Is it always possible to determine the time? If not, how many times a day can you not determine the time?

Chapter 7: Differentiation

Speedometers

Two millennia after Zeno's famous paradoxes, which we discussed in the first section of this text, Isaac Newton and others were trying to solve some fundamental physics problems which also required a deeper understanding of the infinite. For example, he wanted to know the answer to (an old-timey version of) this problem: If a car is driving in a straight line and you know the position of the car at every moment, can you determine the velocity of the car at every moment? Let's make this concrete through a specific example.

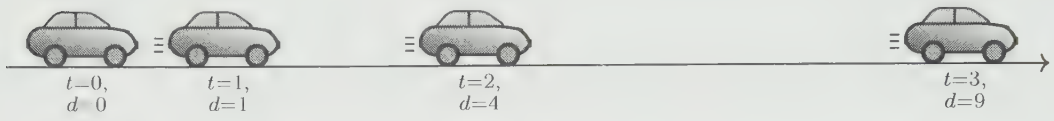
Example 7.1. Suppose you are sitting in your car at a red light. The light turns green and you accelerate (which of course changes your position/location). The position graph is below. The t-axis represents the amount of time (in seconds) since the light turned green, and the d-axis represents the distance (in feet) that the car has traveled since that moment. The same graph is drawn twice – the second is color-coded[1] to represent what color the streetlight is at that moment in time.



The right-hand side of the graph is the function $d(t) = t^2$. In other words, after 1 second the car has traveled $d(1) = 1$ foot. After 2 seconds it has traveled $d(2) = 4$ feet. After 3 seconds it has gone 9 feet, and so on. As you can see, the car is speeding

[1]For black and white viewing: darker=red, lighter=green.

up: in the first second it traveled 1 foot, in the second second it traveled $4 - 1 = 3$ feet, in the third second it traveled $9 - 4 = 5$ feet, and so on.



Here's the question we want to answer: At a fixed time t_0, what is the speed of the car at that moment? This is called the car's *instantaneous velocity*. If you're the driver you can tell your speed by looking at the speedometer. Thus we are asking whether we can figure out a function which reports exactly what the car's speedometer does.

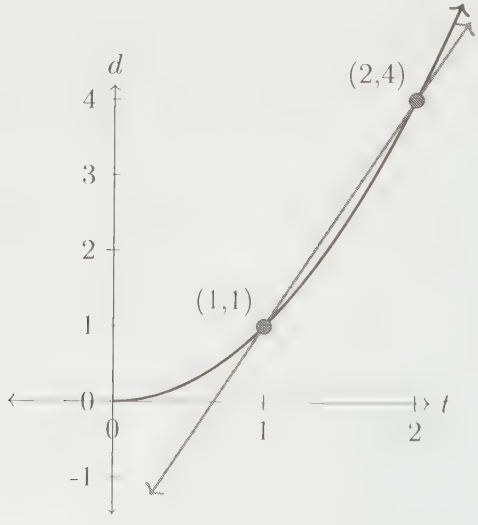
This is not an easy question, and a full course of high school physics is likely insufficient to find an answer for even some simple distance functions. Indeed, it took perhaps the greatest scientist in history—Isaac Newton—to find the answer; and in doing so, calculus was born. □

7.1 Graphical Interpretations of Velocity

One of the principal tenets of calculus is that geometry can be used to solve algebra problems. This is good news—the geometry tends to be pretty basic and visual, so not only is it helpful to solve the problems but also it helps us all see (and thus understand) what's going on. To solve the problem in Example 7.1 we will do this. We begin by reminding you of the geometry that is needed.

Secant Lines

A *secant line* is a line segment from one point on a curve to another. It looks like this:



Every line has a slope, and, just like all lines, the slope of the secant line from a point (x_1, y_1) to another point (x_2, y_2) is "change in y over change in x":

$$\text{slope of secant} = \frac{y_2 - y_1}{x_2 - x_1}.$$

If these points are on the curve of the function d, then $y_1 = d(x_1)$ and $y_2 = d(x_2)$, so the same formula can be written as

$$\text{slope of secant} = \frac{d(x_2) - d(x_1)}{x_2 - x_1}.$$

Average Velocity

Returning to Example 7.1, how would we, for instance, determine the car's velocity after one second? Remember that this is something that you do in your car all the time, as your speedometer is constantly telling you your velocity. We are simply asking what we would see if we looked at the speedometer after 1 second of driving. Well, the one equation you might remember from high school—and a formula Newton knew—was that for *average velocity*:

$$\text{average velocity} = \frac{\text{change in position}}{\text{change in time}}.$$

So, for example, if a car travels 120 miles in 2 hours then its average velocity is $120m/2h = 60m/h$. If Usain Bolt jogs 100 meters in 10 seconds then his average velocity is 10m/s.

Now, you might be tempted to just say "the velocity at $t = 1$ seconds must just be the average velocity from $t = 1$ to $t = 1$." Sadly this doesn't work out, though. From $t = 1$ to $t = 1$ no time has passed and no change in position has occurred, which would give

$$\text{average velocity} = \frac{\text{change in position}}{\text{change in time}} = \frac{0}{0} = \text{undefined}.$$

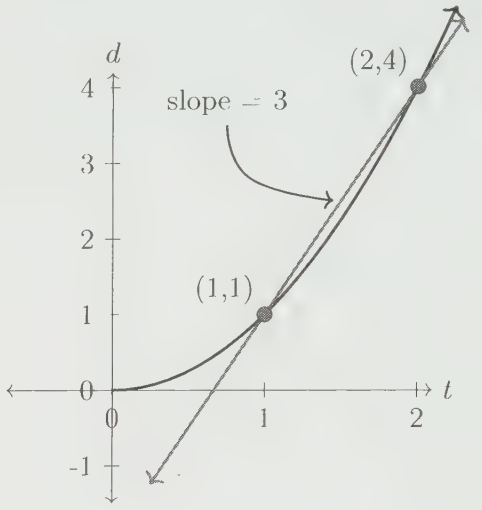
And in this case we would agree with Zeno that something is amiss, and we can not simply apply this formula.

Nevertheless, we will be able to use this formula in a different way. Notice that the formula looks a lot like the formula for the slope of a secant line...

Average Velocity = Slope of Secant Line

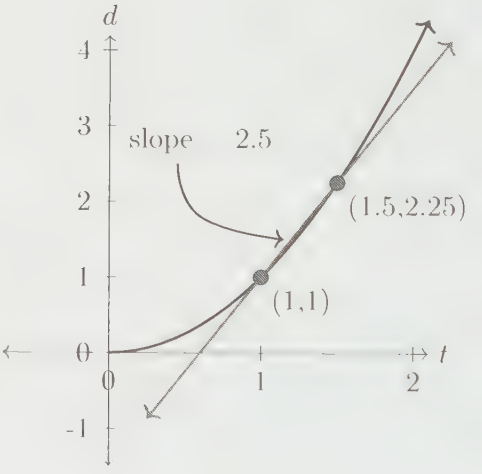
If we want to know a car's velocity at time $t = 1$ second, one thing we could do is find the average velocity of the car from $t = 1$ second to $t = 2$ seconds—the car's velocity won't change tooooo much over just one second, so the average velocity over this time period should be a decent guess for its speed at $t = 1$.

$$\begin{aligned}\text{average velocity from 1s to 2s} &= \frac{d(2) - d(1)}{2 - 1} \\ &= \frac{2^2 - 1^2}{2 - 1} \\ &= \frac{3}{1} \\ &= 3.\end{aligned}$$



The average velocity from $t = 1$ to $t = 2$ is 3 feet per second; note that the average velocity calculation matches the slope of the corresponding secant line. 3 feet per second is thus a decent guess for the velocity at $t = 1$, but let's try to improve it. If we cut down the time range to just $t = 1$ to $t = 1.5$ then the car would have less time to change its velocity, and thus we would have a better estimate. Doing this calculation gives

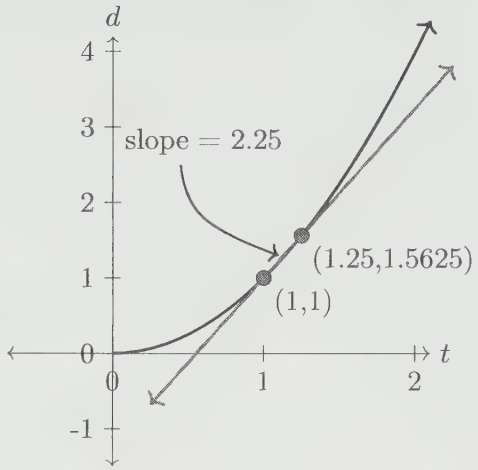
$$\begin{aligned}\text{average velocity from 1s to 1.5s} &= \frac{d(1.5) - d(1)}{1.5 - 1} \\ &= \frac{1.5^2 - 1^2}{1.5 - 1} \\ &= \frac{1.25}{0.5} \\ &= 2.5.\end{aligned}$$



The average velocity from $t = 1$ to $t = 1.5$ is 2.5 feet per second; our last guess of 3 ft/s was clearly not perfect. We must be getting closer to the true velocity, but

why stop here? Let's keep going. An even better estimate for the velocity would be the average velocity from $t = 1$ to $t = 1.25$.

$$\begin{aligned}\text{average velocity from 1s to 1.25s} &= \frac{d(1.25) - d(1)}{1.25 - 1} \\ &= \frac{1.25^2 - 1^2}{1.25 - 1} \\ &= \frac{0.5625}{0.25} \\ &= 2.25.\end{aligned}$$

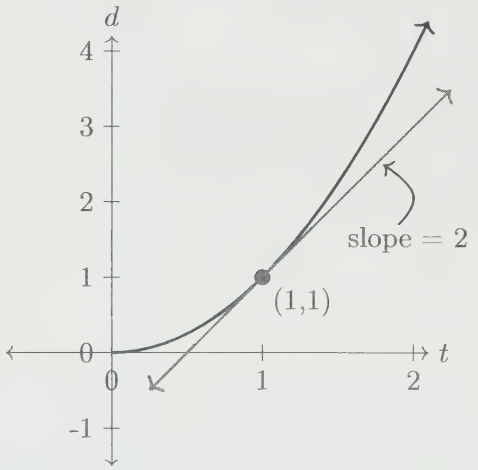


The average velocity from $t = 1$ to $t = 1.25$ is 2.25 ft/s, again with the formula matching the secant's slope. So 2.25 ft/s is an even better estimate, but let's not stop now.

Instantaneous Velocity = Slope of Tangent Line

If we continue to do this for smaller and smaller gaps, the average velocities look like the left-hand side below.

from	to	average velocity
t=1	t=2	3 ft/s
t=1	t=1.5	2.5 ft/s
t=1	t=1.25	2.25 ft/s
t=1	t=1.1	2.1 ft/s
t=1	t=1.01	2.01 ft/s
t=1	t=1.001	2.001 ft/s
t=1	t=1.0001	2.0001 ft/s
⋮	⋮	⋮
t=1	t=1	Inst. Vel. = 2 ft/s?



As the time interval gets smaller and smaller, the average velocity approaches 2 feet per second. And, indeed, as you would probably guess, the car's instantaneous velocity at $t = 1$ second turns out to be exactly 2 feet per second.

And what about the right-hand side of the above? The secant lines approach the tangent line, and the slopes of the secant lines approach the slope of the tangent line.

If $f(x)$ is a continuous at x_0 then the rate of change of f at a point (x_0, y_0) is equal to the slope of the tangent line to f at (x_0, y_0).

In our particular case, this means that the velocity at a time t_0 is equal to the slope of the tangent line to $d(t)$ at the point $(t_0, d(t_0))$.

Moreover—and we will talk later about how important this is—we would get the same conclusion if instead of beginning at $t = 1$ and ending a little past it, we instead started a little before $t = 1$ and ended at it, as you can see below.

from	to	average velocity
t=0	t=1	1 ft/s
t=0.5	t=1	1.5 ft/s
t=0.75	t=1	1.75 ft/s
t=0.9	t=1	1.9 ft/s
t=0.99	t=1	1.99 ft/s
t=0.999	t=1	1.999 ft/s
t=0.9999	t=1	1.9999 ft/s
⋮	⋮	⋮
t=1	t=1	Inst. Vel. = 2 ft/s?

7.2 The Derivative

These average-velocity and slope-of-secant formulas are exactly how the *derivative* is defined.

Definition.

Definition 7.2. Let I be an interval, $f : I \to \mathbb{R}$, and $c \in I$. We say f is *differentiable* at c if

$$\lim_{x \to c} \frac{f(x) - f(c)}{x - c}$$

exists. That is, this limit equals some real number.

If C is the collection of points at which f is differentiable, then the *derivative* of f is the function $f' : C \to \mathbb{R}$ where

$$f'(c) = \lim_{x \to c} \frac{f(x) - f(c)}{x - c}.$$

Notation.

Notation 7.3. Given a function f, we write $\frac{d}{dx}f(x)$ to be its derivative. That is,

$$\frac{d}{dx}f(x) = f'(x).$$

Examples

Example 7.4. Let I be an interval and $f : I \to \mathbb{R}$ be given by $f(x) = x^2$. Then for any $c \in I$,

$$\begin{aligned} f'(c) &= \lim_{x\to c} \frac{f(x) - f(c)}{x - c} \\ &= \lim_{x\to c} \frac{x^2 - c^2}{x - c} \\ &= \lim_{x\to c} \frac{(x - c)(x + c)}{x - c} \qquad (\text{☎}) \\ &= \lim_{x\to c} (x + c) \end{aligned}$$

Recall that since x is a variable and c is a constant, "$x + c$" is a continuous function and hence we were allowed to just plug in "c."

$$\begin{aligned} &= c + c \\ &= 2c. \end{aligned}$$

That's the answer at every fixed c, so we can instead rewrite this using the variable x: $f'(x) = 2x$. Or, writing it out as the derivative of a function:

$$\frac{d}{dx}x^2 = 2x.$$

□

We can actually prove a much more general formula, that for any $n \in \mathbb{N}$ we have $\frac{d}{dx}x^n = nx^{n-1}$. We show that now.

Example 7.5. Let I be an interval and $f : I \to \mathbb{R}$ be given by $f(x) = x^n$ where $n \in \mathbb{N}$. Then for any $c \in I$,

$$\frac{d}{dx}x^n = \lim_{x\to c} \frac{f(x) - f(c)}{x - c} = \lim_{x\to c} \frac{x^n - c^n}{x - c}$$

Remember how we factored $x^2 - c^2$ in (☎)? We can factor $x^n - c^n$ in a similar

way.[2] Note that

$$x^n - c^n = (x-c)(x^{n-1} + x^{n-2}c + x^{n-3}c^2 + \cdots + xc^{n-2} + c^{n-1}),$$

because

$$\begin{aligned}(x-c)(x^{n-1} + x^{n-2}c + x^{n-3}c^2 + \cdots + xc^{n-2} + c^{n-1}) &\\ = x^n + x^{n-1}c + x^{n-2}c^2 + \cdots + x^2c^{n-2} + xc^{n-1} &\\ - x^{n-1}c - x^{n-2}c^2 - \cdots - x^2c^{n-2} - xc^{n-1} - c^n &\\ = x^n - c^n. &\end{aligned}$$

With this factorization, we can now compute the derivative at an arbitrary point c.

$$\begin{aligned} f'(c) &= \lim_{x\to c} \frac{x^n - c^n}{x-c} \\ &= \lim_{x\to c} \frac{(x-c)(x^{n-1} + x^{n-2}c + x^{n-3}c^2 + \cdots + xc^{n-2} + c^{n-1})}{x-c} \\ &= \lim_{x\to c} (x^{n-1} + x^{n-2}c + x^{n-3}c^2 + \cdots + xc^{n-2} + c^{n-1}) \\ &= (c^{n-1} + c^{n-2}c + c^{n-3}c^2 + \cdots + c\cdot c^{n-2} + c^{n-1}) \\ &= n\cdot c^{n-1}. \end{aligned}$$

Since c was arbitrary, $\frac{d}{dx}x^n = nx^{n-1}$ for any natural number n.[3] □

Another good example on this is Example 7.7 below. But it's a useful example to think about in the context of the following theorem, so we start with that.

7.3 Continuity and Differentiability

There are some important questions that we should now ask. Throughout this course we have spent much time studying properties of functions, the biggest being continuity. As we continue our study of differentiability, note that both continuity and differentiability are local conditions. It's therefore quite natural to ask how continuity and differentiability relate.

Do you think a function is continuous if and only if it is differentiable? If not, which of those implications fails? Is every continuous function differentiable, but not every differentiable function continuous? Or vice versa? Or maybe neither implication holds?

If f is continuous, is it the derivative of some function g? And if so, is g continuous? What about if f is instead differentiable? And what about with uniform continuity? Extreme values? Boundedness? Intermediate value property? We have all sorts of notions that we have looked at, and so it would be very interesting to know how this new notion of a derivative fits in with our theory thus far.

[2] Indeed, one way to see that this should be possible is to just note that if you plug c into $x^n - c^n$ you get out zero: $c^n - c^n = 0$, and hence c is a root of $x^n - c^n$, and hence $x - c$ should be a factor.

[3] In fact, this holds for any real number n, but that's harder to prove.

We begin with the first question: Does differentiability imply continuity and vice versa? It turns out that differentiability is the strictly stronger criterion. Thus our goal is to prove that every differentiable function is continuous, and then to find an example of a continuous function that is not differentiable.

Theorem.

Theorem 7.6 (*Differentiable $\Rightarrow$ continuous*). Suppose I is an interval and $f : I \to \mathbb{R}$ is differentiable at $c \in I$. Then f is continuous at $c \in I$.

Proof. By assumption, f's derivative exists at c. That is, there is some $L \in \mathbb{R}$ for which

$$\lim_{x \to c} \frac{f(x) - f(c)}{x - c} = L.$$

We wish to show that f is continuous at c. By Theorem 6.16 part (v), it suffices to show that $\lim_{x \to c} f(x) = f(c)$; i.e., that $\lim_{x \to c} [f(x) - f(c)] = 0$. Since the differentiable limit exists and the limit $\lim_{x \to c}(x - c)$ exists, this follows quickly from the sequence limit laws (Theorem 3.21).

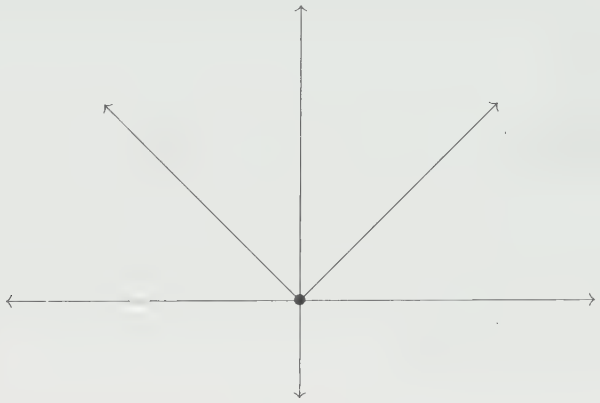
$$\begin{aligned}
\lim_{x \to c} [f(x) - f(c)] &= \lim_{x \to c} \frac{f(x) - f(c)}{x - c} \cdot (x - c) \\
&= \left(\lim_{x \to c} \frac{f(x) - f(c)}{x - c} \right) \cdot \left(\lim_{x \to c} (x - c) \right) \qquad \text{(Limit laws)} \\
&= L \cdot 0 \\
&= 0.
\end{aligned}$$

□

When you have a single-direction implication, it is worthwhile to consider the contrapositive, which is logically equivalent. The contrapositive of Theorem 7.6 is: If f is not continuous at some $c \in I$, then f is also not differentiable at $c \in I$.

The converse, though, is not true. That is, we can find a function which is continuous at some point but not differentiable at that point. (And if you tried to use the above proof in reverse, it would fail because we can not break up the limit like that if we do not already know that it is differentiable.)

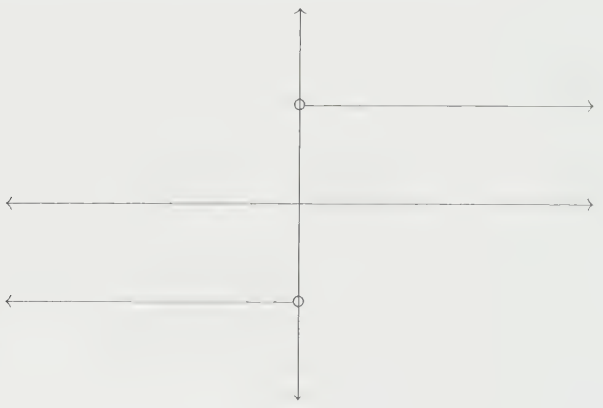
Example 7.7. Let $f(x) = |x|$ and $c = 0$.



Then f is continuous at 0 since clearly $\lim_{x \to c} f(x) = f(c)$.[4] However, f is not differentiable at 0. To see this note that

$$\begin{aligned} \lim_{x \to c} \frac{f(x) - f(c)}{x - c} &= \lim_{x \to 0} \frac{f(x) - f(0)}{x - 0} \\ &= \lim_{x \to 0} \frac{f(x)}{x} \\ &= \lim_{x \to 0} \frac{|x|}{x}, \end{aligned}$$

and this limit does not exist since if $x > 0$ this quotient is 1, but if $x < 0$ this quotient equals -1. Thus, this limit does not exist by Exercise 6.7.



□

[4] To prove it, let $\varepsilon > 0$ and $\delta = \varepsilon$, and apply the definition of a limit.

7.4 Differentiability Rules

First we recall some notation that you have probably seen in previous courses.

Notation.

Notation 7.8. Suppose f and g are functions. Then define

- $(f+g)(x) = f(x) + g(x)$
- $(f-g)(x) = f(x) - g(x)$
- $(fg)(x) = f(x)g(x)$
- $\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$

The first result is that derivatives satisfy the linearity conditions.

Proposition.

Proposition 7.9 (*Linearity of the derivative*). Let I be an interval, let $f, g : I \to \mathbb{R}$ be differentiable at $c \in I$, and let k be a constant. Then

(i) $(f+g)'(c) = f'(c) + g'(c)$.

(ii) $(f-g)'(c) = f'(c) - g'(c)$.

(iii) $(kf)'(c) = k \cdot f'(c)$.

Proof. The proofs of (i) and (ii) are exactly the same, so we will prove them together. Note that we used a limit law to move from the second line to the third line, which was allowed because by assumption both of those limits existed.

$$\begin{aligned}
\lim_{x\to c} \frac{(f\pm g)(x) - (f\pm g)(c)}{x-c} &= \lim_{x\to c} \frac{\big[f(x)\pm g(x)\big] - \big[f(c)\pm g(c)\big]}{x-c} \\
&= \lim_{x\to c} \frac{\big[f(x)-f(c)\big] \pm \big[g(x)-g(c)\big]}{x-c} \\
&= \lim_{x\to c} \frac{f(x)-f(c)}{x-c} \pm \lim_{x\to c} \frac{g(x)-g(c)}{x-c} \\
&= f'(c) \pm g'(c).
\end{aligned}$$

Now we prove (iii), which again just amounts to writing out the definition of the

derivative, moving a couple things around, and applying a limit law.

$$\begin{aligned}\lim_{x\to c}\frac{(kf)(x)-(kf)(c)}{x-c} &= \lim_{x\to c}\frac{k\cdot f(x)-k\cdot f(c)}{x-c}\\ &= k\cdot\lim_{x\to c}\frac{f(x)-f(c)}{x-c}\\ &= k\cdot f'(c).\end{aligned}$$

□

Example 7.10. We have shown that $\frac{d}{dx}x^{12} = 12x^{11}$ and $\frac{d}{dx}x^5 = 5x^4$. And so, by Proposition 7.9 part (i), we now know that

$$\frac{d}{dx}(x^{12}+x^5) = 12x^{11}+5x^4.$$

Also,

$$\frac{d}{dx}(7\cdot x^5) = 7\cdot\frac{d}{dx}x^5 = 7\cdot 5x^4 = 35x^4.$$

Theorem.

Theorem 7.11 (*The product rule*). Let I be an interval and let $f, g : I \to \mathbb{R}$ be differentiable at $c \in I$. Then

$$(fg)'(c) = f'(c)g(c) + f(c)g'(c).$$

Proof. Note that

$$\begin{aligned}(fg)'(c) &= \lim_{x\to c}\frac{(fg)(x)-(fg)(c)}{x-c}\\ &= \lim_{x\to c}\frac{f(x)g(x)-f(c)g(c)}{x-c}\\ &= \lim_{x\to c}\frac{f(x)g(x)-f(c)g(x)+f(c)g(x)-f(c)g(c)}{x-c}\\ &= \lim_{x\to c}\frac{f(x)g(x)-f(c)g(x)}{x-c}+\lim_{x\to c}\frac{f(c)g(x)-f(c)g(c)}{x-c}\\ &= \lim_{x\to c}\frac{f(x)-f(c)}{x-c}\cdot g(x)+\lim_{x\to c}f(c)\cdot\frac{g(x)-g(c)}{x-c}\\ &= \lim_{x\to c}\frac{f(x)-f(c)}{x-c}\cdot\lim_{x\to c}g(x)+f(c)\cdot\lim_{x\to c}\frac{g(x)-g(c)}{x-c}\\ &= f'(c)g(c)+f(c)g'(c).\end{aligned}$$

□

Theorem.

Theorem 7.12 (*The quotient rule*). Let I be an interval and let $f, g : I \to \mathbb{R}$ be differentiable at $c \in I$. Then

$$\left(\frac{f}{g}\right)'(c) = \frac{f'(c)g(c) - f(c)g'(c)}{\big[g(c)\big]^2},$$

provided that $g(c) \neq 0$.

Proof. The proof is similar to the previous proof, and is Exercise 7.3. □

Theorem.

Theorem 7.13 (*The chain rule*). Let I_1 and I_2 be intervals and $g : I_1 \to I_2$ and $f : I_2 \to \mathbb{R}$. If g is differentiable at $c \in I_1$ and f is differentiable at $g(c) \in I_2$, then

$$(f \circ g)'(c) = f'(g(c)) \cdot g'(c).$$

Wrong Proof. Below is NOT a correct proof of the chain rule. That said, it is close. And in fact, if you dust off your old calculus textbook and find where they state the chain rule, if they provide a "proof" (as they usually do) then they probably will give the following false proof.

That all said, here is the false proof:

$$\begin{aligned}
(f \circ g)'(c) &= \lim_{x \to c} \frac{f(g(x)) - f(g(c))}{x - c} \\
&= \lim_{x \to c} \frac{f(g(x)) - f(g(c))}{x - c} \cdot \frac{g(x) - g(c)}{g(x) - g(c)} \\
&= \lim_{x \to c} \frac{f(g(x)) - f(g(c))}{g(x) - g(c)} \cdot \frac{g(x) - g(c)}{x - c} \\
&= \lim_{x \to c} \frac{f(g(x)) - f(g(c))}{g(x) - g(c)} \cdot \lim_{x \to c} \frac{g(x) - g(c)}{x - c} \\
&= f'(g(x)) \cdot g'(x).
\end{aligned}$$

As $x \to c$, note that since g is continuous at c we do have $g(x) \to g(c)$. So the problem is *not* that in the first limit we have a $g(x)$ instead of an x.

The actual problem here is this: When you take, say,

$$\lim_{x \to c} \frac{g(x) - g(c)}{x - c},$$

we are never concerned about dividing by zero, because in the definition of a functional limit (Definition 6.7) we never have $x = c$. Likewise, when we take

$$\lim_{x \to c} \frac{f(g(x)) - f(g(c))}{g(x) - g(c)},$$

we again guarantee that $x \neq c$, but it's important to note that this does *not* guarantee that $g(x) \neq g(c)$. For all we know, we could have $g(x) = g(c)$ for x values getting arbitrarily close to c. And in that case, we have a significant divide-by-zero problem.

To fix this, we define a new function which is equal to this quotient, provided the denominator is non-zero—but whenever we do have a $g(x) = g(c)$ issue, our function just redefines this quotient to be what the derivative ought to be. Doing this right will ensure that we do not change the limit in the meantime. Below is this.

Proof. Consider the function

$$d(y) = \begin{cases} \frac{f(y) - f(g(c))}{y - g(c)} & \text{if } y \neq g(c) \\ f'(g(c)) & \text{if } y = g(c) \end{cases}$$

Note that d is continuous at $g(c)$; i.e., $\lim_{y \to g(c)} d(y) = f'(g(c))$, because f is differentiable at $g(c)$. Now, note that

$$\frac{f(g(x)) - f(g(c))}{x - c} = d(g(x)) \cdot \frac{g(x) - g(c)}{x - c}. \qquad (\text{◕})$$

If $g(x) \neq g(c)$, then (◕) is just

$$\frac{f(g(x)) - f(g(c))}{x - c} = \frac{f(g(x)) - f(g(c))}{g(x) - g(c)} \cdot \frac{g(x) - g(c)}{x - c},$$

which are the same just by multiplying top-and-bottom by $g(x) - g(c)$, and moving a few things around. If $g(x) = g(c)$, then applying the definition of d to (◕) gives

$$\frac{f(g(x)) - f(g(c))}{x - c} = f'(g(c)) \cdot \frac{g(x) - g(c)}{x - c},$$

and then applying $g(x) = g(c)$,

$$\frac{f(g(c)) - f(g(c))}{x - c} = f'(g(c)) \cdot \frac{g(c) - g(c)}{x - c},$$

which is simply $0 = 0$, which is true. So in either case, the above is true. Therefore,

$$\begin{aligned}(f \circ g)'(c) &= \lim_{x \to c} \frac{f(g(x)) - f(g(c))}{x - c} \\ &= \lim_{x \to c} d(g(x)) \cdot \frac{g(x) - g(c)}{x - c} \\ &= \lim_{x \to c} d(g(x)) \cdot \lim_{x \to c} \frac{g(x) - g(c)}{x - c} \\ &= f'(g(c)) \cdot g'(c).\end{aligned}$$

□

7.5 Topologist's Sine Curve Examples

Below are some important examples, which all deal with the following general form of a function.

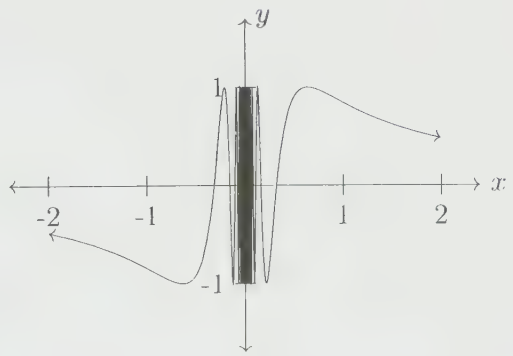
$$g_n(x) = \begin{cases} x^n \sin(1/x) & \text{if } x \neq 0, \\ 0 & \text{if } x = 0. \end{cases}$$

We will focus on the three functions g_0, g_1 and g_2. For each of these we ask: is g_n continuous at 0? Is it differentiable at 0?

Example 7.14. We begin with

$$g_0(x) = \begin{cases} \sin(1/x) & \text{if } x \neq 0, \\ 0 & \text{if } x = 0. \end{cases}$$

It's graph:



As you can see, g_0 is not continuous at 0, because $\lim_{x \to 0} \sin(1/x) =$ does not exist. Indeed, by looking at the graph you can see that the one-sided limits of this limit are equivalent to

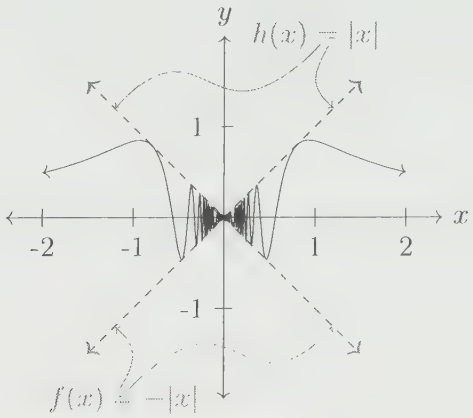
$$\lim_{t \to \infty} \sin(t) = \text{D.N.E.} \qquad \text{and} \qquad \lim_{t \to -\infty} \sin(1/t) = \text{D.N.E.}$$

And then by (the contrapositive to) Theorem 7.6, g_0 being not continuous at 0 implies that it is also not differentiable at 0. □

Example 7.15. Next, consider the function

$$g_1(x) = \begin{cases} x\sin(1/x) & \text{if } x \neq 0, \\ 0 & \text{if } x = 0. \end{cases}$$

This function has the graph



To see that g_1 is continuous at 0, first note that by combining $-1 \leq \sin(1/x) \leq 1$ and $-|x| \leq x \leq |x|$, we have that

$$-|x| \leq x\sin(1/x) \leq |x|.$$

Also,

$$\lim_{x \to 0} -|x| = 0 \qquad \text{and} \qquad \lim_{x \to 0} |x| = 0.$$

So by the Squeeze Theorem (Corollary 6.14),

$$\lim_{x \to 0} x\sin(1/x) = 0.$$

I.e.,

$$\lim_{x \to 0} g_1(x) = g_1(0),$$

which, by Theorem 6.16 part v, means that g_1 is continuous at 0.

However, g_1 is *not* differentiable at 0. To see this, we can simply use the definition of the derivative (Definition 7.2):

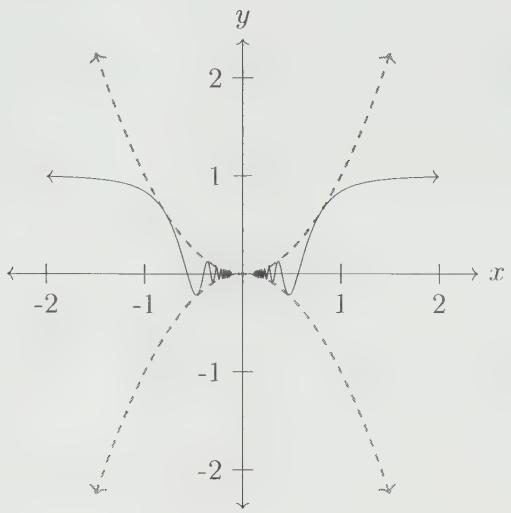
$$\begin{aligned} g'(0) &= \lim_{x \to 0} \frac{g_1(x) - g_1(0)}{x - 0} \\ &= \lim_{x \to 0} \frac{x\sin(1/x) - 0}{x - 0} \\ &= \lim_{x \to 0} \sin(1/x) \\ &= \text{Does not exist.} \end{aligned}$$

Note, the fact that the final limit does not exist is from Example 7.14. □

Example 7.16. Lastly, consider the function

$$g_2(x) = \begin{cases} x^2 \sin(1/x) & \text{if } x \neq 0, \\ 0 & \text{if } x = 0. \end{cases}$$

This function has the graph



Note that g_2 is continuous at 0 by the Squeeze Theorem (by essentially the exact calculation as above).

And g_2 is also differentiable at 0:

$$\begin{aligned} g_2'(0) &= \lim_{x \to 0} \frac{g_2(x) - g_2(0)}{x - 0} \\ &= \lim_{x \to 0} \frac{x^2 \sin(1/x) - 0}{x - 0} \\ &= \lim_{x \to 0} x \sin(1/x) \\ &= 0. \end{aligned}$$

Note that we had already found this final limit (via the Squeeze Theorem) in the previous example above. □

To summarize the conclusions of the last three examples,

	Continuous at 0?	Differentiable at 0?
g_0	no	no
g_1	yes	no
g_2	yes	yes

7.6 Local Minimums and Maximums

Derivatives are not just interesting to study, they are darn-right useful too. When you took calculus, perhaps the most important application was using the derivative to find local maxes and mins. If you recall, all those optimization problems you did came down to finding critical points of functions, which is essentially the same as finding local maxes and mins.

Definition.

Definition 7.17. Let $f : A \to \mathbb{R}$. Then f has a *local maximum* at $c \in A$ if there exists some $\delta > 0$ such that for all $x \in A$ for which $|x - c| < \delta$, we have

$$f(x) \leq f(c).$$

Likewise, f has a *local minimum* at $c \in A$ if there exists some $\delta > 0$ such that for all $x \in A$ for which $|x - c| < \delta$, we have

$$f(x) \geq f(c).$$

An important idea from calculus that you might remember is that a differentiable function's local mins and maxes only happen at places where its derivative equal zeros. Indeed, back in the day you did lots of examples where you took the derivative of a function, set that derivative equal to zero, and solved[5]. We prove now that this does indeed work.

Recall.

Recall 7.18. First, though, quickly recall a basic fact about limits that will be useful: If $s_n \geq 0$ for all n, and $s_n \to L$, then $L \geq 0$.

Proposition.

Proposition 7.19 (*Local maxes/mins happen where $f'(c) = 0$*). Let I be an open interval and suppose $f : I \to \mathbb{R}$ is differentiable at $c \in I$. If f has a local max or min at c, then $f'(c) = 0$.

[5]Typically you solved by factoring, using the quadratic formula, or using some other algebraic properties, like those of exponentials or logs.

Proof. Since f is differentiable at c, we know that

$$\lim_{x \to c} \frac{f(x) - f(c)}{x - c} = f'(c).$$

Suppose $c \in A$ is a local max. Then there exists some $\delta > 0$ where $x \in A$ and $|x - c| < \delta$ implies $f(x) \leq f(c)$ (and hence, $f(x) - f(c) \leq 0$). Consequently, for all such x,

$$\frac{f(x) - f(c)}{x - c} \quad \text{is} \quad \begin{cases} \leq 0 & \text{if } x > c; \\ \geq 0 & \text{if } x < c. \end{cases} \tag{♡}$$

So when we take the limit

$$\lim_{x \to c} \frac{f(x) - f(c)}{x - c} = f'(c),$$

for the x values approaching from the left, (once they are within δ of c) the quotient is at most zero. Conversely, for the x values approaching from the right, (once they are within δ of c) the quotient is at least zero. The only way this can happen is if the limit is exactly 0.

To formalize this, we use the sequential formulation of the above functional limit (cf. Theorem 6.11). That is, we know that given any sequence (a_n) from A, if $a_n \to c$ then

$$\frac{f(a_n) - f(c)}{a_n - c} \to f'(c).$$

We will now pick one sequence approaching from the positive end and one approaching from the negative end; both will be within δ of c. That is, first pick a sequence (a_n) from A in which $c - \delta < a_n < c$ for all n, and also $a_n \to c$. Then by (♡) we know that

$$\frac{f(a_n) - f(c)}{a_n - c} \geq 0$$

for all n, and

$$\frac{f(a_n) - f(c)}{a_n - c} \to f'(c).$$

Therefore $f'(c) \geq 0$. Conversely, if we pick a sequence (b_n) from A for which $c < b_n < c + \delta$ for all n, and also $b_n \to c$, then by (♡) we know that

$$\frac{f(b_n) - f(c)}{b_n - c} \leq 0$$

for all n, and

$$\frac{f(b_n) - f(c)}{b_n - c} \to f'(c).$$

Therefore $f'(c) \leq 0$. And of course, the only way that $f'(c) \geq 0$ and also $f'(c) \leq 0$ is if $f'(c) = 0$. [ ]

Mathematicians knew of this result for about as long as derivatives themselves have been defined. Fermat, for instance, made use of this to solve optimization problems. It took 200 years, though, for someone to notice an important corollary to this result.

Recall that continuous functions have the intermediate value property, but not every function with the intermediate value property (IVP) is in fact continuous. Also, we know that every differentiable function is continuous, but not every continuous function is differentiable. So in terms of how strong a condition is,

$$f \text{ is differentiable } \Rightarrow \ f \text{ is continuous } \Rightarrow \ f \text{ has the IVP,}$$

but none of those implications go the other way.

So now, here's where the fun begins. If f is differentiable, it is certainly *not* the case that f' will also have to be differentiable. Ok, let's go weaker: will f' have to be continuous? Again, the answer is no. But the intermediate value property...that we *can* guarantee! This is known as Darboux's theorem.

Theorem.

Theorem 7.20 (*Darboux's theorem*). Suppose $f : [a, b] \to \mathbb{R}$ is differentiable. If α is between $f'(a)$ and $f'(b)$, then there exists $c \in (a, b)$ where $f'(c) = \alpha$.

First we note that "α between $f'(a)$ and $f'(b)$" means that either $f'(a) < \alpha < f'(b)$, or $f'(b) < \alpha < f'(a)$.

Proof. Without loss of generality, assume $f'(b) < \alpha < f'(a)$. Define $g(x) = f(x) - \alpha x$. Then g is differentiable on $[a, b]$, and $g'(x) = f'(x) - \alpha$. Furthermore,

$$g'(a) = f'(a) - \alpha > 0 \qquad \text{and} \qquad g'(b) = f'(b) - \alpha < 0.$$

If we can find a local maximum in (a, b), then we can apply Proposition 7.19 to show that $g'(c) = 0$ for some $c \in (a, b)$, which will prove that $f'(c) = \alpha$. So that is our goal.

Since $[a, b]$ is compact, by the extreme value theorem (Theorem 6.31) g attains a maximum, which of course is also a local maximum.

Claim: g's maximum is not at a.

To see that this is true, observe that if it were true then $g(x) \leq g(a)$ for all $x \in [a, b]$, but this implies that $\frac{g(x)-g(a)}{x-a} = \frac{\text{negative or } 0}{\text{positive}} \leq 0$ for all $x \in (a, b]$. And hence

$$g'(a) = \lim_{x \to a} \frac{g(x) - g(a)}{x - a} \leq 0.$$

But this contradicts our earlier deduction that $g'(a) > 0$. $\square_{\text{(claim)}}$

Claim: g's maximum is not at b.

To see that this is true, observe that if it were true then $g(x) \geq g(a)$ for all $x \in [a, b]$, but this implies that $\frac{g(x)-g(b)}{x-b} = \frac{\text{negative or 0}}{\text{negative}} \geq 0$ for all $x \in [a, b)$. And hence

$$g'(a) = \lim_{x \to a} \frac{g(x) - g(a)}{x - a} \geq 0.$$

But this contradicts our earlier deduction that $g'(a) < 0$. $\square_{\text{(claim)}}$

Therefore the max is at some $c \in (a, b)$.

Then (taking $\delta = \min\{b - c, c - a\}$), we see that g has a local maximum at c. But then by Proposition 7.19, we know that $g'(c) = 0$. i.e., $f'(c) - \alpha = 0$. i.e., $f'(c) = \alpha$. This completes the proof. $\square$

7.7 The Mean Value Theorems

Theorem.

Theorem 7.21 (*Rolle's theorem*). Let $f : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b). If $f(a) = f(b)$, then there exists $c \in (a, b)$ where $f'(c) = 0$.

Proof. By the extreme value theorem, f attains a maximum at some $c_1 \in [a, b]$, and f attains a minimum at some $c_2 \in [a, b]$. Note that if either $c_1 \in (a, b)$ or $c_2 \in (a, b)$, then by Proposition 7.19 such a point c would have $f'(c) = 0$, as desired.

And so we may assume that c_1 and c_2 are the end points; without loss of generality, assume that $c_1 = a$ and $c_2 = b$. Then for all $x \in [a, b]$,

$$f(a) \geq f(x) \geq f(b).$$

But also by assumption, we have $f(a) = f(b)$. Therefore

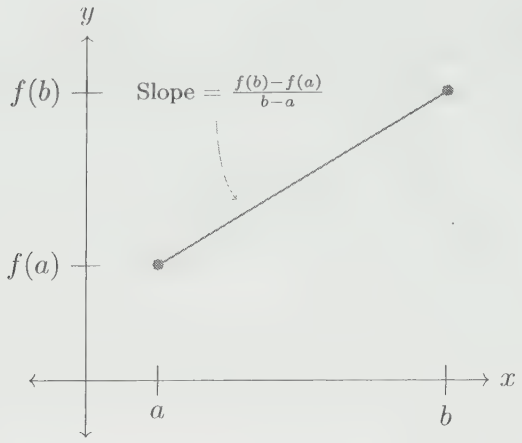
$$f(a) = f(b) = f(x)$$

for all $x \in [a, b]$. So f is constant, and hence $f'(x) = 0$ for all x, so for any $c \in (a, b)$ we have $f'(c) = 0$, completing the proof. $\square$

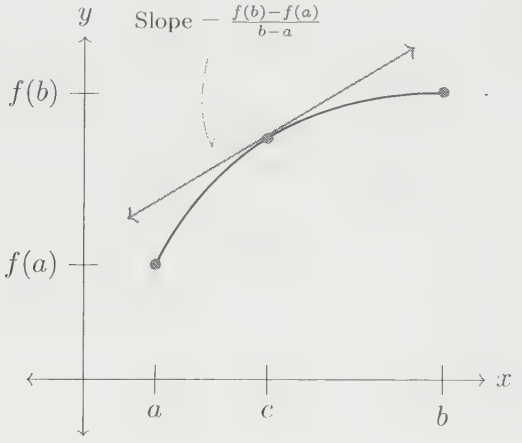
Note that if we are told that f is a continuous function on $[a, b]$, then the most direct route between the points $f(a)$ and $f(b)$ (whatever those points may be) is a straight line between them. This line would have slope

$$\frac{f(b) - f(a)}{b - a}.$$

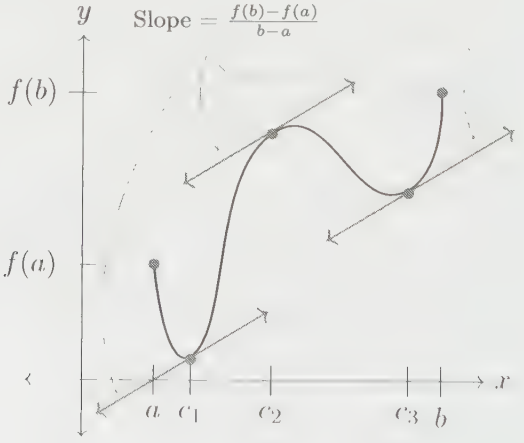
In fact, this line's derivative would be the constant function $f'(x) = \frac{f(b) \quad f(a)}{b - a}$.



Now, if f is still continuous but is not a line, like below, there is still a point c for which the derivative at that point (i.e., the slope of the tangent line at that point) is equal to the "average slope," which is $\frac{f(b)-f(a)}{b-a}$:



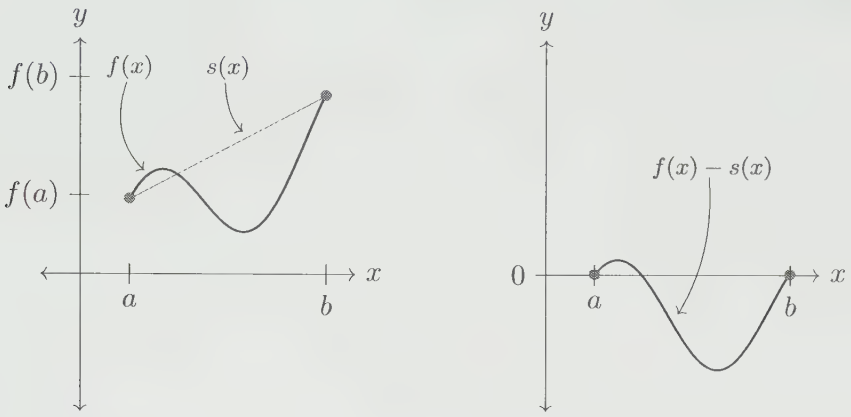
Of course, the function could be even more complicated, and in those situations typically there is more than one such value of c.



Theorem.

Theorem 7.22 (*The (derivative) mean value theorem*). Let $f : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b). Then there exists some $c \in (a, b)$ where

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$



Our plan is to reduce this theorem to Rolle's theorem. If you start with $f(x)$, and then subtract off the secant line from $(a, f(a))$ to $(b, f(b))$ (call this secant line $s(x)$), what you get is a function with $f(a) = f(b) = 0$, which the theorem can handle. The theorem will then give us what we seek. Let's Rolle.

Proof. Note that the equation of the secant line from $f(a)$ to $f(b)$ is

$$s(x) = \left(\frac{f(b) - f(a)}{b - a}\right)(x - a) + f(a).$$

Let $g(x) = f(x) - s(x)$. That is, $g(x)$ is what you get when you subtract off this line from $f(x)$:

$$g(x) = f(x) - \left[\left(\frac{f(b) - f(a)}{b - a}\right)(x - a) + f(a)\right].$$

Note that g is differentiable on $[a, b]$ and continuous on (a, b), since f and s are. Also note that $g(a) = f(a) - s(a) = 0$, and $g(b) = f(b) - s(b)$. So we may apply Rolle's theorem to g to say that there exists some $c \in (a, b)$ where $g'(c) = 0$. I.e., $f'(c) - s'(c) = 0$, or equivalently that $f'(c) = s'(c)$. And since

$$s'(x) = \frac{f(b) - f(a)}{b - a},$$

we have indeed found a $c \in (a, b)$ such that $f'(c) = \dfrac{f(b) - f(a)}{b - a}$. □

You might be surprised by how important this result is. As Edward Mills Purcell put it, "The mean value theorem is the midwife of calculus—not very important or glamorous by itself, but often helping to deliver other theorems that are of major significance."

Indeed, this theorem will be a crucial piece of many major theorems to come. The first of these important results are a trio of direct corollaries.

Corollary.

Corollary 7.23 ($f'(x) = 0 \Rightarrow f(x) = C$)**.** Let I be an interval and $f : I \to \mathbb{R}$ be differentiable. If $f'(x) = 0$ for all $x \in I$, then f is constant on I.

Proof. Pick $x_1, x_2 \in I$ with $x_1 < x_2$. We aim to show that $f(x_1) = f(x_2)$.

Since f is differentiable, it is in particular differentiable on $[x_1, x_2]$, and so by the mean value theorem there exists some $c \in (x_1, x_2)$ such that

$$f'(c) = \frac{f(x_2) - f(x_1)}{x_2 - x_1}.$$

But since by assumption $f'(x) = 0$ for all $x \in I$, we have

$$0 = \frac{f(x_2) - f(x_1)}{x_2 - x_1}.$$

And hence, $0 = f(x_2) - f(x_1)$, which gives $f(x_1) = f(x_2)$, completing the proof. □

The above can also be proved nicely by contradiction. But moving on, do you remember how your calculus teacher would constantly remind not to forget your "$+C$" when taking an antiderivative?[6] Below is where that comes from.

Corollary.

Corollary 7.24 (*"Don't forget your $+C$!"*)**.** Let I be an interval and $f, g : I \to \mathbb{R}$ be differentiable. If $f'(x) = g'(x)$ for all $x \in I$, then $f(x) = g(x) + C$ for some $C \in \mathbb{R}$.

Proof. Let $h(x) = f(x) - g(x)$. Note that $h'(x) = f'(x) - g'(x) = 0$ for all $x \in I$. So by Corollary 7.23, $f(x) - g(x) = C$ for some $C \in \mathbb{R}$. And hence $f(x) = g(x) + C$, as desired. □

[6] Confession: I was terrible at remembering my "$+C$" when I took calculus. And I got burned by it so much that eventually I would go into an exam chanting to myself "$+C, +C, +C, \ldots$" And then as soon as I got my exam I would write a huge "$+C!$" on the top of every page, so I wouldn't forget. It helped.

Corollary.

Corollary 7.25 ($f' \geq 0 \Leftrightarrow f$ *is increasing*). Let I be an interval and $f : I \to \mathbb{R}$ be differentiable.

(i) f is increasing if and only if $f'(x) \geq 0$ for all $x \in I$.

(ii) f is decreasing if and only if $f'(x) \leq 0$ for all $x \in I$.

Proof. We will prove (i), and simply note that (ii) is extremely similar. For (i), first assume that f is increasing, meaning that for all $x_1, x_2 \in I$ where $x_1 < x_2$, we have $f(x_1) \leq f(x_2)$. Note that this implies that for any $x, c \in I$,

$$f(x) - f(c) \qquad \text{and} \qquad x - c$$

are either both positive (or 0) or are both negative (or 0). And therefore, for any $x \neq c$,

$$\frac{f(x) - f(c)}{x - c} \geq 0$$

for all $x, c \in I$. Thus,

$$f'(c) = \lim_{x \to c} \frac{f(x) - f(c)}{x - c} \geq 0$$

for all $c \in I$, as desired.

Now we show the backwards direction of (i). To that end, assume that $f'(x) \geq 0$ for all $x \in I$. Now pick any $x_1, x_2 \in I$ for which $x_1 < x_2$. To conclude that f is increasing, it suffices to show that $f(x_1) \leq f(x_2)$.

By the mean value theorem, there exists some $c \in (x_1, x_2)$ such that

$$f'(c) = \frac{f(x_2) - f(x_1)}{x_2 - x_1},$$

i.e.,

$$f(x_2) - f(x_1) = f'(c)(x_2 - x_1).$$

But by assumption $f'(c) \geq 0$, and since $x_1 < x_2$ we also have $(x_2 - x_1) > 0$. So their product is also positive; that is,

$$f(x_2) - f(x_1) \geq 0,$$

and hence $f(x_2) \geq f(x_1)$, as desired. □

7.8 L'Hôpital's Rule

By the func-y limit laws (Corollary 6.13),

$$\lim_{x \to a} \frac{f(x)}{g(x)} = \frac{\lim_{x \to a} f(x)}{\lim_{x \to a} g(x)},$$

provided both of these limits equal a positive real number. If the denominator is converging to zero and the numerator is either converging to something non-zero or is diverging to $\pm\infty$, then the limit of the quotient either does not exist or equals $\pm\infty$, and with a little thought about the signage one can determine which it is. Likewise if the numerator diverges to $\pm\infty$ and the denominator is convergent.

What happens if both the numerator and denominator converge to zero, or both diverge to $\pm\infty$? Intuitively, it should depend on which one is converging/diverging "fastest." For example, if both are diverging to ∞ but the numerator is doing so much more quickly, then the denominator will not be slowing it down enough to prevent the entire quotient from also diverging to ∞. This suggests that each function's *rate of change* will determine what happens to the limit of the ratio. And since derivatives measure rate of change, it is reasonable to think that the derivative will play a role.

We end this chapter by stating l'Hôpital's rule, without proof. This theorem continues a long tradition in mathematics of being named after someone who did not prove it. Johann Bernoulli first proved the theorem, although it was named after his student Guillaume de l'Hôpital who published it in his book *Analyse des Infiniment Petits pour l'ntelligence des Lignes Courbes*, which was the first ever calculus textbook.[7]

Theorem.

Theorem 7.26 (*L'Hôpital's rule*). Suppose I is an interval containing a point a, and $f : I \to \mathbb{R}$ and $g : I \to \mathbb{R}$ are differentiable on I (except possibly at a). Suppose also $g'(x) \neq 0$ on I. Then if

$$\lim_{x \to a} f(x) = 0 \qquad \text{and} \qquad \lim_{x \to a} g(x) = 0,$$

or if

$$\lim_{x \to a} f(x) = \pm\infty \qquad \text{and} \qquad \lim_{x \to a} g(x) = \pm\infty,$$

then

$$\lim_{x \to a} \frac{f'(x)}{g'(x)} = L \quad \text{implies} \quad \lim_{x \to a} \frac{f(x)}{g(x)} = L.$$

[7] The first book on a subject is soooooooo much harder to write than 20 renditions later. So as a real analysis author, nothing but respect to l'Hôpital for paving the way. **pours one out**

— Notable Exercises —

- In Exercise 7.3 you are asked to prove the quotient rule. This is one of the principle rules for actually computing derivatives, which is why it plays a prominent role in calculus. Its proof is similar to the proof of the product rule.

- In Exercise 7.5 you are asked to prove a special case of the power rule, another highly useful rule for computing actual derivatives, and one which played a truly staring role in your calculus class.

- A function is continuous if it is continuous at each point in its domain. Being continuous at each point c means that, for every $\varepsilon > 0$, there is some $\delta > 0$ such that... yada yada yada.... Intuitively, if your function is really steep at c, then you need a really small δ, whereas if your function is not so steep, then although a small δ also works, a larger δ will work too. But note that for a fixed $\varepsilon > 0$, each point c can have its own δ. The idea behind *uniform continuity* is that a single δ works for *all* the points c; essentially, the function never gets "too steep." Exercise 7.6 puts meat on the bones of this idea. It says this if a derivative is bounded (which is another way to say the function never gets "too steep"), then the function will be uniformly continuous.

- Exercise 7.8 deals with Lipschitz functions, which is another way to make precise this important notion of a function never getting "too steep."

- Exercise 7.14 asks you to prove a generalization of the mean value theorem (Theorem 7.22) called the *Cauchy mean value theorem*, which says this:

  > If f and g are continuous on $[a, b]$ and differentiable on (a, b), then there is a number $x \in (a, b)$ such that
  >
  > $$[f(b) - f(a)] \cdot g'(x) = [g(b) - g(a)] \cdot f'(x).$$

  It is indeed a special case of the mean value theorem (set $g(x) = x$ to see), but this more general form is particularly useful in proving some later theorems, including l'Hôpital's rule.

- In Exercises 7.15 and 7.16 you are asked to prove (nearly) all of L'Hôpital's rule (Theorem 7.26). The former isn't soooo bad, provided you use Exercise 7.14. The latter takes a bit more care.

— Exercises —

Exercise 7.1. Use the definition of the derivative to find the derivatives of the following functions.

(a) $f(x) = \sqrt{x+3}$ (b) $g(x) = \dfrac{1}{x}$ (c) $h(x) = 4x + \dfrac{1}{x^2}$

Exercise 7.2. Prove that there do not exist differentiable function f and g for which $f(0) = g(0) = 0$ and $x = f(x)g(x)$.

Exercise 7.3. Prove the *quotient rule* (Theorem 7.12).

Exercise 7.4. Is

$$f(x) = \begin{cases} \frac{1}{2}x & \text{if } x \in \mathbb{Q}, \\ x & \text{if } x \notin \mathbb{Q} \end{cases}$$

differentiable at 0?

Exercise 7.5. Use induction to prove that $\frac{d}{dx}x^n = nx^{n-1}$ for all $n \in \mathbb{N}$, which is the integer case of the *power rule*.

Exercise 7.6. Assume that I is an interval and $f : I \to \mathbb{R}$ is differentiable. Prove that if f' is bounded then f is uniformly continuous.

Exercise 7.7. Let

$$f_a(x) = \begin{cases} x^a & \text{if } x \geq 0 \\ 0 & \text{if } x < 0. \end{cases}$$

(a) For which values of $a \in \mathbb{R}$ is f continuous at 0?

(b) For which values of $a \in \mathbb{R}$ is f differentiable at 0?

Definition 7.27. A function f is called *Lipschitz* if there exists some $C \in \mathbb{R}$ where

$$|f(x) - f(y)| \leq C \cdot |x - y|$$

for all x and y.

Exercise 7.8. Let $f : [a, b] \to \mathbb{R}$ be differentiable, and assume f' is continuous on $[a, b]$. Prove that f is Lipschitz on $[a, b]$

Exercise 7.9. Let A be an interval and $f : A \to \mathbb{R}$ be differentiable. Show that f is Lipschitz on A if and only if f' is bounded on A.

Exercise 7.10. In this exercise you will explore why each of the three main assumptions in Rolle's theorem is necessary. To that end, give an example of a function f which satisfies of each of the following conditions, and yet $f'(c) \neq 0$ for all $c \in (a, b)$.

(a) A function f which is continuous on $[a, b]$ and differentiable on (a, b) (but we don't assume $f(a) = f(b)$).

(b) A function f which is continuous on $[a, b]$ and $f(a) = f(b)$ (but we don't assume f is differentiable on (a, b)).

(c) A function f which is differentiable on (a, b) and $f(a) = f(b)$ (but we don't assume f is continuous on $[a, b]$).

Exercise 7.11. Use the derivative to find all values of a for which $|x-a| = (x-2)^2$.

Exercise 7.12. Prove that among all rectangles with a given perimeter, the square has the greatest area.

Exercise 7.13. Show that the sum of a positive number and its reciprocal is at least 2.

Exercise 7.14. Prove the *Cauchy mean value theorem*, which says this:

> If f and g are continuous on $[a, b]$ and differentiable on (a, b), then there is a number $x \in (a, b)$ such that
>
> $$[f(b) - f(a)] \cdot g'(x) = [g(b) - g(a)] \cdot f'(x).$$

(Note that the mean value theorem (Theorem 7.22) is a special case of this, in which $g(x) = x$.)

Exercise 7.15. Prove the following "0/0" case of l'Hôpital's rule. Suppose I is an interval containing a point a, and $f : I \to \mathbb{R}$ and $g : I \to \mathbb{R}$ are differentiable on I (except possibly at a). Use Exercise 7.14 to prove that if $f(a) = 0$ and $g(a) = 0$, then

$$\lim_{x \to a} \frac{f'(x)}{g'(x)} = L \quad \text{implies} \quad \lim_{x \to a} \frac{f(x)}{g(x)} = L.$$

Exercise 7.16. Prove the following "∞/∞" case of l'Hôpital's rule. Prove that if $f : (a, b) \to \mathbb{R}$ and $g : (a, b) \to \mathbb{R}$ are differentiable functions and $\lim_{x \to a} f(x) = \lim_{x \to a} g(x) = \infty$, then

$$\lim_{x \to a} \frac{f'(x)}{g'(x)} = L \quad \text{implies} \quad \lim_{x \to a} \frac{f(x)}{g(x)} = L.$$

— Open Question —

Question. Notice that the formula for the area of a circle is πr^2. Differentiating this with respect for r gives $\frac{d}{dr}\pi r^2 = 2\pi r$, the formula for perimeter of the circle. The formula for the area of a square with side length $2r$ (so that its "radius" is r) is $(2r)(2r) = 4r^2$. Differentiating this with respect to r gives $\frac{d}{dr}4r^2 = 8r$, which is the perimeter of the square.

Classify all shapes where this property holds (including the higher-dimensional analogues, where you differentiate the volume formula to get the surface area formula). What other geometric interpretations of "radius" are there that give this property?

Chapter 8: Integration

8.1 Brief Review

There are a handful of ideas from earlier chapters that will be particularly important in this chapter (and I anticipate this chapter being near the start of the second semester of an analysis course, so a review seems fitting). For those reasons, we will briefly review these ideas.

- U is an *upper bound* of a set A if $x \leq U$ for all $x \in A$.
- L is a *lower bound* of A if $x \geq L$ for all $x \in A$.
- A is *bounded* if there exists an upper bound and a lower bound of A. Or, equivalently, if there exists some M such that $|x| \leq M$ for all $x \in A$.
- If $x \leq y$ for all $x \in A$ and $y \in B$, then $\sup(A) \leq \inf(B)$. (Exercise 8.3)
- A function $f : A \to \mathbb{R}$ is *bounded* if $f(A)$ is bounded. Or, equivalently, if there exists some M such that $|f(x)| \leq M$ for all $x \in A$.
- If $f : [a, b] \to \mathbb{R}$ is continuous, then f is bounded.
- $f : [a, b] \to \mathbb{R}$ is continuous if and only if f is uniformly continuous.
- Dirichlet's function
$$f(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q} \\ 0 & \text{if } x \notin \mathbb{Q} \end{cases}$$
is continuous nowhere.
- The modified Dirichlet function
$$g(x) = \begin{cases} x & \text{if } x \in \mathbb{Q} \\ 0 & \text{if } x \notin \mathbb{Q} \end{cases}$$
is continuous only at $x = 0$.

- Thomae's function (i.e. the "popcorn function")

$$h(x) = \begin{cases} 1/n & \text{if } x \neq 0 \text{ and } x = m/n \in \mathbb{Q} \text{ in lowest terms with } n > 0 \\ 0 & \text{if } x \notin \mathbb{Q} \\ 1 & \text{if } x = 0 \end{cases}$$

  is continuous on $\mathbb{R} \setminus \mathbb{Q}$, and discontinuous on $\mathbb{Q}$.

- Consider

$$g_n(x) = \begin{cases} x^n \sin(1/x) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0. \end{cases}$$

  Then,

	Continuous at 0?	Differentiable at 0?
g_0	no	no
g_1	yes	no
g_2	yes	yes

- If f is differentiable, then f is continuous. However, f' need not be continuous.
- If f is differentiable, then f' has the intermediate value property.

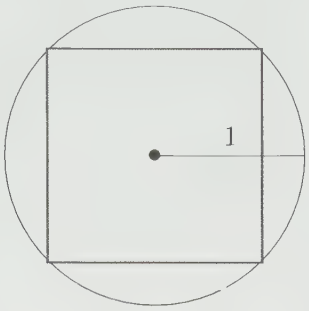
8.2 The Area of a Circle

A couple thousand years ago—give or take—many around the world had the same idea for how to estimate the areas of difficult shapes. What is the area of a unit circle (a circle of radius 1), for instance? They could see that it was somewhere between, say, 3.0 and 3.3, suspiciously like the value π (but we should be careful; at this point, how was π defined? Like so: 2π is defined to be the circumference of a unit circle, and 2π radians a complete rotation of that circle), but getting a more precise answer for the area seemed challenging.

The great Archimedes came up with a clever idea which today is known as the *method of exhaustion*, which is as follows.

We know that the area is smaller than 4, since it is smaller than the 2×2 square that *circumscribes* it:

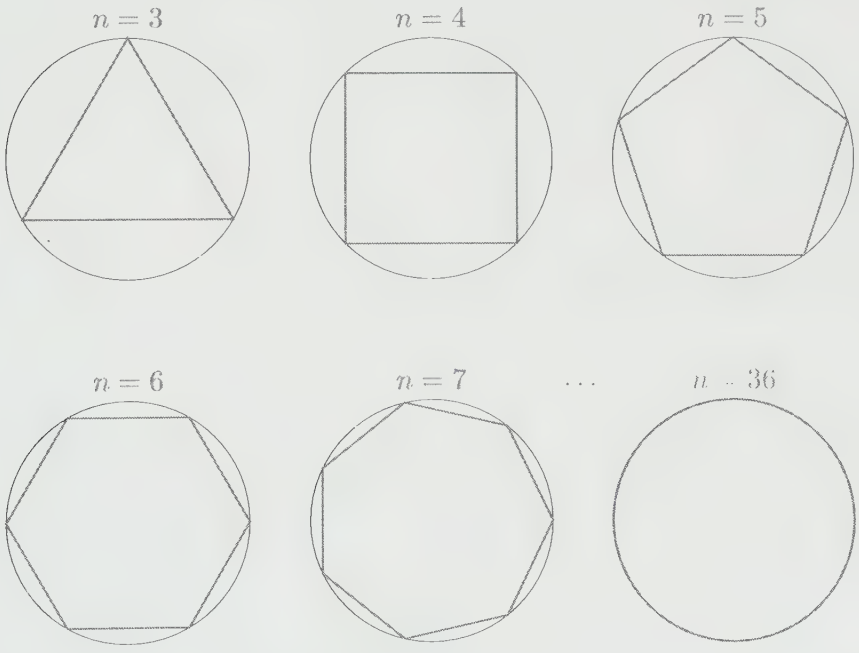
Likewise, you could use a little geometry to figure out the area of the square that *inscribes* a circle:



Indeed, if you realize that the square's upper-right vertex is at 45°, and the rest are 90° rotations from there, then you can quickly deduce that the side length of such a square is $\sqrt{2}$, and hence its area is 2.

So we just proved that the area of a unit circle is between 2 and 4! How can we do better? Well, instead of using squares, let's use polygons with more sides, as those more closely resemble a circle.

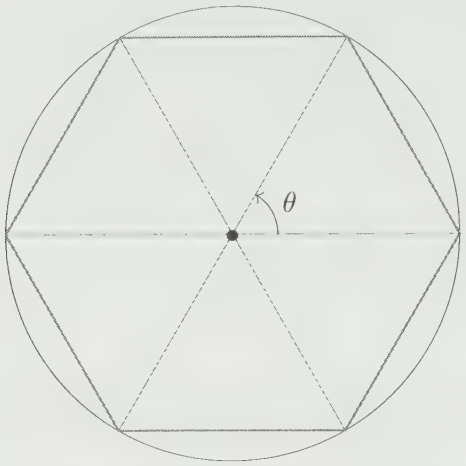
We first focus on the inscribed polygons. A sequence of them looks like so:



For $n = 36$, you have to zoom in quite a bit just to see any bit of the circle, or to see the edges of the 36-gon!

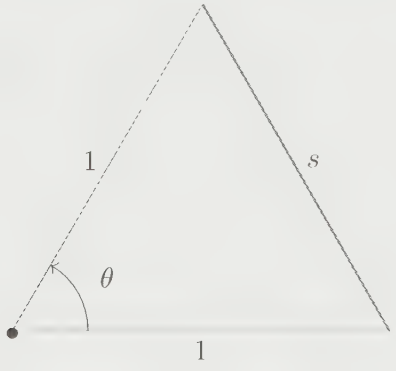
So what is the area of each of these n-gons? Each serves as a lower bound on the actual area of the unit circle, and presumably they are getting closer and closer to the true value. Let's calculate these values and see how they're doing.

Consider a regular n-gon with side length s inscribed in a unit circle. Let $\theta = 2\pi/n$ radians be the measure of a central angle, as shown below (in the picture, $n = 6$):

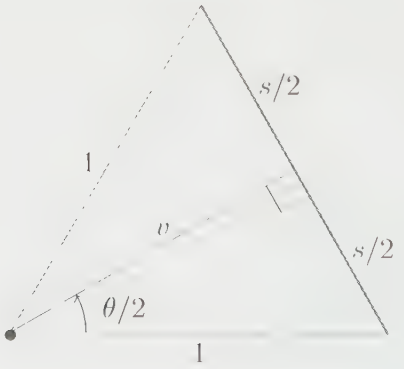


And of course, since the n-gon is regular, every interior angle has this same angle, and all n triangles that are produced are identical. So the area of the n-gon is equal to n times the area of one triangle. Thus, we turn our attention to determining the area of a single triangle.

One such triangle looks like this:



Dropping a vertical from the lower-left vertex to s gives:



We want to write the area in terms of just n. The area formula is going to be $\frac{1}{2}(\text{base})(\text{height}) = \frac{1}{2}sv$, so let's work to convert s and v into a function of n. First, using trig we can relate s and θ, and thus s and n:

$$\begin{aligned} \sin(\theta/2) &= \frac{s/2}{1} \\ \sin(2\pi/2n) &= s/2 \\ s &= 2\sin(\pi/n). \end{aligned}$$

Likewise, we can relate v and θ, and thus v and n:

$$\begin{aligned} \cos(\theta/2) &= \frac{v}{1} \\ \cos(2\pi/2n) &= v \\ v &= \cos(\pi/n). \end{aligned}$$

Thus, the area $A_\triangle$ of this whole triangle is

$$\begin{aligned} A_\triangle &= \frac{1}{2}(\text{base})(\text{height}) \\ &= \frac{1}{2}(s)(v) \\ &= \frac{1}{2}(2\sin(\pi/n))(\cos(\pi/n)) \\ &= \frac{1}{2}(2\sin(\pi/n)\cdot\cos(\pi/n)) \\ &= \frac{1}{2}\sin(2\pi/n), \end{aligned}$$

where the last line is the double angle trig identity $2\sin(a)\cos(a) = \sin(2a)$.

The area A of the entire n-gon is then n times the area of the triangle:

$$A = n \cdot A_\triangle = \frac{n}{2}\sin(2\pi/n).$$

By plugging in values for n, you do indeed get extremely good estimates for π. In fact, we could even evaluate this as $n \to \infty$ by rewriting the limit in a form to which we can apply L'Hôpital's Rule:

$$\begin{aligned} \lim_{n\to\infty} \frac{n}{2}\sin(2\pi/n) &= \lim_{n\to\infty} \frac{\sin(\pi\cdot(2/n))}{2/n} \\ &= \lim_{x\to 0} \frac{\sin(\pi\cdot x)}{x} \\ &= \lim_{x\to 0} \frac{\pi\cdot\cos(\pi\cdot x)}{1} \qquad \text{(L'Hôpital's)} \\ &= \pi\cdot\cos(0) \\ &= \pi. \end{aligned}$$

And we could of course repeat this general idea to find the areas of the regular polygons which *circumscribe* the circle, which would provide *upper* bounds on the area of a circle. And if you did, you would find that the areas of the larger polygons are also shrinking down to π.

So indeed, these n-gons are providing better and better lower bounds for the area of the circle, and the area is π.[1]

#IntegralGoals

What do we want out of an integral? Here are three important goals:

- Integrals find antiderivatives—at least the indefinite integrals should.
- Integrals relate to area under a curve—at least the definite integrals should.
- We want them to be fairly powerful—we want to be able to integrate as many functions as possible.

When integrals were first developed, the goal was simply to find antiderivatives. This is a perfectly fine goal, but developing them in this way significantly limits their power and utility. For instance, consider the very simple step function $s : [0, 2] \to \mathbb{R}$ given by

$$s(x) = \begin{cases} 1 & \text{if } x \in [0, 1] \\ 2 & \text{if } x \in (1, 2] \end{cases}$$

Finding the "area under the curve" of this function is something you could have done when you were 9, but this function is *not* a derivative. Recall that Darboux's theorem (Theorem 7.20) states that every derivative has the intermediate value property—however, $s(x)$ clearly does not satisfy the IVP (for example, nothing maps to 0.5). So if we demand that integrals be antiderivatives, then $\int_0^2 s(x)\,dx$ would not exist, even though we already know in our hearts that the answer should exist and should equal 3.

So in order to also use our area understanding, and in order to have it be fairly powerful (at least powerful enough to integrate $s(x)$), we should develop it from the area perspective, provided the antiderivative angle also works out. Newton and the Boys realized this area interpretation, but it was Cauchy and Riemann who realized that the best approach was to define it *only* using area, and later prove that antiderivatives pop out from a (fundamental) theorem.

More on this later. For now, let's jump right into the familiar pictures of Riemann sums.

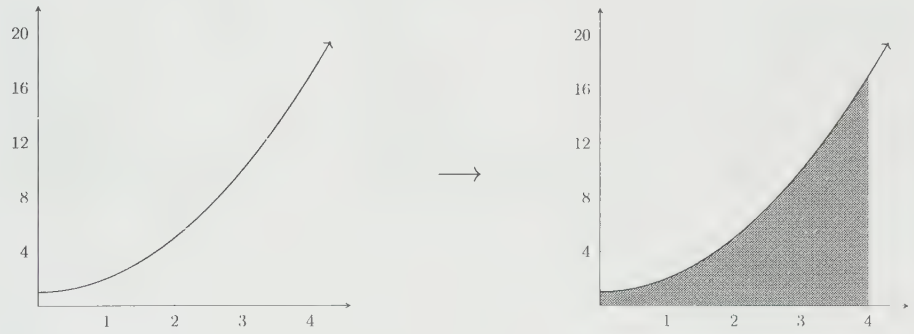
[1] Confession: We sort of cheated in the above, using a few facts that we have not yet officially developed. If we wanted to work harder, we could have been a little more rigorous.

8.3 Simplified Approach

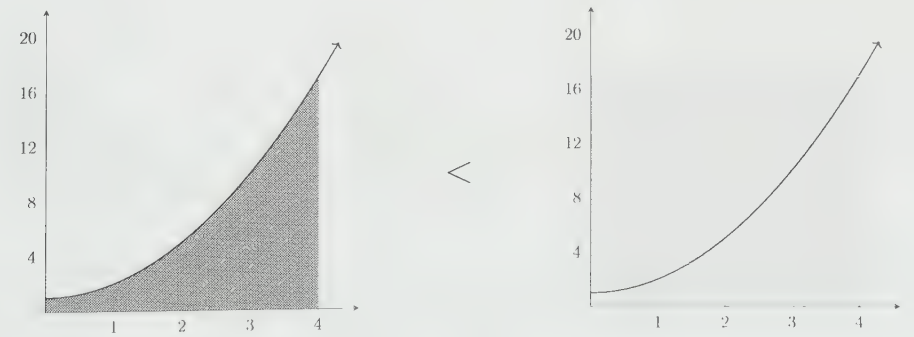
Our approach will be similar to the Method of Exhaustion in which we approached the area of a circle by finding upper bounds using circumscribed n-gons, and lower bounds using inscribed n-gons, and then by taking a limit; if the upper bounds and lower bounds are converging to the same thing, then we have an answer. Likewise, we will find a sequence of upper bound areas and a sequence of lower bound areas, and then take a limit, and if they are converging to the same thing then we have an answer.

– Pictures from Calculus –

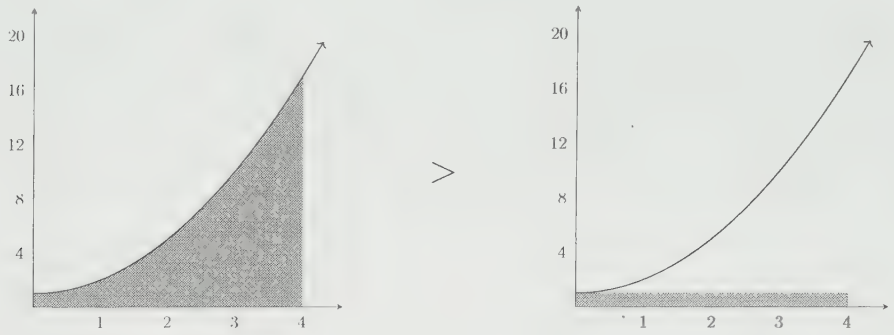
Back in calculus you were likely shown some pictures like those below. For example, below is the function $f(x) = x^2 + 1$.



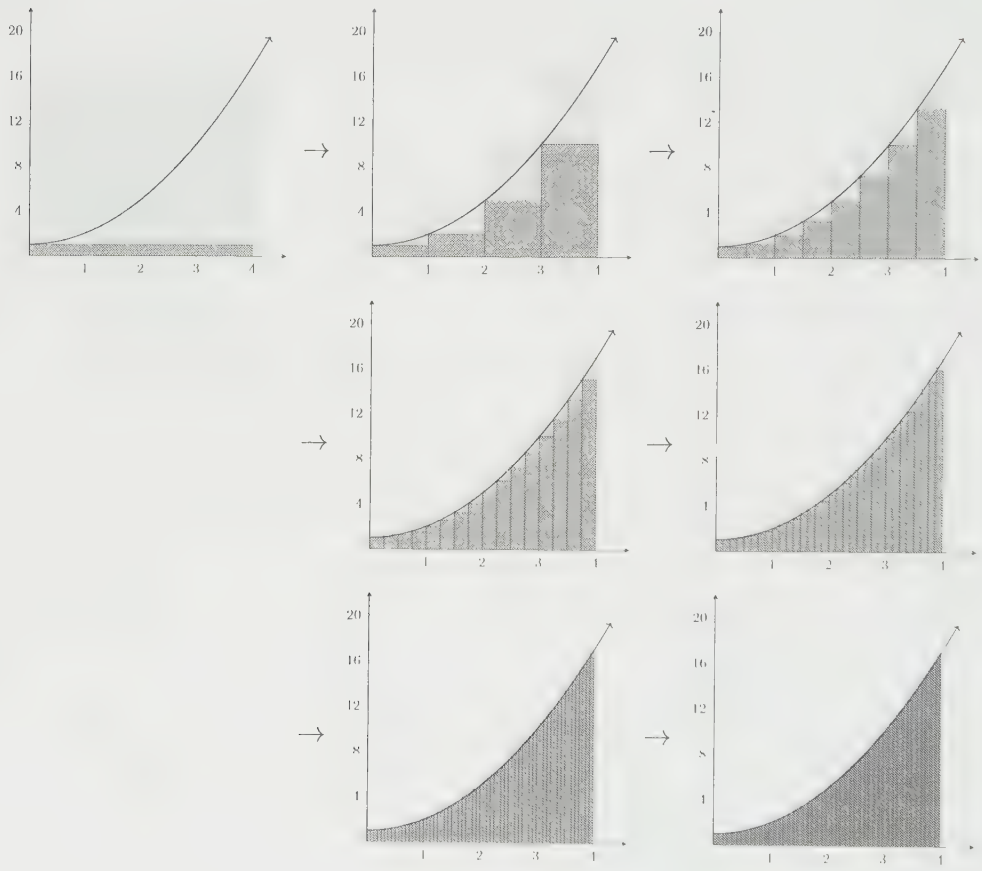
To find this blue/shaded area, you learned that the first thing to do is to find a simple rectangular upper bound, like so:



A lower bound was then found by putting a sinfle (red) rectangle inside the (blue) region under the curve, so that it lies entirely below the curve:



Then you continue. For example, here's a sequence of pictures where the red/dark and green/light boxes are overlaid.



By using more and more rectangles you get a sequence of increasingly better upper and lower bounds. And as you discussed while taking calculus, it certainly is reasonable to wave your hands and say that for a function as simple as this, clearly the sequence of upper bounds and lower bounds are converging to a common

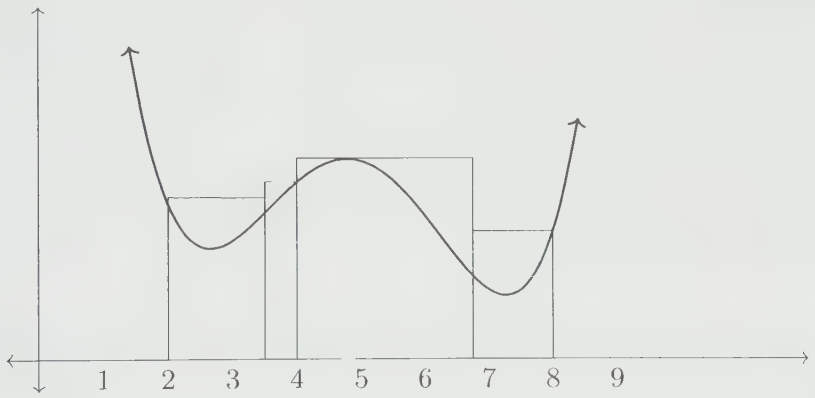
number—the area under the curve that we are trying to find. Of course, to do this there is a limit taking place behind the scenes.

– More General Approach –

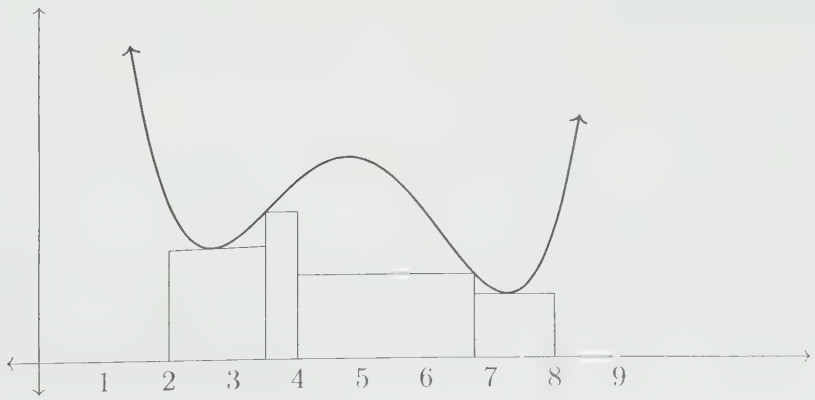
The above works particularly well when the function is continuous and increasing. We will return to the continuity issue in depth later, but two quick changes are:

- We no longer make all the rectangles the same width—we consider all possible widths.
- When the function is increasing, the right endpoint gives the largest function value and the left endpoint gives the smallest function value. If the function is not increasing, then we have to instead find the maximum (or supremum) point on our own, and the minimum (or infimum) point on our own.

For example, here is an upper bound (soon called an "upper sum") where the rectangles have different widths and one of the rectangle's maximum was not an endpoint:



Likewise, here is a lower bound (soon called a "lower sum"):



Types of Integrals

It's worthwhile to take a moment to point out that there are many ways to define an integral. Below are types of integrals that mathematicians have defined:

- Riemann integral
- Generalized Riemann integral
- Riemann-Stieltjes integral
- Lebesgue integral
- Lebesgue-Stieltjes integral
- Darboux integral
- Daniell integral
- Haar integral
- Ito integral
- Statonovich integral
- Young integral

...and *all of these are different*!

We will study Riemann integrals, which are the classic and most fundamental option. This is not the only approach, though, but it is the most important for now.

8.4 The Riemann Integral

First we collect the fundamental definitions to allow us to talk about Riemann sums.

– Three Preliminary Definitions/Notation –

First up, the rectangles can have arbitrary widths. To define these widths, we need only define the set of points which are the left/right endpoints of the rectangles. (Note that this is much easier than trying to define the specific widths of each rectangle while also ensuring those widths add up to $b - a$.)

Definition

Definition 8.1. A *partition* of $[a, b]$ is a finite set

$$P = \{a = x_0, x_1, x_2, \ldots, x_n = b\}$$

such that $x_0 < x_1 < x_2 < \cdots < x_n$.

For example, one partition of $[2, 5]$ is $\{2, 2.7, 3.1, \pi, 4.2, 5\}$. There are, of course, infinitely many others.

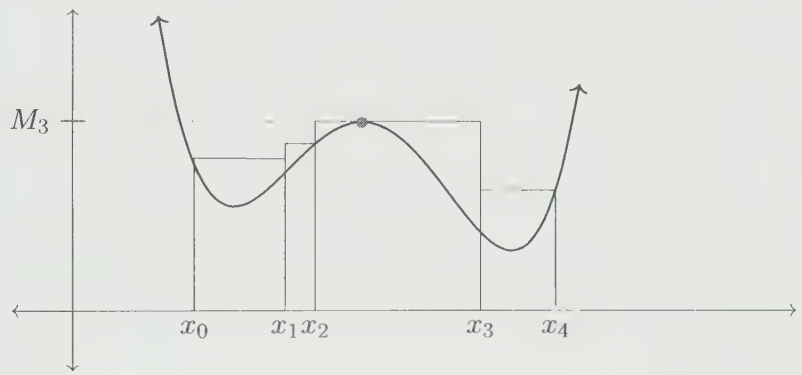
Next, we need to define the heights of the rectangles. The rectangles are either big enough to completely encompass the region under the curve, or are small enough to be completely encompassed by the region. These two possibilities lead to two different notations that we need to define.

Notation

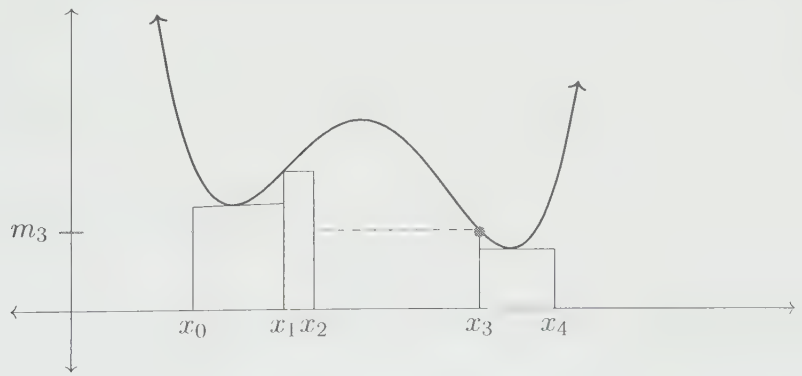
Notation 8.2. Given a partition $\{x_0, \ldots, x_n\}$ of $[a, b]$, consider a subinterval $[x_{i-1}, x_i]$. Denote

- $m_i := \inf\{f(x) : x \in [x_{i-1}, x_i]\}$;
- $M_i := \sup\{f(x) : x \in [x_{i-1}, x_i]\}$.

For example, below is a partition of $[2, 8]$ where we labeled M_3, the height of the third (upper) rectangle.



And here is m_3, the height of the third (lower) rectangle:



Now that we have a way to talk about the widths and heights of both the big rectangles and the small ones, we can write down what we are really interested in: a formula for the upper bound area and the lower bound area. These are called "upper sums" and "lower sums," respectively.

Definition.

Definition 8.3. Consider a function $f : [a, b] \to \mathbb{R}$ and a partition $P = \{x_0, x_1, \ldots, x_n\}$ of $[a, b]$. Define

- The *upper sum*

$$U(f, P) := \sum_{i=1}^{n} M_i \cdot (x_i - x_{i-1}), \text{ and}$$

- The *lower sum*

$$L(f, P) := \sum_{i=1}^{n} m_i \cdot (x_i - x_{i-1}),$$

where M_i and m_i were defined in Notation 8.2.

– Refinements –

Suppose you had a partition of $[a, b]$ that contains 10 points. This would give 9 rectangles, and the upper and lower sums may be ok estimates of the integral. But if instead you had 100 points—giving 99 rectangles—that would presumably give a much better estimate. This isn't necessarily the case, though, since maybe all 10 points in the first partition are nicely spread out, while in the second partition 99 of the 100 points are clumped together on one end. Those 99 points would give a fantastic estimate of the area on that one end, but then there's just one huge rectangle to estimate the rest of the area, which will likely have a huge error.

So you see, it's not just getting more points that we want, we also want them to be distributed nicely. We begin with an intermediate step, called a *refinement*. The idea here is that if you want to ensure that 100 points are better than 10, construct your 100 points by first starting with the 10 points from the first partition and then adding 90 *more* points. If your original 10 points are still there, then you guarantee that you aren't making things any worse, and are usually making the estimates better and better.

Observe that if P is a partition and Q is another partition with all of P's points plus others, then $P \subseteq Q$. Thus we can use subset notation to define a refinement.

Definition.

Definition 8.4. Consider a function $f : [a, b] \to \mathbb{R}$ and a partition $P = \{x_0, \ldots, x_n\}$ of $[a, b]$. A partition Q of $[a, b]$ is called a *refinement* of P if $P \subseteq Q$.

We begin with an intuitive result—a refinement really does refine; that is, a

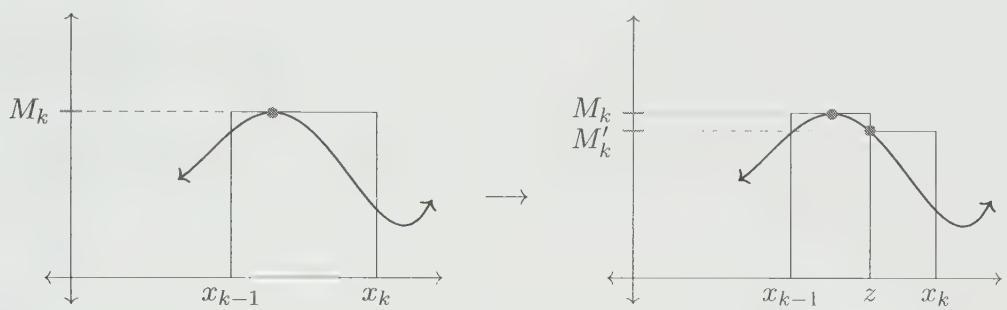
refinement makes upper sums smaller and lower sums bigger, or at worst leaves them unchanged.

Proposition.

Proposition 8.5 (*Refinements refine*). Consider a function $f : [a, b] \to \mathbb{R}$ and a partition $P = \{x_0, \ldots, x_n\}$ of $[a, b]$. If Q is a refinement of P, then

$$U(f, P) \geq U(f, Q) \quad \text{and} \quad L(f, P) \leq L(f, Q).$$

Proof. The idea is fairly intuitive when you consider the following picture, showing an interval from the partition P being broken into two intervals by the refinement Q.



From the picture you can see that adding an additional point did indeed decrease the area of the rectangle(s). Formally, this is because in the upper sum formula the only thing that changes is that the term $M_k \cdot (x_k - x_{k-1})$ is replaced by two terms, representing the sum of the areas of the two next rectangles. If, without loss of generality, the previous supremum height M_k is in the first region, then the two new terms are

$$M_k \cdot (z - x_{k-1}) + M'_k \cdot (x_k - z).$$

Now, M_k was the supremum of f in $[x_{k-1}, x_k]$, and M'_k is the supremum of f in $[z, x_k] \subseteq [x_{k-1}, x_k]$, which by Exercise 8.2 means that $M_k \geq M'_k$. Thus,

$$\begin{aligned} M_k \cdot (x_k - x_{k-1}) &= M_k \cdot (x_k - z + z - x_{k-1}) \\ &= M_k \cdot (x_k - z) + M_k \cdot (z - x_{k-1}) \\ &\geq M'_k \cdot (x_k - z) + M_k \cdot (z - x_{k-1}) \\ &= M_k \cdot (z - x_{k-1}) + M'_k \cdot (x_k - z). \end{aligned}$$

And hence refining the partition by one point did indeed decrease the upper sum. Since partitions are finite sets, Q has finitely many more points than P. Doing the above $|Q \setminus P|$ times (or by applying induction to the above) shows that $U(f, P) \geq U(f, Q)$.

The lower sum result is very similar and is left as an exercise. □

Next we have a two-part proposition that ensures that another basic, intuitive result holds: Any upper sum is larger than any lower sum.

Proposition.

Proposition 8.6 (*Lower sums ≤ Upper sums*). Consider a function $f : [a, b] \to \mathbb{R}$. and a partition $P = \{x_0, \ldots, x_n\}$ of $[a, b]$.

(a) If P is a partition of $[a, b]$ then $L(f, P) \leq U(f, P)$.

(b) If P_1 and P_2 are any partitions of $[a, b]$, then

$$L(f, P_1) \leq U(f, P_2).$$

Part (a) is of course a special case of part (b), but (a) will be particularly useful so it is helpful to state it explicitly. Also, we will use (a) to prove (b).

Proof. Part (a) is immediate: Clearly $m_k \leq M_k$ since the former is the infimum of f over $[x_{k-1}, x_k]$ and the latter is the supremum of f over the exact same interval. And so clearly

$$L(f, P) = \sum_{i=1}^{n} m_i \cdot (x_i - x_{i-1}) \leq \sum_{i=1}^{n} M_i \cdot (x_i - x_{i-1}) = U(f, P).$$

Now we show (b). The basic idea is this: To show that one thing is less than another, find something that is in-between them. The true integral would be ideal, but we don't know yet what an integral even is. Instead, let's use a sufficiently refined partition—if it is finer than both P_1 and P_2, then it should be closer to the value of the "integral" than either one is, which would be good enough. In particular, if we can find a partition Q that refines *both* P_1 and P_2, then Q's upper and lower sums will be squished between $L(f, P_1)$ and $U(f, P_2)$.

Let $Q = P_1 \cup P_2$. Since P_1 and P_2 are partitions, Q is as well. Moreover, Q is a refinement of both P_1 and of P_2! Therefore by Proposition 8.5,

$$L(f, P_1) \leq L(f, Q) \quad \text{and} \quad U(f, Q) \leq U(f, P_2).$$

Moreover, by (a) we have that

$$L(f, Q) \leq U(f, Q).$$

Combining these,

$$L(f, P_1) \leq L(f, Q) \leq U(f, Q) \leq U(f, P_2),$$

giving $L(f, P_1) \leq U(f, P_2)$, as desired. □

8.5 Integrability

As you probably already anticipated, the idea is to take better and better partitions to get smaller and smaller upper sums, and larger and larger lower sums. If the upper sums and the lower sums are converging to the same thing, then this common value is what the *integral* is equal to.

But how do we define the partitions? We don't want to insist on having equally-spaced rectangles—that can lead to issues that we will see later. Our refinement theory is good, but if we simply insist on having the sequence of partitions where each is a refinement of the previous, that can also lead to problems. How do you know you're continuously refining everywhere, rather than refining one region better and better while ignoring another?

The solution is simple: Don't try to be clever; instead, just consider *all* partitions. Indeed, that's what the following definition does.

Definition.

Definition 8.7. Let $f : [a, b] \to \mathbb{R}$ be a function and let $\mathcal{P}$ be the collection of *all* partitions of $[a, b]$. The *upper integral* of f is defined to be

$$U(f) = \inf\{U(f, P) : P \in \mathcal{P}\}.$$

The *lower integral* of f is defined to be

$$L(f) = \sup\{L(f, P) : P \in \mathcal{P}\}.$$

Here's a quick result:

Lemma.

Lemma 8.8 ($m(b-a) \leq L(f) \leq U(f) \leq M(b-a)$)**.** Let $f : [a, b] \to \mathbb{R}$ be a bounded function; say, $m \leq f(x) \leq M$ for all $x \in [a, b]$. Then we have

$$m(b-a) \leq L(f) \leq U(f) \leq M(b-a).$$

Proof. The inequality $L(f) \leq U(f)$ follows immediately from Proposition 8.6 part (b), and Exercise 8.3.

Now we show the bounds on $L(f)$ and $U(f)$, which are a sup/inf over all partitions. The simplest partition possible is the 2-point partition $P_0 = \{a, b\}$ containing only the endpoints (that is, $n = 1$, $x_0 = a$, and $x_1 = b$); this in turn gives a single rectangle in the upper or lower Riemann sum. Thus,

$$\begin{aligned} L(f) &= \sup\{L(f,P) : P \in \mathcal{P}\} \\ &\geq L(f,P_0) \\ &= \sum_{i=1}^{n} m_i \cdot (x_i - x_{i-1}) \\ &= \sum_{i=1}^{1} m_i \cdot (b-a) \\ &\geq m \cdot (b-a). \end{aligned} \qquad \begin{aligned} U(f) &= \inf\{U(f,P) : P \in \mathcal{P}\} \\ &\leq U(f,P_0) \\ &= \sum_{i=1}^{n} M_i \cdot (x_i - x_{i-1}) \\ &= \sum_{i=1}^{1} M_i \cdot (b-a) \\ &\leq M \cdot (b-a). \end{aligned}$$

Collectively,

$$m(b-a) \leq L(f) \leq U(f) \leq M(b-a).$$

□

We are now ready for a biggie: the definition of the (Riemann) integral.

Definition.

Definition 8.9. A bounded function $f : [a,b] \to \mathbb{R}$ is *integrable* if $L(f) = U(f)$. When this happens, we denote $\int_a^b f$ and $\int_a^b f(x)\,dx$ to be this common value. That is,

$$\int_a^b f(x)\,dx = L(f) = U(f).$$

Note: Officially this is the definition of *Riemann-Integrable*, but in this class we will just say "integrable." (The formal distinction is due to the fact that there are approximately a bajillion other integral definitions that mathematicians have invented.)

Example 8.10. Suppose $f : [a,b] \to \mathbb{R}$ is a constant function; say, $f(x) = c$. Then f is integrable and $\int_a^b f(x)\,dx = c(b-a)$.

Proof. To see this, note that if we let $m = c$ and $M = c$ then we have $m \leq f(x) \leq M$ for all $x \in [a,b]$. So by Lemma 8.8,

$$c(b-a) \leq L(f) \leq U(f) \leq c(b-a).$$

Therefore

$$L(f) = U(f) = c(b-a).$$

By definition, we have shown that f is integrable, and moreover that $\int_a^b f(x)\,dx = c(b-a)$, as desired. □

Example 8.11. Let $f : [0,1] \to \mathbb{R}$ be the Dirichlet function

$$f(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q} \\ 0 & \text{if } x \notin \mathbb{Q}. \end{cases}$$

Then f is *not* integrable.

Proof. To see this, let P be an arbitrary partition of $[0,1]$. By the density of both the rationals $\mathbb{Q}$ and of the irrationals $\mathbb{R} \setminus \mathbb{Q}$, any subinterval $[x_{i-1}, x_i]$ contains both a rational and an irrational. Therefore, $m_i = 0$ and $M_i = 1$ for all i. Thus,

$$L(f,P) = \sum_{i=1}^{n} m_i \cdot (x_i - x_{i-1}) = \sum_{i-1}^{n} 0 \cdot (x_i - x_{i-1}) = 0$$

and

$$U(f,P) = \sum_{i=1}^{n} M_i \cdot (x_i - x_{i-1}) = \sum_{i=1}^{n} 1 \cdot (x_i - x_{i-1}) = 1.$$

Since this is true for *any* partition P,

$$L(f) = \sup\{L(f,P) : P \in \mathcal{P}\} = \sup\{0 : P \in \mathcal{P}\} = 0$$

and

$$U(f) = \inf\{U(f,P) : P \in \mathcal{P}\} \sup\{1 : P \in \mathcal{P}\} = 1.$$

Since $L(f) \neq U(f)$, we conclude that f is not integrable. □

Here's a quick proposition.

Proposition.

Proposition 8.12 ($f \geq 0 \Rightarrow \int f \geq 0$)**.** Assume that $f : [a,b] \to \mathbb{R}$ is integrable. If $f(x) \geq 0$ for all $x \in [a,b]$, then

$$\int_a^b f(x)\, dx \geq 0.$$

Proof. Since f is integrable, by the definition of integrability we also know f is bounded. Moreover, by the assumption in the proposition we know 0 is a lower bound. So, there exists some M such that

$$0 \leq f(x) \leq M$$

for all $x \in [a,b]$. Therefore by Proposition 8.8 (and using the fact that f integrable

means that $L(f) = U(f) = \int_a^b f$),

$$0 \cdot (b-a) \leq \int_a^b f(x)\, dx \leq M \cdot (b-a).$$

In particular,

$$0 \leq \int_a^b f(x)\, dx,$$

as desired. □

Unbounded functions

Before proving some theorems about integrability I want to quickly address one subtle point in the definition of *integrable*. In the definition we immediately declared that the function f be bounded. Why is this necessary? If we didn't require boundedness, could an unbounded function be integrable? As you probably remember from calculus, "integrable" essentially means "the integral exists and is finite." This is the same goal in analysis.

So can an unbounded function satisfy this definition? The answer is no. Remember, to be integrable means that $L(f) = U(f)$. Consider for a moment the case where $f : [a, b] \to \mathbb{R}$ is not bounded above. Each upper sum $U(f, P)$ is a finite sum of the form

$$U(f, P) = \sum_{i=1}^{n} M_i \cdot (x_i - x_{i-1}),$$

and each of these M_i is the supremum of f over an interval $[x_{i-1}, x_i]$. But there are only finitely many of these intervals and collectively they equal all of $[a, b]$, so if f is unbounded on $[a, b]$ then it is also unbounded on at least one of these intervals. And if it is unbounded on $[x_{k-1}, x_k]$ then $M_k = \infty$. This would in turn mean that

$$U(f, P) = \sum_{i=1}^{n} M_i \cdot (x_i - x_{i-1})$$

has at least one ∞ in this sum (and certainly no $-\infty$'s), so this sum will be ∞.

Now, given such an f, $L(f)$ may or may not be infinite. But either way, we get a "not integrable" result. If $L(f) < \infty$ then $L(f) \neq U(f)$ and so it's not integrable by our old definition. If $L(f) = U(f) = \infty$ then the only reasonable answer would be to say the integral equals infinity, which again is undesirable.

So the standard operating procedure is to avoid it at the offset by demanding in the definition of "integrable" that f be bounded.

8.6 Integrability Criteria

By definition, f is integrable when $L(f) = U(f)$, where $L(f)$ is a supremum of a bunch of $L(f, P)$'s and $U(f)$ is an infimum of a bunch of $U(f, P)$'s. And for a supremum of one set to equal an infimum of another set, of course that means that elements from the first set are getting closer and closer to elements of the second set. Thus, of course, it makes sense that an analytic condition for this is that f is integrable if and only if for every $\varepsilon > 0$, there exists some partition P such that $U(f, P) - L(f, P) < \varepsilon$. Indeed, this is the content of the following theorem.

Theorem.

Theorem 8.13 (f *integrable* $\Leftrightarrow U(f, P_\varepsilon) - L(f, P_\varepsilon) < \varepsilon$). Let $f : [a, b] \to \mathbb{R}$ be bounded. Then f is integrable if and only if, for all $\varepsilon > 0$ there exists a partition P_ε of $[a, b]$ where

$$U(f, P_\varepsilon) - L(f, P_\varepsilon) < \varepsilon.$$

Proof. First we prove the backwards direction. Let $\varepsilon > 0$ and choose P_ε so that $U(f, P_\varepsilon) - L(f, P_\varepsilon) < \varepsilon$. Recall that $L(f, P_\varepsilon) \leq L(f)$, since $L(f)$ is a supremum over a set containing $L(f, P_\varepsilon)$; and likewise that $U(f) \leq U(f, P_\varepsilon)$. Therefore,

$$|U(f) - L(f)| = U(f) - L(f) \leq U(f, P_\varepsilon) - L(f, P_\varepsilon) < \varepsilon.$$

But since ε was arbitrary, this means that $U(f) - L(f) = 0$, and hence $L(f) = U(f) = \int_a^b f(x)\, dx$, meaning that f is integrable.

Now we prove the forward direction. Assume that f is integrable; that is, assume that $L(f) = U(f)$. Let $\varepsilon > 0$; we wish to find some P_ε such that $U(f, P_\varepsilon) - L(f, P_\varepsilon) < \varepsilon$.

Since $L(f) = \int_a^b f(x)\, dx$, and $L(f) = \sup\{L(f, P) : P \in \mathcal{P}\}$, there must be some P_1 for which $L(f, P_1) > \int_a^b f(x)\, dx - \frac{\varepsilon}{2}$. Likewise, since $U(f) = \int_a^b f(x)\, dx$, and $U(f) = \inf\{U(f, P) : P \in \mathcal{P}\}$, there must be some P_2 for which $U(f, P_2) < \int_a^b f(x)\, dx + \frac{\varepsilon}{2}$. Since each is within $\varepsilon/2$ of $\int_a^b f(x)\, dx$, this implies that

$$U(f, P_2) - L(f, P_1) < \varepsilon.$$

We are almost there, but our goal is to find a *single* partition P_ε for which the difference is less than ε—a refinement should do the trick! Let $P_\varepsilon = P_1 \cup P_2$. Then,

$$L(f, P_\varepsilon) \geq L(f, P_1) > \int_a^b f(x)\, dx - \frac{\varepsilon}{2},$$

and

$$U(f, P_\varepsilon) \leq U(f, P_2) < \int_a^b f(x)\, dx + \frac{\varepsilon}{2},$$

Thus the upper and lower sums with this new partition are still both within $\varepsilon/2$ of the integral, and hence are within ε of each other:

$$U(f, P_\varepsilon) - L(f, P_\varepsilon) < \varepsilon.$$

□

Corollary.

Corollary 8.14 (f *integrable* $\Rightarrow \lim[U(f, P_n) - L(f, P_n)] = 0$). If $f : [a, b] \to \mathbb{R}$ is integrable, then there exists a sequence P_n of partitions of $[a, b]$ for which

$$\lim_{n\to\infty}[U(f, P_n) - L(f, P_n)] = 0.$$

Proof. This is Exercise 8.11. Hint: Remember that corollaries follow quickly from theorems, so you should probably try to use the previous theorem to prove this... □

8.7 Integrability of Continuous Functions

Which classes of functions are integrable? As we have seen, being continuous at a point, or at a dense collection of points, can still give counterintuitive results. If you know a function is continuous *everywhere*, though, then the function is much more tame. And fortunately, this is good enough to guarantee integrability.

This does make some sense when you think about the formulas for an upper sum and a lower sum. In Theorem 8.13 we proved that being integrable means that for each $\varepsilon > 0$ there is a partition $P_\varepsilon = \{x_0, x_1, \ldots, x_n\}$ of $[a, b]$ for which

$$U(f, P_\varepsilon) - L(f, P_\varepsilon) < \varepsilon.$$

Substituting in the definitions of these upper and lower sums,

$$\begin{aligned} U(f, P_\varepsilon) - L(f, P_\varepsilon) &< \varepsilon \\ \sum_{i=1}^{n} M_i \cdot (x_i - x_{i-1}) - \sum_{i=1}^{n} m_i \cdot (x_i - x_{i-1}) &< \varepsilon \\ \sum_{i=1}^{n} (M_i - m_i) \cdot (x_i - x_{i-1}) &< \varepsilon. \end{aligned}$$

We want this to hold; so what sorts of functions give this conclusion? We note now that all that matters now is controlling $M_i - m_i$. This is true because if we can, say, somehow force

$$M_i - m_i < \frac{\varepsilon}{b - a}$$

for all i, then

$$\begin{aligned}\sum_{i=1}^{n}(M_i - m_i)\cdot(x_i - x_{i-1}) &< \sum_{i=1}^{n}\frac{\varepsilon}{b-a}\cdot(x_i - x_{i-1}) \\ &= \frac{\varepsilon}{b-a}\cdot\sum_{i=1}^{n}(x_i - x_{i-1}) \\ &= \frac{\varepsilon}{b-a}\cdot\Big((x_1 - x_0) + (x_2 - x_1) + (x_3 - x_2) + \cdots + (x_n - x_{n-1})\Big)\end{aligned}$$

Note that this sum contains an x_1 and a $-x_1$, so these cancel. It also contains an x_2 and a $-x_2$, and so on, up to the x_{n-1}'s. All that's left is a $-x_0$ and an x_n:

$$= \frac{\varepsilon}{b-a}\cdot(-x_0 + x_n)$$

And by definition of a partition, $x_0 = a$ and $x_n = b$:

$$\begin{aligned}&= \frac{\varepsilon}{b-a}\cdot(-a + b) \\ &= \frac{\varepsilon}{b-a}\cdot(b - a) \\ &= \varepsilon.\end{aligned}$$

So if we can find a partition for which we can guarantee $M_i - m_i$ is small, we can deduce integrability.

But why do continuous functions satisfy this? Consider the partition P_n where each rectangle has width $(b-a)/n$. If n is large, each $x_i - x_{i-1}$ is small. So what we want to say is that whenever x and y are close, that $f(x)$ and $f(y)$ are also close—this is precisely the definition of uniform continuity! And remember that if $f : [a, b] \to \mathbb{R}$ is continuous, then in particular it is a continuous function on a compact set and hence (Proposition 6.39) also uniformly continuous.

This is the proof idea behind the following theorem.

Theorem.

Theorem 8.15 (*Continuous f are integrable*). If $f : [a, b] \to \mathbb{R}$ is continuous then f is integrable.

Proof. Let $\varepsilon > 0$ and note that by the extreme value theorem f is bounded. Thus by Theorem 8.13 it suffices to find a partition P_ε where $U(f, P_\varepsilon) - L(f, P_\varepsilon) < \varepsilon$. This is our goal.

Since f is continuous on a compact set, f is uniformly continuous (by Proposition 6.39). By this and the fact that

$$\frac{\varepsilon}{b-a} > 0,$$

for this "ε" there exists some $\delta > 0$ for which $|x - y| < \delta$ implies that

$$|f(x) - f(y)| < \frac{\varepsilon}{b-a}.$$

Choose $n \in \mathbb{N}$ so that $\frac{b-a}{n} < \delta$, and let $P_\varepsilon = \{x_0, x_1, \ldots, x_n\}$ where $x_i = a + i \cdot \frac{b-a}{n}$. Note that P_ε is a partition of $[a, b]$.

We now claim, for each i, that $M_i - m_i < \frac{\varepsilon}{b-a}$. To see this, recall that since f is continuous on $[x_{i-1}, x_i]$, by the extreme value theorem we know that f attains its supremum M_i and its infimum m_i in this interval. Say, $f(x_{\min}) = m_i$ and $f(x_{\max}) = M_i$. Since $x_{\min}, x_{\max} \in [x_{i-1}, x_i]$, we know that $|x_{\min} - x_{\max}| < \delta$, and hence by the uniform continuity of f we deduce that

$$|f(x_{\max}) - f(x_{\min})| < \frac{\varepsilon}{b-a},$$

completing the claim.

Now, with this partition we have that

$$\begin{aligned}
U(f, P_\varepsilon) - L(f, P_\varepsilon) &= \sum_{i=1}^{n} M_i \cdot (x_i - x_{i-1}) - \sum_{i=1}^{n} m_i \cdot (x_i - x_{i-1}) \\
&= \sum_{i-1}^{n} (M_i - m_i) \cdot (x_i - x_{i-1}) \\
&< \sum_{i=1}^{n} \frac{\varepsilon}{b-a} \cdot (x_i - x_{i-1}) \\
&= \frac{\varepsilon}{b-a} \cdot \sum_{i-1}^{n} (x_i - x_{i-1}) \\
&= \frac{\varepsilon}{b-a} \cdot \Big((x_1 - x_0) + (x_2 - x_1) + (x_3 - x_2) + \cdots + (x_n - x_{n-1}) \Big)
\end{aligned}$$

Note that this sum contains an x_1 and a $-x_1$, so these cancel. It also contains an x_2 and a $-x_2$, and so on, up to the x_{n-1}'s. All that's left is a $-x_0$ and an x_n. And by definition of a partition, $x_0 = a$ and $x_n = b$:

$$\begin{aligned}
&= \frac{\varepsilon}{b-a} \cdot (-x_0 + x_n) \\
&= \frac{\varepsilon}{b-a} \cdot (-a + b) \\
&= \frac{\varepsilon}{b-a} \cdot (b - a) \\
&= \varepsilon.
\end{aligned}$$

□

8.8 Integrability of Discontinuous Functions

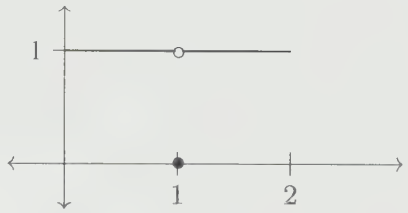
So continuous functions are integrable. Are any discontinuous functions integrable? We have so far only tested one discontinuous function: in Example 8.11 we showed that the *highly* discontinuous Dirichlet function was *not* integrable. Let's go to the other extreme—let's take an extremely simple continuous function and make it discontinuous at just a single point and see if the result is integrable.

Example 8.16. Let $f_1 : [0,2] \to \mathbb{R}$ be defined by

$$f_1(x) = \begin{cases} 1 & \text{if } x \neq 1 \\ 0 & \text{if } x = 1. \end{cases}$$

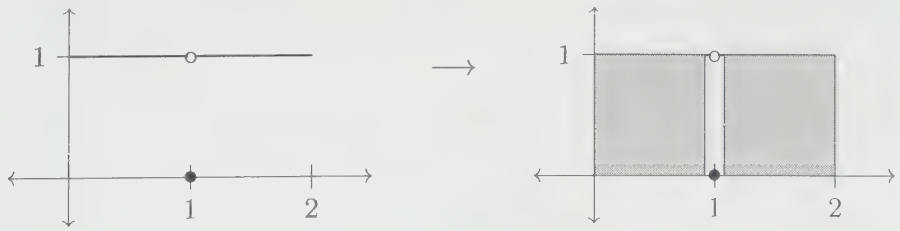
Then f_1 is integrable.

Proof. First, here is f_1's graph:



If it weren't for that one point, it would be clear that the "area under the curve" should equal 2. As it turns out, changing one point doesn't change this answer.

Indeed, we will show that $\int_0^2 f_1(x)\,dx = 2$, and will do so by using Theorem 8.13; to this end, note first that f_1 is clearly bounded. Now let $\varepsilon > 0$. We will find a particular partition P_ε for which $U(f_1, P_\varepsilon) - L(f_1, P_\varepsilon) < \varepsilon$. Here's the idea: The upper sums will always equal 2, and to get the lower sums within $2 - \varepsilon$ we will use 3 rectangles: Two big rectangles will get close to $x = 1$ on each side and will combine to have area $2 - \varepsilon$. Then one really skinny "rectangle" (whose height will be 0), will be around $x = 1$; its height is 0 because for the lower sum the height m_i is the infimum of the function values, and since $f_1(1) = 0$, that infimum will be 0.



More precisely, consider the partition $P_\varepsilon = \{0, 1 - \varepsilon/4, 1 + \varepsilon/4, 2\}$. Then the sups/infs for our sums are:

	$i = 1$	$i = 2$	$i = 3$
M_i	1	1	1
m_i	1	0	1

So,

$$\begin{aligned} U(f_1, P_\varepsilon) &= \sum_{i=1}^{n} M_i \cdot (x_i - x_{i-1}) \\ &= M_1 \cdot (x_1 - x_0) + M_2 \cdot (x_2 - x_1) + M_3 \cdot (x_3 - x_2) \\ &= 1 \cdot [(1 - \varepsilon/4) - 0] + 1 \cdot [(1 + \varepsilon/4) - (1 - \varepsilon/4)] + 1 \cdot [2 - (1 + \varepsilon/4)] \\ &= 1 - \frac{\varepsilon}{4} + 1 + \frac{\varepsilon}{4} - 1 + \frac{\varepsilon}{4} + 2 - 1 - \frac{\varepsilon}{4} \\ &= 2. \end{aligned}$$

And,

$$\begin{aligned} L(f_1, P_\varepsilon) &= \sum_{i=1}^{n} m_i \cdot (x_i - x_{i-1}) \\ &= m_1 \cdot (x_1 - x_0) + m_2 \cdot (x_2 - x_1) + m_3 \cdot (x_3 - x_2) \\ &= 1 \cdot [(1 - \varepsilon/4) - 0] + 0 \cdot [(1 + \varepsilon/4) - (1 - \varepsilon/4)] + 1 \cdot [2 - (1 + \varepsilon/4)] \\ &= 1 - \frac{\varepsilon}{4} + 2 - 1 - \frac{\varepsilon}{4} \\ &= 2 - \frac{\varepsilon}{2}. \end{aligned}$$

Therefore $U(f_1, P_\varepsilon) - L(f_1, P_\varepsilon) = 2 - (2 - \varepsilon/2) = \frac{\varepsilon}{2} < \varepsilon$. And so by Theorem 8.13, f_1 is indeed integrable. And moreover, since ε is arbitrary, we can also see that $\int_0^2 f_1(x)\, dx = 2$. □

A corollary is the direct result of a proposition or a theorem. A *porism* is the direct result of a *proof*. For instance, since the above proof could be done exactly the same regardless of where the point of discontinuity is, we have the following porism.

Porism.

Porism 8.17 (f_1 *integrable for any discontinuity* a). For any $a \in [0, 2]$, if $f_1 : [0, 2] \to \mathbb{R}$ is given by

$$f_1(x) = \begin{cases} 1 & \text{if } x \neq a \\ 0 & \text{if } x = a \end{cases},$$

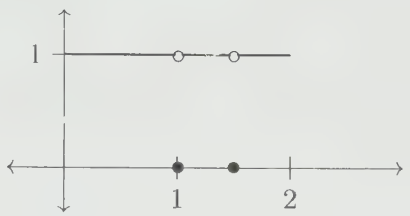
then $f_1(x)$ is integrable.

Two discontinuities

What if this function had two points of discontinuity? For example, what about

$$f_2(x) = \begin{cases} 1 & \text{if } x \neq 1 \text{ or } 1.5 \\ 0 & \text{if } x = 1 \text{ or } 1.5, \end{cases}$$

whose graph looks like

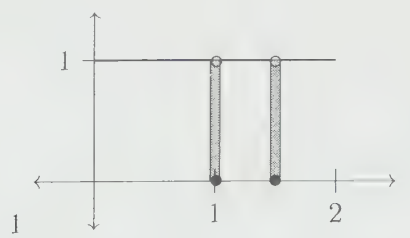


You can probably already imagine that a very similar proof to the above would work the same way. If you used the partition $P = \{0, 1-\varepsilon/8, 1+\varepsilon/8, 1.5-\varepsilon/8, 1.5+\varepsilon/8, 2\}$ then the upper sum would again equal 2 and the lower sum would equal

$$\begin{aligned} L(f_2, P) &= 1 \cdot \left(1 - \frac{\varepsilon}{8}\right) + 0 + 1 \cdot \left(0.5 - 2 \cdot \frac{\varepsilon}{8}\right) + 0 + 1 \cdot \left(0.5 - \frac{\varepsilon}{8}\right) \\ &= 2 - \frac{\varepsilon}{2} \\ &< 2 - \varepsilon. \end{aligned}$$

So the upper and lower sums are within ε of each other, showing that this very similar approach does indeed work.

Note: A good way to think about these lower sums is to think about the *missing* area. The area of the entire rectangle would be $2 \times 1 = 2$. So how much of that is missing? It's the blue/shaded area:



And the rectangle around $x = 1$ has width $\varepsilon/4$ and height 1, and the rectangle around $x = 1.5$ has width $\varepsilon/4$ and height 1. Each has area $\varepsilon/4$, so combined they have area $\varepsilon/2$. Since this is the missing area, the lower sum in total has area $2 - \varepsilon/2$.

But still there is an even better way to handle this, by using the following lemma.

Lemma.

Lemma 8.18 (*f integrable on $[a,c]$ and $[c,b] \Leftrightarrow$ on $[a,b]$*)**.** Assume $f : [a,b] \to \mathbb{R}$ and $a < c < b$. Then f is integrable on $[a,b]$ if and only if f is integrable on both $[a,c]$ and on $[c,b]$

Proof. We will use Theorem 8.13, so note that f is bounded on both of $[a,c]$ and $[c,b]$ if and only if f is bounded on $[a,b]$. So the boundedness condition is satisfied. We now prove the rest.

First assume that f is integrable on $[a,c]$ and on $[c,b]$. Since f is integrable on $[a,c]$, by Theorem 8.13 there is a partition $P_\varepsilon^{(1)}$ of $[a,c]$ such that

$$U(f, P_\varepsilon^{(1)}) - L(f, P_\varepsilon^{(1)}) < \frac{\varepsilon}{2}.$$

Since f is integrable on $[c,b]$, by Theorem 8.13 there is a partition $P_\varepsilon^{(2)}$ of $[c,b]$ such that

$$U(f, P_\varepsilon^{(2)}) - L(f, P_\varepsilon^{(2)}) < \frac{\varepsilon}{2}.$$

Then $Q_\varepsilon := P_\varepsilon^{(1)} \cup P_\varepsilon^{(2)}$ is a partition of $[a,b]$ and

$$\begin{aligned} U(f,Q_\varepsilon) - L(f,Q_\varepsilon) &= \left[U(f,P_\varepsilon^{(1)}) + U(f,P_\varepsilon^{(2)})\right] - \left[L(f,P_\varepsilon^{(1)}) + L(f,P_\varepsilon^{(2)})\right] \\ &= \left[U(f,P_\varepsilon^{(1)}) - L(f,P_\varepsilon^{(1)})\right] + \left[U(f,P_\varepsilon^{(2)}) - L(f,P_\varepsilon^{(2)})\right] \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\ &= \varepsilon. \end{aligned}$$

So f is integrable on $[a,b]$.

Now, conversely, assume that f is integrable on $[a,b]$. Then there is a partition P of $[a,b]$ such that

$$U(f,P) - L(f,P) < \varepsilon.$$

We may assume that $c \in P$, since if not we can add it to P and by Proposition 8.5 the resulting refinement would if anything only bring the upper and lower sums closer together.

Let $\bar{P} = P \cap [a,c]$; that is, P is the partition of $[a,c]$ that is the restriction of P to the first interval $[a,c]$. Then we can write $P = \{x_0, x_1, \ldots, x_T, x_{T+1}, \ldots, x_N\}$, where $\bar{P} = \{x_0, x_1, \ldots, x_T\}$. Intuitively, it's probably clear that the difference in the areas of the upper and lower rectangles from a to c is smaller than that of a to b, and

since the latter difference is less than ε, so is the former. Writing this out formally:

$$\begin{aligned} U(f,\bar{P}) - L(f,\bar{P}) &= \sum_{i=1}^{T} M_i(x_i - x_{i-1}) - \sum_{i=1}^{T} m_i(x_i - x_{i-1}) \\ &= \sum_{i-1}^{T} (M_i - m_i)(x_i - x_{i-1}) \end{aligned}$$

And since each $(M_i - m_i) \geq 0$ and each $(x_i - x_{i-1}) \geq 0$, and $T \leq N$,

$$\begin{aligned} &\leq \sum_{i=1}^{N} (M_i - m_i)(x_i - x_{i-1}) \\ &= \sum_{i=1}^{N} M_i(x_i - x_{i-1}) - \sum_{i=1}^{N} m_i(x_i - x_{i-1}) \\ &= U(f,P) - L(f,P) \\ &< \varepsilon. \end{aligned}$$

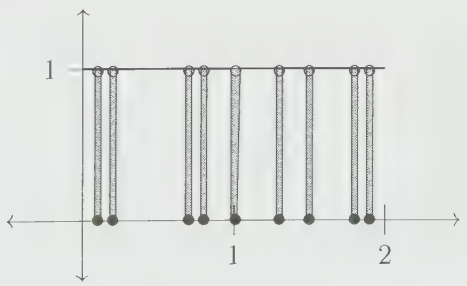
Thus, we have found a partition of $[a,c]$ for which $U(f,\bar{P}) - L(f,\bar{P}) < \varepsilon$, which by Theorem 8.13 implies that f is integrable on $[a,c]$.

An identical argument shows that f is also integrable on $[c,b]$. □

Using this theorem, you can see another reason why having two discontinuities like with f_2 does not prevent integrability. On $[0,1.25]$ f_2 is integrable by the exact same argument as with f_1, and on $[1.25,2]$ f_2 is integrable by the exact same argument as with f_1, and so by the lemma f_2 is integrable on $[0,2]$ as well.

A finite number of discontinuities

You can now imagine that there is nothing special about 1 or 2 discontinuities. If you had a function f_N which is equal to 1 except at N points of discontinuity, where f_N equals 0, then just create a partition where you include the points $x - \frac{\varepsilon}{4N}$ and $x + \frac{\varepsilon}{4N}$ in your partition.[2] Indeed, by doing this each of the N missing rectangles has width $\frac{\varepsilon}{2N}$ and height 1; hence each of the N rectangles has area $\frac{\varepsilon}{2N}$ and so collectively the missing area has a total area of $\varepsilon/2$, giving a lower sum of $2 - \varepsilon/2$.



[2] If one of these is outside $[0,2]$ then you don't include it.

Or, you could apply Lemma 8.18: Break up the domain into $N+1$ closed intervals so that there is just one discontinuity in each interval. Then f will be integrable in each of these intervals, and hence by applying Lemma 8.18 f is integrable on all of $[0, 2]$.

A countable number of discontinuities

As we've seen, the goal is to have the "missing area" add up to less than ε; in the above we always made it up to $\varepsilon/2$. If you have countably many discontinuities you can still make it add up to $\varepsilon/2$, and therefore such a function is still integrable! In particular, if your set of discontinuities is $\{x_1, x_2, x_3, \dots\}$, just include in your partition the points $x - \frac{1}{2^{i+2}}$ for each i. Then the rectangle around the first point will have width $\frac{\varepsilon}{2^2}$, the second will have width $\frac{\varepsilon}{2^3}$, the third will have width $\frac{\varepsilon}{2^4}$, and so on. Adding these up, the total amount of "missing" area is

$$\sum_{i=1}^{\infty} \frac{\varepsilon}{2^{i+1}} = \frac{\varepsilon}{2} < \varepsilon,$$

as desired. For instance, consider the following example.

Example 8.19. The function $f_\infty : [0, 2] \to \mathbb{R}$ where

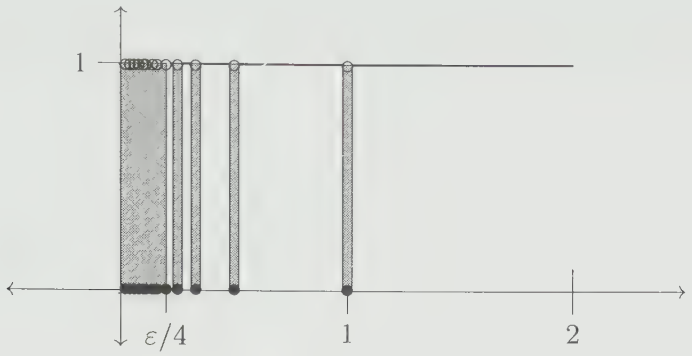
$$f_\infty(x) = \begin{cases} 1 & \text{if } x \neq \frac{1}{n} \text{ for any } n \in \mathbb{N}; \\ 0 & \text{if } x = \frac{1}{n} \text{ for some } n \in \mathbb{N}. \end{cases}$$

is integrable.

Proof Idea. Here is the idea behind the following proof. First, a partition can only be a *finite* set. So we have to find a way to have a finite number of skinny rectangles which contain all the points of discontinuity and their areas collectively add up to less than ε (once again, they will add up to $\varepsilon/2$). And then we will use Theorem 8.13.

So how do we get the "missing" areas to add up to $\varepsilon/2$? To do this, we will utilize the fact that the points of discontinuity are converging to 0. So for any $\varepsilon > 0$, only finitely many of them are outside of $[0, \varepsilon/4]$. So if we pick a partition that includes $\varepsilon/4$ (and, as is required, includes 0), then the rectangle

will capture all but finitely many of the points, and do so in a rectangle of area $(\varepsilon/4) \cdot 1 = \varepsilon/4$. Then we just add skinny rectangles around the finitely many points left, and make sure they are skinny enough that the sum of their areas equals $\varepsilon/4$.



Then collectively the sum of this "missing" area will be $\varepsilon/4 + \varepsilon/4 = \varepsilon/2$, which is less than ε, as required by Theorem 8.13. Ok, now here's the proof.

Proof. Let $\varepsilon > 0$. By the Archimedean principle (Lemma 1.26) there exists some (smallest) $N \in \mathbb{N}$ such[3] that $\frac{1}{N+1} \leq \varepsilon/4$. Now consider the partition containing $0, \varepsilon/4, 2$ and, for each point of discontinuity x outside of $[0, \varepsilon/4]$, also including the points $x \pm \frac{\varepsilon}{8N}$. That is,[4]

$$P = \left\{0, \frac{\varepsilon}{4}, \frac{1}{N} - \frac{\varepsilon}{8N}, \frac{1}{N} + \frac{\varepsilon}{8N}, \frac{1}{N-1} - \frac{\varepsilon}{8N}, \frac{1}{N-1} + \frac{\varepsilon}{8N}, \ldots, \frac{1}{1} - \frac{\varepsilon}{8N}, \frac{1}{1} + \frac{\varepsilon}{8N}, 2\right\}.$$

This is clearly a partition of $[0, 2]$. Moreover, clearly all of the points of discontinuity are contained inside the interval $[0, \varepsilon/4]$ or inside one of the N intervals $\left[x - \frac{\varepsilon}{8N}, x + \frac{\varepsilon}{8N}\right]$, where x is one of the N points of discontinuity outside of $[0, \varepsilon/4]$.

We are getting ready to compute $U(f_\infty, P) - L(f_\infty, P)$. Note that when we do, the intervals that do not contain a point of discontinuity can be ignored since the upper and lower sums are equal in those intervals, as $M_i = m_i$ there. Thus, we will focus only on these $N + 1$ intervals containing points of discontinuity; this is the "missing" area.

Furthermore, note that the intervals $\left[x - \frac{\varepsilon}{8N}, x + \frac{\varepsilon}{8N}\right]$ each have length $\frac{\varepsilon}{4N}$; so if these equal x_{i-1} and x_i, then $x_i - x_{i-1} = \frac{\varepsilon}{4N}$. Lastly, on each of these intervals

[3]Note: The reason why we included a "+1" is so there are N points of discontinuity,

$$\frac{1}{1}, \frac{1}{2}, \frac{1}{3}, \ldots, \frac{1}{N},$$

outside of $[0, \varepsilon/4]$, while the rest,

$$\frac{1}{N+1}, \frac{1}{N+2}, \frac{1}{N+3}, \ldots,$$

are inside of $[0, \varepsilon/4]$. This makes the notation easier.

[4]Technically we should assume that $\varepsilon \leq 8$ to ensure that all of these points are in $[0, 2]$, but this is unimportant. There are many easy fixes, such as just removing such points if they occur, or insisting that $\varepsilon < 8$.

containing points of discontinuity, note that $m_i = 0$ while $M_i = 1$. Putting this all together,

$$\begin{aligned} U(f_\infty, P) - L(f_\infty, P) &= \sum_{i=1}^{N+1} M_i(x_i - x_{i-1}) - \sum_{i=1}^{N+1} m_i(x_i - x_{i-1}) \\ &= \sum_{i=1}^{N+1} (M_i - m_i)(x_i - x_{i-1}) \\ &= (1-0)(\varepsilon/4 - 0) + \sum_{i=2}^{N+1} (1-0)\left(\frac{\varepsilon}{4N}\right) \\ &= \frac{\varepsilon}{4} + N \cdot (1-0)\left(\frac{\varepsilon}{4N}\right) \\ &= \frac{\varepsilon}{4} + \frac{\varepsilon}{4} \\ &= \frac{\varepsilon}{2} \\ &< \varepsilon \end{aligned}$$

And since f_∞ is clearly bounded, Theorem 8.13 now says that f_∞ is integrable. □

Example 8.20. Thomae's function on $[0, 2]$ is integrable.

Proof. This is Exercise 8.13. □

An uncountable number of discontinuities

Amazingly, an integrable function can even have an uncountable number of discontinuities. It requires a weird uncountable set. It requires an uncountable set that is "small" in the visual sense. It requires... The Cantor Set! We won't go into the details, but here is such an example:

$$g(x) = \begin{cases} 1 & \text{if } x \in C \\ 0 & \text{if } x \notin C, \end{cases}$$

where C is the Cantor set. (See Example B.1 in Appendix B.)

Other continuous functions

You can imagine that if you take *any* function f on an interval $[a, b]$ that is discontinuous at just 1 point, a very similar argument would show that that function is also integrable; like above, you would hollow out a tiny region around the point of discontinuity, then the rest of the function's upper and lower bounds get as close as you wish, and the error caused by the point of discontinuity you can make as small as you want by making the tiny region around it as tiny as you need. There's no concern about the function getting too big or small on the subinterval because f being continuous implies that f is bounded.

Then, by Lemma 8.18, we can handle any finite number of discontinuities. And, like above, in certain situations we can even handle an infinite number of discontinuities.

But how can we tell in general that we have an integrable function? The answer takes a little care to write down, but it's awesome. It's called the Measure Zero Criterion.

8.9 The Measure Zero Integrability Criterion

In Example 8.19 we showed in detail that f_∞ was integrable. Why did it work out? Well it was highly convenient that the infinitely many points were all bunching up together around 0. This allowed all but a finite number of the points to be contained inside in a tiny interval. And the remaining finite number of points could be handled easily. This suggests one condition that guarantees integrability:

"f is integrable if for all $\varepsilon > 0$ the points of discontinuity can be covered by a finite number of intervals $I_1, I_2, I_3, \ldots, I_N$ such that the total length of the intervals sums to less than ε: $\sum_{k=1}^{N} |I_k| < \varepsilon$."

Here, by "length" we mean the following (for more on this, see the Example B.1 in Appendix B).

Definition.

Definition 8.21. The *length* of the interval $[a, b]$ is equal to $b-a$, and is denoted $\mathscr{L}([a, b])$. This is also the length of (a, b), $(a, b]$ and $[a, b)$. Intervals that involve $\pm\infty$ are said to have length ∞.

So above we argued informally that if, for every $\varepsilon > 0$, the points of discontinuity can be covered by a finite set of intervals whose lengths add up to less than ε, then f is integrable. In fact, even more is true—a countable collection of sets works too! This is the idea of a set having "measure 0."

Definition.

Definition 8.22. A set A has *measure zero* if for all $\varepsilon > 0$ there exists a countable collection $I_1, I_2, I_3, \ldots$ of intervals such that

$$A \subseteq \bigcup_{k=1}^{\infty} I_k \qquad \text{and} \qquad \sum_{k=1}^{\infty} \mathscr{L}(I_k) < \varepsilon.$$

This leads to a famous theorem sometimes called *Lebesgue's integrability condition*—an insightful and delightful criterion for integrability.

Theorem.

Theorem 8.23 (*f integrable $\Leftrightarrow$ discontinuities have measure zero*). Let $f : [a, b] \to \mathbb{R}$ be bounded and let $\mathcal{D}$ be the set of points where f is discontinuous. Then f is integrable if and only if $\mathcal{D}$ has measure zero.

The proof of this theorem is very long and quite a pain; we will skip it.[5]

8.10 Linearity of the Integral

In this section we will prove many properties that you will be familiar with from calculus. Here's the first result.

Proposition.

Proposition 8.24 ($\int_a^b f = \int_a^c f + \int_c^b f$). Assume that $f : [a, b] \to \mathbb{R}$ is integrable. If $a < c < b$, then

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx.$$

Proof. Recall from Lemma 8.18 that f is integrable on $[a, b]$ if and only if f is integrable on both $[a, c]$ and on $[c, b]$. Thus we only have to show that the asserted sum holds.

For every $\varepsilon > 0$ we can find a partition P_1 of $[a, c]$ and a partition P_2 of $[c, b]$ for which

$$U(f, P_1) - L(f, P_1) < \frac{\varepsilon}{2} \qquad \text{and} \qquad U(f, P_2) - L(f, P_2) < \frac{\varepsilon}{2}.$$

Letting $P = P_1 \cup P_2$ produces a partition of $[a, b]$ where

$$\begin{aligned} U(f, P) - L(f, P) &= [U(f, P_1) + U(f, P_2)] - [L(f, P_1) + L(f, P_2)] \\ &= [U(f, P_1) - L(f, P_1)] + [U(f, P_2) - L(f, P_2)] \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\ &= \varepsilon. \end{aligned}$$

[5] You're welcome.

Lower sums are always less than the integral, and upper sums are always greater than the integral.

$$L(f,P) \leq \int_a^b f(x)\,dx \leq U(f,P)$$

We saw above that $U(f,P) - L(f,P) < \varepsilon$, which is equivalent to $U(f,P) < L(f,P) + \varepsilon$ and to $U(f,P) - \varepsilon < L(f,P)$. Thus,

$$U(f,P) - \varepsilon < L(f,P) \leq \int_a^b f(x)\,dx \leq U(f,P) < L(f,P) + \varepsilon.$$

Simplifying this by removing the extra upper and lower sums,

$$U(f,P) - \varepsilon < \int_a^b f(x)\,dx < L(f,P) + \varepsilon$$

And since $L(f,P) = L(f,P_1) + L(f,P_2)$ and $U(f,P) = U(f,P_1) + U(f,P_2)$,

$$[U(f,P_1) + U(f,P_2)] - \varepsilon < \int_a^b f(x)\,dx < [L(f,P_1) + L(f,P_2)] + \varepsilon$$

Once again, upper sums are bigger than their integrals, and lower sums are smaller than their integrals. Thus,

$$\left[\int_a^c f(x)\,dx + \int_c^b f(x)\,dx\right] - \varepsilon < \int_a^b f(x)\,dx < \left[\int_a^c f(x)\,dx + \int_c^b f(x)\,dx\right] + \varepsilon$$

So we know that $\int_a^c f(x)\,dx + \int_c^b f(x)\,dx$ and $\int_a^b f(x)\,dx$ are within ε of each other. And since ε was arbitrary, this implies that they are in fact equal:

$$\int_a^c f(x)\,dx + \int_c^b f(x)\,dx = \int_a^b f(x)\,dx,$$

as desired. □

Yes, I agree, that proof was fairly boring. And the bad news is that this mostly continues. The theorems about integrals are quite beautiful, but since they are defined in terms of infimums and supremums of sums of other infimums and supremums over arbitrary partitions... the elegance is often lost in the weeds. Nevertheless, we persevere. Here's a cute puppy for motivation. Next up is a proposition on linearity properties of integrals.

Proposition.

Proposition 8.25 (*Linearity of the integral*). Assume that $f : [a, b] \to \mathbb{R}$ and $g : [a, b] \to \mathbb{R}$ are integrable. Then,

(i) For any $k \in \mathbb{R}$, kf is also integrable on $[a, b]$, and

$$\int_a^b k \cdot f(x)\, dx = k \cdot \int_a^b f(x)\, dx.$$

(ii) The function $f + g$ is also integrable on $[a, b]$, and

$$\int_a^b (f(x) + g(x))\, dx = \int_a^b f(x)\, dx + \int_a^b g(x)\, dx.$$

Proof. First we prove (i). If $k > 0$, then intuitively all we are doing is making each rectangle in our upper/lower sums k times bigger. Indeed, each M_i in $U(f, P)$ will turn into $k \cdot M_i$ in $U(kf, P)$. And so,

$$U(k \cdot f, P) = \sum_{i=1}^{n} k \cdot M_i(x_i - x_{i-1}) = k \cdot \sum_{i=1}^{n} M_i(x_i - x_{i-1}) = k \cdot U(f, P).$$

And likewise for the lower sums we have $L(k \cdot f, P) = k \cdot L(f, P)$.

Since f is integrable on $[a, b]$, by Corollary 8.14 there exists a sequence (P_n) of partitions of $[a, b]$ for which

$$\int_a^b f(x)\, dx = \lim_{n\to\infty} U(f, P_n) = \lim_{n\to\infty} L(f, P_n).$$

But using what we argued above and properties of limits,

$$\lim_{n\to\infty} U(kf, P_n) = \lim_{n\to\infty} k \cdot U(f, P_n) = k \cdot \lim_{n\to\infty} U(f, P_n) = k \cdot \int_a^b f(x)\, dx$$

and

$$\lim_{n\to\infty} L(kf, P_n) = \lim_{n\to\infty} k \cdot L(f, P_n) = k \cdot \lim_{n\to\infty} L(f, P_n) = k \cdot \int_a^b f(x)\, dx.$$

So we have shown that

$$\lim_{n\to\infty} U(kf, P_n) = \lim_{n\to\infty} L(kf, P_n) = k \cdot \int_a^b f(x)\, dx,$$

which by Corollary 8.14 concludes (i) in this case where $k > 0$. The case where $k < 0$ is very similar, the only difference is that the upper sums for f become lower sums for kf, and vice versa. In particular, $U(kf, P) = k \cdot L(f, P)$ and $L(kf, P) = k \cdot U(f, P)$.

The rest is almost identical.[6] Finally, the case where $k = 0$ is immediate.

The proof of (ii) can use the same general approach, but is just messier. It's left as an exercise.[7] □

8.11 More Properties of the Integral

The next result is a Corollary of an earlier result.

Recall 8.26. In Proposition 8.25 we proved the following:

Assume that $f : [a, b] \to \mathbb{R}$ is integrable. If $f(x) \geq 0$ for all $x \in [a, b]$, then

$$\int_a^b f(x)\, dx \geq 0.$$

Corollary.

Corollary 8.27 ($f \leq g \Rightarrow \int f \leq \int g$). Assume that $f : [a, b] \to \mathbb{R}$ and $g : [a, b] \to \mathbb{R}$ are integrable. If $f(x) \leq g(x)$ for all $x \in [a, b]$, then

$$\int_a^b f(x)\, dx \leq \int_a^b g(x)\, dx.$$

Proof. Since $f(x) \leq g(x)$ we have $0 \leq g(x) - f(x)$. By Proposition 8.25 part (i), since f is integrable, $(-1) \cdot f = -f$ is also integrable. And since g is also assumed to be integrable, by Proposition 8.25 part (ii) we have $g + (-f) = g - f$ is integrable.

We have shown that $g - f$ is a non-negative, integrable function and thus by Proposition 8.12 we have that

$$\int_a^b (g - f)(x)\, dx \geq 0.$$

[6] Interviewer: "So why did you become a mathematician?" Mathematician: "Because I never liked working with numbers."

[7] I'd assign it as an exercise, but it's not even worth it.

Applying Proposition 8.25 one more time,

$$\int_a^b [g(x) - f(x)]\ dx \geq 0$$
$$\int_a^b g(x)\ dx - \int_a^b f(x)\ dx \geq 0$$
$$\int_a^b g(x)\ dx \geq \int_a^b f(x)\ dx.$$

□

The next result can be thought of as an "integral triangle inequality." When adding up discrete numbers you can use the triangle inequality: $|x + y| \leq |x| + |y|$. More generally, $|\sum x_i| \leq \sum |x_i|$. And in some sense, the way we "add up" a continuous range of numbers is using the integral. So this is the continuous version of the triangle inequality.

Corollary.

Corollary 8.28 ($|\int f| \leq \int |f|$). If $f : [a, b] \to \mathbb{R}$ is integrable then $|f|$ is also integrable, and

$$\left|\int_a^b f(x)\ dx\right| \leq \int_a^b |f(x)|\ dx.$$

Proof. Let $\varepsilon > 0$. Note that since f is integrable, f is bounded and hence $|f|$ is also bounded. Furthermore, since f is integrable there exists a partition $P_\varepsilon = \{x_0, x_1, \ldots, x_n\}$ such that

$$U(f, P_\varepsilon) - L(f, P_\varepsilon) < \varepsilon.$$

Over each interval $[x_{i-1}, x_i]$ we have notation m_i and M_i to represent the infimum and supremum of f over the interval, respectively. Likewise, define

$$\overline{m}_i = \inf\{|f(x)| : x \in [x_{i-1}, x_i]\}$$
$$\overline{M}_i = \sup\{|f(x)| : x \in [x_{i-1}, x_i]\}.$$

Since $|f(x)| \geq f(x)$ for all x, clearly $\overline{m}_i \geq m_i$ and $\overline{M}_i \geq M_i$ for all i. Moreover,

$$M_i - m_i \geq \overline{M}_i - \overline{m}_i$$

since if M_i and m_i are the same sign (both positive or both negative) then these two are equal, while if they are opposite signs then taking the absolute value only makes them closer to each other.

Then,

$$\begin{aligned} U(|f|, P_\varepsilon) - L(|f|, P_\varepsilon) &= \sum_{i=1}^{n} \left(\overline{M}_i - \overline{m}_i\right)(x_i - x_{i-1}) \\ &\leq \sum_{i=1}^{n} (M_i - m_i)(x_i - x_{i-1}) \\ &= U(f, P_\varepsilon) - L(f, P_\varepsilon) \\ &< \varepsilon. \end{aligned}$$

By Theorem 8.13, $|f|$ is then integrable.

To show that the integral inequality holds, first note that using the fact that $-|f(x)| \leq f(x) \leq |f(x)|$ and applying Corollary 8.27 (twice) we have that

$$-\int_a^b |f(x)|\, dx \leq \int_a^b f(x)\, dx \leq \int_a^b |f(x)|\, dx. \qquad (\text{🖷})$$

Finally, simply recall that $-t \leq s \leq t$ if and only if $|s| \leq t$ for any real numbers s and t. Using this, (🖷) is equivalent to

$$\left|\int_a^b f(x)\, dx\right| \leq \int_a^b |f(x)|\, dx. \qquad \square$$

We only defined the integral on intervals $[a, b]$ where $a < b$. Therefore the below have to be stated as definitions, rather than deduced as properties. Note, though, that they are consistent with a natural generalization of our definitions.

Definition.

Definition 8.29.

- For any function f, define

$$\int_a^a f(x)\, dx = 0.$$

- For any integrable $f : [a, b] \to \mathbb{R}$, define

$$\int_a^b f(x)\, dx = -\int_b^a f(x)\, dx.$$

The Mean Value Theorems

Recall the (derivative) mean value theorem.

Theorem.

Theorem 7.22 (*The (derivative) mean value theorem*). Let $f : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b). Then there exists some $c \in (a, b)$ where

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

This theorem will have a practical use very soon, but first it motivates the next result, known as the *integral mean value theorem.*

When looked at just right, this new mean value theorem looks a lot like the derivative mean value theorem, provided you recall the fundamental theorem of calculus from when you took calculus. Instead of viewing the $f'(c)$ as being the derivative f, you could instead think about f as being an *antiderivative* of f' (something we will define soon). Let F be such an antiderivative. Then the Derivative MVT says

$$F'(c) = \frac{F(b) - F(a)}{b - a}.$$

But if $F'(c) = f(c)$, and if $F(b) - F(a) = \int_a^b f(x)\, dx$ as we already expect by the fundamental theorem of calculus, then we essentially get the integral mean value theorem, which we state now.

Proposition.

Proposition 8.30 (*The integral mean value theorem*). If f is a continuous function on $[a, b]$, then there is some $c \in [a, b]$ such that

$$f(c) = \frac{1}{b - a} \int_a^b f(x)\, dx.$$

Proof. Since f is continuous on a closed interval, f is bounded (by the extreme value theorem) and integrable (by Theorem 8.15). The extreme value theorem furthermore implies that f attains a minimum $f(x_{\min}) = m$ and a maximum $f(x_{\max}) = M$ at some points $x_{\min}, x_{\max} \in [a, b]$. Therefore by Lemma 8.8, since $m \leq f(x) \leq M$ for all $x \in [a, b]$, we have

$$m(b - a) \leq \int_a^b f(x)\, dx \leq M(b - a).$$

I.e.,

$$m \leq \frac{1}{b-a} \int_a^b f(x)\, dx \leq M.$$

In summary, f is a continuous function and $\frac{1}{b-a} \int_a^b f(x)\, dx$ is some number between $f(x_{\min})$ and $f(x_{\max})$. Therefore, by the intermediate value theorem, there is some c between $x_{\min}$ and $x_{\max}$ for which

$$f(c) = \frac{1}{b-a} \int_a^b f(x)\, dx.$$

□

8.12 The Fundamental Theorem of Calculus

The fundamental theorem of calculus (FTC) is remarkable. And it's a tragedy that you all already know the grand reveal. If tonight you watch Star Wars for the first time, you probably won't be surprised to learn that Darth Vader is Luke Skywalker's father; that has been common knowledge for over 40 years now, even among those who haven't seen the movies. But for those who learned it in the theaters for the first time on May 21, 1980, this was an astounding development. How can two disparate forces in this grand story be linked in such an intimate way??[8]

Likewise, due to the evils of calculus, you already know what FTC says. But for a moment, pretend you don't. Under this veil of ignorance, we have an amazing plot twist to our story.

In Chapter 7 we developed the theory of derivatives as a limit, motivated by finding slopes of tangent lines, in order to solve physics problems. And we then pushed the theory further and proved lots of nice theorems. During this chapter we did something completely different. We developed integrals as a way to approximate areas under curves, by using infimums and supremums of upper and lower sums. This purely-geometric endeavor also produced some nice results with some fairly intuitive properties.

There is absolutely no reason at all to think these two have anything to do with each other. With the veil over our eyes, we should even be confused why these two topics are contained in the same class. They are not even remotely related, right? Right?? Well with four words with implications as startling as "I am your father," we should be amazed at how the "fundamental theorem of calculus" intimately ties together these seemingly-disparate ideas in the most incredible way imaginable.

So either take off that veil or hold onto your hats, because we are about to state the fundamental theorem of calculus.

[8]Millennials version: How shocked would you have been if Voldemort was Harry Potter's father? And if there was an amazing backstory why? What a twist!

Super Cool Theorem.

Theorem 8.31 (*The fundamental theorem of calculus*).

(i) If $f : [a, b] \to \mathbb{R}$ is integrable, and $F : [a, b] \to \mathbb{R}$ satisfies $F'(x) = f(x)$ for all x, then

$$\int_a^b f(x)\, dx = F(b) - F(a).$$

(ii) Let $g : [a, b] \to \mathbb{R}$ be integrable and define $G : [a, b] \to \mathbb{R}$ by

$$G(x) = \int_a^x g(t)\, dt.$$

Then G is continuous. Moreover, if g is continuous then G is differentiable and $G'(x) = g(x)$. That is, $\dfrac{d}{dx}\displaystyle\int_a^x g(t)\, dt = g(x)$.

Surprisingly, the proof of part (i) just comes down to an application of the derivative mean value theorem and a cute use of telescoping.

Proof of (i). Let $P = \{x_0, \ldots, x_n\}$ be an arbitrary partition of $[a, b]$. Consider the behavior of F on some subinterval $[x_{i-1}, x_i]$. Since F is differentiable, by the derivative mean value theorem (stated above) there must exist some $c_i \in (x_{i-1}, x_i)$ where

$$F'(c_i) = \frac{F(x_i) - F(x_{i-1})}{x_i - x_{i-1}}$$

That is,

$$F(x_i) - F(x_{i-1}) = F'(c_i) \cdot (x_i - x_{i-1})$$

Using now the assumption that $F'(x) = f(x)$,

$$F(x_i) - F(x_{i-1}) = f(c_i) \cdot (x_i - x_{i-1}).$$

Consider now the upper sum $U(f, P)$ and the lower sum $L(f, P)$. These sums contain the sups/infs M_i and m_i which satisfy $m_i \leq f(c_i) \leq M_i$, implying that

$$\sum_{i=1}^{n} m_i(x_i - x_{i-1}) \leq \sum_{i=1}^{n} f(c_i)(x_i - x_{i-1}) \leq \sum_{i=1}^{n} M_i(x_i - x_{i-1}).$$

The left and the right sums are just the lower and upper sums, and the terms of the

middle sum we already noted are just $F(x_i) - F(x_{i-1})$.

$$L(f,P) \leq \sum_{i=1}^{n} \Big(F(x_i) - F(x_{i-1})\Big) \leq U(f,P).$$

But notice that the middle sum telescopes! Indeed,

$$\begin{aligned}
&\sum_{i=1}^{n}\Big(F(x_i) - F(x_{i-1})\Big) \\
&\quad = \Big(\cancel{F(x_1)} - F(x_0)\Big) + \Big(\cancel{F(x_2)} - \cancel{F(x_1)}\Big) + \Big(\cancel{F(x_3)} - \cancel{F(x_2)}\Big) + \dots \\
&\qquad\qquad \dots + \Big(F(x_n) - \cancel{F(x_{n-1})}\Big) \\
&\quad = -F(x_0) + F(x_n) \\
&\quad = F(x_n) - F(x_0) \\
&\quad = F(b) - F(a).
\end{aligned}$$

Using this to simplify the middle term of the previous inequality,

$$L(f,P) \leq F(b) - F(a) \leq U(f,P).$$

But notice that this holds no matter which partition we choose! So we also have that

$$\sup\{L(f,P) : P \in \mathcal{P}\} \leq F(b) - F(a) \leq \inf\{U(f,P) : P \in \mathcal{P}\},$$

where $\mathcal{P}$ is the collection of all partitions of $[a,b]$. That is,

$$L(f) \leq F(b) - F(a) \leq U(f). \qquad (\clubsuit)$$

And this is perfect, since f being integrable means that $L(f) = U(f) = \int_a^b f(x)\,dx$. This and ($\clubsuit$) implies that

$$\int_a^b f(x)\,dx = F(b) - F(a),$$

completing the proof of part (i). $\square_{(i)}$

Proof of (ii). First, assume that g is integrable; we will prove that $G(x) = \int_a^x g(t)\,dt$ is continuous, by using the definition of continuity. First, note that if $g(x) = 0$ for all x then also $G(x) = 0$ on all x, and we are done. Therefore, we may assume that

$$M = \sup\{|g(x)| : x \in [a,b]\}$$

is greater than zero.

Pick any $x_0 \in [a,b]$. To prove that G is continuous at x_0, let $\varepsilon > 0$, and then let $\delta = \frac{\varepsilon}{M}$. Assume now that $|x - x_0| < \delta$; we will prove that $|G(x) - G(x_0)| < \varepsilon$. Note

that

$$\begin{aligned}
|G(x) - G(x_0)| &= \left| \int_a^x g(t)\, dt - \int_a^{x_0} g(t)\, dt \right| \\
&= \left| \int_{x_0}^x g(t)\, dt \right| \\
&\leq \int_{x_0}^x |g(t)|\, dt && \text{(Corollary 8.28)} \\
&\leq M \cdot |x - x_0| && \text{(Lemma 8.8)} \\
&< M \cdot \delta && \text{(Assumption on } \delta\text{)} \\
&= M \cdot \frac{\varepsilon}{M} && \text{(Choice of } \delta\text{)} \\
&= \varepsilon.
\end{aligned}$$

This proves G is continuous at x_0, and since x_0 was arbitrary, it proves that G is continuous on all of $[a, b]$, concluding the first half of (ii).

Next assume that g is also continuous; we will now prove that G is differentiable and that $G'(x) = g(x)$. To prove this, pick an arbitrary $c \in [a, b]$; by definition we should prove

$$\lim_{x \to c} \frac{G(x) - G(c)}{x - c} = g(c).$$

But recall that by the sequential definition of limits that this is equivalent to proving that every sequence (x_n) from $[a, b] \setminus \{c\}$ such that $x_n \to c$ has the property that

$$\lim_{n \to \infty} \frac{G(x_n) - G(c)}{x_n - c} = g(c).$$

We now prove this. To that end, choose such a sequence (x_n). Note that

$$G(x_n) - G(c) = \int_a^{x_n} g(t)\, dt - \int_a^c g(t)\, dt = \int_c^{x_n} g(t)\, dt. \qquad (\ddagger)$$

By the integral mean value theorem which we proved a moment ago (but here we write it with the "$b-a$" moved to the other side), there is some c_n between x_n and c such that $\int_c^{x_n} g(t)\, dt = g(c_n) \cdot (x_n - c)$. Applying this to $(\ddagger)$, $G(x_n) - G(c) = g(c_n) \cdot (x_n - c)$. I.e.,

$$\frac{G(x_n) - G(x_0)}{x_n - c} = g(c_n).$$

Moreover, since c_n was between x_n and c for each n, and $x_n \to c$, we also know that $c_n \to c$. We are ready for the final computation:

$$\begin{aligned}
G'(c) &= \lim_{n \to \infty} \frac{G(x_n) - G(c)}{x_n - c} \\
&= \lim_{n \to \infty} g(c_n)
\end{aligned}$$

And since g is continuous, by Theorem 6.16 (iv),

$$\begin{aligned} &= g\left(\lim_{n\to\infty} c_n\right) \\ &= g(c). \end{aligned}$$ $\square_{(ii)}$

There's also a neat way to prove FTC (i) from FTC (ii). Suppose $f : [a, b] \to \mathbb{R}$ is integrable, and $F : [a, b] \to \mathbb{R}$ satisfies $F'(x) = f(x)$ for all x. We'll show

$$\int_a^b f(x) = \, dx = F(b) - F(a).$$

To that end, let

$$h(x) = \int_a^x f(t)\, dt. \tag{*}$$

Then by FTC (ii), $h'(x) = f(x)$. That is, both h and F are antiderivatives of the function f, which by the "$+C$ corollary" (Corollary 7.24), these two functions can only differ by a constant; that is, $h(x) = F(x) + C$, for some $C \in \mathbb{R}$. Flipping around (*) and plugging $h(x) = F(x) + C$ then gives

$$\int_a^x f(t)\, dt = F(x) + C. \tag{**}$$

Equation (**) holds for any x, so in particular when $x = b$,

$$\int_a^b f(t)\, dt = F(b) + C.$$

We are nearly there! As long as we can show that $C = -F(a)$, we will have it. And why is that true? Well, (**) also holds when $x = a$! And plugging that in gives

$$0 = \int_a^a f(t)\, dt = F(a) + C,$$

implying that $C = -F(a)$, as desired. $\square$

Definition

Definition 8.32. Assume F and f are functions satisfying the conditions of Theorem 8.31, then F is called an *antiderivative* of f.

8.13 Integration Rules

The fundamental theorem of calculus has a number of nice consequences, all of which will be familiar from calculus.

Rule 1. Integration by Parts—The "Inverse Product Rule"

Here is our first integration rule.

Corollary.

Corollary 8.33 (*Integration by parts*). If f and g are differentiable functions which have continuous derivatives on $[a, b]$, then fg' and $f'g$ are integrable and

$$\int_a^b fg' = f(b)g(b) - f(a)g(a) - \int_a^b f'g.$$

Note that in calculus this is commonly written $\displaystyle\int_a^b u\,dv = uv\Big|_a^b - \int_a^b v\,du$.

Integration by parts can be thought of as an "inverse product rule." Indeed, the proof essentially just integrates a product rule and applies FTC.

Proof. We assumed that f and g are differentiable, and hence are continuous. We also assumed that f' and g' are continuous. This all implies that both fg' and $f'g$ are continuous, and hence by Theorem 8.15 are integrable.

Now, by the Product Rule,

$$(fg)' = f'g + fg'.$$

Integrating both sides,

$$\int (fg)' = \int f'g + \int fg'. \qquad (\text{ })$$

We now use FTC to rewrite the left-hand side. Clearly fg is an antiderivative of $(fg)'$, and so by the fundamental theorem of calculus part (i),

$$\int (fg)' = (fg)(b) - (fg)(a) = f(b)g(b) - f(a)g(a).$$

Plugging this into ( ) gives

$$f(b)g(b) - f(a)g(a) = \int f'g + \int fg',$$

which is equivalent to what we are trying to prove. □

Rule 2. w-Substitution—The "Inverse Chain Rule"

Next is what most people call u-substitution, but for various calculus-based pedagogical reasons I call w-substitution.

Corollary.

Corollary 8.34 (*w-substitution*). Suppose g is a function whose derivative g' is continuous on $[a, b]$, and suppose that f is a function that is continuous on $\{g(x) : x \in [a, b]\}$. Then

$$\int_a^b f(g(x))g'(x)\,dx = \int_{g(a)}^{g(b)} f(w)\,dw.$$

This rule can be thought of as an "inverse chain rule"—the substitution essentially collapses down an application of the chain rule. The proof drives this point home by showing that it holds via a combination of the chain rule with FTC. Indeed, the big-picture idea is that if you let $F(x) = \int_{g(a)}^{x} f(t)\,dt$ then we should have

$$\int_a^b f(g(x))g'(x)\,dx = F(g(b)) - F(g(a)) = \int_{g(a)}^{g(b)} f(w)\,dw.$$

Now here is the detailed proof.

Proof. Since g is continuous, note that $g([a, b])$ is either a closed interval or a single point. (For a detailed explanation, use the extreme value theorem to show that g attains a max and a min, and then use the intermediate value theorem to show that it doesn't miss anything in-between.) Thus we have two cases to consider.

Case 1: $g([a, b]) = \{L\}$. In this case, $g(x) = L$ for all $x \in [a, b]$. Since the derivative of constant functions are identically zero, $g'(x) = 0$ for all x. Indeed, the equality in the corollary holds because both sides simply equal 0:

$$\begin{aligned}\int_a^b f(g(x)) \cdot g'(x)\,dx &= \int_a^b f(g(x)) \cdot 0\,dx \\ &= 0 \\ &= \int_L^L f(w)\,dw \\ &= \int_{g(a)}^{g(b)} f(w)\,dw,\end{aligned}$$

proving the equality holds for Case 1.

Case 2: $g([a,b]) = [c,d]$. Define

$$F(x) = \int_{g(a)}^{x} f(t)\ dt.$$

Note that $F(g(b))$ is the right-hand side of the equality we are trying to show, showing that it suffices to prove that $F(g(b)) = \int_a^b f(g(x)) \cdot g'(x)\ dx$. Furthermore, note that $F(g(a)) = \int_{g(a)}^{g(a)} f(t)\ dt = 0$. Thus it is also sufficient to prove that

$$F(g(b)) - F(g(a)) = \int_a^b f(g(x)) \cdot g'(x)\ dx.$$

This is what we will show, and we will combine the chain rule and FTC to do so.

By assumption, $f(g(x))$ and $g'(x)$ are continuous, implying that $f(g(x)) \cdot g'(x)$ is continuous and hence integrable. Applying the chain rule to $F(g(x))$,

$$\frac{d}{dx}F(g(x)) = F'(g(x)) \cdot g'(x) = f(g(x)) \cdot g'(x).$$

Said differently, $F(g(x))$ is an antiderivative of $f(g(x)) \cdot g'(x)$. And by FTC part (i), this means that

$$\int_a^b f(g(x)) \cdot g'(x)\ dx = F(g(b)) - F(g(a)),$$

as desired. □

— Notable Exercises —

- Exercise 8.2 and 8.3 may seem misplaced, given that they are on supremums. They are included in this chapter due to their renewed importance.
- In Chapter 6 on continuity, Theorem 6.11 gave a method to convert theorems on sequences into theorems on functions. Indeed, it was able to quickly turn the sequence limit laws and the sequence squeeze theorem into the func-y limits laws and the func-y squeeze theorem. Exercise 8.11 is in some ways similar: It (halfway) turns an esoteric integrability criterion of Definitions 8.7 and 8.9 into a concrete version involving a specific sequence of partitions. However, unlike with the Chapter 6 examples, this one is not biconditional. But nevertheless, from a practical standpoint it is still quite nice.
- You are asked for two proofs that Thomae's function is integrable, one in Exercise 8.13 and one in Exercise 8.19. Knowing multiple proofs of a theorem is an underrated way to understand some material. A minor reason for proving things is so that we know the theorem is true. They are primarily to allow us to understand *why* it is true. Digesting multiple proofs will richen this understanding.
- Continuing the train of thought from the above bullet point, Exercise 8.21 asks for a second proof of the intermediate value theorem (Theorem 6.37).
- In more general integration theory, *step functions* play a crucial role. Exercises 8.28 and 8.29 begin this journey.

— Exercises —

Exercise 8.1. Lakhveer, Morgan and Quin are splitting an Uber to get to their homes; they all live on the same straight street, which is also the same street they are on now. After driving 10 miles on this street they reach Lakhveer's house and she gets out, after another 10 miles they reach Morgan's house and she gets out, after another 10 miles they reach Quin's house and he gets out.

The cost of the Uber is \$1.50 per mile, so in total they have to pay \$45. They discuss how much should each person have to pay for it to be fair. Lakhveer sends her reasoning, and it looks like this:

$$\begin{cases} \text{Lakhveer} & = \frac{1}{3} \cdot 15 = \$5 \\ \text{Morgan} & = \frac{1}{3} \cdot 15 + \frac{1}{2} \cdot 15 = \$5 + \$7.50 = \$12.50 \\ \text{Quin} & = \frac{1}{3} \cdot 15 + \frac{1}{2} \cdot 15 + 15 = \$5 + \$7.50 + \$15 = \$27.50 \end{cases}$$

But Quin sends his reasoning and it looks like this:

$$\Longrightarrow \quad \begin{cases} \text{Lakhveer} & = \frac{1}{6} \cdot 45 = \$7.50 \\ \text{Morgan} & = \frac{2}{6} \cdot 45 = \$15 \\ \text{Quin} & = \frac{3}{6} \cdot 45 = \$22.50 \end{cases}$$

Explain each person's reasoning. Why do they get different answers? Who do you think is correct? Explain why theirs is better than the other.

Exercise 8.2. Prove that if A and B are nonempty sets and $A \subseteq B$, then $\sup(A) \leq \sup(B)$ and $\inf(B) \leq \inf(A)$.

Exercise 8.3. Suppose A and B are nonempty sets of real numbers such that for any $x \in A$ and $y \in B$ we have $x \leq y$. Prove that $\sup(A) \leq \inf(B)$.

Exercise 8.4. Let $f : [0, 2] \to \mathbb{R}$ be defined by $f(x) = x^2 - x$, and let $P = \{0, 1, \frac{3}{2}, 2\}$. Compute $U(f, P)$ and $L(f, P)$ with respect to this particular partition P.

Exercise 8.5. Suppose $a \leq b < c \leq d$ and that f is integrable on $[a, d]$. Prove that f is also integrable on $[b, c]$.

Exercise 8.6. Prove that $\displaystyle\int_0^b x \, dx = \frac{b^2}{2}$ by considering partitions into n equal subintervals. Hint: First prove by induction that $\displaystyle\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$.

Exercise 8.7. Prove that $\int_0^b x^2\,dx = \frac{b^3}{3}$ by considering partitions into n equal subintervals. Hint: First prove by induction that $\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$.

Exercise 8.8. Prove that $\int_0^b x^3\,dx = \frac{b^4}{4}$ by considering partitions into n equal subintervals. Hint: First prove by induction that $\sum_{k=1}^{n} k^3 = \frac{n^2(n+1)^2}{4}$.

Exercise 8.9. Consider the function $s : [0,2] \to \mathbb{R}$ given by

$$s(x) = \begin{cases} 1 & \text{if } x \in [0,1) \\ 5 & \text{if } x = 1 \\ 2 & \text{if } x \in (1,2] \end{cases}$$

Prove that s is integrable on $[0,2]$.

Exercise 8.10. For the following question you do not need to prove that your example works. Give an example of a function $f : [0,2] \to \mathbb{R}$ which has the following two properties:

- When restricted to the domain $[0,1]$ (so now $f : [0,1] \to \mathbb{R}$), we have

  $$L(f,P) < U(f,P)$$

  for every partition P of $[0,1]$.

- But when restricted to the domain $[1,2]$ (so now $f : [1,2] \to \mathbb{R}$), we have

  $$L(f,P) = U(f,P)$$

  for every partition P of $[1,2]$.

Exercise 8.11. Prove Corollary 8.14. That is, prove that if $f : [a,b] \to \mathbb{R}$ is integrable, then there exists a sequence P_n of partitions of $[a,b]$ for which

$$\lim_{n\to\infty} [U(f,P_n) - L(f,P_n)] = 0.$$

Exercise 8.12. A function f is *strictly increasing* on $[a,b]$ if for any x_1 and x_2 from $[a,b]$ where $x_1 < x_2$, we have that $f(x_1) < f(x_2)$. Prove that if f is strictly increasing on $[a,b]$, then f is integrable.

Exercise 8.13. Consider Thomae's function h restricted to $[0,2]$. That is,

$h : [0, 2] \to \mathbb{R}$ is given by

$$h(x) = \begin{cases} 1/n & \text{if } x \neq 0 \text{ and } x = m/n \in \mathbb{Q} \text{ in lowest terms with } n > 0 \\ 0 & \text{if } x \notin \mathbb{Q} \\ 1 & \text{if } x = 0 \end{cases}$$

We proved in class that h is continuous at every irrational number and discontinuous at every rational number. In this question you will prove that Thomae's function is integrable. To do this, complete the following steps:

(a) Prove that $L(h, P) = 0$ for any partition P of $[0, 2]$.

(b) Let $\varepsilon > 0$. And note why h is bounded.

(c) Determine whether there are finitely many points x such that $h(x) > \varepsilon/4$, or infinitely many such points. Explain your answer.

(d) Explain how to construct a partition P_ε of $[0, 2]$ where $U(h, P_\varepsilon) < \varepsilon$. And prove that partition works.

(e) State which theorem completes this proof.

Exercise 8.14. Give an example of a function $f : [0, 1] \to \mathbb{R}$ for which f is *not* integrable, but $(f(x))^2$ *is* integrable.

Exercise 8.15. Recall that the *modified Dirichlet function* is defined to be

$$g(x) = \begin{cases} x & \text{if } x \in \mathbb{Q} \\ 0 & \text{if } x \notin \mathbb{Q}. \end{cases}$$

(a) Let P be a partition of $[0, 4]$. Compute $L(g, P)$.

(b) Find $\inf\{U(g, P) : P \text{ a partition of } [0, 4]\}$.

Exercise 8.16.

(a) Suppose f is continuous on $[a, b]$ and $\int_a^x f(t)dt = 0$ for all $x \in [a, b]$. Prove that $f(x) = 0$ for all $x \in [a, b]$.

(b) Suppose f is continuous on $[a, b]$ and $\int_a^x f(t)dt = \int_x^b f(t)dt$ for all $x \in [a, b]$. Prove that $f(x) = 0$ for all $x \in [a, b]$. Hint: One slick proof uses part (a) and properties of integrals.

Exercise 8.17. Give an example of numbers a and b, and of integrable functions $f, g : [a, b] \to \mathbb{R}$, where

$$U(f + g, P) < U(f, P) + U(g, P)$$

for every partition P of $[a, b]$. Make sure to prove your answer.

Exercise 8.18. For each function f_k below, find a formula for $F_k(x) = \int_{-1}^{x} f_k(t)\,dt$. Where is F_k continuous? Where is F_k differentiable? Where does $F_k' = f_k$? You do not need to prove your answers.

(a) $f_1 : [-1, 1] \to \mathbb{R}$ given by $f(x) = |x|$.

(b) $f_2 : [-1, 1] \to \mathbb{R}$ given by

$$f_2(x) = \begin{cases} 1 & \text{if } x < 0 \\ 2 & \text{if } x \geq 0. \end{cases}$$

(c) $f_3 : [-1, 1] \to \mathbb{R}$ given by

$$f_3(x) = \begin{cases} 1 & \text{if } x \neq 0 \\ 2 & \text{if } x = 0. \end{cases}$$

Exercise 8.19. Use Theorem 8.23 to provide a second proof (easier than the way outlined in Exercise 8.13) that Thomae's function is integrable.

Exercise 8.20.

(a) Assume f is integrable on $[a, b]$, and $f(x) \geq 0$ for all x. Moreover, assume that $\int_a^b f(x)\,dx > 0$. Prove that there are infinitely many points x for which $f(x) > 0$.

(b) Assume g is integrable on $[a, b]$ and $g(a) \geq 0$ for all x. If $g(x) > 0$ for an infinite number of vales of x from $[a, b]$, must it be the case that $\int_a^b g(x)\,dx > 0$?

Exercise 8.21. Use the fundamental theorem of calculus (Theorem 8.31) and Darboux's theorem (Theorem 7.20) to give a second proof of the intermediate value theorem (Theorem 6.37).

Exercise 8.22. Suppose f is continuous, $f(x) \geq 0$ for all x, and for some $x_0 \in [a, b]$ we have $f(x_0) > 0$. Prove that $\displaystyle\int_a^b f(x)\,dx > 0$.

Exercise 8.23. Prove that

$$\int_1^a \frac{1}{x}\,dx + \int_1^b \frac{1}{x}\,dx = \int_1^{ab} \frac{1}{x}\,dx.$$

Exercise 8.24. Prove that

$$\int_{ca}^{cb} f(x)\,dx = c \cdot \int_a^b f(cx)\,dx$$

Exercise 8.25. Suppose that $f : [a, b] \to \mathbb{R}$ is bounded. Suppose further that f is continuous at every point in $[a, b]$ with the exception of a single point $x_0 \in (a, b)$. Prove that f is integrable on $[a, b]$.

Exercise 8.26. Suppose $f : [a, b] \to \mathbb{R}$ is integrable.

(a) Prove that there is some $x_0 \in [a, b]$ such that

$$\int_a^{x_0} f(t)\ dt = \int_{x_0}^b f(t)\ dt.$$

(b) Give an example showing that it's not always the case that $x_0 \in (a, b)$.

Exercise 8.27.

(a) Prove that if f is integrable on $[a, b]$ and $m \leq f(x) \leq M$ for all x in $[a, b]$, then

$$\int_a^b f(x)\ dx = (b - a) \cdot \gamma$$

for some γ with $m \leq \gamma \leq M$.

(b) Prove that if f is continuous on $[a, b]$, then

$$\int_a^b f(x)\ dx = (b - a) \cdot f(x_0)$$

for some $x_0 \in [a, b]$.

(c) Show by example that (b) need not hold if f is not continuous.

(d) Prove the following more general result (Part (c) is the special case $g(x) = 1$ for all x.), known as the *general mean value theorem for integrals*: Suppose that f is continuous on $[a, b]$ and g is integrable and nonnegative on $[a, b]$. Prove that

$$\int_a^b f(x)g(x)\ dx = f(x_0) \cdot \int_a^b g(x)\ dx$$

for some $x_0 \in [a, b]$.

Exercise 8.28. A function $s : [a, b] \to \mathbb{R}$ is called a *step function* if there is a partition $P = \{x_0, x_1, \ldots, x_n\}$ of $[a, b]$ such that s is constant on each interval (x_{i-1}, x_i).

(a) Prove that if f is integrable on $[a, b]$, then for any $\varepsilon > 0$ there is a pair of step functions s_1 and s_2 where $s_1(x) \leq f(x) \leq s_2(x)$ for all x, and

$$\int_a^b [f(x) - s_1(x)]\ dx < \varepsilon \qquad \text{and} \qquad \int_a^b [s_2(x) - f(x)]\ dx.$$

(b) Suppose that for any $\varepsilon > 0$ there is a pair of step functions s_1 and s_2 where $s_1(x) \leq f(x) \leq s_2(x)$ for all x, and

$$\int_a^b [f(x) - s_1(x)]\ dx < \varepsilon \qquad \text{and} \qquad \int_a^b [s_2(x) - f(x)]\ dx.$$

Prove that f is integrable.

Exercise 8.29. Define a *step function* as in Exercise 8.28. Prove that if $f : [a, b] \to \mathbb{R}$ is integrable, then for any $\varepsilon > 0$ there is a continuous function g where $g(x) \leq f(x)$ for all x, and

$$\int_a^b f(x)\ dx - \int_a^b g(x)\ dx < \varepsilon.$$

Hint: Use Exercise 8.28 to obtain a step function with this property, and then draw a picture and experiment how to turn this into a continuous function.

— Open Questions —

Question 1. Under what conditions does the inequality

$$\int_a^b f^{\alpha+\beta}(x)\ dx \geq \left(\int_a^b (x-a)^\alpha f^\beta(x)\ dx \right)^\lambda$$

hold for α, β and λ?

Question 2. Under what conditions does the inequality

$$\frac{\displaystyle\int_a^b f^{\alpha+\beta}(x)\ dx}{\displaystyle\int_a^b f^{\alpha+\gamma}(x)\ dx} \geq \frac{\displaystyle\left(\int_a^b (x-a)^\alpha f^\beta(x)\ dx \right)^\delta}{\displaystyle\left(\int_a^b (x-a)^\alpha f^\gamma(x)\ dx \right)^\lambda}$$

hold for $\alpha, \beta, \gamma, \delta$ and λ?

Chapter 9: Sequences and Series of Functions

9.1 Introduction to Pointwise Convergence

Recall.

Recall 9.1.

- We have looked at sequences of numbers. For example, if $a_k = k^2$, then (a_k) is notation for the sequence $a_1, a_2, a_3, a_4, \ldots$. For this particular a_k, this is the sequence $1, 4, 9, 16, \ldots$.
- $a_k \to a$ if for every $\varepsilon > 0$ there is some $N \in \mathbb{N}$ for which $|a_k - a| < \varepsilon$ for all $k \geq N$.

Similarly, we now consider sequences of *functions*. First, here's our notation:

Notation.

Notation 9.2. If, for each $k \in \mathbb{N}$, we have a function $f_k : A \to \mathbb{R}$, then

$$f_1, f_2, f_3, f_4, \ldots$$

is a *sequence of functions* and is denoted (f_k).

And like with sequences, we have a notion of convergence. (Note: This is one of two types of functional convergence we will study.)

Definition.

Definition 9.3. Suppose (f_k) is a sequence of functions, each defined on $A \subseteq \mathbb{R}$. The sequence (f_k) of functions *converges pointwise* to a function $f : A \to \mathbb{R}$ if, for each $x_0 \in A$,

$$\lim_{k\to\infty} f_k(x_0) = f(x_0).$$

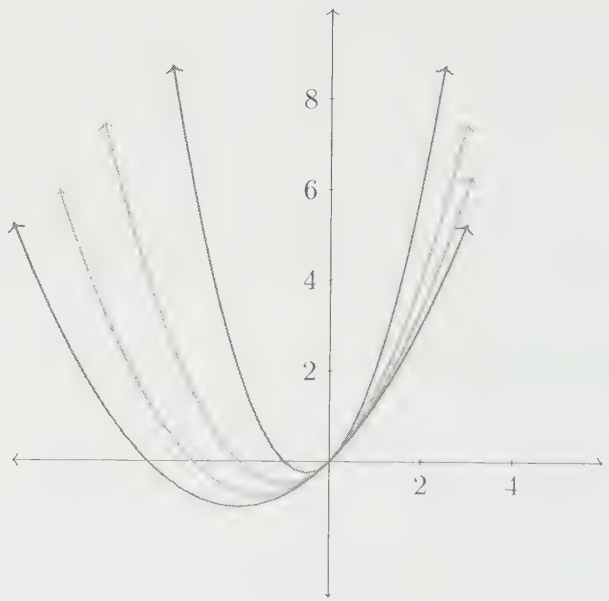
In this case, we write "$f_k \to f$ pointwise."

It is important to note that since x_0 is fixed, the sequence $f_k(x_0)$ is a sequence of *numbers*, it's not a sequence of functions: $f_1(x_0) \in \mathbb{R}$, $f_2(x_0) \in \mathbb{R}$, $f_3(x_0) \in \mathbb{R}$, ... So the sequence (f_k) of functions *converges pointwise* to a function $f : A \to \mathbb{R}$ if for each $x_0 \in A$ the sequence of real numbers $f_k(x_0)$ converges to the real number $f(x_0)$.

Consider the following example.

Example 9.4. For each $k \in \mathbb{N}$, let $f_k : \mathbb{R} \to \mathbb{R}$ be given by $f_k = \dfrac{x^2 + kx}{k}$. That is, for each fixed $n \in \mathbb{N}$ we have defined a function of x. Collectively we have defined a *sequence of functions*:

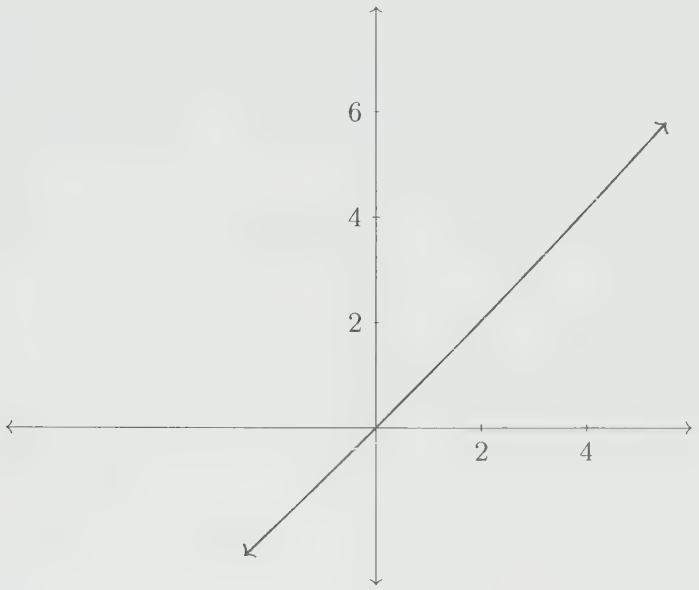
$$f_1 = x^2 + x$$
$$f_2 = \frac{x^2 + 2x}{2}$$
$$f_3 = \frac{x^2 + 3x}{3}$$
$$f_4 = \frac{x^2 + 4x}{4}$$
$$\vdots$$



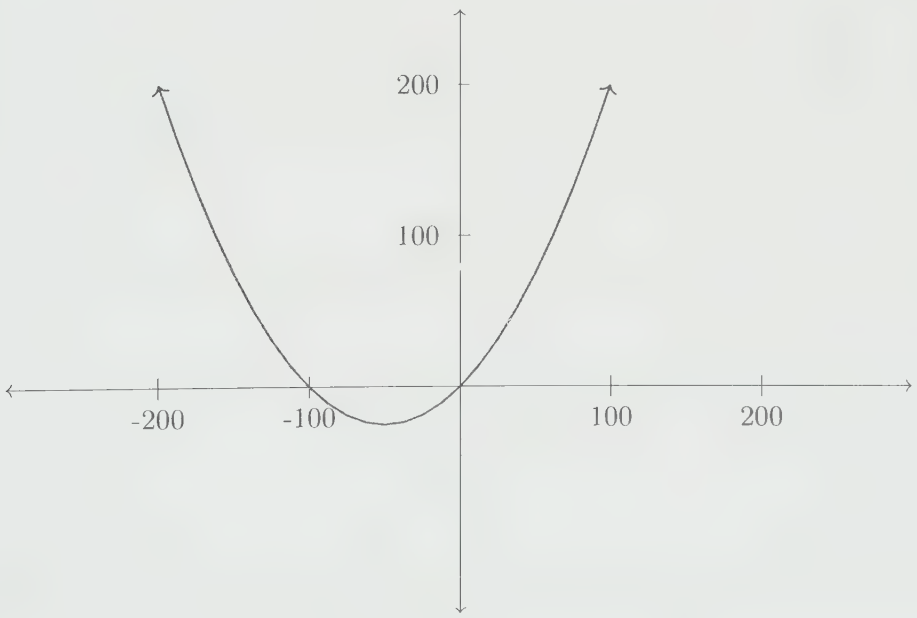
The first four iterations (f_1, f_2, f_3 and f_4) are graphed[1] above on the right.

[1] Geogebra can demonstrate the whole sequence well, with an enormously satisfying slider bar!

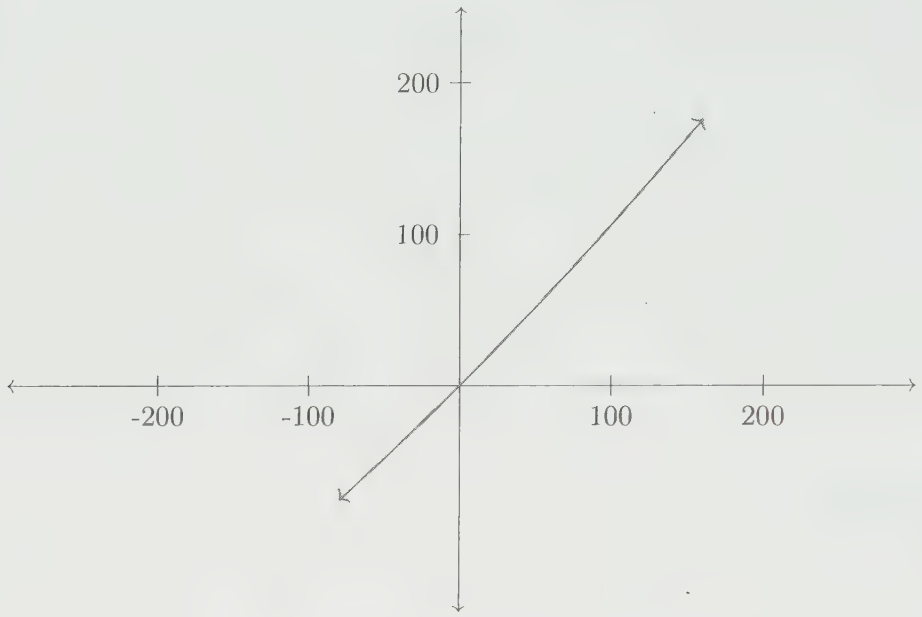
These functions look like nice parabolas, but take a look what happens when n gets large. Here's f_{100}:



Why does it look more like a line than a parabola? Well, we are just too zoomed in! Zooming out,



But if we then plot $f_{1,000}$ on these same x-values, it again looks like a line:



So you see, for any particular range of x-values like this, we can crank up n high enough to make the parabola look like a line (even though if you zoom out far enough it will look like a parabola again).

The way we think about this is that, for any particular $x_0 \in \mathbb{R}$, if you let $k \to \infty$ we have

$$\lim_{k\to\infty} f_k(x_0) = \lim_{k\to\infty} \frac{{x_0}^2 + kx_0}{k} = \lim_{k\to\infty} \left(\frac{{x_0}^2}{k} + x_0 \right) = 0 + x_0 = x_0.$$

That is, in general we have $\lim_{k\to\infty} f_k(x) = x$, meaning that (f_k) converges pointwise to $f(x) = x$. □

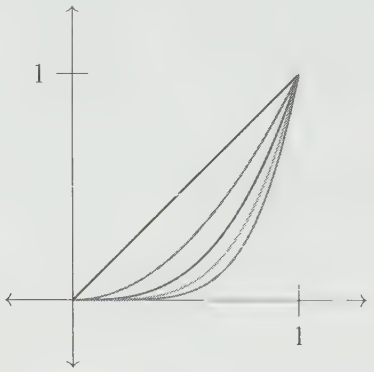
9.2 Continuity and Functional Convergence

Suppose we know $f_k \to f$ pointwise. We want to start investigating which properties of f_k will carry over to f. Here's a basic question:

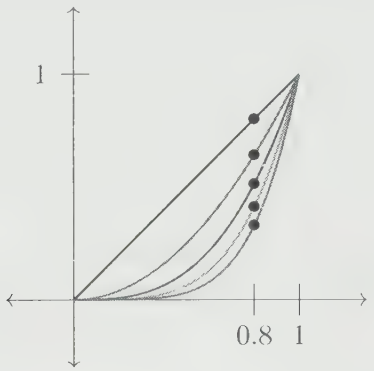
Question 9.5. If $f_k \to f$ pointwise, and each f_k is continuous, must f also be continuous?

After some thought you might realize that the answer is 'no.' What's a counterexample? One simple way to have a discontinuous function is to have a jump discontinuity at just a single point, as the next example shows.

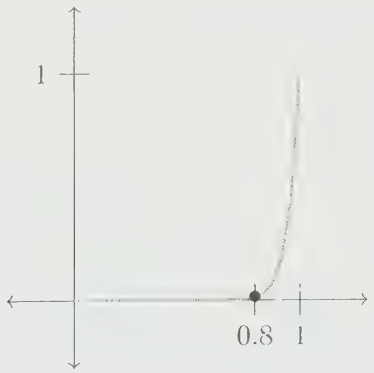
Example 9.6. Let $f_k : [0,1] \to \mathbb{R}$ where $f_k(x) = x^k$. This gives us a sequence of functions (f_k). The first five iterations ($k = 1, 2, 3, 4, 5$) look like this:



As you can see, for any fixed $x \in [0,1)$, as k gets large we see that x^k is tending to zero. For example, fix $x = 0.8$; the point $f_k(0.8)$ is marked on each of the graphs, and you can tell it's tending towards zero:



For x-values close to 1 it takes longer, but all will eventually get arbitrarily close to 0. For instance, below is $f_{20}(x)$'s graph (with $f_{20}(0.8)$ again marked):



Plugging a few points into the function, we have $f_{20}(0.8) = 0.01$, $f_{20}(0.9) = 0.12$, and $f_{20}(0.99) = 0.82$, so really all but the last 10% of the domain is becoming quite close to 0 by $n = 20$. And $f_{500}(0.99) = 0.01$, so by $n = 500$ far less than 1% of interval remains far from 0.

There is of course one point that will never get close to 0: the point $x = 1$. For any k, $f_k(1) = 1^k = 1$. To summarize:

$$\begin{cases} \lim_{k\to\infty} f_k(x) = 0 & \text{if } x \in [0,1), \\ \lim_{k\to\infty} f_k(x) = 1 & \text{if } x = 1. \end{cases}$$

So we have shown that the sequence of continuous functions $f_k = x^k$ converge pointwise to the discontinuous function

$$f(x) = \begin{cases} 0 & \text{if } x \in [0,1), \\ 1 & \text{if } x = 1. \end{cases}$$

□

This is kind of a bummer. We might have hoped that pointwise convergence would preserve the continuity property, but sadly it is not strong enough. Now, there is another, stronger notion of a functional limit that *is* strong enough to maintain continuity in the limit. To understand where it comes from, let's try to prove that pointwise convergence preserves continuity, and see where the proof breaks down (and hence how to fix it).

***Almost*-correct proof that pointwise convergence preserves continuity.** Assume that each $f_k : A \to \mathbb{R}$ is continuous, and that $f_k \to f$ pointwise. Fix a point $c \in A$ and an $\varepsilon > 0$. To show that f is continuous at c we would have to show that there is some $\delta > 0$ such that

$$|x - c| < \delta \qquad \Longrightarrow \qquad |f(x) - f(c)| < \varepsilon.$$

Rewriting $|f(x) - f(c)|$ in a tricky way and applying the triangle inequality twice,

$$\begin{aligned} |f(x) - f(c)| &= |f(x) - f_k(x) + f_k(x) - f_k(c) + f_k(c) - f(c)| \\ &\leq |f(x) - f_k(x)| + |f_k(x) - f_k(c)| + |f_k(c) - f(c)| \\ &= |f_k(x) - f(x)| + |f_k(x) - f_k(c)| + |f_k(c) - f(c)|. \end{aligned}$$

Notice how this works for *any* choice of k. If we can make each term less than $\varepsilon/3$, then we would have a complete proof; so what goes wrong? You might think:

- We can make the first term small because $f_k \to f$ pointwise.
- We can make the second term small because each f_k is continuous.
- We can make the third term small because $f_k \to f$ pointwise.

One of these doesn't work though... Can you spot it...? It's actually the first term, but to really see why, let's look at the others first, in reverse order:

- We can indeed make the third term small: Since c is fixed we can choose some k_0 such that $|f_{k_0}(c) - f(c)| < \varepsilon/3$.

- We can then make the second term small: We had assumed that each f_k is continuous, and so by f_{k_0}'s continuity there exists some $\delta_0 > 0$ such that $|f_{k_0}(x) - f_{k_0}(c)| < \varepsilon/3$ for all x where $|x - c| < \delta_0$.

- But this is a problem for the first term. For each particular x, we can apply pointwise convergence to produce a big enough k_0 for which $|f_{k_0}(x) - f(x)| < \varepsilon/3$. However, each x will likely have a different "k_0" value, and so we have no guarantee that any particular k_0 is big enough for *all* of the infinitely many x for which $|x - c| < \delta_0$.

  But in the event that our k_0 is not big enough, why don't we just go back and pick a large enough k_0 so that it does work for all these x? Well first, maybe no k_0 is big enough for them all (there are infinitely many x-values and each requires a large k, so maybe there's no upper bound on the needed k's). But more to the point, even if there was a max k-value (call it k_1) that was required and we went back and switched out k_0 for this k_1, then suddenly δ_0 likely changes (to some δ_1), since it is now dependent on the function f_{k_1} and not f_{k_0}. So we found a new k_1 that works for all the x-values for which $|x - c| < \delta_0$, but with our new k_1 we instead need it to work for all the x-values for which $|x - c| < \delta_1$!

  See? Because x depends on δ_0, which in turn depends on k_0, we can't just change k_0 without messing everything up.

So the problem is in controlling $|f_k(x) - f(x)|$ for all x. Indeed, in Example 9.6, each $f_k(x) = x^k$ was continuous, but as k got large, x-values close to 1 required much smaller δ-values than did the x away from 1. This shows why we were unable to control the $|f_k(x) - f(x)|$ term in this sequence.

Pointwise convergence allows us to control each individual x, but we want to control them all at the same time; that is, we want *uniform* control. This directly motivates the following definition of *uniform convergence*.

Uniform Convergence

Definition.

Definition 9.7. Let (f_k) be a sequence of functions defined on a set A. We say (f_k) *converges uniformly on* A to a function f if, for every $\varepsilon > 0$, there exists an $N \in \mathbb{N}$ such that $|f_k(x) - f(x)| < \varepsilon$ for all $k \geq N$ and for all $x \in A$.

When this happens we write "$f_k \to f$ uniformly."

We constructed this form of convergence so that it will preserve continuity, so of course our first result is that it does indeed do this.

Proposition.

Proposition 9.8 (*f_k continuous, $f_k \to f$ uniformly $\Rightarrow$ f continuous*)**.** Assume that each $f_k : A \to \mathbb{R}$ is continuous at some $c \in A$. If (f_k) converges uniformly to f, then f is also continuous at c.

Proof. Fix a $c \in A$ and let $\varepsilon > 0$. By the uniformity assumption, there exists some $N \in \mathbb{N}$ such that

$$|f_k(x) - f(x)| < \frac{\varepsilon}{3}$$

for all $k \geq N$ and for all $x \in A$. Note that this in particular means that $|f_N(c) - f(c)| < \frac{\varepsilon}{3}$. We also assumed that each f_k is continuous at c. In particular, since f_N is continuous at c there must exist a $\delta > 0$ for which

$$|x - c| < \delta \qquad \text{implies} \qquad |f_N(x) - f_N(c)| < \frac{\varepsilon}{3}.$$

And so, by rewriting $|f(x) - f(c)|$ in a tricky way, applying the triangle inequality twice and using the above,

$$\begin{aligned} |f(x) - f(c)| &= |f(x) - f_N(x) + f_N(x) - f_N(c) + f_N(c) - f(c)| \\ &\leq |f(x) - f_N(x)| + |f_N(x) - f_N(c)| + |f_N(c) - f(c)| \\ &= |f_N(x) - f(x)| + |f_N(x) - f_N(c)| + |f_N(c) - f(c)| \\ &< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} \\ &= \varepsilon \end{aligned}$$

for all $x \in A$ for which $|x - c| < \delta$. Thus by the definition of continuity, f is continuous at c. □

9.3 Other Properties with Functional Convergence

We want to investigate which properties are preserved by pointwise and uniform convergence. We aim to complete the following chart.

Assume that each f_k has the below property	If $f_k \to f$ pointwise, must f satisfy property?	If $f_k \to f$ uniformly, must f satisfy property?
Continuous	No, by Example 9.6	Yes, by Proposition 9.8
Bounded	???	???
Unbounded	???	???
Uniformly Continuous	???	???
Differentiable	???	???
Integrable	???	???

We won't get to all of these, but ticking off as many as we can will help us better understand the differences between pointwise and uniform convergence. It's also pretty fun to think about finding a proof of counterexample of each of these. Let's start at the top—boundedness.

Boundedness

We thought about whether boundedness carries over. We discovered that for pointwise convergence it might not, but with uniform convergence it does. We begin with an example showing the pointwise claim.

Example 9.9. Assume that each f_k is bounded and $f_k \to f$ pointwise. It need not be the case that f is also bounded. Here is an example of that:

Define $f_k : (0,1] \to \mathbb{R}$ by

$$f_k(x) = \begin{cases} \frac{1}{x} & \text{if } x \in [1/k, 1] \\ 0 & \text{if } x \in (0, 1/k). \end{cases}$$

Clearly $0 \leq f_k(x) \leq k$ for each x, so each f_k is bounded.[2]

However, for any $x \in (0,1]$, by the Archimedean principle there is some $N \in \mathbb{N}$ for which $\frac{1}{N} < x$. And thus for all $k \geq N$ we have $f_k(x) = 1/x$. So clearly for each x we have

$$\lim_{k\to\infty} f_k(x) = \frac{1}{x}.$$

That is, by letting $f(x) = 1/x$ we know that each f_k is bounded, f is unbounded, and $f_k \to f$ pointwise on $(0,1]$. So we are done. □

But while pointwise convergence does not preserve boundedness, we prove next that uniform convergence does.

Proposition.

Proposition 9.10 (*f_k bounded, $f_k \to f$ uniformly $\Rightarrow$ f bounded*)**.** Assume that each $f_k : A \to \mathbb{R}$ is bounded. If (f_k) converges uniformly to f, then f is also bounded.

Proof. Since $f_k \to f$ uniformly, then (by letting $\varepsilon = 1$) we know that there exists some $N \in \mathbb{N}$ such that

$$|f_k(x) - f(x)| < 1$$

for all $k \geq N$ and $x \in A$. In particular,

$$|f_N(x) - f(x)| < 1$$

[2]Recall again that for each particular f_k, the number k is fixed. So, eg., $0 \leq f_{100}(x) \leq 100$. If we knew that $g : \mathbb{R} \to \mathbb{R}$ had the property $g(x) \leq x$, then this would not mean that g is bounded, because x is variable. But saying $f_k(x) \leq k$ does mean that f_k is bounded above, since k is fixed.

for all $x \in A$. I.e.,

$$f_N(x) - 1 < f(x) < f_N(x) + 1$$

for all $x \in A$. And since f_N was assumed to be bounded, there exist real numbers L and U such that $L \leq f_N(x) \leq U$ for all $x \in A$. Combining this with above, we see that

$$L - 1 < f(x) < U + 1$$

for all $x \in A$, proving that f is bounded. □

Unboundedness

These will be left for Exercise 9.3.

Uniform continuity

In case you need a reminder, below is the definition of *uniform continuity.*

Recall.

Recall 9.11. Let $f : A \to \mathbb{R}$. We say f is *uniformly continuous* if for all $\varepsilon > 0$ there exists some $\delta > 0$ such that, for all $x, y \in A$,

$$|x - y| < \delta \qquad \text{implies} \qquad |f(x) - f(y)| < \varepsilon.$$

This section will be brief. I will tell you one example where each f_k is uniformly continuous but yet f is not uniformly continuous, where $f_k \to f$ pointwise. For uniform convergence, though, it does hold. I will state that as a proposition, but leave it to you to prove in the exercises. First up, the example.

Example 9.12. Assume that each f_k is uniformly continuous and $f_k \to f$ pointwise. It need not be the case that f is also uniformly continuous. In fact, our sequence from when we discussed continuity still works:

Define $f_k : [0, 1] \to \mathbb{R}$ by $f_k(x) = x^k$. Since each f_k is a continuous function on a closed interval, by Proposition 6.39 we know that each f_k is uniformly continuous. However, as we saw in Example 9.6, $f_k \to f$ pointwise where

$$f(x) = \begin{cases} 0 & \text{if } x \in [0, 1) \\ 1 & \text{if } x = 1 \end{cases}.$$

And note that f is not uniformly continuous since it is not even continuous. □

Proposition.

Proposition 9.13 (*f_k unif. cts., $f_k \to f$ uniformly $\Rightarrow$ f unif. cts.*). Assume that each $f_k : A \to \mathbb{R}$ is uniformly continuous. If (f_k) converges uniformly to f, then f is also uniformly continuous.

Proof. This is Exercise 9.12. □

As a quick reminder, here is where we are in our efforts to complete the table from earlier:

Assume that each f_k has the below property	If $f_k \to f$ pointwise, must f satisfy property?	If $f_k \to f$ uniformly, must f satisfy property?
Continuous	No, by Example 9.6	Yes, by Proposition 9.8
Bounded	No, by Example 9.9	Yes, by Proposition 9.10
Unbounded	Exercise 9.3	Exercise 9.3
Uniformly Continuous	No, by Example 9.12	Yes, by Prop 9.13 and Ex 9.12
Differentiable	???	???
Integrable	???	???

9.4 Convergence of Derivatives and Integrals

Let's keep going. Next up, differentiability and integrability. As you may have guessed, the weakness of pointwise convergence persists. For these more sophisticated properties, though, it is less obvious right off the bat whether uniform convergence is strong enough to preserve the properties.

Differentiability

Assume each f_k is differentiable at a point c. It turns out that (f_k) converging to some f pointwise does not guarantee that f is differentiable. But even stronger, it's also the case that $f_k \to f$ uniformly does not guarantee that f is differentiable. So for this subsection, we can give just one example that shows that neither convergence is sufficient.

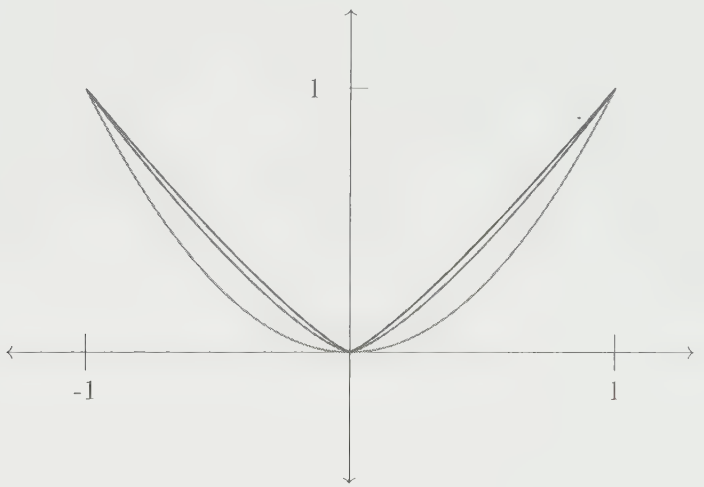
Example 9.14. Assume that each f_k is differentiable and $f_k \to f$ uniformly (and hence also pointwise). It need not be the case that f is also differentiable. Here is an example of that:

We will consider the functions $f_k : [-1, 1] \to \mathbb{R}$, defined by $f_k(x) = x^{1+\frac{1}{2k-1}}$; note that, indeed, f_k is defined on $[-1, 1]$, since $2k - 1$ is always odd. For any $k \in \mathbb{N}$ the function f_k is differentiable at $x = 0$. (In fact, each f_k is differentiable on all of $[-1, 1]$ by the power rule.) To understand the limit of this function as $k \to \infty$, note that

$$\lim_{k\to\infty} x^{\frac{1}{2k-1}} = \begin{cases} 1 & \text{if } x \in (0, 1] \\ -1 & \text{if } x \in [-1, 0) \\ 0 & \text{if } x = 0 \end{cases}$$

And so $f_k(x) = x^{1+\frac{1}{2k-1}} = x \cdot x^{\frac{1}{2k-1}} \to |x|$. So $f_k \to f$ pointwise where $f(x) = |x|$, which is not differentiable at $x = 0$, despite each f_k being differentiable at 0. (Challenge: Show that we also have $f_k \to f$ uniformly.) □

The first three functions in Example 9.14 are graphed below (note how they are already looking very much like $f(x) = |x|$).



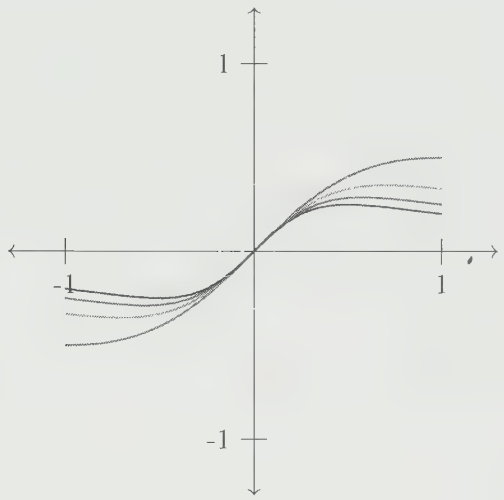
By looking at those first few graphs it seems pretty convincing that the functions do converge uniformly. Showing it requires a little finesse, though.

What's particularly amazing, though, is that there even exists a function g which has this:

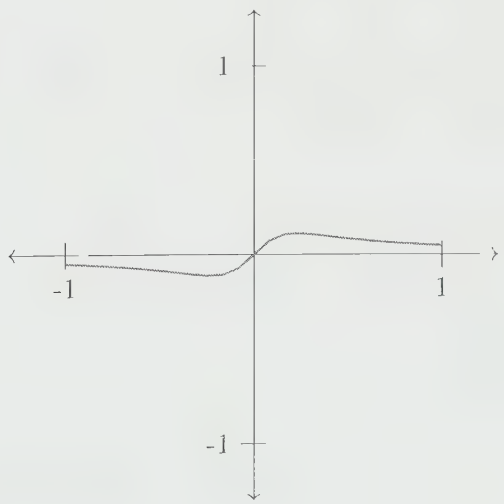
1. A sequence (g_k) of functions such that $g_k \to g$ uniformly,
2. Each g_k is differentiable at a point c,
3. g is also differentiable at c,
4. And the limit $\lim_{k\to\infty} g_k'(c)$ exists,
5. But yet $\lim_{k\to\infty} g_k'(c) \neq g'(c)$.

Weird! The idea, though, is that you can be getting really close to a function but yet at a particular point, by using the ε-high wiggle room, you can achieve any slope you want. In the below, for instance, the functions are getting closer and closer to 0 (whose derivative is 0 everywhere). However, right at zero they re-angle themselves so that they go through the origin with a slope of 1, giving $f_k'(0) = 1$ for each n. Here's that:

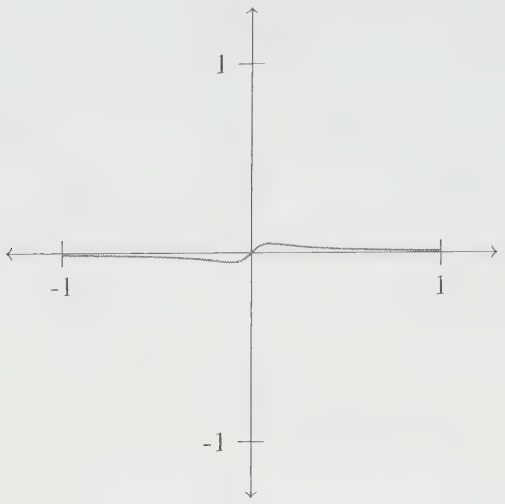
Example 9.15. Let $g_k : [-1, 1] \to \mathbb{R}$ be defined by $g_k(x) = \dfrac{x}{1 + nx^2}$, The first few of four functions looks like this:



The later functions are closer to the x-axis. For instance, here is g_{20}:



And here is g_{100}:



So as you can see, the sequence (g_k) is converging both pointwise and uniformly to the function $g(x) = 0$. However, despite this, each $g_k'(0) = 1$, since right at $x = 0$ each of the functions will reorient itself so that if passes through the origin with a slop of 1. Indeed, by the quotient rule,

$$\frac{d}{dx}\left(\frac{x}{1+kx^2}\right) = \frac{1\cdot(1+kx^2) - x\cdot(2kx)}{(1+kx^2)^2} = \frac{1-kx^2}{(1+kx^2)^2}.$$

So $g_k'(0) = \dfrac{1-k(0)^2}{(1+k(0)^2)^2} = 1.$

So we have shown that each $g_k'(0) = 1$, but meanwhile the g_k's converge to the zero function $g(x) = 0$ which clearly has $g'(0) = 0$. □

Integrability

Example 9.16. Assume that each f_k is integrable and $f_k \to f$ pointwise. It need not be the case that f is also integrable. For Exercise 9.6 you will give such an example. □

What about for uniform limits? We showed in the last section that uniform limits do not preserve differentiability. For integrability, though, this property is preserved.

Proposition.

Proposition 9.17 (*f_k integrable, $f_k \to f$ uniformly $\Rightarrow$ f integrable*)**.** Assume that each $f_k : [a, b] \to \mathbb{R}$ is integrable. If (f_k) converges uniformly to f, then f is also integrable.

Before we begin, recall that if $I = [x_{i-1}, x_i]$ is a subinterval in a Riemann sum,

$$\sup(g(x) + h(x) : x \in I) \leq \sup(g(x) : x \in I) + \sup(h(x) : x \in I)$$
$$\inf(g(x) + h(x) : x \in I) \geq \inf(g(x) : x \in I) + \inf(h(x) : x \in I)$$

We will use these in the following proof. As for the second, we will in particular use the equivalent statement

$$-\inf(g(x) + h(x) : x \in I) \leq -\inf(g(x) : x \in I) - \inf(h(x) : x \in I).$$

Now we prove the proposition

Proof. We will prove that f is integrable by using the definition of integrability. That is, we will show that f is bounded and $L(f) = U(f)$. For the former, simply note that since each f_k is integrable and $f_k \to f$ uniformly, by Proposition 9.10 we may conclude that f is also bounded.

Now we prove that $L(f) = U(f)$. Let $\varepsilon > 0$. Since $f_k \to f$ uniformly there exists some $N \in \mathbb{N}$ such that

$$|f_k(x) - f(x)| < \frac{\varepsilon}{2(b-a)}$$

for all $k \geq N$ and for all $x \in [a, b]$. Then,

$$U(f) - L(f) = U(f - f_N + f_N) - L(f - f_N + f_N).$$

So by the recall right before this proof,

$$\leq U(f - f_N) + U(f_N) - L(f - f_N) - L(f_N).$$

And since f_N is integrable we have $U(f_N) = L(f_N)$,

$$= U(f - f_N) - L(f - f_N).$$

By our choice of N above, $|f(x) - f_N(x)| < \frac{\varepsilon}{2(b-a)}$. That is, $\frac{-\varepsilon}{2(b-a)} < f(x) - f_N(x) < \frac{\varepsilon}{2(b-a)}$. Thus,

$$\leq U\left(\frac{\varepsilon}{2(b-a)}\right) - L\left(\frac{-\varepsilon}{2(b-a)}\right)$$
$$= \frac{\varepsilon}{2} + \frac{\varepsilon}{2}$$
$$= \varepsilon.$$

So we have that $0 \leq U(f) - L(f) \leq \varepsilon$ for all $\varepsilon > 0$. This implies that $U(f) = L(f)$, and hence that f is integrable. □

Completed Table

Assume that each f_k has the below property	If $f_k \to f$ pointwise, must f satisfy property?	If $f_k \to f$ uniformly, must f satisfy property?
Continuous	No, by Example 9.6	Yes, by Proposition 9.8
Bounded	No, by Example 9.9	Yes, by Proposition 9.10
Unbounded	Exercise 9.3	Exercise 9.3
Uniformly Continuous	No, by Example 9.12	Yes, by Prop 9.13 and Ex 9.12
Differentiable	No, by Example 9.14	No, by Example 9.14
Integrable	Exercise 9.6	Yes, by Proposition 9.17

Table 9.1: Properties of the limit of (f_k)

Some Final Results

In this section I want to list some final results about sequences of functions. The first will be quite useful when studying series of functions. The second is important in its own right. Both are generalizations of analogous results about numerical sequences that we learned in Chapter 4.

Proposition.

Proposition 9.18 (*Cauchy criterion for sequences of functions*). Let $f_k : A \to \mathbb{R}$. The sequence (f_k) converges uniformly if and only if for every $\varepsilon > 0$ there exists some $N \in \mathbb{N}$ such that $|f_m(x) - f_n(x)| < \varepsilon$ for all $m, n \geq N$ and all $x \in A$.

Proof. This is Exercise 9.8. □

In Chapter 3 we studied sequences of numbers. There are some theorems about numerical sequences that have analogous results for functional sequences. Recall the Bolzano-Weierstrass theorem:

Theorem.

Theorem 3.37 (*The Bolzano-Weierstrass theorem*). Every bounded sequence has a convergent subsequence.

There is an analogous theorem for sequences of functions (f_k), where the conclusion is that there exists a subsequence of (f_k) that converges uniformly. It's a famous

result known as the *Arzela-Ascoli theorem.* Bolzano-Weierstrass requires that the numerical sequences are bounded (there exists some M such that $|a_k| \leq M$ for all k), and likewise, Arzela-Ascoli requires that the functions be bounded. However note that if $f_k(x) = k$ for all x then each f_k is bounded but yet clearly (f_k) is not converging to any particular f. So we need more. It is not enough for each f_k to be bounded; we need the whole sequence of functions to be bounded. Instead, we need there to exist an M such that $|f_k(x)| \leq M$ for all x and for all k; this is called being *uniformly bounded.*

Unfortunately, being uniformly bounded is alone not enough to get a convergent subsequence (either pointwise or uniform). Convince yourself that every subsequence of $(f_k = x^k)$ on $[0, 1]$ converges to the same limit, and no subsequence converges uniformly to that limit. The problem is that the functions are getting steeper and steeper; each is uniformly continuous, but for later and later functions in the sequence, the "ε" in the definition of uniform continuity requires a smaller and smaller "δ."

What we need is that, for every $\varepsilon > 0$, there exists a single $\delta > 0$ that works for *all* functions f_k. Then, for this δ, we have that $|x-y| < \delta$ implies that $|f_k(x)-f_k(y)| < \varepsilon$ for all $x, y \in A$ and for all $k \in \mathbb{N}$. This is called being *equicontinuous.*

Theorem.

Theorem 9.19 (*The Arzela-Ascoli theorem*)**.** If (f_k) is uniformly bounded and equicontinuous on A, then (f_k) contains a uniformly convergent subsequence.

The proof is a diagonalization-type argument, that we omit.

Brief Review of Series of Numbers

After studying sequences of numbers in Chapter 3, we moved on to studying series of numbers in Chapter 4. Likewise, now that we have studied sequences of functions, we are about to move on to *series of functions*. There are a handful of ideas from Chapter 4 that will be particularly important as we move forward. We will now briefly review those ideas.

- In Chapter 4 we had a definition for what it means for a series to converge, in terms of the sequence of partial sums (Definition 4.1). In the below we write that, plus state a pair of different ways to write that definition, and finally at the end we state the Cauchy Criterion for convergence, which is equivalent to converging.

$$\begin{aligned}
\sum_{k=1}^{\infty} a_k \text{ converges } &\Leftrightarrow \text{Sequences of partial sums converges} \\
&\Leftrightarrow (S_n) \text{ converges, where } S_n = \sum_{k=1}^{n} a_k \\
&\Leftrightarrow \lim_{n\to\infty} \sum_{k=1}^{n} a_k \text{ exists} \\
&\Leftrightarrow \forall \varepsilon > 0 \;\; \exists N \;\; \text{s.t.} \;\; \left| \sum_{j=m}^{n} a_j \right| < \varepsilon \quad \forall n \geq m \geq N.
\end{aligned}$$

- We also proved many tests for convergence, including the k^{th}-term test, geometric series test, comparison test, series p-test and alternating series test (respectively, Propositions 4.5, 4.9, 4.12, 4.16, and 4.17).

- In particular, for $\sum_{k=1}^{\infty} a_k$ to converge, it must be the case that $a_k \to 0$. The converse, though, is not true. For example, the harmonic series $\sum_{k=1}^{\infty} \frac{1}{k}$ diverges to infinity even though $\frac{1}{k} \to 0$ (Proposition 4.15).

9.5 Series of Functions

We will define both pointwise and uniform convergence of series, although almost all of our focus will be on uniform convergence. The reason is, as we have seen, pointwise convergence is so weak that it does almost nothing worthwhile.

Definition.

Definition 9.20. Let (f_n) be a sequence of functions defined on a set A and let $s_n = \sum_{k=1}^{n} f_k$ be the n^{th} partial sum.

(i) The series $\sum_{k=1}^{\infty} f_k$ *converges pointwise* to a function $f : A \to \mathbb{R}$ if the sequence (s_n) converges pointwise to f.

(ii) The series $\sum_{k=1}^{\infty} f_k$ *converges uniformly* to a function $f : A \to \mathbb{R}$ if the sequence (s_n) converges uniformly to f.

Note that if each f_k is continuous, then $s_n = \sum_{k=1}^{n} f_k$ is also continuous. And, likewise, finite sums also preserve boundedness, uniform continuity, differentiability and integrability. Therefore, if you know that each f_n satisfies one of these properties, then s_n will too. This fact, along with Definition 9.20, allows us to recast the results of Table 9.1

Assume that each f_k has the below property	Must each s_n satisfy property?	If $\sum_{k=1}^{\infty} f_k$ converges uniformly to f, must f satisfy property?
Continuous	Yes	Yes, by Proposition 9.8
Bounded	Yes	Yes, by Proposition 9.10
Uniformly Continuous	Yes	Yes, by Prop 9.13 and Ex 9.12
Differentiable	Yes	No
Integrable	Yes	Yes, by Proposition 9.17

Table 9.2: Properties of $\sum_{k=1}^{\infty} f_k$

Since a series is just a sequence of partial sums, by rewriting each "$\sum_{k=1}^{\infty} f_k$" as "$\lim_{n\to\infty} s_n$" we can apply all of our previous results to series to complete the above table. The one exception to this is with differentiation; we won't go into details here, but to guarantee that the sum is differentiable you need a couple further assumptions.

Next we generalize the Cauchy criterion to series of functions.

Proposition.

Proposition 9.21 (*Cauchy criterion for series of functions*). Let $f_k : A \to \mathbb{R}$. The series $\sum_{k=1}^{\infty} f_k$ converges uniformly on A if and only if for every $\varepsilon > 0$ there exists some $N \in \mathbb{N}$ such that

$$\left| \sum_{k=m}^{n} f_k(x) \right| < \varepsilon$$

for all $n \geq m \geq N$ and all $x \in A$.

Proof. This is Exercise 9.14. □

As with every other use of the Cauchy criterion, the advantage to using Cauchyness is that you can show convergence without knowing what the sequence/series is converging to. That said, this criterion is still usually hard to apply. Its corollary, though, is much easier to apply.

The above gives a condition that guarantees that a series of functions converges uniformly, but oftentimes it is a cumbersome condition to apply. The following corollary will give another such condition. It is weaker and so it does not always work, but when it does it is often much easier to apply. We will build up to it by discussing how "tame" such functions will have to be for their sum to converge uniformly.

- If each $f_k(x) = k$, then the functions are all fairly "tame," since each is a constant. But *collectively* they are not tame. Indeed, certainly $\sum_{k=1}^{\infty} f_k$ does not converge uniformly; it diverges to ∞ everywhere! So having functions which are each individually "nice" is not enough. And certainly they can't go off to infinity.

- But what does "collectively tame" mean? If $f_k(x) = 1$ for all k, then certainly $\sum_{k=1}^{\infty} f_k$ does not converge uniformly; it again diverges to ∞ everywhere! So having functions that don't go off to infinity is not enough, and also having

them be close together is not enough, because their sum could still go off to infinity.

- Let $f_k(x) = \frac{x^k}{k}$ on $[0,1]$. The function $f_1(x) = x^k$ is already pretty small for most $x \in [0,1]$, and then for larger k the functions f_k are getting even smaller and smaller. For instance, f_{100} only takes values between 0 and 0.01. Nevertheless, $\sum f_k$ is still not converging to any f. The reason is that at $x = 1$ we have $f_k(1) = 1/k$ for each k. And so at this x-value,

$$\sum_{k=1}^{\infty} f_k(1) = \sum_{k=1}^{\infty} \frac{1}{k} = \infty,$$

as this is the harmonic series which in Chapter 4 we saw diverges. So it is *not* true that the sum converges for *all* $x \in [0,1]$, and hence the sum $\sum_{k=1}^{\infty} f_k$ does *not* converge to an f. Indeed, if it did then we would have to have $f(1) = \infty$, which is not allowed. One way to phrase an issue with the above is that $|f_k(x)| \leq \frac{1}{k}$, but $\sum \frac{1}{k}$ does not converge.

- So not only do we need bounds on the f_k, but we also need those bounds to converge. The corollary is called the Weierstrass M-test.

Corollary.

Corollary 9.22 (*Weierstrass M-test*). Let $f_k : A \to \mathbb{R}$ and suppose that, for each $k \in \mathbb{N}$, there exists $M_k \in \mathbb{R}$ such that $|f_k(x)| \leq M_k$ for all $x \in A$. If $\sum_{k=1}^{\infty} M_k$ converges then $\sum_{k=1}^{\infty} f_k$ converges uniformly on A.

Proof Sketch. The general outline of the proof is this:

$$\begin{aligned} \textstyle\sum M_k \text{ converges } &\Rightarrow M_k \text{ Cauchy} \\ &\Rightarrow \textstyle\sum f_k \text{ Cauchy} \\ &\Rightarrow \textstyle\sum f_k \text{ converges.} \end{aligned}$$

Now we prove it in detail.

Proof. Let $\varepsilon > 0$. Since $\sum_{k=1}^{\infty} M_k$ converges, the partial sums of $\sum_{k=1}^{\infty} M_k$ are Cauchy. Thus, there exists some $N \in \mathbb{N}$ where for all $n \geq m \geq N$ we have

$$\sum_{k=m}^{n} M_k < \varepsilon.$$

Thus, for this same N, by the triangle inequality we have

$$\left| \sum_{k=m}^{n} f_k(x) \right| \leq \sum_{k=m}^{n} |f_k(x)| \leq \sum_{k=m}^{n} M_k < \varepsilon,$$

for all $n \geq m \geq N$. So by the Cauchy criterion for series of functions (Proposition 9.21), $\sum_{k=1}^{\infty} f_k$ converges uniformly on A. □

Here's a quick example of using the Weierstrass M-test on a specific function.

Example 9.23. Let $f_k(x) = \dfrac{1}{x^4 + 3x^2k + k^2 + 7}$ on $\mathbb{R}$. We will show that (f_k) converges uniformly on $\mathbb{R}$.

Solution. To see this, note that every term in the denominator of f_k is positive, and hence

$$|f_k(x)| = \frac{1}{x^4 + 3x^2k + k^2 + 7} \leq \frac{1}{k^2}.$$

Let $M_k = \frac{1}{k^2}$, and note that $\sum_{k=1}^{\infty} M_k = \sum_{k=1}^{\infty} \dfrac{1}{k^2}$ converges by the p-test. So by the above we have that $|f_k(x)| \leq M_k$ for all k, and that the sum of the M_k's converge. Therefore by the Weierstrass M-test, $\sum f_k$ converges uniformly on $\mathbb{R}$. □

9.6 Power Series

We have seen that a series of functions $\sum f_k$ can converge to another function f. But if you are given the f first, are there "simple" functions f_k which sum to f? Working towards this end, we begin with some of the simplest functions: polynomials. We first define *power series*, which essentially are polynomials with an infinite number of terms.

Definition

Definition 9.24. A *power series* is an expression that can be written as an infinite polynomial. That is,

$$f(x) = \sum_{k=0}^{\infty} a_k x^k = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \dots$$

where each $a_k \in \mathbb{R}$.

Note that the sum has to start at $k = 0$, since we need a constant term in the sum. Also note that, unlike finite polynomials, for each x-value you plug in, the resulting sum might converge or might not converge. So the first important question to ask is, for a specific sequence (a_k), for which x's does f's sum converge? Of course, for any (a_k), at $x = 0$ we have that $f(0) = a_0$ converges. But where else?

Example 9.25. The geometric series

$$\sum_{k=0}^{\infty} x^k = 1 + x + x^2 + x^3 + \dots$$

is a power series where each $a_k = 1$. And since (in the lingo of geometric series) the common ratio $r = x$, and saw in Proposition 4.9 that this power series converges precisely on the interval $(-1, 1)$. □

As it happens, the points for which a power series converges always look somewhat like this. First, they always for an interval. Second, that interval is always centered at zero. (However, note that we do include the extreme cases as an option, like the "interval" that is just the point $x = 0$, as well as all of $\mathbb{R}$, in addition to standard intervals like $(-1, 1)$.)

Intervals of Convergence

Lemma.

Lemma 9.26 (*P.S. converges on* $(-x_0, x_0) \Rightarrow$ *uniformly on closed subintervals*). If a power series $\sum_{k=0}^{\infty} a_k x^k$ converges at some point $x_0 \in \mathbb{R}$ and $0 < c < |x_0|$, then it converges uniformly on $[-c, c]$.

In particular, if a power series converges at some x_0 and $|x| < |x_0|$, then it also converges at x.

The proof is a fun tour through many of the important results we have proven throughout this text.

Proof. Since $\displaystyle\sum_{k=0}^{\infty} a_k x_0^k$ converges, by the k^{th}-term test (Proposition 4.5) we know that the terms $a_k x_0^k \to 0$, which in turn implies that these terms are bounded by some $M > 0$ (Proposition 3.20); that is, $|a_k x_0^k| \leq M$ for all $k \in \mathbb{N}$. And so, for any $x \in [-c, c]$ (implying $|x| < c < |x_0|$), we have

$$|a_k x^k| \leq |a_k c^k| = |a_k x_0^k| \left|\frac{c}{x_0}\right|^k \leq M \left|\frac{c}{x_0}\right|^k.$$

And we know that $\displaystyle\sum_{k=0}^{\infty} M \left|\frac{c}{x_0}\right|^k$ converges by the geometric series test (Proposition 4.9, with $r = \left|\frac{c}{x_0}\right| < 1$). So by the Weierstrass M-test (Corollary 9.22, with $M_k = M \cdot \left|\frac{c}{x_0}\right|^k$), we have that $\displaystyle\sum_{k=0}^{\infty} |a_k x^k|$ converges uniformly on $[-c, c]$, and so we are done. □

Note that this lemma implies that if a power series converges at some $x_0 > 0$ then it converges on all of $(-x_0, x_0]$. And if it converges at some $x_0 < 0$ then it converges on all of $[x_0, -x_0)$. The lemma has nothing to say about $-x_0$, though. It may or it may not converge at this point. "Most" examples will have it converge at $-x_0$, but one power series where it does not converge is

$$\sum_{k=1}^{\infty} \frac{(-1)^k}{k} x^k.$$

At $x_0 = 1$ this series converges by the alternating series test, but at $-x_0 = -1$ this is just the harmonic series, which diverges. This power series converges on $(-1, 1]$.

Continuing radii of convergence

Theorem.

Theorem 9.27 (*P.S. converge on an interval centered at 0*). The set of points that the power series $\sum_{k=0}^{\infty} a_k x^k$ converges at is

- $\{0\}$,
- $\mathbb{R}$, or
- One of $(-R, R), (-R, R], [-R, R)$ or $[-R, R]$, for some $R \in \mathbb{R}$.

Proof. Let C be the collection of points for which the power series converges. We pointed out earlier that $0 \in C$. If this is all there is, then we are done. If not, then C is some nonempty set with at least one non-zero element and hence $\{|x| : x \in C\}$ has a positive supremum.

If $\sup(\{|x| : x \in C\}) = \infty$, then for any $x \in \mathbb{R}$ there exists some $x_0 \in C$ such that $|x| < |x_0|$. But then by Lemma 9.26 we have that $x \in C$, giving $C = \mathbb{R}$, and we are done.

If, however, $\sup(\{|x| : x \in C\}) = R$ for some $R \in \mathbb{R}$ (implying that if $|x| > R$ then $x \notin C$), then we know by the lemma that every $x \in (-R, R)$ is also in C. That is, the only possibilities for C are $(-R, R), (-R, R], [-R, R)$, and $[-R, R]$, as desired. □

Definition.

Definition 9.28. The R in the above theorem is called the *radius of (pointwise) convergence*. If the set of points of convergence is $\{0\}$, then we say that the radius of convergence is 0, and if the set of convergence is $\mathbb{R}$, then we say that the radius of convergence is ∞.

What about uniform convergence? When is that guaranteed? Turns out the answer has to do with *compact* sets. For simplicity, though, in the below we will state it just in terms of closed intervals.

Theorem.

Theorem 9.29 (*P.S. converges at $c \Rightarrow$ uniformly on $[-c, c]$*). If a power series $\sum_{k=0}^{\infty} a_k x^k$ converges absolutely at some $x_0 \in \mathbb{R}$, then it converges uniformly on the closed interval $\big[-|x_0|, |x_0|\big]$.

Proof. This is Exercise 9.16. □

It is worthwhile to note that even if you know that a power series converges uniformly on *any* closed interval within the interval of convergence, that does *not* imply that it converges uniformly on the entire interval of convergence. For example, the following geometric series converges (but not uniformly) on its interval of convergence, $(-1, 1)$.

$$1 + x + x^2 + x^3 + \cdots = \frac{1}{1 - x}$$

9.7 Properties of Power Series

In the following table, we are considering a function $f_k(x) = a_k x^k$ on some closed interval $[-c, c]$. This allows us to apply Theorem 9.29 and the results from Table 9.2 to learn properties about $\sum_{k=0}^{\infty} f_k$.

$f_k(x) = a_k x^k$ on some $[-c, c]$ have the below property	Each s_n has property?	If $\sum_{k=1}^{\infty} a_k x^k$ converges uniformly, to f, must f satisfy property?
Continuous	Yes	Yes, by Proposition 9.8
Bounded	Yes	Yes, by Proposition 9.10
Uniformly Continuous	Yes	Yes, by Prop 9.13 and Ex 9.12
Integrable	Yes	Yes, by Proposition 9.17

Table 9.3: Properties of $\sum_{k=1}^{\infty} a_k x^k$

Note that the above table is somewhat redundant. Since the power series was only defined on a finite, closed interval $[-c, c]$, once we deduced that it was also

continuous we immediately got that it was also bounded (Corollary 6.30), uniformly continuous (Proposition 6.39) and integrable (Theorem 8.15).

Derivatives and Integrals of Power Series

Are power series differentiable and integrable? The answer is nice—it says you can differenti'te or integrate term-by-term, just like with (finite) polynomials—but the proofs (given what we have developed thus far) would be long and tedious. Therefore I will tell you the answer and give a quick sketch of one of the proofs, but that's it.

Theorem.

Theorem 9.30 (*Derivatives and integrals of P.S.*). Let $\sum_{k=0}^{\infty} a_k x^k$ be a power series with radius of convergence R.

(i) If $R > 0$ and $x \in (-R, R)$, then

$$\frac{d}{dx}\left(\sum_{k=0}^{\infty} a_k x^k\right) = \sum_{k=0}^{\infty}\left(\frac{d}{dx} a_k x^k\right) = \sum_{k=0}^{\infty} a_k \cdot k x^{k-1}.$$

(ii) If $[a, b] \subseteq (-R, R)$, then

$$\int_a^b \left(\sum_{k=0}^{\infty} a_k x^k\right) dx = \sum_{k=0}^{\infty}\left(\int_a^b a_k x^k \, dx\right) = \sum_{k=0}^{\infty}\left(a_k \cdot \frac{1}{k+1}\left[b^{k+1} - a^{k+1}\right]\right)$$

Proof Sketch. Here is a quick sketch of how (ii) is proved.

$$\sum_{k=0}^{\infty} \int_a^b a_k x^k \, dx = \lim_{n\to\infty} \sum_{k=0}^{n} \int_a^b a_k x^k \, dx = \lim_{n\to\infty} \int_a^b \sum_{k=0}^{n} a_k x^k \, dx$$

$$= \lim_{n\to\infty} \int_a^b s_n \, dx = \int_a^b \lim_{n\to\infty} s_n \, dx = \int_a^b \sum_{k=0}^{\infty} a_k x^k \, dx.$$

Needs justification

□

9.8 New Power Series from Old

Theorem 9.30 affords us a valuable tool to turn old power series into new power series. Indeed, some of the most important power series can be obtained by manipulating geometric series using this theorem and some algebra. The next example is a growing collection of such series.

Example 9.31. By the geometric series test, we know that, for $|r| < 1$,

$$\frac{a}{1-r} = a + ar + ar^2 + ar^3 + \dots.$$

In particular,

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \dots$$

with interval of convergence $(-1, 1)$. By replacing "x" with "x^3" we get the power series

$$\begin{aligned} \frac{1}{1-x^3} &= 1 + x^3 + \left(x^3\right)^2 + \left(x^3\right)^3 + \dots \\ &= 1 + x^3 + x^6 + x^9 + \dots, \end{aligned} \qquad (\text{☞})$$

with interval of convergence $(-1, 1)$.

We can now find a power series for $\dfrac{x^2}{1-x^3}$ simply by multiplying both sides of (☞) by x^2:

$$\begin{aligned} x^2\left(\frac{1}{1-x^3}\right) &= x^2\left(1 + x^3 + x^6 + x^9 + \dots\right) \\ \frac{x^2}{1-x^3} &= x^2 + x^5 + x^8 + x^{11} + \dots. \end{aligned}$$

Likewise, we can find a power series for $\dfrac{1}{1-x^3} - 1$ by using (☞) and subtracting 1 from both sides of the power series for $\dfrac{1}{1-x^3}$:

$$\begin{aligned} \frac{1}{1-x^3} - 1 &= \left(1 + x^3 + x^6 + x^9 + \dots\right) - 1 \\ \frac{1}{1-x^3} - 1 &= x^3 + x^6 + x^9 + \dots. \end{aligned}$$

Finally, we can use calculus to find a new power series. A power series representation

of $f(x) = \dfrac{3x^2}{(1-x^3)^2}$ can be obtained in the following way. First note that

$$\begin{aligned}\frac{d}{dx}\left(\frac{1}{1-x^3}\right) &= \frac{d}{dx}\left(1-x^3\right)^{-1} \\ &= -\left(1-x^3\right)^{-2}\cdot(-3x^2) \\ &= \frac{3x^2}{(1-x^3)^2}.\end{aligned}$$

This is helpful since we already showed that

$$\frac{1}{1-x^3} = 1 + x^3 + x^6 + x^9 + \dots$$

Differentiating both sides (term-by-term, by Theorem 9.30)

$$\begin{aligned}\frac{d}{dx}\left(\frac{1}{1-x^3}\right) &= \frac{d}{dx}\left(1 + x^3 + x^6 + x^9 + \dots\right) \\ \frac{3x^2}{(1-x^3)^2} &= \left(0 + 3x^2 + 6x^5 + 9x^8 + \dots\right)\end{aligned}$$

Thus we have a power series representation of $\dfrac{3x^2}{(1-x^3)^2}$. □

9.9 Taylor and Maclaurin series

Recall.

Recall 9.32. The k^{th} derivative of f is denoted $f^{(k)}$.

Here is a section that is truly Taylor-made. I recommend you begin by watching an excellent video motivating *Taylor series*. It is a video from the inimitable Grant Sanderson, through his 3Blue1Brown channel on YouTube. Here is a link to the video: `https://youtu.be/3d6DsjIBzJ4`.

The video explains intuitively how, for "nice" functions, the function equals its Taylor series where it converges. That is,

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(c)}{k!}(x-c)^k$$

for every x in some interval of convergence I centered at $x = c$.

To get into our formal development of this, let's define a Taylor series, and as a special case a *Maclaurin series*.

Definition.

Definition 9.33. The *Taylor series* of f about $x = c$ is

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(c)}{k!}(x-c)^k.$$

If $c = 0$ then the series is called a *Maclaurin series.*

What we do not know yet, is when a function is equal to its Taylor/Maclaurin. (The answer is nice, we just aren't there yet.) Before that, though, here is an example of finding a Taylor/Maclaurin series.

Example 9.34. Here we find the Maclaurin series of $f(x) = \cos(x)$.

$f(x) = \cos(x)$	$f(0) = 1$	$a_0 = 1$
$f'(x) = -\sin(x)$	$f'(0) = 0$	$a_1 = 0$
$f''(x) = -\cos(x)$	$f''(0) = -1$	$a_2 = -\frac{1}{2!}$
$f'''(x) = \sin(x)$	$f'''(0) = 0$	$a_3 = 0$
$f^{(4)}(x) = \cos(x)$	$f^{(4)}(0) = 1$	$a_4 = \frac{1}{4!}$
$f^{(5)}(x) = -\sin(x)$	$f^{(5)}(0) = 0$	$a_5 = 0$

It is evident that

$$f^{(k)}(0) = \begin{cases} 0, & \text{if } k \text{ is odd,} \\ (-1)^{k/2}, & \text{if } k \text{ is even.} \end{cases}$$

So the Maclaurin series for $f(x) = \cos(x)$ is

$$\begin{aligned} \sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!}(x)^k &= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots \\ &= \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} x^{2k} \end{aligned}$$

□

It is worth repeating that at this point we have *not* shown that $\cos(x)$ actually equals its Maclaurin series. This will turn out to be true, but we still have work to do to show this.

Indeed, only nice functions equal their Maclaurin series. So what does "nice" mean here? Well, the formula itself requires that we are able to differentiate f arbitrarily many times; that is, they are *infinitely differentiable.* So "nice" certainly must include this property. What else? Let's investigate.

Define the *Taylor polynomial of degree n* at c to be

$$T^n_{x=c}(f) = \sum_{k=0}^{n} \frac{f^{(k)}(c)}{k!}(x-c)^k.$$

By construction, the first n derivatives of $T^n_{x=c}(f)$ agree with the first n derivatives of f at the point $x = c$. They are likely not equal, though, and in fact they differ by

$$E_n(x) = f(x) - T^n_{x=c}(f),$$

called the *error function*. And, as probably makes intuitive sense, f equals its Taylor series on I if and only if this error tends to 0.

Theorem.

Theorem 9.35 (*f equals its Taylor series $\Leftrightarrow$ $E_n \to 0$*)**.** Assume that f is infinitely differentiable in an interval I, and $c \in I$. Then

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(c)}{k!}(x-c)^k \qquad \text{if and only if} \qquad E_n(x) \to 0 \quad \text{pointwise.}$$

So one way to determine whether a function f equals its Taylor series is by checking its error function. There are many ways to express this error. We begin with an integral expression for it.

Theorem.

Theorem 9.36 (*Integral form of the error function*)**.** Assume that f is infinitely differentiable in an interval I, and $c \in I$. Then for $x \in I$ we have

$$E_n(x) = \frac{1}{n!}\int_c^x (x-t)^n f^{(n+1)}(t)\, dt.$$

Proof Sketch. This proof is fairly long and technical, but here is a sketch. The proof is by induction on n, and uses FTC and Integration by Parts along the way. The $(n = 1)$ base case alone gives a nice feel for the important elements of the proof. In this case we wish to show that $E_1(x) = \int_c^x (x-t)f''(t)\, dt$. We will now show this.

First note that by the definition of the error function,

$$\begin{aligned} E_1(x) &= f(x) - T^n_{x=c}(x) = f(x) - \sum_{k=0}^{1} \frac{f^{(k)}(c)}{k!}(x-c)^k \\ &= f(x) - f(c) - f'(c)(x-c) \end{aligned}$$

By the fundamental theorem of calculus (twice),

$$\begin{aligned} &= \int_c^x f'(t)\ dt - \int_c^x f'(c)\ dt \\ &= \int_c^x \big(f'(t) - f'(c)\big)\ dt \end{aligned}$$

Now we use Integration by Parts $\left(\int u\ dv = uv - \int v\ du\right)$. In it, recall that when finding v from dv, we can pick *any* antiderivative.

$$\begin{aligned} u &= f'(t) - f'(c) & \qquad v &= t - x \\ du &= f''(t)\ dt & \qquad dv &= dt \end{aligned}$$

This allows us to continue our string of equations for $E_1(x)$.

$$\begin{aligned} &= \big(f'(t) - f'(c)\big)(t-x)\Big|_c^x - \int_c^x (t-x)f''(t)\ dt \\ &= 0 + \int_c^x (t-x)f''(t)\ dt, \end{aligned}$$

as desired. This completes the base case, $n = 1$. The rest of the proof is by induction. By a little algebra one can show that

$$E_{k+1}(x) = E_k(x) - \frac{f^{(k+1)}(c)}{(k+1)!}(x-c)^{k+1}.$$

Using this, the inductive hypothesis, and one more application of Integration by Parts, one can finish the proof. □

There is one more representation for the error function that we will discuss, and it is known as the *Lagrange form*, after its founder. The theorem says that at any particular point x_0, f is equal to its $(n-1)^{\text{st}}$ Taylor polynomial evaluated at x_0, plus the n^{th} term of the the Taylor series evaluated at some other nearby point.

Theorem.

Theorem 9.37 (*Lagrange form of the error function*). Assume that f is infinitely differentiable in an interval I, and $c \in I$. Then for any other $x_0 \in I$ there exists some α_n between x_0 and c (which depends on x_0 and c) such that

$$f(x_0) = \left(\sum_{k=0}^{n-1} \frac{f^{(k)}(c)}{k!} (x_0 - c)^k \right) + \frac{f^{(n)}(\alpha_n)}{n!} (x_0 - c)^n.$$

In essence, this theorem asserts that $E_{n-1}(x_0) = \dfrac{f^{(n)}(\alpha_n)}{n!}(x_0 - c)^k$, for some α_n.

The proof follows from the integral form of the error function (Theorem 9.36), but also make use of two other important theorems from earlier in the text.

Proof. Since $f^(n)$ is continuous on $[c, x_0]$ it is bounded (Corollary 6.30). So we may set

$$m : \inf_{u \in [c,x_0]} f^n(u), \qquad M : \inf_{u \in [c,x_0]} f^n(u)$$

And so, by the integral form of the remainder (Theorem 9.36),

$$\frac{m}{n!} \int_c^{x_0} (x_0 - u)\, du \leq R_{n-1,c}(x_0) \leq \frac{M}{n!} \int_c^{x_0} (x_0 - u)\, du.$$

Therefore there must be some $m \leq \beta \leq M$ such that

$$R_{n-1,c}(x_0) = \frac{\alpha}{n!}(x_0 - c)^n.$$

And since $f^{(n)}$ is differentiable, it's continuous (Theorem 7.6). And so by the intermediate value theorem (Theorem 6.37), there must exist some $\alpha_n \in [c, x_0]$ such that $f^{(n)}(\alpha_n) = \beta$, concluding the proof. □

— Notable Exercises —

- Exercise 9.8 asks you to prove the *Cauchy Criterion for Uniform Convergence*:

  > Let $f_n : A \to \mathbb{R}$. The sequence (f_n) converges uniformly if and only if for every $\varepsilon > 0$ there exists some $N \in \mathbb{N}$ such that $|f_m(x) - f_n(x)| < \varepsilon$ for all $m, n \geq N$ and all $x \in A$.

  As with numerical sequences from Chapter 3, the advantage to a Cauchy-criterion is that you can show something is converging (uniformly) without knowing what it converges to.

- Exercise 9.16 essentially says that Taylor series always converge in a region cetered at 0. It does this by asing you to prove that if a Taylor series converges at a point x_0, then it in fact converges on the closed interval $\big[-|x_0|, |x_0|\big]$.

— Exercises —

Exercise 9.1. Let $f_n(x) = \dfrac{x \sin(x)}{n}$ on $[0, 10]$. Prove that $f_n(x)$ converges uniformly.

Exercise 9.2. Let $f_n(x) = x/n$ on $\mathbb{R}$, and let $f(x) = 0$ on $\mathbb{R}$. Prove that $f_n(x)$ converges pointwise to f, but does not uniformly to f.

Exercise 9.3. Assume that, for each $n \in \mathbb{N}$, the function $f_n : A \to \mathbb{R}$ is unbounded.

(a) If $f_n \to f$ pointwise, must f be unbounded? Either prove that f must be unbounded or give a counterexample.

(b) If $f_n \to f$ uniformly, must f be unbounded? Either prove that f must be unbounded or give a counterexample.

Exercise 9.4. Assume that, for each $n \in \mathbb{N}$, the function $f_n : A \to \mathbb{R}$ is uniformly continuous. Also assume that $f_n \to f$ uniformly. Prove that f is uniformly continuous.

Exercise 9.5. Give an example of a sequence (f_n) of non-continuous functions for which $f_n \to f$ uniformly, for some continuous f.

Exercise 9.6. Give an example of a sequence (f_n) of integrable functions for which $f_n \to f$ pointwise, but yet f is not integrable.

Exercise 9.7. Give an example of a sequence (f_n) of integrable functions for which $f_n \to f$ pointwise, and f is integrable, but yet

$$\lim_{n\to\infty} \int_a^b f_n(x)\, dx \neq \int_a^b f(x)\, dx.$$

Exercise 9.8. Prove the *Cauchy Criterion for Uniform Convergence*: Let $f_n : A \to \mathbb{R}$. The sequence (f_n) converges uniformly if and only if for every $\varepsilon > 0$ there exists some $N \in \mathbb{N}$ such that $|f_m(x) - f_n(x)| < \varepsilon$ for all $m, n \geq N$ and all $x \in A$.

Exercise 9.9. Suppose $f_n \to f$ pointwise and $g_n \to g$ pointwise, both on a set $A \subseteq \mathbb{R}$. Prove that $(f_n + g_n) \to (f + g)$ pointwise on A.

Exercise 9.10. Suppose $f_n \to f$ uniformly and $g_n \to g$ uniformly, both on a set $A \subseteq \mathbb{R}$. Prove that $(f_n + g_n) \to (f + g)$ uniformly on A.

Exercise 9.11. Give an example of a sequence (f_n) of functions where

(i) (f_n) converges uniformly on $[-10, 10]$, and

(ii) (f_n) converges pointwise on $\mathbb{R}$, but does not converge uniformly on $\mathbb{R}$.

Exercise 9.12. Given an example of (1) a sequence of functions f_n on $[0, 1]$, (2) a function f on $[0, 1]$, and (3) a sequence (a_n) where each $a_n \in [0, 1]$ and $a_n \to a$ for some a, such that the following conditions hold.

(i) Each f_n is continuous;

(ii) $f_n \to f$ pointwise; and

(iii) $f(a_n) \not\to f(a)$.

Exercise 9.13. Suppose (f_n) is a sequence of functions on $[a, b]$, each of which is monotone increasing. Suppose further that $f_n \to f$, for some f on $[a, b]$. Prove that f is monotone increasing on $[a, b]$.

Exercise 9.14. Prove the *Cauchy criterion for series of functions.* That is, let $f_k : A \to \mathbb{R}$, and prove that the series $\sum_{k=1}^{\infty} f_k$ converges uniformly on A if and only if for every $\varepsilon > 0$ there exists some $N \in \mathbb{N}$ such that

$$\left| \sum_{k=m}^{n} f_k(x) \right| < \varepsilon$$

for all $n \geq m \geq N$ and all $x \in A$.

Exercise 9.15. Let $f_n : A \to \mathbb{R}$. A sequence of functions (f_n) is said to be *uniformly bounded* if there exists some $C \in \mathbb{R}$ such that $|f_n(x)| \leq C$ for all $n \in \mathbb{N}$ and all $x \in A$.

(a) Explain why the property "each f_n is bounded" is different than the property "(f_n) is uniformly bounded."

(b) Give an example of a set A and a sequence (f_n) for which each f_n is bounded on A but the sequence (f_n) is not uniformly bounded.

Exercise 9.16. Prove that if a power series $\sum_{k=0}^{\infty} a_k x^k$ converges absolutely at some $x_0 \in \mathbb{R}$, then it converges uniformly on the closed interval $\big[-|x_0|, |x_0|\big]$.

Exercise 9.17. Find the Taylor series for the following functions about the given point.

(a) $f(x) = \cos(x)$ about $x = \pi/3$.

(b) $g(x) = x^3$ about $x = 2$.

Exercise 9.18. Let $f(x) = \ln(x + 1)$.

(a) Find the Maclaurin series for f.

(b) State where this series converges.

(c) Use this to find a series representation for $\ln(2)$.

— Open Question —

Question. Let $R(x) = \dfrac{P(x)}{Q(x)}$ where $P(x)$ and $Q(x)$ are polynomials with integer coefficients and $Q(0) \neq 0$. Is there an algorithm where, given $P(x)$ and $Q(x)$ as input, always halts and correctly decides if the Taylor series of $R(x)$ at $x = 0$ has a constant coefficient of 0?

Appendices

Appendix A: Construction of $\mathbb{R}$

"Far out in the uncharted backwaters of the unfashionable end of the western spiral arm of the Galaxy lies a small unregarded yellow sun. Orbiting this at a distance of roughly ninety-two million miles is an utterly insignificant little blue green planet whose ape-descended life forms are so amazingly primitive that they still think digital watches are a pretty neat idea." This is how Douglas Adams began his classic book *The Hitchhiker's Guide to the Galaxy.* He began its sequel, "The story so far: In the beginning the Universe was created. This has made a lot of people very angry and been widely regarded as a bad move." Douglas Adams can certainly set a tone.

"It was a pleasure to burn" is the first sentence of Ray Bradbury's *Fahrenheit 451.* "I'm pretty much fucked" is the start to Andy Weir's *The Martian.* And, "All happy families are alike; each unhappy family is unhappy in its own way" is how Leo Tolstoy began *Anna Karenina.*

If the first chapter of the *Harry Potter* series were a dry introduction to wand movements and cauldron thickness, few would read on. Yes, the spells and potions rely on those details, but it's not where the magic really is. Therefore J.K. Rowling's first chapter contained exhibitions of magic, mention of a dark wizard so terrifying that most fear to speak his name, allusions to this wizard's sadistic reign which mysteriously and abruptly just ended, and a significant baby boy, dropped on the doorstep of the least magical people possible. Lots of questions, few answers.

Where am I going with this? The material in Appendix A is the 'wand movements' and 'cauldron thickness' part of real analysis. The spells and potions we developed in this text rely on the foundational aspects we are about to discuss, but if I started the book with them you might have fallen asleep before Chapter 2. And just like how, after reading (and rereading[1]) the books, hardcore Harry Potter fans crave to know the small—but fundamental—details about the world they entered, perhaps now you will be motivated to learn the underpinnings of the world of real analysis.

Everything we proved relied on the existence of the real numbers. They needed to be there and to have the properties we discussed, and the rigor of mathematics demands we don't simply assume their existence. Which axioms are necessary to give us the world we want, and why? Doing this thoroughly (in long-form style) might take 50 pages, and even hardcore analysis fans would likely fall asleep. This appendix is just a 7 page sketch of it. Its goal is simply to illustrate for you the meticulous process which needed to be done to formally begin the theory of real analysis.[2]

[1] And rereading

[2] And to make you appreciate the brave soul who took one for the team, and painstakingly did it.

A.1 Axioms of Set Theory

It is hard to get more simple than the notion of a set, but the mathematical world did a double take in 1901 when Bertrand Russell discovered a paradoxical set. Recalling that the elements of a set can themselves be sets, he considered the set

$$R = \{x : x \notin x\}.$$

Symbolically, there was no reason to disallow such a set definition, but yet you can work out the strange contradiction that $R \in R$ if and only if $R \notin R$; indeed, if R is not a member of itself, then according to its definition it must be a member of itself, and if it does contain itself, then it contradicts its own definition.[3]

When mathematicians reach a logical impasse, they often trace their reasoning backwards; the longer they trace, the more nervous they get, as they fear for the foundations of their field. These nerves caused some great mathematicians of the early twentieth century to rework set theory from the ground up. In the end, *Zermelo-Fraenkel+Choice set theory* (ZFC) was born—nine axioms for the field of set theory, and in turn for much of mathematics.

1. Axiom of extensionality: Two sets are equal if they have the same elements.
2. Axiom of regularity: Every non-empty set A contains an element B such that A and B are disjoint sets.
3. Axiom schema of specification: You can create subsets using set builder notation.
4. Axiom of pairing: If A and B are sets, there exists a set which contains A and B as elements.
5. Axiom of union: You can take union over the elements of a set. (E.g., the union of the elements of the set $\{\{1,3\},\{3,\smiley\}\}$ is $\{1,3,\smiley\}$.
6. Axiom schema of replacement: Given a function f whose domain is a set A, $f(A)$ is also a set.
7. Axiom of infinity: There exists a set having infinitely many members.
8. Axiom of power set: For any set A, there is a set B that contains every subset of A.
9. Axiom of Choice:[4] First, a *choice function* is a function f, defined on a collection X of nonempty sets, such that for every set A in X, $f(A)$ is an element of A. Now, the axiom is: For any set X of nonempty sets, there exists a choice function f defined on X.

These are the axioms of set theory, and on their shoulders the reals are built.

[3] A classic riddle works along the same lines: "A barber cuts the hair of everyone in his town who does not cut their own hair. Does the barber cut his own hair?"

[4] Axioms 1-8 you probably think are completely obvious and fair to assume, while Axiom 9 is a bit confusing. Indeed, due to its less-than-painstakingly-obvious stature, it has been blamed for all the paradoxical-sounding theorems of set theory—the final example in Appendix B, most notably.

A.2 Constructing $\mathbb{N}$

Everything we build is going to rely on nothing more than the axioms of set theory. Indeed, we begin by constructing the natural numbers—which are themselves *sets*.

We represent the number 0 as the empty set, $\emptyset$; this may seem weird, but bear with me here. We then define a *successor function* $S(x)$ of a set x to be the set

$$S(x) = x \cup \{x\}.$$

In the naturals, the successor to 0 is 1, and indeed we are going to represent 1 as the set $S(0) = S(\emptyset) = \emptyset \cup \{\emptyset\} = \{\emptyset\}$; note that this equals $\{0\}$. Then, 2 is going to be represented as the set $S(1) = S(\{\emptyset\}) = \{\emptyset\} \cup \{\{\emptyset\}\} = \{\emptyset, \{\emptyset\}\}$; note that this equals $\{0, 1\}$.

Continuing in this way, we get a collection of sets which are going to represent the natural numbers for us.

$$\begin{aligned}
0 &= \emptyset \\
1 &= \{\emptyset\} = \{0\} \\
2 &= \{\emptyset, \{\emptyset\}\} = \{0, 1\} \\
3 &= \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\} = \{0, 1, 2\} \\
4 &= \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}, \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\}\} = \{0, 1, 2, 3\} \\
&\vdots
\end{aligned}$$

We define $\mathbb{N}_0$ to be the set of all these numbers (represented as sets).

Of course, the natural numbers are not just isolated objects that we need to make up representations for—they also have an order to them and an algebraic structure. Beginning with the former, if we are going to describe the natural numbers as sets, we must also describe what "$n < m$" means in terms of these sets. One reason why the naturals were constructed like the above is that it is now easy to describe their order. Notice how, as sets, each number (represented as a set) is *an element of* each later number (represented as a set). Indeed, we say

$$n < m \qquad \text{if} \qquad n \in m.$$

For example, $1 < 3$ since it is true that $\{\emptyset\} \in \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\}$. We now have an ordering on $\mathbb{N}_0$!

Another important property of the numbers in $\mathbb{N}_0$ that we need to capture is that we can add them together. How should we define the addition $2 + 5$ by using only the tools of set theory applied to the sets representing 2 and 5? This one is a little more complicated to write down, but the basic idea is to think about "$m + n$" as $m+1+1+\cdots+1$, where you have n copies of 1. With this frame of mind, "$m+n$" is defined to be $S(S(S(\ldots S(m)\ldots)))$, where you apply the successor function n times. The only problem being that, at this point 'n' is a set, so "n times" doesn't make sense. Instead, we must create a recursive function S_n that equals a composition of

'n' successor functions.

$$\begin{aligned} S_1(x) &:= S(x) \\ S_2(x) &:= S(S(x)) \\ S_3(x) &: \quad S(S(S(x))) \\ S_4(x) &:= S(S(S(S(x)))) \\ S_5(x) &:= S(S(S(S(S(x))))) \\ &\vdots \end{aligned}$$

Then, "$m + n$" can be defined to be $S_n(m)$.

As for multiplication, we define $n \cdot 0 = 0$, and for other multiplication problems we iteratively turn them into an addition problem by defining $n \cdot (m + 1) = n \cdot m + n$.

One can then spend 15 pages proving the following, where $m, n, k \in \mathbb{N}_0$.

- $(m + n) + k = m + (n + k)$
- $m + n = n + m$
- $n + 0 = n$
- $(m \cdot n) \cdot k = m \cdot (n \cdot k)$
- $m \cdot n = n \cdot m$
- $n \cdot 1 = n$
- $m \cdot (n + k) = (m \cdot n) + (m \cdot k)$
- If $m < n$, then $m + k < n + k$
- If $m < n$ and $0 < k$, then $k \cdot m < k \cdot n$
- If $m < n$ and $n < k$, then $m < k$
- Exactly one of the following holds: $m < n$, $m = n$, or $n < m$.

As you'll see, I am beginning to really lean into the fact that this is a "sketch" of how to construct $\mathbb{R}$. Properties like the above are a pain and a half to prove, and even harder to type up in LaTeX without making a typo. But, it can be done.

Once completed, you can proudly claim to have shown that $(\mathbb{N}_0, +, \cdot, <)$ is what is called an *ordered semi-ring*. This is a completed construction of the natural numbers (and 0), but we still have a ways to go. An ordered semi-ring is good, but to remove the "semi," we construct negative numbers.

A.3 Constructing $\mathbb{Z}$

Axiomatically, $\mathbb{Z}$ is the smallest ordered ring (containing $\mathbb{N}_0$), but to construct it explicitly, it suffices to introduce subtraction. The difference between two numbers is going to have to be something new that breaks the mold established by the successor function. Indeed, we define the difference "$m - n$" simply to be the ordered pair $(m, n) \in \mathbb{N}_0 \times \mathbb{N}_0$.

If we stopped here, we would have the problem where we would consider "$6 - 2$" to be different from "$7 - 3$"; after all, as ordered pairs $(6, 2)$ and $(7, 3)$ are certainly different. To get around this, we define an *equivalence relation* on ordered pairs, where we want $(m, n) \sim (k, \ell)$ provided "$m - n = k - \ell$." But since we haven't defined subtraction yet, we move the 'n' and 'ℓ' over, giving the condition we want: $m + \ell = k + n$. That is, we say

$$(m, n) \sim (k, \ell) \qquad \text{if and only if} \qquad m + \ell = k + n.$$

One would then prove that this is indeed an equivalence relation. To make it easier to read along, though, let's write the ordered pair (m, n) as $[m - n]$. This will vastly help one follow along —just don't forget that these are technically ordered pairs, and that minus sign is just notation.

Of course, one would have to formally prove that $\sim$ did indeed form an equivalence relation... but it does. You can also do a mental check that the following make sense as definitions of addition, multiplication and ordering.

Define the following,[5]

$$\begin{aligned}
&[m - n] + [k - \ell] := [(m + k) - (n + \ell)] \\
&[m - n] \cdot [k - \ell] := [(m \cdot k + n \cdot \ell) - (m \cdot \ell + n \cdot k)] \\
&[m - n] < [k - \ell] \quad \text{if and only if} \quad m + \ell < k + n
\end{aligned}$$

But here, the plot annoyingly thickens again: Given that $[m - n]$ represents an entire equivalence class, for our definitions of '$+$', '$\cdot$' and '$<$' to be well-defined they have to be playing nicely with this equivalence relation. The proofs of these are actually reasonably short (under a page for all three), but are enormously boring and so we will again skip them.

Negatives also distribute, so we define $-[m - n]$ to be $[n - m]$. Moreover, we can simplify our notation of these elements, by defining $\boldsymbol{n} := [n - 0]$. The set of all such $\boldsymbol{n}$ we define to be the set of *integers*, $\mathbb{Z}$.

Then, for instance, we can say $\boldsymbol{n}$ is *positive* if $\mathbf{n} > 0$. And is *negative* if $\mathbf{n} < 0$. Moreover, one can prove that every integer is either of the form $\boldsymbol{m}$ or $-\boldsymbol{m}$, and that multiplication handles the signs as expected. And, finally, at this point one could struggle through a 3-page proof that $(\mathbb{Z}, +, \cdot, <)$ satisfies the additive, multiplicative and order properties to be an ordered ring.

And with that, it has been fully sketched how to construct the integers.

[5] Technically it's an abuse of notation to cavalierly reuse '$+$', '$\cdot$' and '$<$' without ensuring that doing so is consistent with our previous uses of it... but with some elbow grease it does work out.

A.4 Constructing $\mathbb{Q}$

Axiomatically, $\mathbb{Q}$ is the smallest ordered field (containing $\mathbb{Z}$), but we again describe its construction, which has many similarities to the construction of $\mathbb{Z}$.

The construction of $\mathbb{Z}$ was focused on incorporating in subtraction; here we focus on incorporating in division. To describe this division, we again use ordered pairs. The fraction $\frac{m}{n}$ that we are after will be given by the ordered pair (m, n). Since we do not want the denominator to be zero, and we only have to allow either the numerator or denominator to be negative, we begin by considering the ordered pairs $\mathbb{Z} \times \mathbb{N}$, where $\mathbb{N} = \mathbb{N}_0 \setminus \{0\}$.

We now need another equivalence relation. Intuitively, $(m, n) = (k, \ell)$ when $\frac{m}{n} = \frac{k}{\ell}$, which we can write without division as $m \cdot \ell = k \cdot n$. So we define the equivalence relation $\sim$ on $\mathbb{Z} \times \mathbb{N}$ by

$$(m, n) \sim (k, \ell) \qquad \text{if and only if} \qquad m \cdot \ell = k \cdot n.$$

One can again prove that this is an equivalence relation, and then can define $\mathbb{Q}$ to be the set of all such equivalence classes. Again, to make it easier to read, we will write the ordered pair (m, n) as $[m/n]$, understanding that this is just notation. You can check by rewriting them as fractions that the below definitions make sense.[6]

- $[m/n] + [k/\ell] := [(m \cdot \ell) + (k \cdot n) \ / \ n \cdot \ell]$
- $[m/n] \cdot [k/\ell] := [m \cdot k \ / \ n \cdot \ell]$
- $[m/n] < [k/\ell]$ if and only if $m \cdot \ell < k \cdot n$

Once again, we are applying a definition to an equivalence class but only stating the definition via a representative of that class—therefore one must prove that the above operations are well-defined. (They are.)

This set contains a copy of $\mathbb{Z}$, where you identify $[n/1]$ with $\boldsymbol{n}$.

One can then define positive and negative numbers similar to before, and the additive inverse of $[m/n]$ is $[-m/n]$.

The multiplicative inverse (for "non-zero" classes) must be defined a little more carefully, since we demand that any negative sign be in the numerator (which is the first coordinate of the ordered pair and in the bracket notation).

$$[m/n]^{-1} := \begin{cases} [n/m] & \text{if } m > 0; \\ [-n/-m] & \text{if } m < 0. \end{cases}$$

And, finally, one can work out a long proof that $(\mathbb{Q}, +, \cdot, <)$ satisfies the algebraic properties to form an ordered field. These mostly go through similar to with $\mathbb{Z}$.

And with that, $\mathbb{Q}$ is constructed.

[6] Again, with an abuse of notation.

A.5 Constructing $\mathbb{R}$

Here we are, the moment you've been waiting for. We have the ordered field of rational numbers, but they aren't complete—there are holes everywhere, and to get to $\mathbb{R}$ we must fill in these gaps. There are several ways to do this. Cantor used Cauchy sequences: if you demand that Cauchy sequences always converge to something, and pairs of sequences whose difference is a Cauchy sequence are converging to the same thing, then by adding these limits into your set $\mathbb{Q}$ (by identifying them with an equivalence class of Cauchy sequences converging to them), you in effect complete $\mathbb{Q}$, giving $\mathbb{R}$. Lots of details missing, but that's the big picture idea.

There is also a non-standard approache using so-called ultrafilters, another using hyperrationals, and several more beyond that.

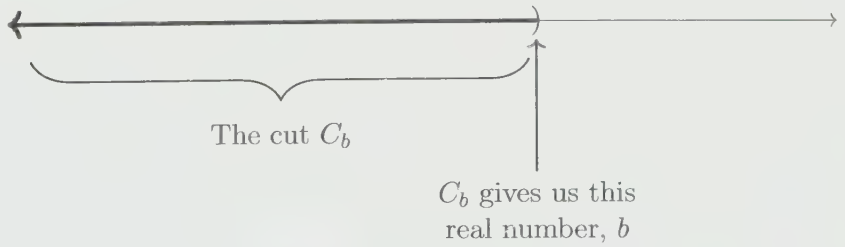
But the most common method, which we discuss now, uses *Dedkind cuts.*

Just like with all our previous constructions, each real number is going to be a set; at this point we have the rationals constructed, so each real number is going to be represented by a set of rationals. The way you want to think about it is this: the real number x is going to be represented by the set of all rational numbers strictly less than x. These sets are going to be called *cuts*, and while we discuss them you can start convincing yourself that each real number will indeed correspond to a unique cut, and each cut corresponds to a unique real number. But first, let's formally discuss cuts.

A *cut* should be thought of as the set $(\infty, b) \cap \mathbb{Q}$. That is, all rational numbers up to a certain point. Formally, it is defined as any set C_b satisfying the following three conditions.

1. $C_b \subseteq \mathbb{Q}$, but $C_b \neq \emptyset$ and $C_b \neq \mathbb{Q}$;
2. If $p \in C_b$ and $q \notin C_b$, then $p < q$;
3. If $p \in C_b$, then there exists some $q \in C_b$ where $p < q$.

And then $\mathbb{R}$ is defined as the set of all cuts.[7]



Although we have an intuitive picture of a cut, at the moment it is just a set satisfying the above properties. And we again are interested in putting an algebraic

[7]So real numbers are infinite sets of rational numbers, which themselves are pairs of integers, which themselves are pairs of natural numbers, which themselves are unions of empty sets and sets containing empty sets and so on. That's what a single real number is constructed to be.

structure on this collection of cuts. Addition and order work quite smoothly. For a pair of cuts C_a and C_b, define the following.[8]

- $C_a + C_b := \{p + q : p \in C_a, q \in C_b\}$
- $C_a < C_n$ if and only if $C_a \subset C_b$

One can verify that the addition of two cuts is still a cut, and that addition is commutative and associative, that the 0 cut behaves as it should (in turn giving *positive* and *negative* cuts) and that additive inverses exist. The inequality has the property that, given any two cuts C_a and C_b, exactly one of the following holds: $C_a < C_b$, $C_a = C_b$ or $C_b < C_a$.

Defining multiplication is trickier, because if you simply multiply the two sets together you'll have massive negative numbers multiplying against each other, creating massive positive numbers. Intuitively, you want $C_a \cdot C_b = C_{ab}$. That is, the cut $(-\infty, a) \cap \mathbb{Q}$ times the cut $(-\infty, b) \cap \mathbb{Q}$ should equal the cut $(-\infty, a \cdot b) \cap \mathbb{Q}$; but cuts aren't defined with such a and b—they produce a and b. One way around this is to first define multiplication for positive cuts. That is, if C_a and C_b are both positive (larger than the cut $\{q \in \mathbb{Q} : q < 0\}$), then define

$$C_a \cdot C_b : \quad \{p \cdot q : p \in C_a, q \in C_b \text{ with } p, q \geq 0\} \cup \{q \in Q : q < 0\}.$$

With this you can then define a product of two negative cuts by setting the product equal to the product of the two corresponding positive cuts. To define multiplication between a positive cut and a negative cut, you know the product should be negative so one approach is to consider the multiplication when both are positive, and then translate the result to the corresponding negative cut. It's a hassle to write out, but that's the idea.

One can then check that the product of two cuts is still a cut, that multiplicative inverses exist, and that multiplication is commutative and associative, as well as the remaining multiplicative/additive distributive and order properties. These would be quite annoying to work out in detail, but you can smile knowing someone carefully checked them.

With our set built, and with the algebraic properties defined and their properties verified, we now know that $\mathbb{R}$ is an ordered field. All that is left is to show that it is complete. This is done by showing that cuts satisfy the *least upper bound property*. That is, if $\mathcal{C}$ is a collection of cuts which is bounded above (meaning there exists some cut D such that $C \leq D$ for all $C \in \mathcal{C}$; i.e., $C \subseteq D$ for all $C \in \mathcal{C}$), then there exists a least upper bound (meaning there is a cut M such that $M \leq D$ for all upper bounds D). The proof is this: Let $M = \bigcup_{C \in \mathcal{C}} C$, and show that

1. M is a cut and therefore $M \in \mathbb{R}$;
2. M is an upper bound of $\mathcal{C}$, but is smaller than all other upper bounds.

With those, $\mathbb{R}$ is a complete ordered field, and hence the real numbers are constructed.[9]

[8] One last time with a slight abuse of notation.

[9] *drops mic*

Appendix B: Peculiar and Pathological Examples

The world of mathematics can be a strange place, and when your mind is exploring this world and finds itself in some dark, foreign region, you risk falling into pits, being eaten by monsters, and missing a beautiful forest for the boring trees.

It's been said that one of the most important goals of learning real analysis is to collect as many bizarre examples as you can, and to keep them in your back pocket. From a practical standpoint they will inform your conjectures and guide your proofs, but they will also help to demonstrate why real analysis is such a great subject.

When little kids learn about the world, many are fascinated by dinosaurs because they show how strange and large and monstrous life on Earth can be. Flowers are also nice. I like flowers. Your course abstract algebra, for instance, is filled with theorems which are small and pretty and I am happy to see them and learn about them. But I prefer the monsters.[1] We have seen many awesome monsters already in this book, but if your back pocket has some room left, this appendix has a few more.

For instructors/students looking to insert these these examples into the earlier chapters, below is where each example can be inserted, consistent with the material.

P&P Example	Suggested Chapter
Menger's Matterless Material	1
The Curious Case of the Cantor Set	3
An Abundant Addition Aboundingly Ascends	4
Peculiar and Pathological Perimeters	8
A Devilish De-Descent	7
A Smooth and Spiky Solution	7
Structuring Stuff from its Shadows' Shapes	8
A Composition Conundrum	8
A Prestigious Proof of a Primal Puzzle	4
Turning the Tables on your Teetering Troubles	6
Finding ϕ for First-place Finishes	3
Modern Measuring's Misfit Member	8
Tarski's Teriffic Talents Times Two	8

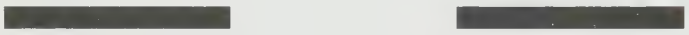
[1] Don't @ me, researchers of the Monster group.

B.1 The Curious Case of the Cantor Set

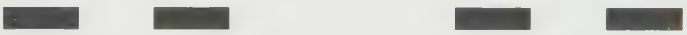
The *Cantor set* has some fascinating properties. To describe this set, the construction is typically done in stages.

Stage 1 is the interval $[0, 1]$, represented by this bar "of length 1."

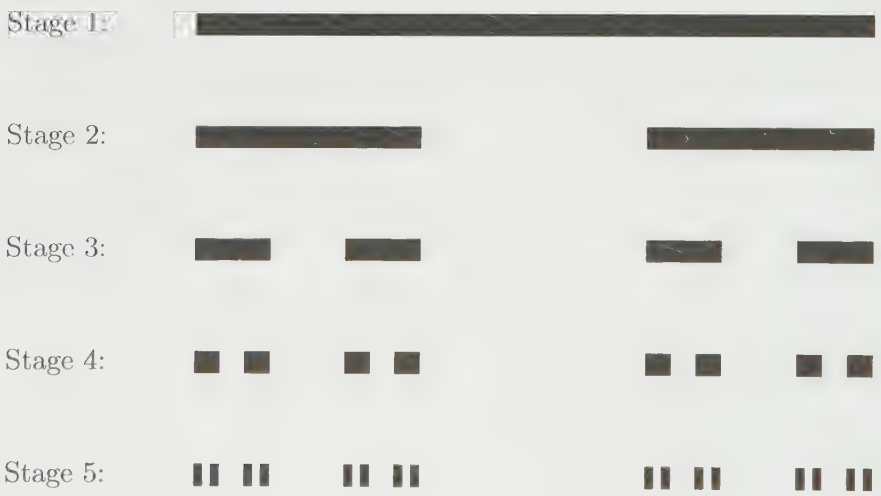
Stage 2 is found by taking the above and removing the middle third. This gives $[0, 1/3] \cup [2/3, 1]$.



Next you take each of the intervals $[0, 1/3]$ and $[2/3, 1]$ and remove the middle third of each of those. This gives $[0, 1/9] \cup [2/9, 1/3] \cup [2/3, 7/9] \cup [8/9, 1]$.



Continuing in this way gives a sequence of sets.



As you can see, we are removing more and more points. The *Cantor set* is defined to be the limit of this procedure. That is, the Cantor set is the set of points in $[0, 1]$ which are never going to be removed by any of the above stages, even if you continue this cutting-out procedure forever.

A very reasonable question would be: are there *any* points in the Cantor set? At first glance it seems possible that *all* the points will eventually be removed! This is

not the case, though. For instance, you can quickly verify that 0 and 1 will always be there. Also 1/3 and 2/3 will always be there. Indeed, if a point is ever the end point of one of the intervals, then it will never be removed.

Here is another way to think about it. Given a number $x \in [0,1]$, that number is typically written in base 10. For instance 0.2748 means this:

$$0.2748 = 2 \cdot 10^{-1} + 7 \cdot 10^{-2} + 4 \cdot 10^{-3} + 8 \cdot 10^{-4}.$$

But you could easily write numbers in other bases too. For example, you can write 7/9 in base 3:

$$7/9 = 2 \cdot 3^{-1} + 1 \cdot 3^{-2}.$$

An alternative perspective: Since this number is in $[0,1]$, the first digit after the decimal tells you if your number is between 0/3 and 1/3, or 1/3 and 2/3, or 2/3 and 3/3. Once you know which length-$\frac{1}{3}$ interval you are in, then divide that interval into 3 subintervals, each of length 1/9. Which one of those your number falls in determines the next digit, and so on.

Fact B.1. The elements in the Cantor set are precisely the numbers in $[0,1]$ where, if you write them in base 3, you get only 0s and 2s.

So the number whose base-3 representation is 0.02202022 is in the Cantor, but the number whose base-3 representation is 0.21012021 is not in the Cantor set.

Now, we mentioned that the endpoints of each interval are in there. So is that it? If so, then among other things we could prove that the Cantor set is countable: go through each stage, enumerating the new endpoints; in this way you list out all the endpoints one after another, proving that the collection of endpoints is countable.

But, no, there are numbers in the Cantor set which are *not* endpoints! For example you can show that 1/4 is in the Cantor set; 3/10 is also in there. (1/4, for instance, has base-3 representation 0.02020202....) And it's not just a few points either—there are *uncountably many* points in the Cantor set. So looking just at the endpoints, we missed essentially all of them!

Fact B.2. There are uncountably many points in the Cantor set.

One way to prove this is to find a bijection between the Cantor set and all the numbers in $[0,1]$, since we know the latter set is uncountably infinite. The bijection is pretty straightforward too: Take any base-3 representation of a number and change all the 2s to 1s; note that the result is a string of 0s and 1s. If you interpret this string as a number in base-2, then this procedure turns a base-3 number into a unique base-2 representation of a number in $[0,1]$. Conversely, take any base-2 representation of a number and by switching the 2s to 3s you get a base-3 representation of a number and hence a number in the Cantor set. In this way, you can show that the Cantor set is uncountable.

At the start of this conversation we wondered whether there were any points at all in the Cantor set, and now we are saying that there are as many in the Cantor set as there are in all of $[0,1]$?? Indeed, one of the amazing things about the set is that it is both "big" and "small" at the same time. It's big in that there are uncountably many points in it, but yet it is small in that there are not any "clumps" of points anywhere![2] The way this is usually stated is in terms of "length."

The length of $[a,b]$ is $b-a$. It's just the length of the interval. And the set $[0,2]\cup[5,6]$ has length $2+1=3$. So Stage 1 of the Cantor set is $[0,1]$ and hence has length 1. Stage 2 of the Cantor set is $[0,1/3]\cup[2/3,1]$ and hence has length $2/3$; another way to think about it is that we removed $1/3$ of the set, leaving 2/3rds left. Stage 3 has length $(2/3)^2$, since again we found each interval and cut out $1/3$ of it. Stage 4 has length $(2/3)^3$ because again we cut $1/3$ out of each interval we had.

Continuing like this we see that Stage n of the Cantor set has length $(2/3)^n$. And hence the Cantor set has length

$$\lim_{n\to\infty}\left(\frac{2}{3}\right)^n = 0.$$

So in this "length" sense, the Cantor set is tiny!

Fact B.3. The Cantor set is both big and small at the same time. It is big because it has uncountable many points, but tiny in that it has zero length.

And that's not all!

Fact B.4. The Cantor set is *totally disconnected*, but yet is *perfect*.

Totally disconnected means that given any two points in the set, there are numbers between these which are not in the Cantor set. The reason is that at some point a "middle third" got removed between these two. And perfect means that each point in the set is a limit of a sequence of other points in the set. So even though *every* point is in some sense "separated" from any other point, with an interval of missing points in-between them, still every point is the limit point of points in the set!

Finally, it is of course fractal-like, which is usually called being *self-similar*. So if you zoom in on any interval from any stage of the Cantor set, what you are looking at looks exactly like the original Cantor set.

What a wacky set.

A set is never big. Nor is it small. It is precisely the size it means to be.

[2]

B.2 An Abundant Addition Aboundingly Ascends

In Chapter 4 we saw many examples in which we summed up *infinitely* many positive numbers and yet got a *finite* number out. For example,

$$\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\frac{1}{16}+\cdots=1.$$

Notice, though, that each of our examples only involved a *countably* infinite collection of numbers, and we have yet to talk about what happens when you sum up an *uncountably* infinite collection of numbers. Can such a sum also add up to a finite number?

First, you might ask what we even mean by this. How can you sum up uncountably many things when sums like

$$\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\frac{1}{16}+\cdots$$

are inherently countable—we are adding up one thing, then the next, then the next, and so on, and we are asking what that sequence of partial sums converges to. But doing one thing after another guarantees that the procedure has only countably many steps.

As it turns out, we don't even have to get into the details about how to generalize our notion of countable sums to uncountable sums. The answer to our original question is 'no'—every sum of uncountably many positive numbers must equal ∞, and in fact something even stronger always holds: The uncountable set is even guaranteed to contain a countable subset which itself sums to ∞!

To see this, let S be an arbitrary set of uncountably many positive numbers, which we want to add up. Since everything in S is positive, we can group these elements into the following sets. Let S_1 be the numbers in S which are at least 1. Let S_2 be the elements of S which are between $1/2$ and 1. Let S_3 be the elements of S which are between $1/3$ and $1/2$. Let S_4 be the elements of S which are between $1/4$ and $1/3$. And so on. In general, for $n \geq 2$ let

$$S_n = \left\{ x \in S : \frac{1}{n} \leq x < \frac{1}{n-1} \right\}.$$

Now simply note that we just put uncountably many numbers into countably many sets. Clearly at least one of these sets must have infinitely many numbers in it (in fact, at least one will have uncountably many numbers in it, but this is stronger than we need). Suppose S_N has infinitely many numbers in it. Adding up any countable collection from this set will be at least

$$\frac{1}{N}+\frac{1}{N}+\frac{1}{N}+\frac{1}{N}+\frac{1}{N}+\cdots,$$

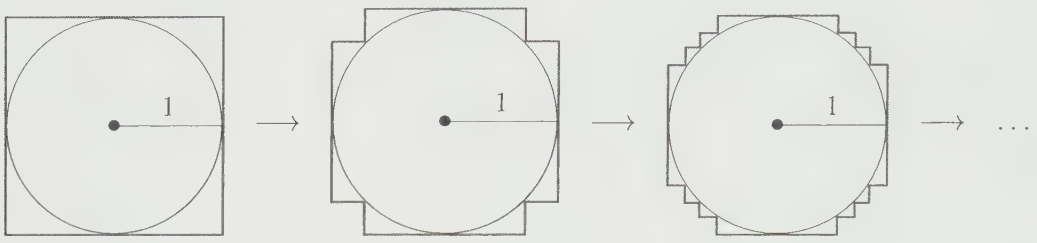
which equals ∞, completing the proof.[3] □

[3] *Shhh... Don't tell anyone, but this is also essentially a solution to Exercise 6.35.*

B.3 Peculiar and Pathological Perimeters

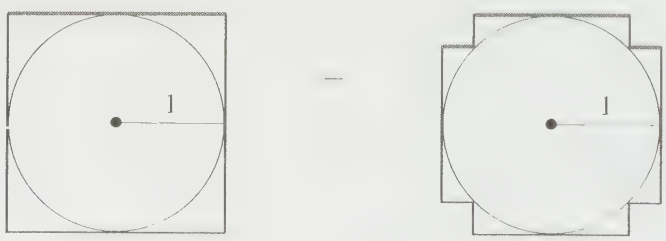
In Chapter 8 we looked at a procedure in which the area of polygons are converging to the area of a circle. First, even though most of our attention was on the inscribed n-gons, it was quite important that we noted then that the areas of the inscribed n-gons *and* the areas of the circumscribed n-gons were *both* converging to the same thing.

Interestingly, perimeter and area do not behave well together. If I have a sequence of shapes containing a circle whose areas are getting closer and closer to the area of a circle, it need not be the case that the perimeters are getting closer and closer to the perimeter of the circle. Indeed, consider the following procedure to converge to a circle:



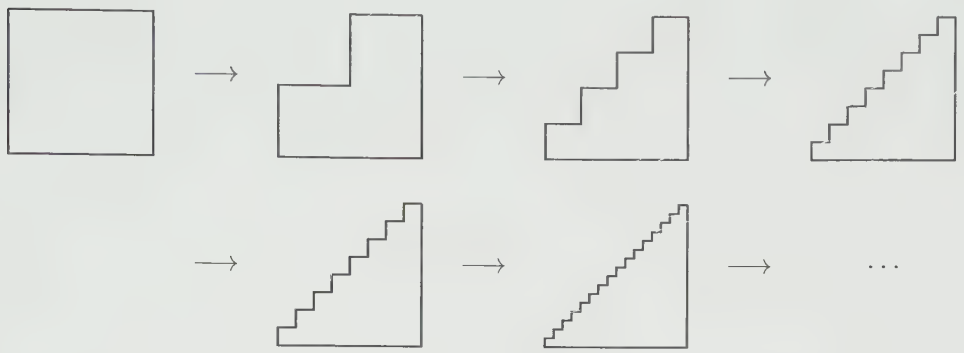
We drew a circumscribed square around the circle. Then, each of the four corners we cut off, instead having straight line segments to the circle. This gave a new shape containing the circle, and this new shape now has 8 corners sticking out. We now repeat, cutting off all 8 of these corners in the same way. Repeating this indefinitely, one can see that the zig-zag shape is, in some way, *converging* to the circle. If you calculated the areas of the these zig-zag shapes, for instance, you would find that they are indeed converging to the area of the circle.

But notice now that the perimeter of the first square is 8. The perimeter of the next zig-zag shape... is also 8! The straight line on the top of the square, for instance, is still there, it's just now in three pieces:



So you have a sequence of shapes, each with perimeter 8, that are "converging" to a shape with perimeter 2π... so $8 = 2\pi$? And so $\pi = 4$? No! You have to be really careful about this—perimeter does not converge in usual ways.

Here's another example. Note that each drawing below has the same perimeter. Let's say the first drawing is a 1×1 square. Then every drawing has perimeter 4. In particular, the stair-step portion of each has length 2.



However, as the steps get increasingly tiny, it sure looks like it is approaching the hypotenuse of a right triangle...

...but in the limit, the length-2 stair steps turn into a straight line hypotenuse of length $\sqrt{2}$. This time, if perimeters and limits behaved nicely, this would imply that $\sqrt{2} = 2$! Again, something is wrong.

Similar phenomena exist in functional analysis. For instance, it is possible for a pair of functions f and g to be very close to each other, while f' and g' are very far from each other. Likewise, it's possible for two functions to be very close while their arc lengths are far apart. Dog owners know this from personal experience, though. If you go on a walk with an energetic dog, even if the leash is pretty short (ensuring you two are "close" together at every moment), still the dog is hopping from side to side, smelling this, peeing on that... the dog's path length is much longer than yours! I bet my mom's dog walks twice the distance she does when they go on a walk together, despite being on a short leash.

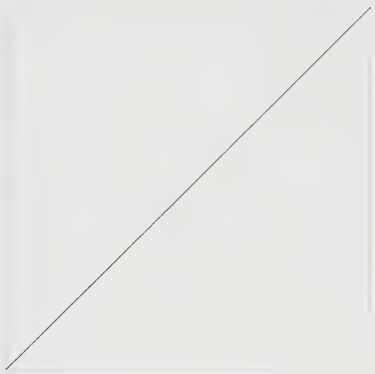
There is one final interesting manifestation of this that I'd like to mention, called *the coastline paradox*. If you fly above a coastline and take pictures of it, and then return home, print them out, and use them to measure the coastline's length, the answer you'll get depends on how good of a camera you have. If you have a better camera your picture will pick out more zigs and zags and consequently you will measure a longer distance than if your resolution is low, and the details run together.[4]

[4] ...Now, if you want to know how the coastline paradox affects Gerrymandering, take my future class on the Math of Politics.

B.4 A Devilish De-Descent

Here's one more example that in some ways relates to the weirdness of the Cantor set. It's a function called *The Devil's Staircase*. And like the Cantor set, it's best to describe it in stages.

Consider the straight-line function from (0,0) to (1,1); this is Stage 1 of the creation of the Devil's Staircase.

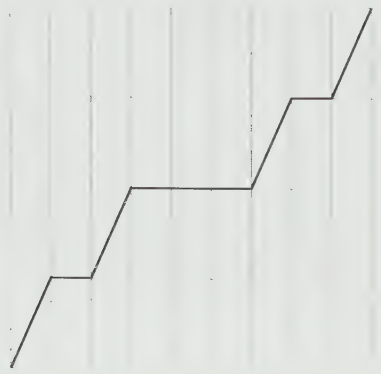


Stage 1

In Stage 2, you take the middle third of the function and replace it with a horizontal line (halfway up; so at $y = 1/2$), and then redraw the diagonal lines in the first and last third to meet this horizontal line.

Stage 2

In Stage 3 you do the same thing to each diagonal line. That is, you take the middle third of the line in $[0, 1/3]$ and the line in $[2/3, 1]$, and replace each with a horizontal line (halfway up; so at $y = 1/4$ and $y = 3/4$, respectively), and then redraw the diagonal lines in the first and last third to meet this horizontal line.



Stage 3

Now continue this. For each diagonal line, replace it with a horizontal line halfway up, and redraw the diagonal lines to make it continuous. Below are the next 4 iterations.

Stage 4

Stage 5

Stage 6

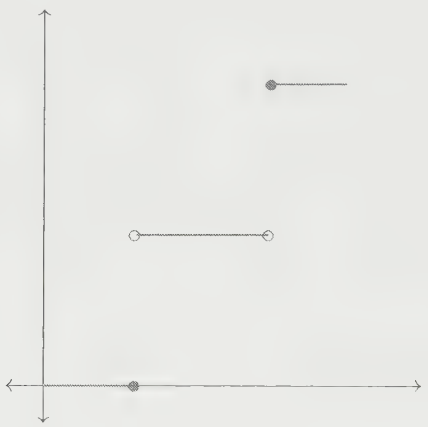
Stage 7

Continuing this process forever, the resulting sequence of functions converges to the function which is called *The Devil's Staircase*. And this function has some fascinating properties.

- Note first that if you get a flat spot at any stage, it remains a flat spot forever. And the flat spot after the first stage is $[1/3, 2/3]$. The flat spots in Stage $n+1$ are those in Stage n, as well as this: Given any diagonal region in Stage n, the middle third becomes flat in Stage $n+1$. That is, the flat spots are exactly the regions removed in the construction of the Cantor set! And therefore, the only points which will never be part of a flat spot are precisely the points in the Cantor set.

- And since the Cantor set has length 0, the Devil's staircase is flat almost everywhere. Said differently, the Devil's Staircase has a derivative of 0 at almost every point. But this is odd, because despite being horizontal nearly everywhere... somehow it still manages to climb from $(0,0)$ to $(1,1)$!

  Now, you might say that you can think of a function that is flat nearly everywhere but still manages to climb from $(0,0)$ to $(1,1)$. A (stair-)step function, for instance:



  But this function is clearly not continuous, and to make it continuous we would have to have to connect the steps together, and however we did so, we would end up with an interval on which the function is increasing (with a positive derivative).

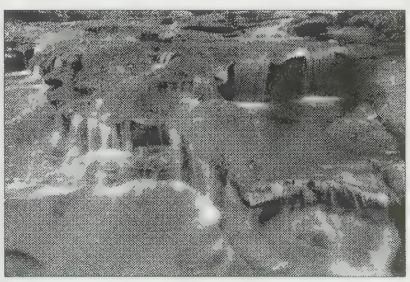
  And yet the Devil's Staircase somehow manages to never have an interval where it is increasing, and at no point does its derivative take a positive value (in fact, its derivative is zero almost everywhere)... and yet it still manages to somehow climb from the origin to $(1,1)$. Amazing.

- And it's worth reiterating that amazing point, for a continuous function to get bigger, you would think that there would have to be some interval in which it is getting bigger. But alas, that's not the case. You can get bigger, be continuous, and yet have no interval in which you are strictly rising. WHAAAAA?!?!?!?

- Another interesting fact about this function is its length. The straight-line function from $(0,0)$ to $(1,1)$ has a length of $\sqrt{2}$. Meanwhile, the line segment from $(0,0)$ to $(1,0)$, plus the segment from $(1,0)$ to $(1,1)$, also makes it from $(0,0)$ to $(1,1)$, but has length 2. So what's the length of the Devil's staircase? You might imagine that since it seems to be a mix of these two that its length is somewhere in between, but in fact its length is also 2. Indeed, it starts at Stage 1 with a length of $\sqrt{2}$, but with each successive iteration its length gets closer and closer to 2, where it ends up.

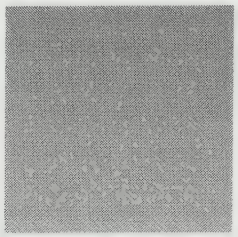
- Final thoughts: This function is also commonly called the Cantor function. Note that since the function is continuous, it has the intermediate value property. Hence not only does it climb from $(0,0)$ to $(1,1)$, but it hits every y-value in between.

- P.S. While researching about the Devil's Staircase I learned about a series of waterfalls in the coastal rainforest of Oregon that are called The Devil's Staircase. It is a gorgeous area that has been labeled an "area of critical environmental concern" and as former congressman Jim Weaver says, "[the area] deserves no less than the highest protection that can be afforded under law," and I gotta say I completely agree. Here are two angles of the Devil's Staircase in the wild:

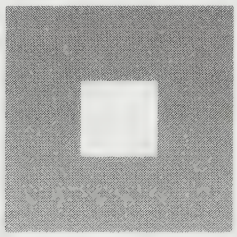


B.5 Menger's Matterless Material

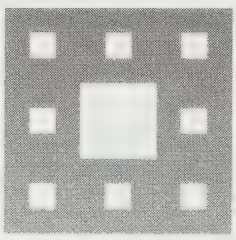
The *Sierpinski carpet* is a fractal-like set constructed as follows. Begin with a square.



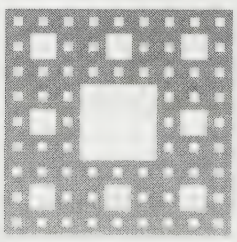
Cut this square into 9 equally-sized smaller squares, and remove the middle piece:



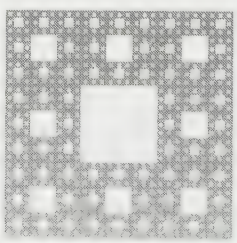
Now consider the remaining 8 squares, cut each of these into 9 equally-sized pieces, and remove each middle piece:



And continue.



Do this forever. The *Sierpinski carpet* is the set of points that will never be removed. Here's what one more stage looks like:

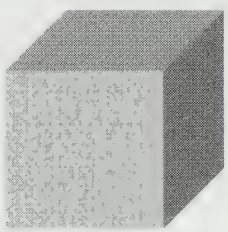


The Sierpinski carpet has some wonderful properties. As one example, you might wonder whether forever poking holes in the carpet will, in the limit, leave nothing left. This isn't true though—like with the Cantor set, lots and lots of points remain. So what's the area of what's left? Notice that at each step, each square chunk of area is reduced to 8/9. And if each piece of area shrinks by 8/9 at each stage, then the whole area also shrinks by 8/9 at each stage. If the area starts out as 1, then the area after stage 1 is 8/9. After stage 2 it is $(8/9) \cdot (8/9) = (8/9)^2$. After stage 3 it is $(8/9)^2 \cdot (8/9) = (8/9)^3$. And so on. Indeed, after stage n, the area is $(8/9)^n$. So what's the area[5] after all the stages? It's

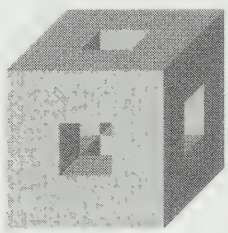
$$\lim_{n\to\infty} (8/9)^n = 0.$$

So as we have pointed out, the carpet does not completely disappear uncountably many points will never be removed. And yet, as the above calculation shows, no area remains![6]

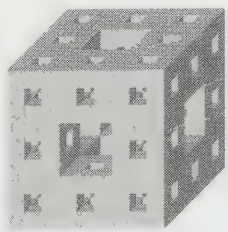
We could discuss the Sierpinski carpet much more, but for now let's turn to the 3-dimensional version of this, called the *Menger sponge*. The idea is similar, except instead of beginning with a square we begin with a cube.



Now divide each face into 9 equally-sized squares and punch-out each middle one.



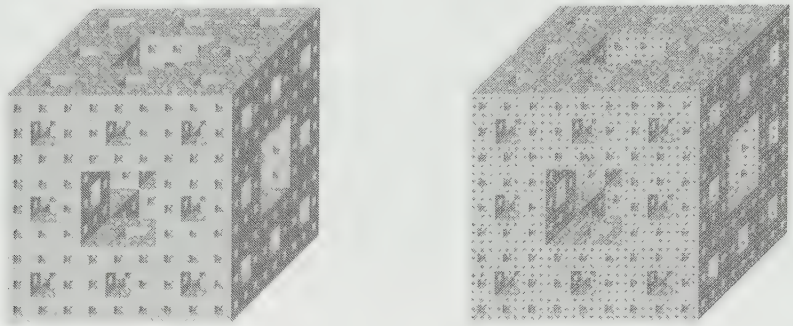
And, like before, continue this procedure.



[5]"Area" in standard Lebesgue measure.

[6]This is particularly convenient if you plan to sew a Sierpinski carpet for yourself, since you won't need any thread.

Here are levels three and four of the sponge:



The *Menger sponge* is the limiting shape of this process. There are many fascinating facts about the Menger sponge, but one of my favorites is what happens when you cut the sponge diagonally. Here is the video showing the awesome answer: `https://youtu.be/fWsmq9E4YC0`, courtesy of the Simon's Foundation. (Make sure to pause the video to guess what shapes you're going to see.)

There are many self-similar, fractal-like geometric constructions (See: *Iterated function systems* and the amazing consequences of the *chaos game*). One shape that I particularly enjoy is called *Barnsley's fern*.

Such shapes are easily and quickly created graphically using just a few basic rules. Such shapes are used in computer graphics to create natural, beautiful shapes with little processing power, and I love how such things mimic nature. Barnsley's fern resembles a common real-world fern called *black spleenwort*, which is native to all sorts of places, from Norway to Egypt to Hawaii to Turkey.

B.6 A Prestigious Proof of a Primal Puzzle

Back in the third century BC, Euclid proved that there are infinitely many primes, and today his argument is perhaps the world's most widely known proof. If we let $\mathcal{P}$ be the collection of all prime numbers, then his proof began by assuming $|\mathcal{P}|$ is finite. If it is, and we let

$$Q = \prod_{p \in \mathcal{P}} p,$$

then we obtain an integer which contains every prime as a factor. The contradiction then comes by considering $Q + 1$, which according to the *fundamental theorem of arithmetic* must also be prime or product of primes. But since $Q + 1 \equiv 1 \pmod{p}$ for every $p \in \mathcal{P}$, our collection of primes must be incomplete. Ta-da!

One consequence of the above reasoning is the following fact.

Fact B.5. If Q is any finite product of primes and $n \in \mathbb{N}$, then $nQ+1$ is not divisible by any of those primes.

Before preceding, recall the following two results from Chapter 4.

- **Proposition 4.9** (*Geometric series test*). Assume that a and r are non-zero real numbers. Then $\displaystyle\sum_{k=0}^{\infty} a \cdot r^k = \begin{cases} \dfrac{a}{1-r} & \text{if } |r| < 1 \\ \text{diverges} & \text{if } |r| \geq 1 \end{cases}$

- **Proposition 4.15.** The harmonic series $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n}$ diverges.

There are infinitely many primes, but they become sparse as you move to larger and larger integers. Indeed, the celebrated *prime number theorem* says that the number of primes in $\{1, 2, 3, \ldots, N\}$ is about $N/\log(N)$. In particular, if you pick a large N, the probability that it is prime is about $1/\log(N)$. Since $1/\log(N) \to 0$ as $N \to \infty$, this means that fewer and fewer numbers are prime.

Now, we saw in Chapter 4 that not only does the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ diverge, but in some sense it just barely diverges –according to the series p-test, $\sum_{n=1}^{\infty} \frac{1}{n^r}$ will converge for *any* $r > 1$.

Here is an interesting question: what happens if instead of considering

$$\sum_{k=1}^{\infty} \frac{1}{k} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8} + \frac{1}{9} + \frac{1}{10} + \frac{1}{11} + \frac{1}{12} + \frac{1}{13} + \ldots$$

we only considered reciprocals of primes?

$$\sum_{p \in \mathcal{P}} \frac{1}{p} = \frac{1}{2} + \frac{1}{3} + \frac{1}{5} + \frac{1}{7} + \frac{1}{11} + \frac{1}{13} + \frac{1}{17} + \frac{1}{19} + \frac{1}{23} + \frac{1}{31} + \frac{1}{37} + \frac{1}{41} + \ldots$$

Notice that even at the start a lot of terms are missing, and once you go way down the sum there are a *lot* fewer numbers. Since the harmonic series just barely diverged,

and the sum of reciprocals of primes seems to be growing much slower than the harmonic series, it would be reasonable to guess that this second sum might converge. Interestingly, as the great Leonard Euler proved, it does not.

Theorem B.6. $\displaystyle\sum_{p\in\mathcal{P}} \frac{1}{p} = \infty.$

There are many known proofs of this. The one I will show you is quite nice and also combines some important tools we developed in Chapter 4.

Proof. Assume for a contradiction that $\sum_{p\in\mathcal{P}} \frac{1}{p}$ converges to some real number L. Since each $1/p > 0$, the sequence of partial sums is monotonically increasing to L, and by the definition of sequence convergence, at some point the partial sums get above $L - \frac{1}{2}$; said differently, at some point the "tail" of this sum is less than $1/2$. That is, if we enumerate the primes in order, $\mathcal{P} = \{p_1, p_2, p_3, \dots\}$, then there must exist some k for which

$$\sum_{i>k} \frac{1}{p_i} < \frac{1}{2}. \qquad (*)$$

Let

$$Q = \prod_{i\le k} p_i.$$

Being the product of all primes at most p_k, by Fact B.5 we see that $nQ+1$ is not divisible by any of $\{p_1, p_2, \dots, p_k\}$, and so must either be a prime in $\{p_{k+1}, p_{k+2}, p_{k+3}, \dots\}$, or a product of primes from this set.

Neat Observation B.7. Recall that $(x+y)^2 = (x+y)(x+y) = (xx+xy+yx+yy)$. Likewise, $(x+y+z)^2 = (x+y+z)(x+y+z) = (xx+xy+xz+yx+yy+yz+zx+zy+zz)$; in particular, the answer is a sum of all ways to pick one thing from $(x+y+z)$ multiplied with one thing from $(x+y+z)$. More generally, $(p_{k+1}+p_{k+2}+p_{k+3}+p_{k+4}+\dots)^2$ is equal to a sum of all numbers which are a product of two (not necessarily distinct) primes from $\{p_{k+1}, p_{k+2}, p_{k+3}, \dots\}$ (with some repetition). Even more generally, $(p_{k+1}+p_{k+2}+p_{k+3}+p_{k+4}+\dots)^j$ is equal to a sum of all numbers which are a product of j (not necessarily distinct) primes from $\{p_{k+1}, p_{k+2}, p_{k+3}, \dots\}$ (with some repetition).

We are almost done. Next note that $(\frac{1}{p_{k+1}} + \frac{1}{p_{k+2}} + \frac{1}{p_{k+3}} + \frac{1}{p_{k+4}} + \dots)^j$ is equal to a sum of all numbers of the form $\frac{1}{m}$ where m is product of j primes from $\{p_{k+1}, p_{k+2}, p_{k+3}, \dots\}$ (with some repetition). And, lastly, that

$$\sum_{j=1}^{\infty} \left(\sum_{i>k} \frac{1}{p_i} \right)^j$$

is the sum of all numbers of the form $\frac{1}{m}$ where m is product of *any number* of (not necessarily distinct) primes from $\{p_{k+1}, p_{k+2}, p_{k+3}, \dots\}$ (with some repetition).

That neat observation is useful because we also know that

$$\sum_{j=1}^{\infty} \left(\sum_{i>k} \frac{1}{p_i} \right)^j$$

converges by the geometric series test. Indeed, if we let $\tilde{r} = \sum_{i>k} \frac{1}{p_i}$, then combining its positivity with ( ) we have that

$$0 < \tilde{r} < \frac{1}{2},$$

and by the geometric series test $\sum_{j=1}^{\infty} r^j$ converges whenever $|r| < 1$. Since it converges, let's say that it converges to $L \in \mathbb{R}$:

$$\sum_{j=1}^{\infty} \left(\sum_{i>k} \frac{1}{p_i} \right)^j = L.$$

To review:

- Since $Q = \prod_{i \leq k} p_i$, by Fact B.5 we know that for any $n \in \mathbb{N}$, $nQ+1$ is a product only of primes in the set $\{p_{k+1}, p_{k+2}, p_{k+3}, \dots\}$. Therefore $\sum_{n=1}^{\infty} \frac{1}{nQ+1}$ is the sum of *some* of the reciprocals of products of primes from $\{p_{k+1}, p_{k+2}, p_{k+3}, \dots\}$.
- By Neat Observation B.7, $\sum_{j=1}^{\infty} \left(\sum_{i>k} \frac{1}{p_i} \right)^j$ is the sum of *all* reciprocals of products of primes from $\{p_{k+1}, p_{k+2}, p_{k+3}, \dots\}$. Moreover, this sum converges to L.

Putting this all together,

$$\sum_{n=1}^{\infty} \frac{1}{nQ+1} \leq \sum_{j=1}^{\infty} \left(\sum_{i>k} \frac{1}{p_i} \right)^j = L. \qquad (\ )$$

But since $0 < nQ + 1 \leq n(Q+1)$ for all $n \in \mathbb{N}$, we see that $\frac{1}{n(Q+1)} < \frac{1}{nQ+1}$. Furthermore, since the harmonic series diverges, we have that

$$\sum_{n=1}^{\infty} \frac{1}{n(Q+1)} = \frac{1}{Q+1} \sum_{n=1}^{\infty} \frac{1}{n} = \infty. \qquad (\ )$$

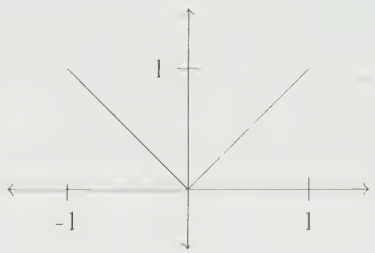
Combining ( ) and ( ) gives the contradiction that $\infty \leq L$:

$$\infty = \frac{1}{Q+1} \sum_{n=1}^{\infty} \frac{1}{n} = \sum_{n=1}^{\infty} \frac{1}{n(Q+1)} \leq \sum_{j=1}^{\infty} \left(\sum_{i>k} \frac{1}{p_i} \right)^j = L.$$

□

B.7 A Smooth and Spiky Solution

Recall that Theorem 7.6 says that if f is differentiable, then f is also continuous. So differentiability is a *stronger* condition; differentiability implies continuity. And in fact, it is *strictly* stronger as well, since there are functions which are continuous at every point, but are not differentiable at every point. The absolute value function $A(x) = |x|$ on $[-1, 1]$ is a classic example.



This function is continuous everywhere, but there is one point where it is not differentiable: at the origin. Indeed, applying the definition of the derivative of A_0 at the point $x = 0$, the limit

$$A_0'(0) = \lim_{h \to 0} \frac{A_0(0+h) - A_0(0)}{h}$$

equals two different values depending on whether you are approaching from the left or from the right. From the left, $A_0(x) = -x$ and so

$$\lim_{h \to 0^-} \frac{A_0(0+h) - A_0(0)}{h} = \lim_{h \to 0} \frac{-h - 0}{h} = \lim_{h \to 0} -1 = -1.$$

While from the right, $A_0(x) = x$ and so

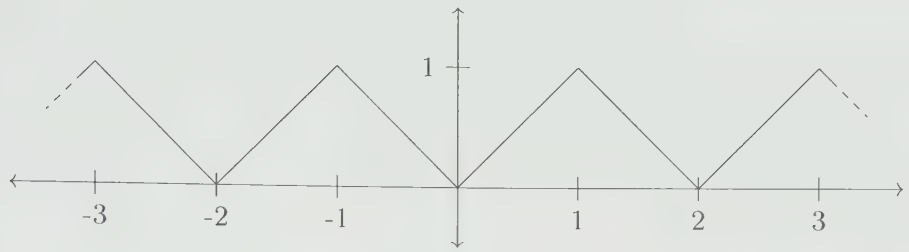
$$\lim_{h \to 0^-} \frac{A_0(0+h) - A_0(0)}{h} = \lim_{h \to 0} \frac{h - 0}{h} = \lim_{h \to 0} 1 = 1.$$

Both of these make sense, as the line approaching zero from the left is $y = -x$ which has slope -1, while the line approaching zero from the right is $y = x$ which has slope 1. And because those are different, the two-sided limit

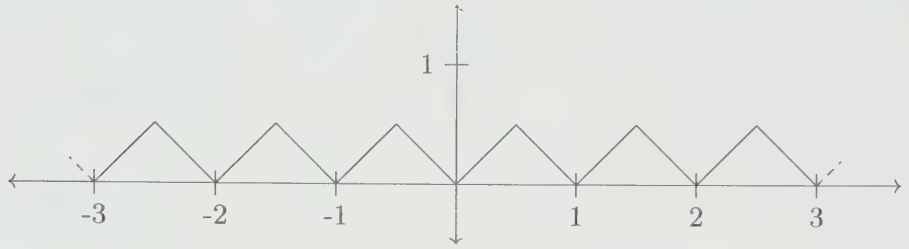
$$A_0'(0) = \lim_{h \to 0} \frac{A_0(0+h) - A_0(0)}{h}$$

does not exist.

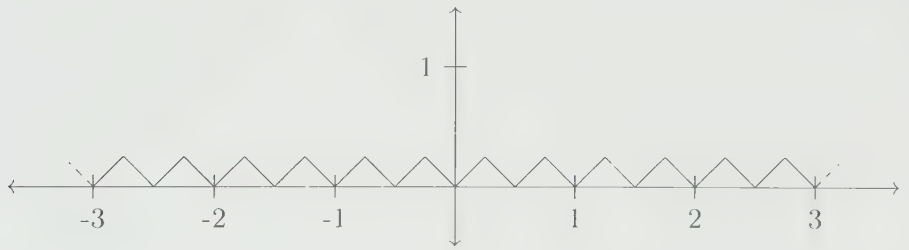
So we can indeed find a function A_0 where there is a point which is continuous but not differentiable. Moreover, if we extend A_0 to be periodic on all of $\mathbb{R}$ then we get a function with "points" at every integer, and hence a function which is non-differentiable at every integer:



But how non-differentiable can our function be? We could of course make the non-differentiable points occur often. Consider the function which we'll call A_1:



Note that A_1 is discontinuous at every point of the form $\frac{m}{2}$, where m is an integer. Better yet, here's A_2:



Note that A_2 is discontinuous at every point of the form $\frac{m}{2^2}$, where m is an integer. Indeed, you could keep doing this and A_k would have the property that it is discontinuous at every point of the form $\frac{m}{2^k}$.

In some sense this is making things weirder, but in another it's not doing much: Is it so different to have every 8^{th} term be discontinuous compared to every 64^{th} term? It's still a discrete set where you have fixed gap-sizes between the points.

The gold standard example of course would be to find a continuous function that is differentiable *nowhere*. Now take a minute to appreciate how peculiar such a function would be. In all our examples thus far, being discontinuous at a point was obtained by having two lines (which are not only continuous, but also differentiable) meet at a single point, forming a peak. The only other techniques I can think of (like having a jump discontinuity) can't be used since they're not continuous. Indeed, it seems reasonable that having a peak like this could be only way to be continuous at a point but also not differentiable at that point.

And it is also reasonable to think that in order to have a peak, you need to have a region around that peak where you have two nice, differentiable lines meeting each

other. And of course, it is true that you need to have a region of differentiability around each non-discontinuous point, then we would be able to conclude that it is not possible to be both continuous at every point and yet discontinuous at no point.

And yet! And yet! It is possible! There do exist functions which are continuous everywhere and differentiable nowhere. Moreover, we are already close to writing down an example...

Consider the function

$$f(x) = \sum_{k=0}^{\infty} A_k(x).$$

This is a good example to introduce series of functions, so I won't assume that you're an expert in the material from Chapter 9. Moreover, that material is quite deep and nuanced, so we will only scratch the surface here. Briefly, though: for any number x_0, and each fixed k, you can clearly plug this number into A_k and get out some other number $A_k(x_0)$; furthermore, note that they are all non-negative outputs. What is $f(x_0)$? It's what you get when you add up all these numbers: $A_1(x_0) + A_2(x_0) + \ldots$. That is, recall from Chapter 4 that a series of non-negative numbers can either add up to some number or to ∞. In this case one can show that the sum is always finite (since it's always less that $\sum_{k=0}^{\infty} \frac{1}{2^k}$).

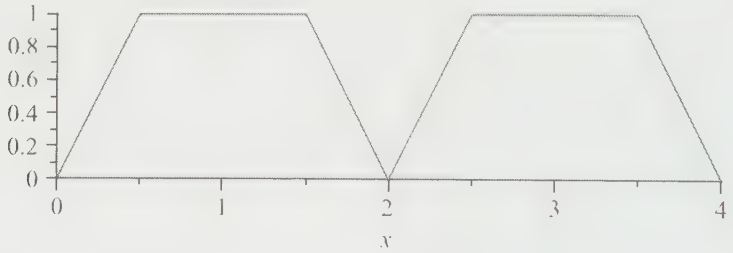
Moreover, one can show that f would also be continuous. And yet—and this is more challenging—one can also prove that f is differentiable nowhere.

In short: If x_0 is of the form $\frac{m}{2^n}$ for integers m and n, then $f(x_0)$ is more intuitively peak-like and still non-differentiable even in the sum. If x_0 is not of this form, then you should instead consider a sequence of points x_n converging to x_0 from the left, each of which *is* of the form $\frac{m}{2^n}$, and a second sequence y_n converging from the right, each of which is of that form. If you do so such that

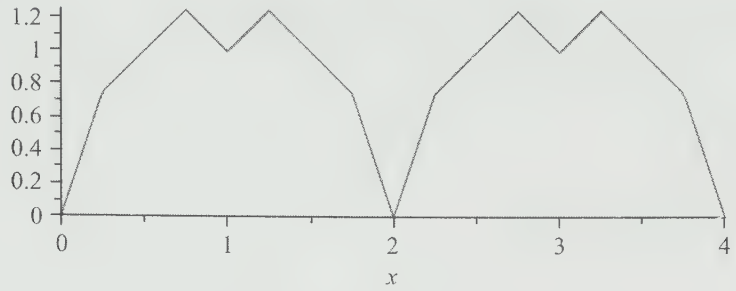
$$x_n = \frac{p_n}{2^n} < x_0 < \frac{p_n + 1}{2^n} = y_n$$

for each n, then the derivative limit along these two sequence are not equal, implying that $f'(x_0)$ does not exist.

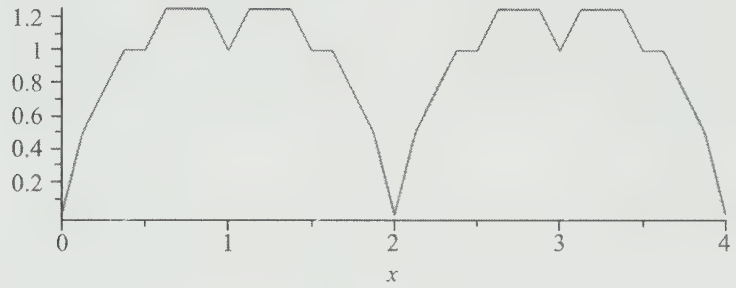
Finally, in closing, here is a look at the graph of f, which is a more discrete version of an example published by Weierstrass in 1872. We begin by showing partial sums of its definition. First, here is $A_0 + A_1$:



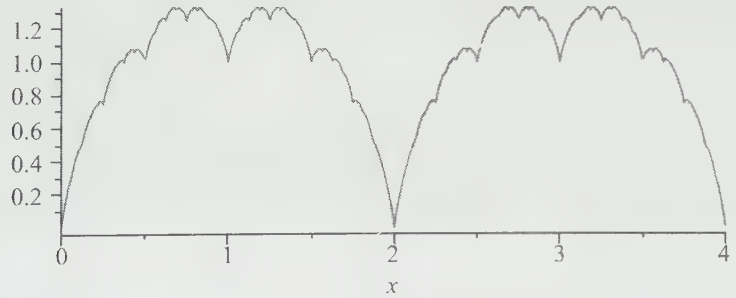
And now here is $A_0 + A_1 + A_2$:



Here is $A_0 + A_1 + A_2 + A_3$:



Skipping ahead a bit, here is $A_0 + A_1 + A_2 + A_3 + A_4 + A_5 + A_6$:



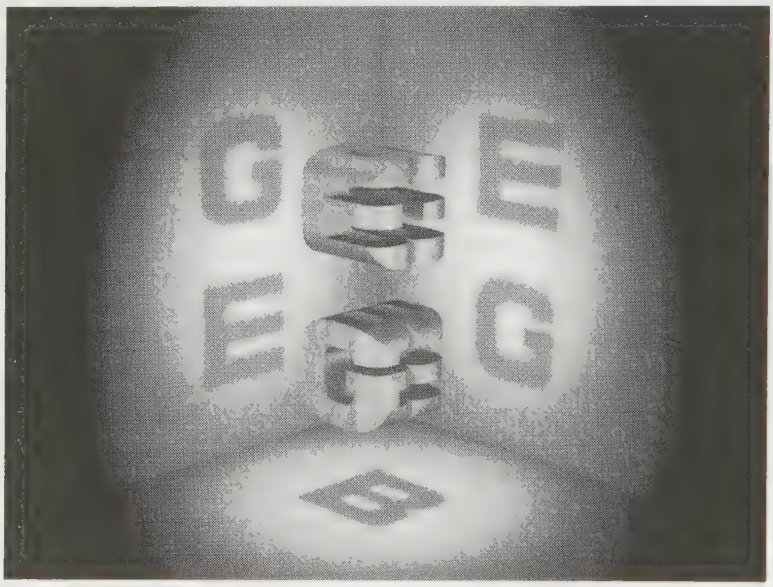
Note that A_7 is a function with amplitude $\frac{1}{2^7} \approx .0078$, and it only gets smaller from there. So the above is about what the final graph looks like. The later sums play an extremely important role, but you'd have to zoom in quite a bit to see how they change the graph.

B.8 Structuring Stuff from its Shadows' Shapes

There are technologies now where you can take a bunch of pictures of something, and it can piece them together to create a 3D rendering of that thing—turning 2D perspectives into a 3D object.

CAT scans also work on this basic principle. Take many different x-ray "pictures" at a given level to generate a 2D cross-section at that level. A computer then pieces these all together to form a 3D rendering of a patient's organ.

But think now about shadows. Given the shadows of something, can you construct the object? Or, more to the point, given a collection of *desired* shadows, does there exist an object that has those shadows? If you want the shadows to be "G", "E" and "B" then the answer is yes (as the book cover to *Godel, Escher, Bach* shows):



In fact, there is a theorem that says that this is *almost*-always nearly possible.

Theorem (2D version). Let $G_\theta \subseteq L_\theta$, $\theta \in [0, \pi)$ be a family of any sets such that $\bigcup_\theta G_\theta$ is a measurable set in the plane. Then there exists a set $F \subset \mathbb{R}^2$ such that

- $G_\theta \subseteq \text{proj}_\theta F$;
- $\text{proj}_\theta F \setminus G_\theta$ has measure 0 for almost all $\theta \in [0, \pi)$.

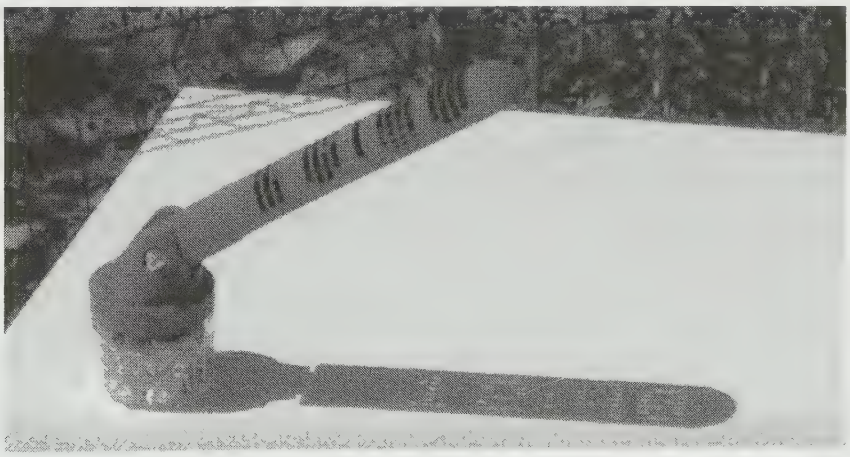
This theorem can also be generalized to three dimensions: Given a collection of 2D sets (which will be our shadows), there is a 3D set for which essentially all of its shadows are what you want—and in order too! This generalization has a fantastic

consequence; suppose the shadows that you want look like times:

1:00
1:01
1:02
1:03
1:04
⋮

Then there exists a 3D object whose shadows are these times—and you can even pick each angle you want for each time. Which means... that you could make a sundial whose shadows—as the sun moves across the sky—are precisely the time! In fact, you could even have it show the seconds too, and each second, as the sun moves, a new shadow is produced representing the new time at that precise second. It's a digital sundial!!

The technology to put this theory into practice is impressive, but admittedly is nowhere near refined enough produce readable per-second changes to the shadows. That said, one can now 3D print a digital sundial which shows the time rounded to the nearest 10-minute mark.



So in the picture above, the time will switch from 12:40 to 12:50 to 1:00 to 1:10 and so on, and will change every 10 minutes. Pretty amazing stuff.

B.9 A Composition Conundrum

Here is an example of integrable functions f and g, where their composition $f \circ g$ is *not* integrable.

At least some of the functions involved are going to have to be pretty weird. Since $f \circ g$ is not integrable, already we know that $f \circ g$ has to be weird, and therefore f and g can't both be super nice.

But we do have a few weird functions at our disposal that we know how to handle. The Chapter 8 exercises (twice) ask you to prove that Thomae's function

$$t(x) = \begin{cases} 1/n & \text{if } x \neq 0 \text{ and } x = m/n \in \mathbb{Q} \text{ in lowest terms with } n > 0 \\ 0 & \text{if } x \notin \mathbb{Q} \\ 1 & \text{if } x = 0 \end{cases}$$

is integrable. And there's another function that isn't so different than this which we know is *not* integrable: Dirichlet's function. Indeed, in Example 8.11 we showed that

$$d(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q} \\ 0 & \text{if } x \notin \mathbb{Q} \end{cases}$$

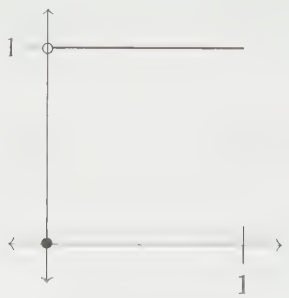
is *not* integrable. So can we choose t to be either f or g, and pick the other one cleverly so that their composition equals the Dirichlet function? There is a way!

Let $g(x)$ be Thomae's function on $[0, 1]$. How do we use f to turn g into the Dirichlet function? The irrationals are already good-to-go. So when we take $f(g(x))$, if $g(x) = 0$ we don't want f to change this fact. That is, we want $f(0) = 0$.

What else? Well, all the rationals are getting mapped to values $1/n$, for some n. And Dirichlet wants those value to all equal 1. That is, if $g(x) > 0$, we want $f(g(x))$ to equal 1. So for any $x > 0$, we want $f(x) = 1$. Make sense? This suggests that we define $f : [0, 1] \to \mathbb{R}$ to be

$$f(x) = \begin{cases} 1 & \text{if } x > 0 \\ 0 & \text{if } x = 0. \end{cases}$$

This f has a very simple form:



And by Porism 8.17, f is integrable. So we have that f and g are both integrable. We now simply note that $f \circ g$ is the Dirichlet function, and hence is not integrable.

$$f(g(x)) = \begin{cases} f(1/n) & \text{if } x \in \mathbb{Q} \\ f(0) & \text{if } x \notin \mathbb{Q} \end{cases}$$

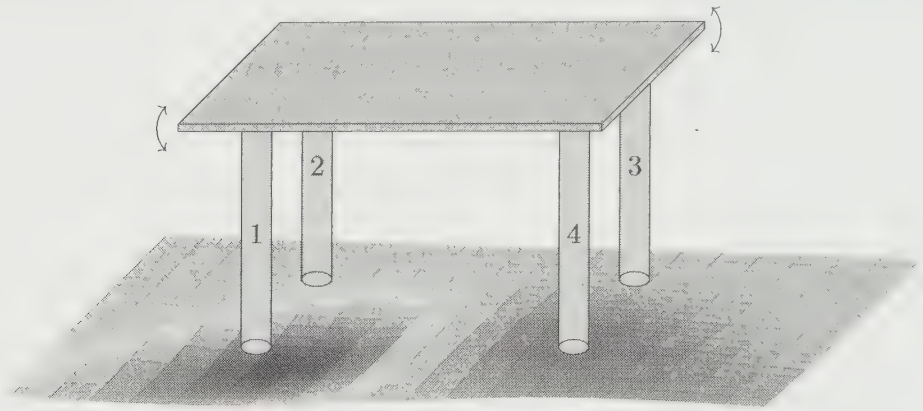
$$= \begin{cases} 1 & \text{if } x \in \mathbb{Q} \\ 0 & \text{if } x \notin \mathbb{Q} \end{cases}$$

So indeed, despite f and g both being integrable, $f \circ g$ wound up being the Dirichlet function which is *not* integrable. □

B.10 Turning the Tables on your Teetering Troubles

You're at a café, you've just ordered a delicious drink and you head to the patio to enjoy the beautiful day while you learn some mathematics. You sit down at the last open table and quickly discover why no one had taken it—the ground there is a little uneven, and so the table wobbles. For a non-mathematician, their entire day would have been utterly ruined; even if they try the classic strategy of stuffing a napkin under one leg, before long it will compress a bit and the wobble will return. Lucky for you, though, there is an easy, mathematically-verified solution—just rotate the table slightly, and like magic you will immediately find a spot where all four legs are completely stable on the ground. Allow me to explain.

Suppose first that we have a square table whose legs are all the same length but due to some uneven ground, the table wobbles. Let's say that legs 1 and 3 wobble, while legs 2 and 4 are on the ground.



Push down near the wobbly leg 3 so that legs 2, 3, and 4 are on the ground, while leg 1 is suspended above the ground. Let's say leg 1 is d inches above the ground.

Now rotate the table like a disc, moving leg 1 towards where leg 2 is, leg 2 towards where leg 3 is, and so on. We claim that before this 90° turn is complete, the table will reach a point where it is completely stable.

When you rotate, keep legs 2, 3 and 4 flush to the ground, and let $d(t)$ be the distance above the ground of leg 1 at each moment of time—therefore we want to prove that at some point $d(t) = 0$. When discussing the proof, it's helpful to imagine that leg 1 is ghost leg, which is able to pass into the ground. With this perspective, every turn is possible, and $d(t)$ will be negative if leg 1 moves into the ground.

Suppose the 90° turn takes 1 second. At $t = 0$ (the moment before the rotation begins) we have said that $d(t) = d$. After the 90° rotation, can you see what $d(t)$ will be? Legs 1 and 3 (previously wobbly) have switched spots with legs 2 and 4 (previously sturdy). So legs 2 and 4 are now on wobbly ground, but remember that we are insisting on keep legs 2, 3 and 4 on the ground—and the only way to take two wobbly legs and make them stable on the ground, is to force one of the other legs

into the ground. Leg 1 must therefore be inside the ground after the turn. That is, $d(1) < 0$. We have shown this:

$$d(0) = d > 0 \qquad \text{while} \qquad d(1) < 0. \tag{B.1}$$

And since this is a café patio ground in the real world (not egregiously bumpy, no steps, etc.), $d(t)$ will be a continuous function. So we have a continuous function which is positive at $t = 0$ and negative at $t = 1$, and hence by the Intermediate Value Theorem (in particular by Proposition 6.36) there must exist some $c \in (0, 1)$ such that $d(c) = 0$. And therefore, at $t = c$, your table will be perfectly stable. You're welcome.

If a table has a circular top but its legs form a square, then of course the above works the same way. What about other table shapes? If the legs form a rectangle, it works almost just the same. Unlike with square tables where after a 90° turn the legs will have collectively returned to their original location, with rectangular legs you have to turn the table 180° to get that property—and it takes up to a 180° turn to stabilize the table. The proof is a little more complicated as well, since after that 180° turn, legs 1 and 3 occupy the same locations (as a pair), rather than the locations of legs 2 and 4.

The proof in this rectangular case is by contradiction,[7] and goes something like this: If legs 1 and 3 were always hovering and legs 2 and 4 were always on the ground, then at some point the table top is closest to the ground. This is a contradiction, though, since we could then rotate the table until legs 2 and 4 are where legs 1 and 3 were—but at this point we have replaced hovering legs with legs touch the ground, which lowers the table top further, and shows that our purported "closest to the ground" location was not actually minimal. And that's a contradiction.

I have used this table-turning technique dozens of times over the years and it has always worked. That said, it is *possible* that the ground is just too uneven. Indeed, if the ground is super wavy, and at points the ground's steepness is larger than $\arctan\left(\frac{1}{\sqrt{2}}\right) \approx 35°$, then it could fail. But that is *reeeally* steep. Even the steepest streets in the world —including Baldwin Street in Dunedin, New Zealand and Filbert Street in San Francisco, United States—only slope at about 20°. In practice you will rarely, if ever, have a problem.

This of course works for things other than tables. For instance, a wobbly step ladder should never be climbed without rotating to find a stable footing.

One final application of this reasoning is finding "hugging squares." Given any shape, one can add a vertical rectangle around it.

[7] For details, check out the paper *Mathematical Table Turning Revisited* by Baritompa, Löwen, Polster, and Ross. See here: `https://arxiv.org/pdf/math/0511490.pdf`

B.11 Finding ϕ for First-Place Finishes

The recurrence $a_{n+1} = a_n + 2$, where $a_1 = 1$, grow linearly. The sequence begins $1, 3, 5, 7, \ldots$. But if the sequence is a recurrence involving a sum of the last *two* terms, it grows *much* faster. If $a_{n+2} = a_{n+1} + a_n + 2$, where $a_1 = a_2 = 1$, then the sequence begins $1, 1, 4, 7, 13, 22, 37, \ldots$. It is hard to spot at this point, but this sequence is growing faster than any polynomial grows—it is growing *exponentially*. The classic exponential function is 2^n, and if you think this function looks innocent, think again. Here's my favorite example of that:

- A standard piece of paper is about 0.05 millimeters thick. If you fold the paper in half, the thickness is now 0.1 millimeters thick, being two pieces thick. A third fold and it is 0.2 millimeters thick.
- With 13 folds you would set a new world record for number of folds.[8] If you accomplished this, the thickness would (theoretically) be about 16 inches thick (although about twice this in practice).
- If you fold it 22 times it'll be about a mile thick.
- If you fold it 43 times it would be as thick as the distance to the moon.
- If you fold it 54 times it would be as thick as the distance to the sun.
- If you fold it 103 times it would be as thick as *the width of the observable universe.*

The moral: Repeated doubling amounts to exponential growth, and exponential growth is super, super fast growth. And indeed, one can show that an innocent looking recurrence like $a_{n+2} = a_{n+1} + a_n + 2$, where $a_1 = a_2 = 1$, is growing exponentially.

Slightly more simply, the Fibonacci sequence obeys the recurrence $F_{n+2} = F_{n+1} + F_n$, where $a_1 = a_2 = 1$. It begins $1, 1, 2, 3, 5, 8, 13, 21, 34, \ldots$. This sequence presents itself in the real world. First introduced to enumerate forms of poetry, it has since found numerous applications in plant biology and elsewhere.

As claimed, it is also true that this sequence is approximated by an exponential function ϕ^n, for some number ϕ. Let's find ϕ. Notice that for exponential functions like 2^n, the base number is the ratio of consecutive terms: $\frac{2^{n+1}}{2^n} = 2$. For functions which grow like an exponential but aren't precisely of this form, the ratio of consecutive terms *approaches* some fixed number. That is,

$$\phi = \lim_{n\to\infty} \frac{F_{n+1}}{F_n}.$$

But how do we use this to find ϕ? It's really clever. First note that since clearly

$$\lim_{n\to\infty} \frac{F_{n+1}}{F_n} = \lim_{n\to\infty} \frac{F_n}{F_{n-1}}, \qquad \text{we also have} \qquad \phi = \lim_{n\to\infty} \frac{F_n}{F_{n-1}},$$

[8] You may use a very large piece of paper, but it must still be 0.05 millimeters thick

which by the limit laws implies that

$$\frac{1}{\phi} = \lim_{n\to\infty} \frac{F_{n-1}}{F_n}.$$

Why is this helpful? Because by applying the Fibonacci recurrence to F_{n+1},

$$\phi = \lim_{n\to\infty} \frac{F_{n+1}}{F_n} = \lim_{n\to\infty} \frac{F_n + F_{n-1}}{F_n} = \lim_{n\to\infty} \left(1 + \frac{F_{n-1}}{F_n}\right) = 1 + \frac{1}{\phi}.$$

That is, by using a little by of trickery with limits we were able to show that whatever ϕ is, it must satisfy the equation

$$\phi = 1 + \frac{1}{\phi}.$$

And here is one of those wonderful moments when the key to finishing off a problem relies only on the math you learned in high school. If ϕ must satisfy the above equation, then we can solve that equation to determine what ϕ can possibly be! Rewriting the above,

$$\phi^2 = \phi + 1.$$

Moving everything to the left,

$$\phi^2 - \phi - 1 = 0.$$

And how do you solve a equation like this that doesn't easily factor? The Pythagorean Theorem! The solutions are

$$\frac{-(-1) \pm \sqrt{(-1)^2 - 4(1)(-1)}}{2(1)} = \frac{1 \pm \sqrt{5}}{2}.$$

Clearly the Fibonacci sequence is increasing which means ϕ has to be at least 1, and so we must have

$$\phi = \frac{1 + \sqrt{5}}{2}.$$

This number might look familiar. It's better known as *the golden ratio.* It too has popped up in unsuspecting places, including prominently in architecture. Howeer, my favorite application of ϕ is to distance. First, note that the ratio of consecutive Fibonacci terms approach ϕ pretty quickly:

$$\frac{3}{2} = 1.5 \quad , \quad \frac{5}{3} = 1.66 \quad , \quad \frac{8}{5} = 1.6 \quad , \quad \frac{13}{8} = 1.625 \quad , \quad \frac{21}{13} = 1.615 \quad , \quad \ldots$$

while $\phi = 1.618\ldots$. And here's just a silly coincidence: 1 mile is about 1.61 kilometers. Said differently, 1 mile is roughly ϕ kilometers. So if you want to convert from miles to kilometers, you can use the Fibonacci sequence! For example, 3 miles is about 5 kilometers (as runners know), and 5 miles is about 8 kilometers, and 8 miles is about 13 kilometers, and 13 miles is about 21 kilometers, and so on. Golden.

B.12 Modern Measuring's Misfit Member

Measurability was introduced in Chapter 8 as a generalization of the *length* of an interval. In Chapter 8 we focused only one what it means for a set to have measure zero. Here's a reminder of how we defined those.

- The *length* of the interval $[a, b]$ is equal to $b-a$, and is denoted $\mathscr{L}([a, b])$. This is also the length of (a, b), $(a, b]$ and $[a, b)$. E.g., $\mathscr{L}([-2, 4]) = 6$, $\mathscr{L}([3, \infty)) = \infty$.
- A set A has *measure zero* if for all $\varepsilon > 0$ there exists a countable collection $I_1, I_2, I_3, \ldots$ of intervals such that

$$A \subseteq \bigcup_{k=1}^{\infty} I_k \qquad \text{and} \qquad \sum_{k=1}^{\infty} \mathscr{L}(I_k) < \varepsilon.$$

Intuitively, having measure zero means that you can cover the set with a countable collection of intervals whose collective length is as close to 0 as you want. Sets can also have measures larger than zero. For instance, the measure of $[1, 4]$ (which we denote $\mu([1, 4])$) is equal to 3, which is its length. Also, $\mu([2, 3] \cup [7, 9]) = 3$. More complicated sets can also have a measure equal to 3. Intuitively, this occurs when you can cover your set with a countable collection of intervals whose collective length is as close to 3 as you want, *and also* you can find a countable collection of intervals which are covered *by* the set and whose collective length is as close to 3 as you want.[9]

In the end, one obtains a non-negative function μ on sets which does indeed generalize the notion of length, and has the following four important properties.

1. It gives the correct answer for intervals: If A is an interval, then $\mu(A) = \mathscr{L}(A)$.
2. Bigger sets get bigger measures: $A \subseteq B$, then $\mu(A) \leq \mu(B)$.
3. Shifting a set does not change its measure: If $x_0 \in \mathbb{R}$, then $\mu(A + x_0) = \mu(A)$. ($A + x_0$ means that we add x_0 to everything in A; $A + x_0 = \{a + x_0 : a \in A\}$.)
4. If a set can be broken into pieces, the measure of the set equals the sum of the measures of its parts: If A and B are disjoint, then $\mu(A \cup B) = \mu(A) + \mu(B)$. And more generally, if $A_1, A_2, A_3, \ldots$ are pairwise disjoint, then

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i).$$

These are all properties that you would demand if μ is sensibly measuring the sizes of sets. One might hope that every set under the sun can be measured. Sadly, this is not possible, but the good news is that only really weird sets are non-measurable. Below we construct such a set. The argument works best if we assume for a contradiction that every set is measurable and then construct a set that can not possible satisfy the above four properties.

[9] For details, go to grad school.

The idea is to create a (countably) infinite collection of disjoint sets $E_1, E_2, E_3, \ldots$, each set having the same measure (by using the third property), and whose union has the property $[0,1] \subseteq \bigcup_{i=1}^{\infty} E_i \subseteq [-1,2]$ (which by properties 1 and 2 then imply $1 \leq m\left(\bigcup_{i=1}^{\infty} A_i\right) \leq 3$). If each $\mu(E_i) = \alpha$ (some $\alpha \geq 0$), then this will imply

$$1 \leq m\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i-1}^{\infty} \mu(A_i) = \sum_{i=1}^{\infty} \alpha \leq 3,$$

which you might already be able to see is impossible, giving the contradiction.

That's where we are headed. The main challenge is to construct these disjoint sets E_i which all have the same measure and whose union is between $[0,1]$ and $[-1,2]$.

Assume for a contradiction that every subset of $\mathbb{R}$ is measurable. Define an equivalence relation $\sim$ on $\mathbb{R}$ by $x \sim y$ if $x - y \in \mathbb{Q}$. Each equivalence class is then of the form $\{x + r : r \in \mathbb{Q}\}$, for some x. For example, the equivalence class containing π also contains $\pi - 3$ and $\pi + 24.85$, but doesn't contain $\pi - \sqrt{2}$ or $e + 7$.

Note that each equivalence class contains many numbers in $[0,1]$. For each equivalence class, pick one such number, and let E be the set of these numbers. There are of course lots of options for E. For instance, $\pi - 3$ might be in there (but if so, then $\pi - 2.874$ and $\pi - 3.1$ are not), and $\sqrt{2} - 0.8$ and $e - 2.4$ might be in there too. This all implies a few things:

- Suppose $\{r_1, r_2, r_3, \ldots\}$ is an enumeration of the rationals in $[-1,1]$, and $E_i = E + r_i$. Then,
    - $E_i \subseteq [-1,2]$ (smallest possible number is 0+(-1), biggest is 1+1);
    - $\mu(E) = \mu(E_i)$ (by Property 3);
    - $E_i \cap E_j = \emptyset$ for all $i \neq j$ (by definition of E and E_i);
    - $[0,1] \subseteq \bigcup_{i=1}^{\infty} E_i$ (any $x \in [0,1]$ is in some equivalence class, and so is some $r_i \in [-1,1]$ away from an element of E).

Summarizing,

$$[0,1] \subseteq \bigcup_{i=1}^{\infty} E_i \subseteq [-1,2],$$

which by Properties 2 and 4 mean that

$$\mu([0,1]) \leq \sum_{i=1}^{\infty} \mu(E_i) \leq \mu([-1,2]).$$

So by Property 1,

$$1 \leq \sum_{i=1}^{\infty} \mu(E) \leq 3.$$

That is, we have a sum of infinitely many copies of the same non-negative number, which can only possibly equal 0 or ∞, giving the contradiction. □

B.13 Tarski's Teriffic Talents Times Two

The great Ian Stewart imagined a *hyperdictionary*, which contains not only every word, but also contains every *possible* word. Sure, it contains "math" and "is" and "fun", but it also contains "ydac" and "faqir" and "galbhepvnx".[10] These are called *hyperwords*. The hyperdictionary contains infinitely many words, since any combination of finitely many letters counts as a hyperword. The hyperdictionary begins:

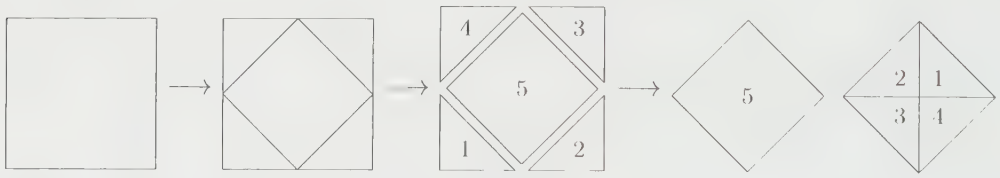
> a, aa, aaa, aaaa, aaaaa, ... <INFINITELY MANY HYPERWORDS> ..., aaab, aaaba, aaabaa, ... <INFINITELY MANY HYPERWORDS> ..., aab, aaba, ... <INFINITELY MANY HYPERWORDS> ..., ab, aba, abaa, ... <INFINITELY MANY WORDS> ..., b, ba, baa, baaa, ...

The words do form what is called a total ordering—meaning that any two of them can be compared to determine which comes first in the hyperdictionary—but things are still a little weird in that most pairs have infinitely many words between them.

The hyperdictionary has another amazing property—in some sense it contains itself. This is what I mean: Suppose the hyperdictionary were divided into 26 volumes, based on each word's first letter. If you want a copy of the hyperdictionary but don't have enough money to buy all 26 volumes, do this: Buy Volume A and a lot of white-out. Every word in Volume A starts with an 'a'—now go through this volume and white out that first 'a' from each word. After doing so what you're left with is the entire 26 volume hyperdictionary! For example, we don't need Volume K to get the word "kdalghsdh", because the word "akdalghsdh" was in Volume A, and after removing the first 'a' with white-out, you're left with "kdalghsdh"! This is an interesting phenomenon, and it's only possible with an infinite dictionary. It might remind you of Chapter 2 where, for instance, we showed that a set (like $\mathbb{N}$) can be in bijection to a proper subset of it (like $2\mathbb{N}$).

Now we move on to an amazing theorem, named after Stefan Banach and Alfred Tarski who in 1924 gave a construction that shocked the mathematical word.[11]

We begin by noting that a square can be divided up into five pieces, and those five pieces can then be rearranged to form two other squares:



These new squares are, of course, smaller—each has half the area of the original square. However, of course, you should always be careful when saying "of course" in

[10] Actually, one of those is a real word!

[11] If I could have either a construction, lemma or theorem named after me, I think my preferences would be in that order. To me, proving something is cool, but proving something that is fundamental enough and used often enough to be called a lemma is even cooler, but the best would be to construct a concrete mathematical object that shocks and awes.

mathematics. The Banach-Tarski theorem is amazing. It is incredible. It says that you can take a solid sphere, break it up into 5 pieces, rearrange those pieces, put those pieces together in a different way, and the result is two separate spheres *both of which are the same size as the original sphere.* Seriously. Throughout this book I've tried to point out moments where you should not just breeze by some fact, but appreciate it. Here, if you're not *marveling*, you're doing it wrong.

This theorem is often called the Banach-Tarski *paradox*, not because there isn't a proof, but because it's just so freaking unbelievable. It's the kind of result where, if it was announced tomorrow that the proof had a mistake that for some reason had been missed for a century, we'd all nod along and wonder how anyone could have ever believed something so obviously false. There are many amazing results when studying the infinite, but Banach-Tarski reigns king over them all.

The first thing to note is that when you break the sphere into 5 pieces, it's not simple cuts like with the square example above. The cuts are *weird.* Some of the resulting pieces are non-measurable (see PP Example B.12), which helps get around the apparent preservation-of-volume contradiction. In fact, the cuts essentially take the sphere and turn it into a *hyperdictionary*!

Doing it in detail goes beyond the scope of this appendix, but the below should at least give you a feel for the proof.

First, when dividing the sphere into 5 pieces, our pieces will have this property: If a point on the surface is in one of the pieces, then all the points between that surface point and the sphere's center are also in the same piece. Basically, this allows us to only focus on the surface of the sphere, because if we know how to divide up those points, then all the internal points are in turn divided up too.

Pick a point s on the surface of the unit sphere, which we call the *starting point.* We consider a hyperdictionary using just the letters F, B, L and R, which stand for 'forward', 'backward', 'left' and 'right'. We call, for example, a point on the sphere $FLBLB$ if we can reach that point by beginning at the starting point s (facing in some direction that we call the "forward direction"), moving one step forward, one step left, one step backward, one step left, and then one step backward (here, each step is of distance $\arccos(1/3) \approx 1.23$). In a sense, each hyperword acts as a set of directions. Consider the hyperdictionary of all such directions, which in turn correspond to a bunch of points on the surface of the sphere. (But throw out directions which include, say, FB or LR, since those cancel each other out.) Notice that unlike when one moves on a plane, when on the surface of a sphere the point FRB and the point R are not the same. Indeed, every[12] such $F/B/L/R$ sequence in this hyperdictionary corresponds to distinct point on the sphere.

Not every point can be reached in this fashion from our starting point, though. So now pick a point that wasn't hit, use that as a new starting point, and go again. To get *all* the points, we must do this with uncountably many starting points, each of which uses the hyperdictionary to produce countably many more points.

Let Collection R be the collection of all points which are reached from their starting point by a sequence of steps that ends with an R (a right step). Likewise,

[12] Well, nearly every. See below.

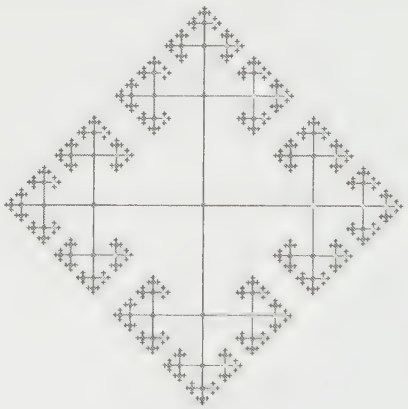
let Collection L be the collection of all points that are reached by a sequence ending in an L. And likewise for Collection F and Collection B. Finally, let Collection S be the set of all starting points. These five sets are nearly the five pieces that we cut the sphere in to. (There is one final issue, which is that there is a small collection of points which actually can be reached in multiple ways, but they are inconsequential in the end so we ignore them here.[13])

So where does the magic happen? Here's the important question: What happens that if you take Collection R, which are the points that ended with a right step, and you simply rotate this collection one step to the left? Notice how that's the same as taking the sequences in Collection R and adding an L to the end! But adding an L after an R simply cancels out that R—almost like applying white-out to the last letter!

• R		• RL		• <A starting point>
• $LFFLBRR$	$\longrightarrow$	• $LFFLBRRL$	$\longrightarrow$	• $LFFLBR$
• BBR		• $BBRL$		• BB
• $FRRFR$		• $FRRFRL$		• $FRRF$

And what's left afterwards? Simply by rotating Collection R to the left one notch we get all of Collection R back, as well as Collections S, B and F! It's the same set of points, just rotated, but somehow this rotation turned one collection into four! This is the magic that allows us to take certain sets, move them around in space, and have them form not only a copy of themselves, but also copies of some of the other sets. If you do this carefully, you can form two copies of everything, which when pushed together just right give two copies of the original sphere. Amazing.

It's kind of like how the right branch of the fractal below, if rotated 90° and scaled properly, gives back itself, as well as the top and left branches.



Following this amazing theorem, I'll end this book with my favorite math joke. Q: What's an anagram of Banach-Tarski? A: Banach-Tarski Banach-Tarski.

[13] You'll know what points I'm referring to if you've ever heard the riddle "A bear walks 1 mile north, then 1 mile east, and then 1 mile south, and ends where it started. What color is the bear?"

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Jay Cummings is a professor in the Department of Mathematics and Statistics at California State University, Sacramento. He received his Ph.D. from UC San Diego under Ron Graham. Most of his published work is in various areas of combinatorics. He believes that learning math has become far too expensive, and is striving to write textbooks which are enjoyable to read, highlight the beauty in mathematics, and are significantly more affordable than the others on the market.

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